Lecture Notes on

Differential Equations for Engineers

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Introduction to differential equations

PART

I

Section 1

Introduction

Mathematical methods of physical or "real life" problems are necessary tools in order to understand and study such problems. We must formulate the problem in mathematical terms; that is, we must construct a mathematical model for the problem. Many physical problems concern relationships between changing quantities. Since rates of change are represented mathematically by derivatives, mathematical models often involve equations relating an unknown function and one or more of its derivatives. Such equations are *differential equations*.

The title of this course suggests two words, **Differential** and **Equation**, and before we can introduce the topic of Differential Equations, I would like to define each of these words:

Definition 1 **Equation:** An equation is a mathematical expression that contains an equals symbol.

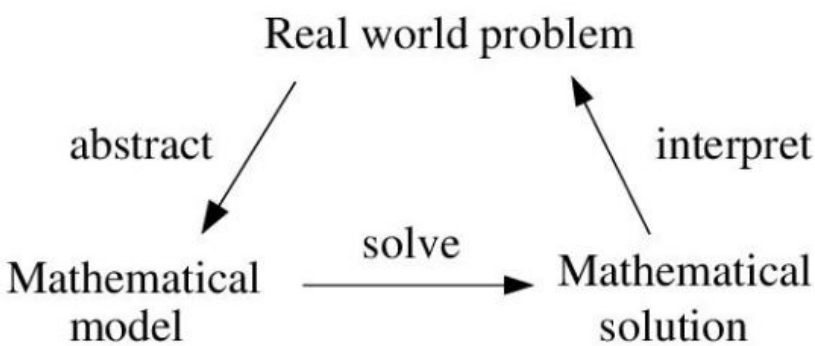
Definition 2 **Differentiation:** In mathematics, differential calculus is a subfield of calculus that studies the rates at which quantities change. It is one of the two traditional divisions of calculus, the other being integral calculus.

Therefore, we can define a differential equation as follows:

Definition 3 **Differential equation:** is any equation (mathematical expression that contains an equal symbol) that contains derivative terms on one or both sides.

Remark Differential equations, normally, represent the dynamics of a given system, where it connects different levels of *behaviors* of a given system into one equation. Most engineering and physical systems can be described based on such a relation. The art and practice of differential equations involves the following sequence of steps: one "models" a system (physical, chemical, biological, economic, or even mathematical) by means of a differential equation; one then attempts to gain information about solutions of this equation; and one then translates this mathematical information back into the scientific context.

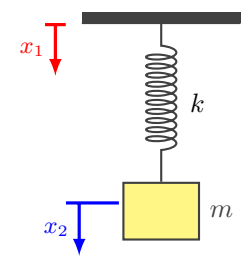
This process can be illustrated as follows:



Here, we will present some examples from different domains to demonstrate the importance of differential equations before we learn how to solve them.

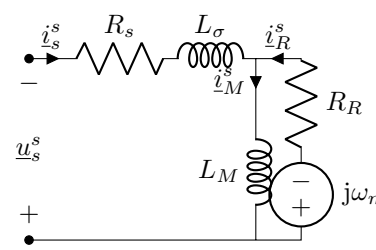
Mechanical Systems: For instance, the spring mass damper system shown below, which commonly used in mechanical systems, is fully described by the following *second order* differential equation

$$F(t) = k\,x(t) + m\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + b\frac{\mathrm{d}x}{\mathrm{d}t}$$



Electrical Systems: Another example where the differential equations are very important is the electrical circuit design. The differential equation in the electrical circuits are used to model the flow of the currant and the difference in the Voltage in the circuit. The most governing equation in electrical circuit design, which is Kirchhoff's voltage law is simply a differential equation.

$$i(t)R + L\frac{di(t)}{dt} - E(t) = 0$$



Biological Systems One more example for a very simple first order differential equation

$$\dot{y} = k\,y,$$

which could model population growth in some biological system, like the growth of yeast cells. From calculus we know that the family of functions

$$y = ce^{k\,t}$$

is the general solution for the differential equation.

Spread of Epidemics: One model for the spread of epidemics assumes that the number of people infected changes at a rate proportional to the product of the number of people already infected and the number of people who are susceptible, but not yet infected. Therefore, if S denotes the total population of susceptible people and $I = I(t)$

denotes the number of infected people at time t, then $S - I$ is the number of people who are susceptible, but not yet infected. Thus,

$$I' = rI(S - I),$$

where r is a positive constant. Assuming that $I(0) = I_0$, the solution of this equation is

$$I = \frac{SI_0}{I_0 + (S - I_0)e^{-rSt}}$$

Since $\lim_{t\to\infty} I(t) = S$, this model predicts that all the susceptible people eventually become infected.

Newton's Second Law of Motion:
According to *Newton's second law of motion*, the instantaneous acceleration a of an object with constant mass m is related to the force F acting on the object by the equation $F = ma$. For simplicity, let's assume that $m = 1$ and the motion of the object is along a vertical line. Let y be the displacement of the object from some reference point on Earth's surface, measured positive upward. In many applications, there are three kinds of forces that may act on the object:

- A force such as gravity that depends only on the position y, which we write as $-p(y)$, where $p(y) > 0$ if $y \geq 0$.
- A force such as atmospheric resistance that depends on the position and velocity of the object, which we write as $-q(y, y')y'$, where q is a nonnegative function and we've put y' "outside" to indicate that the resistive force is always in the direction opposite to the velocity.
- A force $f = f(t)$, exerted from an external source (such as a towline from a helicopter) that depends only on t.

In this case, Newton's second law implies that

$$y'' = -q(y, y')y' - p(y) + f(t),$$

which is usually rewritten as

$$y'' + q(y, y')y' + p(y) = f(t).$$

Since the second (and no higher) order derivative of y occurs in this equation, we say that it is a *second order differential equation*.

Section 2

Solution of DEs

The main aim of this set of lectures is to describe techniques that can solve a small variety of different types of differential equations.

Definition 4 A *solution* of a differential equation is a function that satisfies the differential equation on some open interval; thus, y is a solution if y is n times differentiable and

$$y^{(n)}(x) = f(x, y(x), y'(x), \ldots, y^{(n-1)}(x))$$

for all x in some open interval (a, b). In this case, we also say that y *is a solution of a DE on* (a, b).

Functions that satisfy a differential equation at isolated points are not interesting. For example, $y = x^2$ satisfies

$$xy' + x^2 = 3x$$

if and only if $x = 0$ or $x = 1$, but it's not a solution of this differential equation because it does not satisfy the equation on an open interval.

Remark In practice, some DEs can be solved analytically, and some can't; when we can't solve them analytically, we can sometimes find a numerical solution that gives us an answer that we can use. But the first step is always to try and find an exact, analytic solution, and it is these techniques which we will demonstrate in these notes.

Here are some examples to illustrate the idea of solutions of given DE

Example 1 Verify that

$$y = \frac{x^2}{3} + \frac{1}{x} \tag{2.1}$$

is a solution of

$$xy' + y = x^2 \tag{2.2}$$

on $(0, \infty)$ and on $(-\infty, 0)$.

Solution Substituting (2.1) and

$$y' = \frac{2x}{3} - \frac{1}{x^2}$$

into (2.2) yields

$$xy'(x) + y(x) = x\left(\frac{2x}{3} - \frac{1}{x^2}\right) + \left(\frac{x^2}{3} + \frac{1}{x}\right) = x^2$$

for all $x \neq 0$. Therefore y is a solution of (2.2) on $(-\infty, 0)$ and $(0, \infty)$. However, y isn't a solution of the differential equation on any open interval that contains $x = 0$, since y is not defined at $x = 0$.

Example 2 Show that if c_1 and c_2 are constants then

$$y = (c_1 + c_2 x)e^{-x} + 2x - 4 \tag{2.3}$$

is a solution of

$$y'' + 2y' + y = 2x \tag{2.4}$$

on $(-\infty, \infty)$.

Solution Differentiating (2.3) twice yields

$$y' = -(c_1 + c_2 x)e^{-x} + c_2 e^{-x} + 2$$

and

$$y'' = (c_1 + c_2 x)e^{-x} - 2c_2 e^{-x},$$

so

$$\begin{aligned} y'' + 2y' + y &= (c_1 + c_2x)e^{-x} - 2c_2e^{-x} \\ &\quad +2\left[-(c_1 + c_2x)e^{-x} + c_2e^{-x} + 2\right] \\ &\quad +(c_1 + c_2x)e^{-x} + 2x - 4 \\ &= (1 - 2 + 1)(c_1 + c_2x)e^{-x} + (-2 + 2)c_2e^{-x} \\ &\quad +4 + 2x - 4 = 2x \end{aligned}$$

for all values of x. Therefore y is a solution of (2.4) on $(-\infty, \infty)$.

Example 3 Find all solutions of

$$y^{(n)} = e^{2x}. \tag{2.5}$$

Solution Integrating (2.5) yields

$$y^{(n-1)} = \frac{e^{2x}}{2} + k_1,$$

where k_1 is a constant. If $n \geq 2$, integrating again yields

$$y^{(n-2)} = \frac{e^{2x}}{4} + k_1x + k_2.$$

If $n \geq 3$, repeatedly integrating yields

$$y = \frac{e^{2x}}{2^n} + k_1\frac{x^{n-1}}{(n-1)!} + k_2\frac{x^{n-2}}{(n-2)!} + \cdots + k_n, \tag{2.6}$$

where $k_1, k_2, \ldots, k_n$ are constants. This shows that every solution of (2.5) has the form (2.6) for some choice of the constants $k_1, k_2, \ldots, k_n$. On the other hand, differentiating (2.6) n times shows that if $k_1, k_2, \ldots, k_n$ are arbitrary constants, then the function y in (2.6) satisfies (2.5).
Since the constants $k_1, k_2, \ldots, k_n$ in (2.6) are arbitrary, so are the constants

$$\frac{k_1}{(n-1)!}, \frac{k_2}{(n-2)!}, \cdots, k_n.$$

Therefore Example 3 actually shows that all solutions of (2.5) can be written as

$$y = \frac{e^{2x}}{2^n} + c_1 + c_2x + \cdots + c_nx^{n-1},$$

where we renamed the arbitrary constants in (2.6) to obtain a simpler formula. As a general rule, arbitrary constants appearing in solutions of differential equations should be simplified if possible. You'll see examples of this throughout the text.

Section 3

Initial value problem

An initial value problem (abbreviated as IVP) is a differential equation combined with one or more initial conditions. In Example 3 we saw that the differential equation $y^{(n)} = e^{2x}$ has an infinite family of solutions that depend upon the n arbitrary constants $c_1, c_2, \ldots, c_n$. In the absence of additional conditions, there's no reason to prefer one solution of a

differential equation over another. However, we'll often be interested in finding a solution of a differential equation that satisfies one or more specific conditions.

For example,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y, \qquad y(0) = 1$$

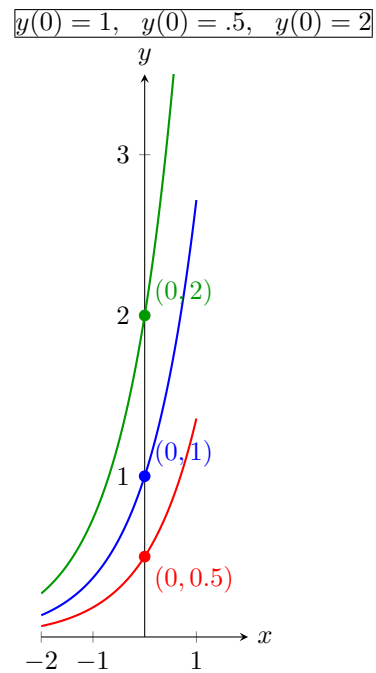
is an initial value problem. The differential equation is solved by $y = Ce^x$, for any value of C. However, the only way to satisfy

$$y(0) = 1$$

with the solution $y = Ce^x$ is to set $C = 1$. That is, the solution to the initial value problem is

$$y = e^x,$$

since it satisfies both the differential equation and the initial conditions.



Initial value problems can also be posed for higher order differential equations.

For example,

$$y'' - 2y' + 3y = e^x, \quad y(0) = 1, \quad y'(0) = 2 \tag{3.1}$$

is an initial value problem for a second order differential equation where y and y' are required to have specified values at $x = 0$. In general, an initial value problem for an n-th order differential equation requires y and its first $n-1$ derivatives to have specified values at some point x_0. These requirements are the *initial conditions*.

Section 4

Existence and Uniqueness of Solution

Although there are methods for solving some nonlinear equations, it's impossible to find useful formulas for the solutions of most. Whether we're looking for exact solutions or numerical approximations, it's useful to know conditions that imply the existence and uniqueness of solutions of initial value problems for nonlinear equations.

Remark Two fundamental questions arise in considering an initial-value problem:

- Does a solution of the problem exist?
- If a solution exists, is it unique?

In this section we state such a condition and illustrate it with examples.

Remark Some terminology: an *open rectangle R* is a set of points (x, y) such that

$$a < x < b \quad \text{and} \quad c < y < d$$

. We'll denote this set by $R : \{a < x < b, c < y < d\}$. "Open" means that the boundary rectangle (indicated by the dashed lines in isn't included in R .

The next theorem gives sufficient conditions for existence and uniqueness of solutions

of initial value problems for first order nonlinear differential equations. We omit the proof, which is beyond the scope of this course.

Theorem 1

- If f is continuous on an open rectangle

$$R : \{a < x < b, c < y < d\}$$

that contains (x_0, y_0) then the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \tag{4.1}$$

has at least one solution on some open subinterval of (a, b) that contains x_0.

- If both f and f_y are continuous on R then (4.1) has a unique solution on some open subinterval of (a, b) that contains x_0.

Remark

It's important to understand exactly what Theorem 1 says.

- part(a): is an *existence theorem*. It guarantees that a solution exists on some open interval that contains x_0, but provides no information on how to find the solution, or to determine the open interval on which it exists. Moreover, part (a) provides no information on the number of solutions that (4.1) may have. It leaves open the possibility that (4.1) may have two or more solutions that differ for values of x arbitrarily close to x_0. We will see in Example 6 that this can happen.

- part(b): is a *uniqueness theorem*. It guarantees that (4.1) has a unique solution on some open interval (a,b) that contains x_0. However, if $(a, b) \neq (-\infty, \infty)$, (4.1) may have more than one solution on a larger interval that contains (a, b). For example, it may happen that $b < \infty$ and all solutions have the same values on (a, b), but two solutions y_1 and y_2 are defined on some interval (a, b_1) with $b_1 > b$, and have different values for $b < x < b_1$; thus, the graphs of the y_1 and y_2 "branch off" in different directions at $x = b$. (See Example 7). In this case, continuity implies that $y_1(b) = y_2(b)$ (call their common value $\overline{y}$), and y_1 and y_2 are both solutions of the initial value problem

$$y' = f(x, y), \quad y(b) = \overline{y} \tag{4.2}$$

that differ on every open interval that contains b. Therefore f or f_y must have a discontinuity at some point in each open rectangle that contains $(b, \overline{y})$, since if this were not so, (4.2) would have a unique solution on some open interval that contains b. We leave it to you to give a similar analysis of the case where $a > -\infty$.

Consider the initial value problem

$$y' = \frac{x^2 - y^2}{1 + x^2 + y^2}, \quad y(x_0) = y_0. \tag{4.3}$$

Since

$$f(x, y) = \frac{x^2 - y^2}{1 + x^2 + y^2} \quad \text{and} \quad f_y(x, y) = -\frac{2y(1 + 2x^2)}{(1 + x^2 + y^2)^2}$$

are continuous for all (x, y), Theorem 1 implies that if (x_0, y_0) is arbitrary, then (4.3) has a unique solution on some open interval that contains x_0.

Consider the initial value problem,

$$y' = \frac{x^2 - y^2}{x^2 + y^2}, \quad y(x_0) = y_0. \tag{4.4}$$

Here

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2} \quad \text{and} \quad f_y(x, y) = -\frac{4x^2 y}{(x^2 + y^2)^2}$$

are continuous everywhere except at $(0, 0)$. If $(x_0, y_0) \neq (0, 0)$, there's an open rectangle R that contains (x_0, y_0) that does not contain $(0, 0)$. Since f and f_y are continuous on R, Theorem 1 implies that if $(x_0, y_0) \neq (0, 0)$ then (4.4) has a unique solution on some open interval that contains x_0.

Consider the initial value problem

$$y' = \frac{x + y}{x - y}, \quad y(x_0) = y_0. \tag{4.5}$$

Here

$$f(x, y) = \frac{x + y}{x - y} \quad \text{and} \quad f_y(x, y) = \frac{2x}{(x - y)^2}$$

are continuous everywhere except on the line $y = x$. If $y_0 \neq x_0$, there's an open rectangle R that contains (x_0, y_0) that does not intersect the line $y = x$. Since f and f_y are continuous on R, Theorem 1 implies that if $y_0 \neq x_0$, (4.5) has a unique solution on some open interval that contains x_0.

Example 4 Assume that we can show that the solutions of

$$y' = 2xy^2 \tag{4.6}$$

are

$$y \equiv 0 \quad \text{and} \quad y = -\frac{1}{x^2 + c},$$

where c is an arbitrary constant. In particular, this implies that no solution of (4.6) other than $y \equiv 0$ can equal zero for any value of x. Show that Theorem 1 part(b) implies this.

Solution We'll obtain a contradiction by assuming that (4.6) has a solution y_1 that equals zero for some value of x, but isn't identically zero. If y_1 has this property, there's a point x_0 such that $y_1(x_0) = 0$, but $y_1(x) \neq 0$ for some value of x in every open interval that contains x_0. This means that the initial value problem

$$y' = 2xy^2, \quad y(x_0) = 0 \tag{4.7}$$

has two solutions $y \equiv 0$ and $y = y_1$ that differ for some value of x on every open interval that contains x_0. This contradicts Theorem 1(b), since in (4.6) the functions

$$f(x, y) = 2xy^2 \quad \text{and} \quad f_y(x, y) = 4xy.$$

are both continuous for all (x, y), which implies that (4.7) has a unique solution on some open interval that contains x_0.

Example 5 Consider the initial value problem

$$y' = \frac{10}{3}xy^{2/5}, \quad y(x_0) = y_0. \tag{4.8}$$

- For what points (x_0, y_0) does Theorem 1 part(a) imply that (4.8) has a solution?
- For what points (x_0, y_0) does Theorem 1 part (b) imply that (4.8) has a unique solution on some open interval that contains x_0?

Solution Part a Since

$$f(x, y) = \frac{10}{3}xy^{2/5}$$

is continuous for all (x, y), Theorem 1 implies that (4.8) has a solution for every (x_0, y_0).
Part b Here

$$f_y(x, y) = \frac{4}{3}xy^{-3/5}$$

is continuous for all (x, y) with $y \neq 0$. Therefore, if $y_0 \neq 0$ there's an open rectangle on which both f and f_y are continuous, and Theorem 1 implies that (4.8) has a unique solution on some open interval that contains x_0.
If $y = 0$ then $f_y(x, y)$ is undefined, and therefore discontinuous; hence, Theorem 1 does not apply to (4.8) if $y_0 = 0$.

Example 6 Example 5 leaves open the possibility that the initial value problem

$$y' = \frac{10}{3}xy^{2/5}, \quad y(0) = 0 \tag{4.9}$$

has more than one solution on every open interval that contains $x_0 = 0$. Show that this is true.

Solution By inspection, $y \equiv 0$ is a solution of the differential equation

$$y' = \frac{10}{3}xy^{2/5}. \tag{4.10}$$

Since $y \equiv 0$ satisfies the initial condition $y(0) = 0$, it's a solution of (4.9).
Now suppose y is a solution of (4.10) that isn't identically zero. Separating variables in (4.10) yields

$$y^{-2/5}y' = \frac{10}{3}x$$

on any open interval where y has no zeros. Integrating this and rewriting the arbitrary constant as $5c/3$ yields

$$\frac{5}{3}y^{3/5} = \frac{5}{3}(x^2 + c).$$

Therefore

$$y = (x^2 + c)^{5/3}. \tag{4.11}$$

Since we divided by y to separate variables in (4.10), our derivation of (4.11) is legitimate only on open intervals where y has no zeros. However, (4.11) actually defines y

for all x, and differentiating (4.11) shows that

$$y' = \frac{10}{3}x(x^2+c)^{2/3} = \frac{10}{3}xy^{2/5},\ -\infty < x < \infty.$$

Therefore (4.11) satisfies (4.10) on $(-\infty,\infty)$ even if $c \le 0$, so that $y(\sqrt{|c|}) = y(-\sqrt{|c|}) = 0$. In particular, taking $c = 0$ in (4.11) yields

$$y = x^{10/3}$$

as a second solution of (4.9). Both solutions are defined on $(-\infty,\infty)$, and they differ on every open interval that contains $x_0 = 0$ In fact, there are *four* distinct solutions of (4.9) defined on $(-\infty,\infty)$ that differ from each other on every open interval that contains $x_0 = 0$. Can you identify the other two?

Example 7 From Example 5, the initial value problem

$$y' = \frac{10}{3}xy^{2/5}, \quad y(0) = -1 \tag{4.12}$$

has a unique solution on some open interval that contains $x_0 = 0$. Find a solution and determine the largest open interval (a,b) on which it's unique.

Solution Let y be any solution of (4.12). Because of the initial condition $y(0) = -1$ and the continuity of y, there's an open interval I that contains $x_0 = 0$ on which y has no zeros, and is consequently of the form (4.11). Setting $x = 0$ and $y = -1$ in (4.11) yields $c = -1$, so

$$y = (x^2-1)^{5/3} \tag{4.13}$$

for x in I. Therefore every solution of (4.12) differs from zero and is given by (4.13) on $(-1,1)$; that is, (4.13) is the unique solution of (4.12) on $(-1,1)$. This is the largest open interval on which (4.12) has a unique solution.

Example 8 From Example 5, the initial value problem

$$y' = \frac{10}{3}xy^{2/5}, \quad y(0) = 1 \tag{4.14}$$

has a unique solution on some open interval that contains $x_0 = 0$. Find the solution and determine the largest open interval on which it's unique.

Solution Let y be any solution of (4.14). Because of the initial condition $y(0) = 1$ and the continuity of y, there's an open interval I that contains $x_0 = 0$ on which y has no zeros, and is consequently of the form (4.11). Setting $x = 0$ and $y = 1$ in (4.11) yields $c = 1$, so

$$y = (x^2+1)^{5/3} \tag{4.15}$$

for x in I. Therefore every solution of (4.14) differs from zero and is given by (4.15) on $(-\infty,\infty)$; that is, (4.15) is the unique solution of (4.14) on $(-\infty,\infty)$.

Section 5

Classification of differential equations

Differential equations can be classified into two main categories,

- An *ordinary differential equation* (ODE) involves derivatives of a function of *only one* variable.
$$F(t) = k\,x(t) + m\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + b\frac{\mathrm{d}x}{\mathrm{d}t}$$
- A *partial differential equation* (PDE) involves *partial derivatives* of a *multi-variable* function.
$$\frac{\partial Q}{\partial t} = \frac{\partial s}{\partial t}$$

We will be focusing on the ODEs for the majority of this course. Later, we will touch off the PDEs and how they are solved.

One important aspect that is normally used to classify the DEs is the *order* of the DE, which is defined as the highest n such that the nth derivative of the function appears.

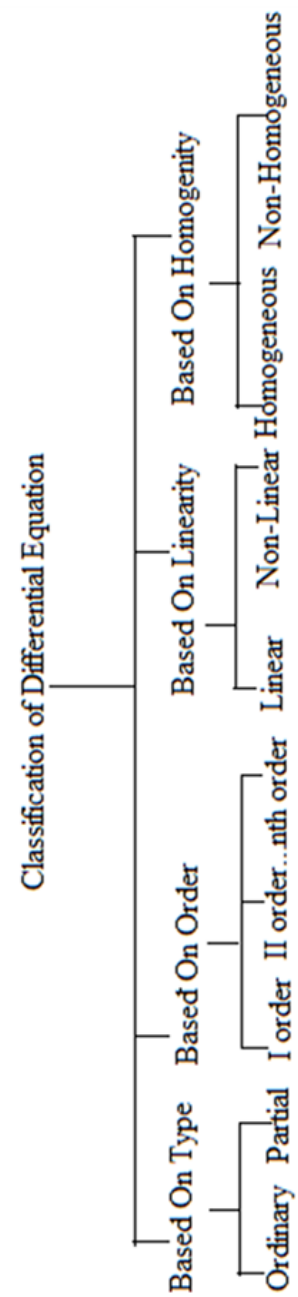
Notation for higher derivatives of a function $y(t)$:

$$\begin{array}{rlll}
\text{first derivative:} & \dot{y} & y' & \dfrac{dy}{dt} \\
\text{second derivative:} & \ddot{y} & y'' & \dfrac{d^2y}{dt^2} \\
\text{third derivative:} & y^{(3)} & & \dfrac{d^3y}{dt^3} \\
 & \vdots & & \\
n^{th}\text{derivative:} & y^{(n)} & & \dfrac{d^ny}{dt^n}
\end{array}$$

Here are some examples:

$$\begin{aligned}
\frac{dy}{dx} - x^2 &= 0 && \text{(first order)},\\
\frac{dy}{dx} + 2xy^2 &= -2 && \text{(first order)},\\
\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y &= 2x && \text{(second order)},\\
xy''' + y^2 &= \sin x && \text{(third order)},\\
y^{(n)} + xy' + 3y &= x && (n\text{-th order}).
\end{aligned}$$

The ***order*** of a differential equation will be our guide to classify the *ordinary differential equation*. In particular, we will study First Order Differential Equations and Second Order Differential Equations in detail in the next sections.



Section 6

Exercises

First order differential equations

PART

II

Section 7

First Order Differential Equations

A first-order differential equation can be defined as an equation:

$$\frac{dy}{dx} = f(x, y), \tag{7.1}$$

of two variables x and y with its function $f(x, y)$ defined on a region in the XY-plane.

Remark

Understanding first-order differential equations is crucial because:

1. They model many real-world phenomena, such as:
   - Population growth (e.g., logistic equation)
   - Radioactive decay
   - Newton's law of cooling
   - Simple electrical circuits
2. They form the basis for understanding higher-order differential equations and systems of differential equations.
3. Many numerical methods for solving differential equations start with techniques for first-order equations.

The first-order differential equation can be classified into several categories based on the way we solve them. For instance,

- First-order *Separable* differential equations
- First-order *Linear* differential equations
- First-order *Bernoulli* differential equations
- First-order *Exact* differential equations
- First-order *Non-Exact* differential equations

In the following subsections, we are going to study in detail some of these categories.

Subsection 7.1

Separable Differential Equations

From the name, **separable equations are differential equations** are these differential equations which can be written as follows:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = h(y) \cdot g(x), \tag{7.2}$$

i.e., the two variables can be separated to a product of two functions.

To **solve** this type of differential equation, from equation (7.2), one can **split** all $y's$ and its derivative on one side and all $x's$ and its derivative on the other side. For instance; (7.2) becomes;

$$\int \frac{1}{h(y)}\, dy = \int g(x)\, dx,$$

and then all we need to do is find the integral and solve for y.

To illustrate this, let's do some examples:

Example 9 Solve the following DE

$$y\frac{\mathrm{d}y}{\mathrm{d}x} = \sin(x) \tag{7.3}$$

with initial condition $y(0) = 1$.

Solution Separating and integrating,

$$\int y\, dy = \int \sin(x)\, dx,$$

yields

$$\frac{y^2}{2} = -\cos(x) + C$$

where C is a constant that will be determined by the initial conditions.
We can use the initial condition $y(0) = 1$ to eliminate one. Since $y = 1$ and $x = 0$, it follows that

$$\begin{aligned} \frac{1}{2} &= C - \cos(0), \\ 2C - 2 &= 1 \\ C &= 3/2 \end{aligned}$$

Example 10 Solve the following DE.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = xy, \quad y(0) = 2$$

Solution First, we separate the variables,

$$\frac{\mathrm{d}y}{y} = x\mathrm{d}x$$

Now, lets integrate both sides,

$$\int \frac{\mathrm{d}y}{y} = \int x \, \mathrm{d}x$$
$$\ln y = \frac{x^2}{2} + C$$

Or the general solution $y(x) = Ce^{\frac{1}{2}x^2}$; initial conditions imply $C = 2$.

.

Example 11 Solve the following differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y \; e^{2x}$$

Solution First, we separate the variables,

$$\frac{\mathrm{d}y}{y} = e^{2x}\mathrm{d}x$$

Now, lets integrate both sides,

$$\int \frac{\mathrm{d}y}{y} = \int e^{2x}\mathrm{d}x$$
$$\ln y = \frac{1}{2}\, e^{2x} + c$$

Some examples will need a bit of effort to separate them. For instance, consider the following example

Example 12 Solve the following differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y \; \cos(x) + \; \cos(x) + y + 1$$

Solution To separate this variables, lets do the following;

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(x)(y+1) + y + 1$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = (y+1)(\cos(x)+1)$$
$$\frac{\mathrm{d}y}{y+1} = (\cos(x)+1)\mathrm{d}x$$

Now, lets integrate both sides,

$$\int \frac{\mathrm{d}y}{y+1} = \int (\cos(x)+1) \; \mathrm{d}x$$
$$\ln|y+1| = \sin(x) + x + c$$

Example 13 Solve the following differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y - 5 = y \; \tan^2(2x) + \; 5\tan^2(2x)$$

Solution To separate this variables, lets do the following;

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \tan^2(2x)(y+5)+y+5\\ \frac{\mathrm{d}y}{\mathrm{d}x} &= (y+5)(\tan^2(2x)+1)\\ \frac{\mathrm{d}y}{y+1} &= (\tan^2(2x)+1)\mathrm{d}x\end{aligned}$$

Now, lets integrate both sides,

$$\begin{aligned}\int \frac{\mathrm{d}y}{y+1} &= \int (\tan^2(2x)+1)\ \mathrm{d}x\\ \int \frac{\mathrm{d}y}{y+1} &= \int (\sec^2(2x))\ \mathrm{d}x\\ \ln|y+1| &= \frac{1}{2}\tan(2x)+c\end{aligned}$$

Subsection 7.2

Exercises

Solve the following DEs:

1. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 5x^2y$
2. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = (x+2)(2y-3)$
3. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2\,y + xy$
4. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^{(x+y)}$
5. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + x\,e^y = 0$
6. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x\sqrt{y}$
7. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x}{y}, \quad y(0)=0$
8. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x\,\sin x}{y}, \quad y(0)=-1$
9. $(e^y-1)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2+\cos x$
10. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\ln x}{xy}$
11. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2xy = x^2y$
12. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y\sin(x)}{x\cos(y)}$
13. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2y^2+y}{xy+1}$
14. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{xe^y}{1+y^2}$
15. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^3y}{\sqrt{1-y^2}}$
16. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x\cos(y)+y\sin(x)}{y\cos(x)-x\sin(y)}$
17. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{xe^{-y}}{y-\ln(x)}$

Subsection 7.3

Linear First Order Differential Equations

As you may have noticed, not every differential equation will be separable. There are other forms that can help us to solve these non-separable differential equations. One of these forms is called *linear differential equation*. The **linearly** here refers to the independent variable, y, i.e., there is no nonlinear terms of any kind associated with the variable y.

Remark

Linear equations have the following form,

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x). \tag{7.4}$$

Note that in this form, the coefficient of $\frac{\mathrm{d}y}{\mathrm{d}x}$ is always 1. Also, there is one y in the equation, which is the one multiplied by the function $P(x)$.
To solve these equations, we first define the *integrating factor* $I(x)$ as follows;

$$I = e^{\int P(x)\,dx}$$

Then, the solution goes as follows;
First, we multiply both sides by the integration factor I,

$$\begin{aligned} I\,\frac{\mathrm{d}y}{\mathrm{d}x} + I\;P(x)\,y &= I\;Q(x) \\ \frac{\mathrm{d}}{\mathrm{d}x}(I\,y) &= I\;Q(x) \\ \mathrm{d}\,(I\,y) &= I\;Q(x)\;\mathrm{d}x \end{aligned}$$

Now, integrate both sides, we get

$$\begin{aligned} \int \mathrm{d}\,(I\,y) &= \int I\;Q(x)\;\mathrm{d}x\;+C \\ I\,y &= \int I\;Q(x)\;\mathrm{d}x\;+C \\ y &= \frac{1}{I}\left(\int I\;Q(x)\;\mathrm{d}x\;+C\right) \end{aligned}$$

You can always do the these steps while solving linear first order DE, or use the final form, which is

$$y = \frac{1}{I}\left(\int I\;Q(x)\;\mathrm{d}x\;+C\right),$$

where $I = e^{\int P(x)\,dx}$.

Let us do some examples to illustrate how we can solve the first order linear DE:

Example 14

Solve the following differential equation the equation

$$x\,\frac{\mathrm{d}y}{\mathrm{d}x} + y = 2\,x\,e^{x}.$$

Solution

First step: we rewrite the equation in the linear form as follows

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = 2\,e^{x}. \tag{7.5}$$

Second step: we identify $P(x)$ and $Q(x)$;

- $P(x) = \frac{1}{x}$
- $Q(x) = 2\,e^x$

Third step: compute the integrating factor

$$I = e^{\int \frac{1}{x}\,\mathrm{d}x} = e^{\ln x} = x.$$

Then, multiply the integration factor into the equation (7.12), we get

$$\begin{aligned} x\,\frac{\mathrm{d}y}{\mathrm{d}x} + y &= 2\,x\,e^x \\ \frac{\mathrm{d}(xy)}{\mathrm{d}x} &= 2\,x\,e^x \\ \mathrm{d}(xy) &= 2\,x\,e^x\,\mathrm{d}x. \end{aligned}$$

integrating this gives

$$xy = \int 2xe^x\,dx + C \tag{7.6}$$

$$y = \frac{1}{x}\left(\int 2xe^x\,dx\right). \tag{7.7}$$

To solve this, we use integration by parts. That is, $\int u\,dv = uv - \int v\,du$. We choose $u = x$ and $dv = e^x\,dx$, so $du = dx$, and $v = e^x$. Thus,

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x.$$

This implies that

$$y = 2e^x(1 - \frac{1}{x}) + \frac{C}{x}.$$

Example 15 Solve the following differential equation:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = x^2.$$

Solution First step: we rewrite the equation in the linear form as follows

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = x^2. \tag{7.8}$$

Second step: we identify $P(x)$ and $Q(x)$;

- $P(x) = \frac{2}{x}$
- $Q(x) = x^2$

Third step: compute the integrating factor

$$I = e^{\int \frac{2}{x},\mathrm{d}x} = e^{2\ln x} = x^2.$$

Then, multiply the integrating factor into the equation (7.8), we get

$$\begin{aligned} x^2, \frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= x^4 \\ \frac{\mathrm{d}\left(x^2 y\right)}{\mathrm{d}x} &= x^4 \\ \mathrm{d}\left(x^2 y\right) &= x^4 \mathrm{d}x. \end{aligned}$$

Integrating this gives

$$x^2 y = \int x^4, dx + C \; y = \frac{1}{x^2}\left(\frac{x^5}{5} + C\right). \tag{7.9}$$

Finally, simplifying the expression, we obtain:

$$y = \frac{x^3}{5} + \frac{C}{x^2}.$$

Example 16 Solve the following differential equation:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y\sin(x) = \cos(x).$$

Solution First step: we rewrite the equation in the linear form as follows

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y\sin(x) = \cos(x). \tag{7.10}$$

Second step: we identify $P(x)$ and $Q(x)$;

- $P(x) = \sin(x)$
- $Q(x) = \cos(x)$

Third step: compute the integrating factor

$$I = e^{\int \sin(x), \mathrm{d}x} = e^{-\cos(x)}.$$

Then, multiply the integrating factor into the equation (7.10), we get

$$\begin{aligned} e^{-\cos(x)}\frac{\mathrm{d}y}{\mathrm{d}x} + ye^{-\cos(x)}\sin(x) &= e^{-\cos(x)}\cos(x) \\ \frac{\mathrm{d}\left(ye^{-\cos(x)}\right)}{\mathrm{d}x} &= e^{-\cos(x)}\cos(x) \\ \mathrm{d}\left(ye^{-\cos(x)}\right) &= e^{-\cos(x)}\cos(x), \mathrm{d}x. \end{aligned}$$

Good luck with the last integration! ☺ ☺ ☺

Integrating this gives

$$ye^{-\cos(x)} = \int e^{-\cos(x)}\cos(x), dx + C. \tag{7.11}$$

Example 17 Solve the following differential equation the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2}y = x^3$$

Solution First step: we make sure that the equation is in the linear form,

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2}y = x^2. \tag{7.12}$$

Second step: we identify $P(x)$ and $Q(x)$;

- $P(x) = \frac{1}{2}$
- $Q(x) = x^2$

Third step: compute the integrating factor

$$I = e^{\int \frac{1}{2}\,\mathrm{d}x} = e^{\frac{1}{2}\int \mathrm{d}x} = e^{\frac{1}{2}x}.$$

Fourth step: substitute into the form

$$y = \frac{1}{I}\left(\int I\ Q(x)\ \mathrm{d}x\ + C\right),$$

which implies,

$$y = \frac{1}{e^{\frac{1}{2}x}}\left(\int x^2 e^{\frac{1}{2}x}\,dx\right). \tag{7.13}$$

To solve this, we use integration by parts. That is, $\int u\,dv = uv - \int v\,\mathrm{d}u$. We choose $u = x^2$ and $\mathrm{d}v = e^{\frac{1}{2}x}\,\mathrm{d}x$, so $du = 2\,x\,\mathrm{d}x$, and $v = 2\,e^{\frac{1}{2}x}$. Thus,

$$\int x^2 e^{\frac{1}{2}x}\,\mathrm{d}x = 2\,x^2 e^{\frac{1}{2}x} - \int 4\,x\,e^{\frac{1}{2}x}\,dx.$$

The last integral is another integration by parts (Home work).

Subsection 7.4

Exercises

Solve the following differential equations:

1. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{1}{x}\,y = 1$
2. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = y - x$
3. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x + 5y$
4. $x\dfrac{\mathrm{d}y}{\mathrm{d}x} - 2y = x^2$
5. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{1}{x}\,y = x^2$
6. $x\dfrac{\mathrm{d}y}{\mathrm{d}x} - 3y = x^4 e^x$
7. $2x\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = 2\sqrt{x}$
8. $x^2\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2xy = \ln x,\quad$ y(1)=2

Subsection 7.5

Bernoulli Equations

Bernoulli Equation is a different form of differential equations that can be easily converted or *transformed* to a linear first order differential equation. The general form of the Bernoulli equation is as follows;

$$y' + P(x)y = Q(x)y^n \tag{7.14}$$

Remark Note that the coefficient of y' is one!. If it is not, we will need to divide on whatever is there!

As you may have noticed, the difference between this form and the linear form is the y^n, which multiplied by $Q(x)$

To convert the Bernoulli Equation (7.14) into the linear differential equation, we use the following substitution

$$z = y^{1-n} \tag{7.15}$$

$$\frac{\mathrm{d}z}{\mathrm{d}x} = (1-n)\, y^{-n} \frac{\mathrm{d}y}{\mathrm{d}x} \tag{7.16}$$

Now, divide the equation (7.14) by y^n. This will result in the following equation:

$$Y^{-n}\, y' + P(x)y^{1-n} = Q(x). \tag{7.17}$$

Substitute from (7.15) into (7.17),

$$\begin{aligned} \frac{1}{1-n}\frac{\mathrm{d}z}{\mathrm{d}x} + P(x)\, z &= Q(x) \\ \frac{\mathrm{d}z}{\mathrm{d}x} + (1-n)\, P(x)\, z &= (1-n)\, Q(x). \end{aligned}$$

The last equation is a linear differential equation with

$$\begin{aligned} P_{new}(x) &= (1-n)\, P(x) \\ Q_{new}(x) &= (1-n)\, Q(x) \end{aligned}$$

Similarly, you can always do these steps while transforming Bernoulli Equation into linear first order DE, or use the final form, which is

$$\frac{\mathrm{d}z}{\mathrm{d}x} + P_{new}(x)z = Q_{new}(x),$$

where, $P_{new}(x) = (1-n)\, P(x)$ *and* $Q_{new}(x) = (1-n)\, Q(x)$

Let us now do some examples!

Example 18 Solve the following differential equation the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = e^x\, y^3.$$

Solution First step: This is Bernoulli Equation and $n = 3$, therefore, we can rewrite the equation as follows;

$$y^{-3}\frac{\mathrm{d}y}{\mathrm{d}x} + y^{-2} = e^x,$$

and

$$z = y^{1-3} = y^{-2}$$
$$\frac{\mathrm{d}z}{\mathrm{d}x} = -2y^{-3}\frac{\mathrm{d}y}{\mathrm{d}x}$$

This implies that the new linear equation will be as follows;

$$\frac{\mathrm{d}z}{\mathrm{d}x} - 2\,z = -2\,e^{x},$$

Second step: we identify $P(x)$ and $Q(x)$;

- $P_{new}(x) = -2$
- $Q_{new}(x) = -2\,e^{x}$

Third step: compute the integrating factor

$$I = e^{\int -2\,\mathrm{d}x} = e^{-2x}.$$

It follows that integrating this gives

$$\begin{aligned} e^{-2x}y &= \int -2e^{-2x}\,e^{x}\,dx + c \\ z &= \frac{1}{e^{-2x}}\left(\int -2e^{-2x}e^{x}\,dx + c\right) \\ z &= \frac{1}{e^{-2x}}\left(-2\int e^{-x}\,dx + c\right) \\ y^{-2} &= \frac{1}{e^{-2x}}\left(2e^{-x} + c\right) \end{aligned}$$

As the original equation in y and we're now solving in z, do NOT forget to replace $z = y^{1-n}$ in the end.

Example 19 Solve the following differential equation the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{x}y = x\,y^{2}.$$

Solution First step: This is Bernoulli Equation and $n = 2$, therefore, we can rewrite the equation as follows;

$$y^{-2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{x}y^{-1} = x,$$

and

$$z = y^{1-2} = y^{-1}$$
$$\frac{\mathrm{d}z}{\mathrm{d}x} = -y^{-2}\frac{\mathrm{d}y}{\mathrm{d}x}$$

This implies that the new linear equation will be as follows;

$$-\frac{\mathrm{d}z}{\mathrm{d}x} - \frac{1}{x}\,z = x,$$
$$\frac{\mathrm{d}z}{\mathrm{d}x} + \frac{1}{x}\,z = -x,$$

This is a linear DE.
Second step: we identify $P_{new}(x)$ and $Q_{new}(x)$;

- $P_{new}(x) = \frac{1}{x}$
- $Q_{new}(x) = -x$

Third step: compute the integrating factor

$$I = e^{\int \frac{1}{x}\,\mathrm{d}x} = e^{\ln x} = x.$$

It implies that

$$z = \frac{1}{x}\left(\int -x^2\,dx + c\right)$$
$$y^{-1} = \frac{1}{x}\left(\frac{-x^3}{3} + c\right)$$

Example 20 Solve the following differential equation the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y = y^2.$$

Solution First step: This is Bernoulli Equation and $n = 2$, therefore, we can rewrite the equation as follows;

$$y^{-2}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y^{-1} = 1,$$

and

$$z = y^{1-2} = y^{-1}$$
$$\frac{\mathrm{d}z}{\mathrm{d}x} = -y^{-2}\frac{\mathrm{d}y}{\mathrm{d}x}$$

This implies that the new linear equation will be as follows;

$$-\frac{\mathrm{d}z}{\mathrm{d}x} + \frac{1}{x}\,z = 1,$$
$$\frac{\mathrm{d}z}{\mathrm{d}x} - \frac{1}{x}\,z = -1,$$

This is a linear DE.
Second step:solve as a linear DE. We identify $P_{new}(x)$ and $Q_{new}(x)$;

- $P_{new}(x) = -\frac{1}{x}$
- $Q_{new}(x) = -1$

Third step: compute the integrating factor

$$I = e^{\int \frac{-1}{x}\,\mathrm{d}x} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1}.$$

It implies that

$$\begin{aligned} z &= \frac{1}{x^{-1}}\left(\int -\frac{1}{x}\,dx + c\right) \\ y^{-1} &= x\,(-\ln x\, + c) \end{aligned}$$

Example 21 Solve the following differential equation the equation:

$$e^{-x}\ \frac{\mathrm{d}y}{\mathrm{d}x} + e^{-x}\ y = y^3.$$

Solution First, we need to make the equation into Bernoulli form by dividing by e^{-x}

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = y^3 e^x.$$

Now this is a Bernoulli Equation and $n = 3$, therefore, we can rewrite the equation as follows;

$$y^{-3}\frac{\mathrm{d}y}{\mathrm{d}x} + y^{-2} = e^x,$$

and

$$\begin{aligned} z &= y^{1-3} = y^{-2} \\ \frac{\mathrm{d}z}{\mathrm{d}x} &= -2y^{-3}\frac{\mathrm{d}y}{\mathrm{d}x} \end{aligned}$$

This implies that the new linear equation will be as follows;

$$\frac{\mathrm{d}z}{\mathrm{d}x} - 2\,z = -2\,e^x,$$

- $P(x) = -2$
- $Q(x) = -2\,e^x$

Compute the integrating factor

$$I = e^{\int -2\,\mathrm{d}x} = e^{-2x}.$$

It follows that integrating this gives

$$\begin{aligned} e^{-2x}y &= \int -2e^{-2x}\, e^{x}\, dx + c \\ z &= \frac{1}{e^{-2x}}\left(\int -2e^{-2x}e^{x}\, dx + c\right) \\ z &= \frac{1}{e^{-2x}}\left(-2\int e^{-x}\, dx + c\right) \\ y^{-2} &= \frac{1}{e^{-2x}}\left(2e^{-x} + c\right) \end{aligned}$$

Subsection 7.6

Exercises

Solve the following differential equations:

1. $y' = 5y + e^{-2x}\, y^{-2}, \quad y(0) = 2$
2. $6y' - 2y = xy^4, \quad y(0) = -2$
3. $y' + \dfrac{y}{x} - \sqrt{y} = 0, \quad y(1) = 0$
4. $y' - \dfrac{3}{x}y = \cos(x)y^3$
5. $y' + \dfrac{1}{3}y = e^{x}\, y^4$

Subsection 7.7

Exact first order differential equations

Exact differential equation is another form of first order differential equation. In fact, you may consider it the most general one.

Any first order differential equations can be rearranged to take the following form,

$$\frac{\partial F(x,y)}{\partial x}dx + \frac{\partial F(x,y)}{\partial y}dy = 0$$

Here we may replace $\frac{\partial F(x,y)}{\partial x}$ by $M(x,y)$. and $\frac{\partial F(x,y)}{\partial y}$ by $M(x,y)$, which implies the following equation:

$$M(x,y)dx + N(x,y)dy = 0$$

We say the above equation is an **exact** first order differential equation if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Remark Note that the solution of the **exact** first order differential equation is indeed

$$F(x,y) = c.$$

In order to find that solution, we are going to use the fact that

$$\frac{\partial F(x,y)}{\partial x} = M(x,y),$$

and

$$\frac{\partial F(x,y)}{\partial y} = M(x,y).$$

Having the above, one may integrate

$$\frac{\partial F(x,y)}{\partial x} = M(x,y),$$

according to x, however, the constant of the integration will be a function of y! Do you know why?

Or, one may integrate

$$\frac{\partial F(x,y)}{\partial y} = N(x,y),$$

according to y, however, the constant of the integration will be a function of x! Do you know why?

The solution then can go as follows:

1. First step: integrate,

$$\frac{\partial F(x,y)}{\partial x} = M(x,y),$$

which gives

$$F(x,y) = \int M(x,y)dx + h(y)$$

. We are almost done, however, we need to find the function $h(y)$. These leads to the second step.

2. Second step: take the derivative with respect to y, i.e., $\dfrac{\partial F(x,y)}{\partial y}$ and this should equal to $N(x,y)$. i.e., $\dfrac{\partial F(x,y)}{\partial y} = N(x,y)$. This will result in having the term $\frac{\mathrm{d}h}{\mathrm{d}y}$ in the equation.
3. Third step: simplify and integrate to solve for $h(y)$.
4. Final step: Substitute the result for $h(y)$ in the expression for $F(x,y)$ from step 1 and then set $F(x,y) = C$.

Let us now do some examples to illustrate what we have explained.

Example 22 Solve the following differential equation:

$$2x\,y - 9\,x^2 + (2y + x^2 + 1)y' = 0$$

Solution First, we should think a bit if we can use the separable method or the linear method, which we explained in the previous sections. Unfortunately, none of them will work! Now, is it exact?
To answer this question, we have to rewrite the equation as follows:

$$(2x\,y - 9\,x^2)dx + (2y + x^2 + 1)dy = 0$$

then,

1. First, define $M(x,y) = 2x\,y - 9\,x^2$ and $N(x,y) = 2y + x^2 + 1$, and $\dfrac{\partial M}{\partial y} = M_y$ has to equal $\dfrac{\partial N}{\partial x} = N_x$, i.e.,

$$\frac{\partial M}{\partial y} = M_y = 2x$$
$$\frac{\partial N}{\partial x} = N_x = 2x$$

   This implies that it is exact!

   Now, the solution goes as follows,

2. First step: define

$$\begin{aligned} F(x,y) &= \int M(x,y)\mathrm{d}x + h(y) \\ &= \int (2x\,y - 9\,x^2)\mathrm{d}x + h(y) \\ &= x^2y + x^3 + h(y) \end{aligned} \tag{7.18}$$

3. Second step: take the derivative with respect to y, i.e., $\dfrac{\partial F(x,y)}{\partial y}$ and this should

equal to $N(x, y)$. i.e.,

$$\frac{\partial F(x,y)}{\partial y} = x^2 + h'(y) = 2y + x^2 + 1$$

This implies that

$$h'(y) = 2y + 1$$

4. Third step: Integrate to find $h(y)$.

$$h(y) = \int (2y+1)\mathrm{d}y = y^2 + y + C$$

5. Final step: Substitute the result for $h(y)$ in the expression for $F(x, y)$ in (7.18) from step 1 and then set $F(x, y) = 0$, i.e.,

$$F(x,y) = x^2y + x^3 + y^2 + y + C = 0$$

Note that we can do the same exact process to obtain $F(x, y)$ by integrating $N(x, y)$ according to y and make the constant as a function of x. Please do it and prove that the solution is the same. It is a good practice.

Example 23 Solve the following differential equation:

$$(ye^{2x} + 8)\mathrm{d}x + (\frac{1}{2}e^{2x} - 7y)\mathrm{d}y = 0$$

Solution Similarly,

1. First, define $M(x, y) = ye^{2x} + 8$ and $N(x, y) = \frac{1}{2}e^{2x} - 7y$, and $\frac{\partial M}{\partial y} = M_y$ has to equal $\frac{\partial N}{\partial x} = N_x$, i.e.,

$$\frac{\partial M}{\partial y} = M_y = e^{2x}$$
$$\frac{\partial N}{\partial x} = N_x = e^{2x}$$

Now, the solution goes as follows,

2. First step: define

$$\begin{aligned} F(x,y) &= \int M(x,y)\mathrm{d}x + h(y) \\ &= \int (ye^{2x} + 8)\mathrm{d}x + h(y) \\ &= \frac{1}{2}ye^{2x} + 8x + h(y) \end{aligned} \tag{7.19}$$

3. Second step: take the derivative with respect to y, i.e., $\frac{\partial F(x,y)}{\partial y}$ and this should equal to $N(x, y)$. i.e.,

$$\frac{\partial F(x,y)}{\partial y} = \frac{1}{2}e^{2x} + h'(y) = \frac{1}{2}e^{2x} - 7y$$

This implies that

$$h'(y) = -7y$$

4. Third step: Integrate to find $h(y)$.

$$h(y) = \int -7x\mathrm{d}y = -\frac{7}{2}y^2 + C$$

5. Final step: Substitute the result for $h(y)$ in the expression for $F(x, y)$ in (7.19) from step 1 and then set $F(x, y) = 0$, i.e.,

$$F(x, y) = \frac{1}{2}ye^{2x} + 8x - \frac{7}{2}y^2 + C = 0$$

Example 24 Solve the following differential equation:

$$(\sin x + 4xy)dx + \left(2x^2 - \cos y\right) dy = 0$$

Solution Same thing, let

$$\begin{aligned} M &= \sin x + 4xy \\ \tfrac{\partial M}{\partial y} &= 4x \\ N &= 2x^2 - \cos y \\ \tfrac{\partial N}{\partial x} &= 4x \end{aligned}$$

$$F(x, y) = \int (\sin x + 4xy)dx = -\cos x + 2x^2y + h(y)$$

Second step: take the derivative with respect to y, i.e., $\dfrac{\partial F(x, y)}{\partial y}$ and this should equal to $N(x, y)$. i.e.,

$$\frac{\partial F(x, y)}{\partial y} = 2x^2 + h'(y) = 2x^2 - \cos y$$

This implies that

$$h'(y) = -\cos(y)$$

Third step: Integrate to find $h(y)$.

$$h(y) = \int -\cos y\mathrm{d}y = -\sin(y) + C$$

Final step: Substitute the result for $h(y)$ in the expression for $F(x, y)$ from step 1 and then set $F(x, y) = 0$, i.e.,

$$F(x, y) = 2x^2y - \cos x - \sin y + C = 0$$

Example 25 Solve the following differential equation:

$$(ye^{xy}\cos 2x - 2e^{xy}\sin 2x + 2x) + (xe^{xy}\cos 2x - 3)\frac{dy}{dx} = 0$$

Solution

Let,

1. First, define $M(x,y) = ye^{xy}\cos 2x - 2e^{xy}\sin 2x + 2x$ and $N(x,y) = (xe^{xy}\cos 2x - 3)$, and $\dfrac{\partial M}{\partial y} = M_y$ has to equal $\dfrac{\partial N}{\partial x} = N_x$, i.e.,

$$\begin{aligned}\tfrac{\partial M}{\partial y} &= e^{xy} + xye^{xy} - 2xe^{xy}\sin 2x\\ \tfrac{\partial N}{\partial x} &= e^{xy} + xye^{xy} - 2xe^{xy}\sin 2x\end{aligned}$$

   Now, the solution goes as follows,

2. First step: define

$$\begin{aligned}F(x,y) &= \int (xe^{xy}\cos 2x - 3)\,dy + h(x)\\ F(x,y) &= e^{xy}\cos 2x - 3y + h(x)\end{aligned}$$

3. Second step: take the derivative with respect to y, i.e., $\dfrac{\partial F(x,y)}{\partial x}$ and this should equal to $M(x,y)$. i.e.,

$$\frac{\partial F}{\partial x} = ye^{xy}\cos 2x - 2e^{xy}\sin 2x + h'(x)$$

   This implies that

$$h'(x) = 2x$$

4. Third step: Integrate to find $h(x)$.

$$h(x) = \int 2x\mathrm{d}x = x^2 + C$$

5. Final step: Substitute the result for $h(x)$ in the expression for $F(x,y)$ from step 1 and then set $F(x,y) = 0$, i.e.,

$$F(x,y) = e^{xy}\cos 2x - 3y + x^2 + C = 0$$

Have you noticed that we obtained $F(x,y)$ by integrating $N(x,y)$ according to y and make the constant as a function of x, instead of what we were doing in the previous examples? Think about it! It is much easier this way. Think why!

Subsection 7.8

Integrating Factors or non-Exact Equations

In the previous section we saw that if M, N, M_y and N_x are continuous and $M_y = N_x$ on an open rectangle R then

$$M(x, y)\, dx + N(x, y)\, dy = 0 \tag{7.20}$$

is exact on R.

But what if $M_y \neq N_x$? is there anything that can be done?

The answer is yes, sometimes an equation that isn't exact can be made exact by multiplying it by an appropriate function.

Think of the following:
Assume that there was a function, say, $F(x, y) = C$ and we decided to do a full derivative for the function, which gives the following:

$$\frac{\partial F(x,y)}{\partial x}dx + \frac{\partial F(x,y)}{\partial y}dy = 0$$

Also, assume that it happens that there is a common factor, say $\mu(x)$ or $\mu(y)$ between $\frac{\partial F(x,y)}{\partial x}$ and $\frac{\partial F(x,y)}{\partial y}$, Which we can say that

$$\mu(x)M(x,y)dx + \mu(x)N(x,y)dy = 0$$

Also, assume that someone removed that common factor $\mu(x)$ and give us the equation as the following,

$$M(x,y)dx + N(x,y)dy = 0,$$

and ask us to solve it. Obviously, we will start by checking either

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

or not, which in this case is not!
The question now is:
Can we retrieve the canceled common factor $\mu(x)$ from the given information?
The answer is yes, in some cases we can
For example,

$$(3x + 2y^2)\, dx + 2xy\, dy = 0 \tag{7.21}$$

is not exact, since $M_y(x, y) = 4y \neq N_x(x, y) = 2y$ in (7.21). However, multiplying (7.21) by $\mu(x) = x$ yields

$$(3x^2 + 2xy^2)\, dx + 2x^2y\, dy = 0, \tag{7.22}$$

which is exact, since the new $M_y(x, y) = N_x(x, y) = 4xy$ in (7.22). Solving (7.22) by the procedure given in Section 2.5 yields the implicit solution

$$x^3 + x^2y^2 = c.$$

A function $\mu = \mu(x, y)$ is an *integrating factor* for (7.20) if

$$\mu(x, y)M(x, y)\,dx + \mu(x, y)N(x, y)\,dy = 0 \tag{7.23}$$

is exact. If we know an integrating factor μ for (7.20), we can solve the exact equation (7.23) as it was explained in the previous section.

7.8.1 Finding Integrating Factors

The next theorem gives simple sufficient conditions for the given equation to have an integrating factor that depends on only one of the independent variables x and y, and for finding an integrating factor in this case.

Theorem 2 Let M, N, M_y, and N_x be continuous on an open rectangle R. Then:

- If $(M_y - N_x)/N$ is independent of y on R and we define

$$p(x) = \frac{M_y - N_x}{N}$$

then

$$\mu(x) = \pm e^{\int p(x)\,dx} \tag{7.24}$$

is an integrating factor for

$$M(x, y)\,dx + N(x, y)\,dy = 0 \tag{7.25}$$

on R.

- If $(N_x - M_y)/M$ is independent of x on R and we define

$$q(y) = \frac{N_x - M_y}{M},$$

then

$$\mu(y) = \pm e^{\int q(y)\,dy} \tag{7.26}$$

is an integrating factor for (7.25) on R.

The next two examples show how to apply Theorem 2.

Example 26 Find an integrating factor for the equation

$$(2xy^3 - 2x^3y^3 - 4xy^2 + 2x)\,dx + (3x^2y^2 + 4y)\,dy = 0 \tag{7.27}$$

and solve the equation.

Solution In (7.27)

$$M = 2xy^3 - 2x^3y^3 - 4xy^2 + 2x,\ \ N = 3x^2y^2 + 4y,$$

and

$$M_y - N_x = (6xy^2 - 6x^3y^2 - 8xy) - 6xy^2 = -6x^3y^2 - 8xy,$$

so (7.27) isn't exact. However,

$$\frac{M_y - N_x}{N} = -\frac{6x^3y^2 + 8xy}{3x^2y^2 + 4y} = -2x$$

is independent of y, so Theorem 2
part (a) applies with $p(x) = -2x$. Since

$$\int p(x)\,dx = -\int 2x\,dx = -x^2,$$

$\mu(x) = e^{-x^2}$ is an integrating factor. Multiplying (7.27) by μ yields the exact equation

$$e^{-x^2}(2xy^3 - 2x^3y^3 - 4xy^2 + 2x)\,dx + e^{-x^2}(3x^2y^2 + 4y)\,dy = 0. \tag{7.28}$$

To solve this equation, we must find a function F such that

$$F_x(x, y) = e^{-x^2}(2xy^3 - 2x^3y^3 - 4xy^2 + 2x) \tag{7.29}$$

and

$$F_y(x, y) = e^{-x^2}(3x^2y^2 + 4y). \tag{7.30}$$

Integrating (7.30) with respect to y yields

$$F(x, y) = e^{-x^2}(x^2y^3 + 2y^2) + \psi(x). \tag{7.31}$$

Differentiating this with respect to x yields

$$F_x(x, y) = e^{-x^2}(2xy^3 - 2x^3y^3 - 4xy^2) + \psi'(x).$$

Comparing this with (7.29) shows that $\psi'(x) = 2xe^{-x^2}$; therefore, we can let $\psi(x) = -e^{-x^2}$ in (7.31) and conclude that

$$e^{-x^2}\left(y^2(x^2y + 2) - 1\right) = c$$

is an implicit solution of (7.28). It is also an implicit solution of (7.27).

Example 27 Find an integrating factor for

$$2xy^3\,dx + (3x^2y^2 + x^2y^3 + 1)\,dy = 0 \tag{7.32}$$

and solve the equation.

Solution In (7.32),

$$M = 2xy^3, \quad N = 3x^2y^2 + x^2y^3 + 1,$$

and

$$M_y - N_x = 6xy^2 - (6xy^2 + 2xy^3) = -2xy^3,$$

so (7.32) isn't exact. Moreover,

$$\frac{M_y - N_x}{N} = -\frac{2xy^3}{3x^2y^2 + x^2y^2 + 1}$$

is not independent of y, so Theorem 2(a) does not apply. However, Theorem 2(b) does

apply, since

$$\frac{N_x - M_y}{M} = \frac{2xy^3}{2xy^3} = 1$$

is independent of x, so we can take $q(y) = 1$. Since

$$\int q(y)\,dy = \int dy = y,$$

$\mu(y) = e^y$ is an integrating factor. Multiplying (7.32) by μ yields the exact equation

$$2xy^3e^y\,dx + (3x^2y^2 + x^2y^3 + 1)e^y\,dy = 0. \tag{7.33}$$

To solve this equation, we must find a function F such that

$$F_x(x, y) = 2xy^3e^y \tag{7.34}$$

and

$$F_y(x, y) = (3x^2y^2 + x^2y^3 + 1)e^y. \tag{7.35}$$

Integrating (7.34) with respect to x yields

$$F(x, y) = x^2y^3e^y + \phi(y). \tag{7.36}$$

Differentiating this with respect to y yields

$$F_y = (3x^2y^2 + x^2y^3)e^y + \phi'(y),$$

and comparing this with (7.35) shows that $\phi'(y) = e^y$. Therefore we set $\phi(y) = e^y$ in (7.36) and conclude that

$$(x^2y^3 + 1)e^y = c$$

is an implicit solution of (7.33).

Remark Theorem 2 does not apply in the next example, but the more general argument that led to Theorem 2 provides an integrating factor.

Example 28 Find an integrating factor for

$$(3xy + 6y^2)\,dx + (2x^2 + 9xy)\,dy = 0 \tag{7.37}$$

and solve the equation.

Solution In (7.37)

$$M = 3xy + 6y^2,\ N = 2x^2 + 9xy,$$

and

$$M_y - N_x = (3x + 12y) - (4x + 9y) = -x + 3y.$$

Therefore

$$\frac{M_y - N_x}{M} = \frac{-x + 3y}{3xy + 6y^2} \quad \text{and} \quad \frac{N_x - M_y}{N} = \frac{x - 3y}{2x^2 + 9xy},$$

so Theorem 2 does not apply.
There are other methods and tricks we can do to solve this example, but it is out of the scope of this course.

Subsection 7.9

Exercises

Solve the following exact equations

1. $(2xy + x^2)\,dx + (x^2 + y^2 + 1)\,dy = 0$
2. $x^5 + y^5 \frac{dy}{dx} = 0$
3. $e^x + y^3 + 3xy^2 \frac{dy}{dx} = 0$
4. $(x + y)\cos(x) + \sin(x) + \sin(x)y' = 0$
5. Find the integrating factor for the following equations making them into exact equations:
6. $e^{xy}\,dx + \frac{y}{x}e^{xy}\,dy = 0$
7. $\frac{e^x + y^3}{y^2}\,dx + 3x\,dy = 0$
8. $4(y^2 + x)\,dx + \frac{2x+2y^2}{y}\,dy = 0$
9. $2\sin(y)\,dx + x\cos(y)\,dy = 0$
10. Suppose that we have the equation $f(x)\,dx - dy = 0$. Is this equation exact?
11. Find the general solution using a definite integral.
12. Find the potential function $F(x, y)$ of the exact equation $\frac{1+xy}{x}\,dx + \left(\frac{1}{y} + x\right)dy = 0$ in two different ways.

    Integrate M in terms of x and then differentiate in y and set to N.

    Integrate N in terms of y and then differentiate in x and set to M.

Section 8

General first order differential equation problems

Solve the following differential equations:

1. $y' = 5y + e^{-2x}\, y^{-2}, \quad y(0) = 2$
2. $6y' - 2y = xy^4, \quad y(0) = -2$
3. $y' + \dfrac{y}{x} - \sqrt{y} = 0, \quad y(1) = 0$
4. $y' - \dfrac{3}{x}y = \cos(x)y^3$
5. $y' + \dfrac{1}{3}y = e^x\, y^4$
6. $y' + \left(\dfrac{1+x}{x}\right) y = 0, \quad y(1) = 1$
7. $xy' + \left(1 + \dfrac{1}{\ln x}\right) y = 0, \quad y(e) = 1$
8. $xy' + (1 + x\cot x)y = 0,\ y\left(\dfrac{\pi}{2}\right) = 2$
9. $y' - \left(\dfrac{2x}{1+x^2}\right) y = 0, \quad y(0) = 2$
10. $y' + \dfrac{k}{x}y = 0,\ y(1) = 3$ (k= constant)
11. $y' + (\tan kx)y = 0, \quad y(0) = 2$ (k = constant)
12. $(x-2)(x-1)y' - (4x-3)y = (x-2)^3$
13. $y' + 7y = e^{3x},\ y(0) = 0$
14. $(1+x^2)y' + 4xy = \dfrac{2}{1+x^2},\ y(0) = 1$
15. $xy' + 3y = \dfrac{2}{x(1+x^2)},\ y(-1) = 0$
16. $y' + (\cot x)y = \cos x, \quad y\left(\dfrac{\pi}{2}\right) = 1$
17. $y' + \dfrac{1}{x}y = \dfrac{2}{x^2} + 1,\ y(-1) = 0$
18. $xy' + 2y = 8x^2,\ y(1) = 3$
19. $xy' - 2y = -x^2,\ y(1) = 1$
20. $y' + 2xy = x,\ y(0) = 3$
21. $(x^2-1)y' - 2xy = x(x^2-1),\ y(0) = 4$
22. $(x^2-5)y' - 2xy = -2x(x^2-5),\ y(2) = 7$
23. $y' + 2xy = x^2,\ y(0) = 3$

Section 9

Applications of first order differential equations

First-order differential equations are powerful mathematical tools that model various real-world phenomena. They describe how a quantity changes with respect to another, typically in time. Understanding the applications of these equations is crucial as it bridges the gap between abstract mathematics and practical problem solving.

First-order differential equations are ubiquitous in the sciences, engineering, and social sciences. They help us model and predict:

- Population growth and decay
- Heat transfer and temperature changes
- Radioactive decay
- Chemical reaction rates
- Economic growth models
- Fluid dynamics
- Electrical circuits

Subsection 9.1

Population Growth (Separable Equation)

9.1.1 Modeling

The logistic growth model describes population growth with limited resources:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right) \tag{9.1}$$

Where:

- $P(t)$ is the population size at time t
- r is the intrinsic growth rate
- K is the carrying capacity

This model accounts for two factors:

1. Growth factor: rP represents exponential growth
2. Limiting factor: $\left(1 - \frac{P}{K}\right)$ decreases as P approaches K

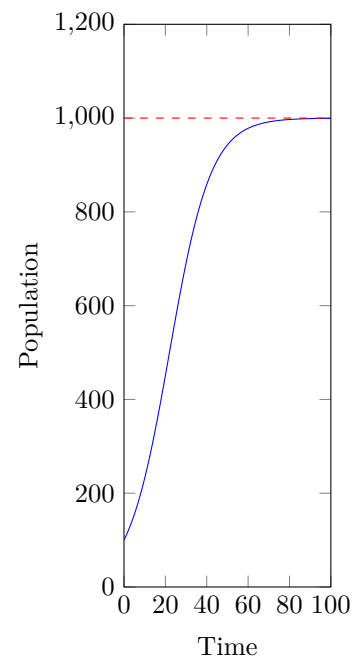


Figure 1. Logistic Growth Model

Example 29

Let $r = 0.1$ and $K = 1000$. The equation becomes:

$$\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{1000}\right) \tag{9.2}$$

To solve this separable equation:

$$\frac{dP}{P(1-P/1000)} = 0.1dt \tag{9.3}$$

$$\int \frac{dP}{P(1-P/1000)} = \int 0.1dt \tag{9.4}$$

$$\ln|P| - \ln|1000-P| = 0.1t + C \tag{9.5}$$

Applying the initial condition $P(0) = 100$:

$$\ln(100) - \ln(900) = C \tag{9.6}$$

The solution is:

$$P(t) = \frac{1000}{1+9e^{-0.1t}} \tag{9.7}$$

This S-shaped curve shows rapid growth initially, slowing as it approaches the carrying capacity.

Subsection 9.2

Newton's Law of Cooling (Linear Equation)

Modeling

Newton's Law of Cooling describes how an object's temperature changes over time:

$$\frac{dT}{dt} = -k(T - T_a) \tag{9.8}$$

Where:

- $T(t)$ is the object's temperature at time t
- T_a is the ambient temperature
- k is the cooling constant

This model assumes that the rate of temperature change is proportional to the difference between the object's temperature and the ambient temperature.

Example 30

Suppose $k = 0.1$, $T_a = 20$řC, and the initial temperature $T(0) = 100$řC. The equation becomes:

$$\frac{dT}{dt} = -0.1(T - 20) \tag{9.9}$$

To solve this linear equation:

$$\frac{dT}{T-20} = -0.1dt \tag{9.10}$$

$$\int \frac{dT}{T-20} = \int -0.1dt \tag{9.11}$$

$$\ln|T-20| = -0.1t + C \tag{9.12}$$

Applying the initial condition:

$$\ln(80) = C \tag{9.13}$$

The solution is:

$$T(t) = 20 + 80e^{-0.1t} \tag{9.14}$$

This exponential decay shows the temperature approaching the ambient temperature over time.

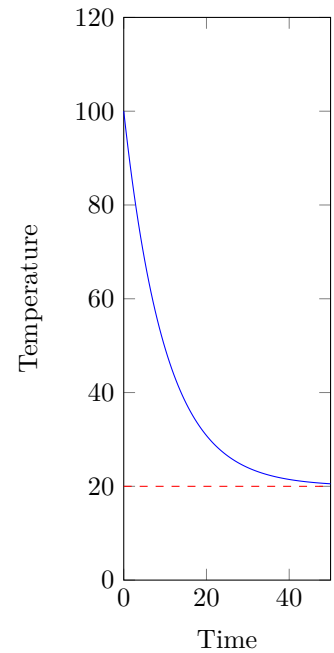


Figure 2. Newton's Law of Cooling

Subsection 9.3

Radioactive Decay (Exponential Equation)

Modeling

Radioactive decay is described by an exponential decay equation:

$$\frac{dN}{dt} = -\lambda N \tag{9.15}$$

Where:

- $N(t)$ is the amount of radioactive material at time t
- λ is the decay constant

This model assumes that the rate of decay is proportional to the amount of material present.

Example 31 For Carbon-14, with a half-life of 5730 years, $\lambda = \frac{\ln(2)}{5730} \approx 0.00012$ per year. The equation becomes:

$$\frac{dN}{dt} = -0.00012N \tag{9.16}$$

To solve this exponential equation:

$$\frac{dN}{N} = -0.00012dt \tag{9.17}$$

$$\int \frac{dN}{N} = \int -0.00012dt \tag{9.18}$$

$$\ln|N| = -0.00012t + C \tag{9.19}$$

Assuming an initial amount $N(0) = N_0$:

$$N(t) = N_0 e^{-0.00012t} \tag{9.20}$$

This exponential decay shows the amount of Carbon-14 decreasing over time, with half of it remaining after 5730 years.

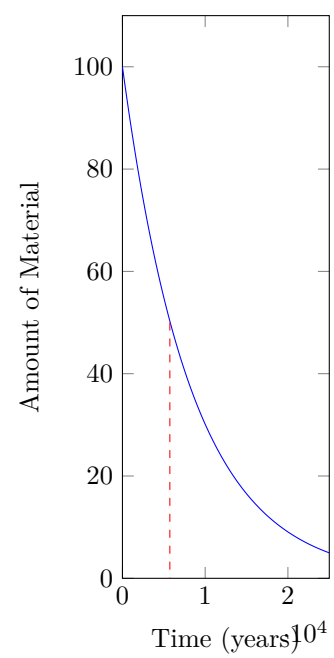


Figure 3. Radioactive Decay of Carbon-14

Second order differential equations

PART III

Section 10

Introduction

Second-order differential equations play a crucial role in various fields of science and engineering due to their ability to model complex dynamic systems. These equations, characterized by the presence of second derivatives, arise naturally in many physical phenomena and are essential tools for describing and predicting real-world behavior.

A second-order differential equation is generally expressed in the form:

$$\frac{d^2y}{dx^2} = f\left(x, y, \frac{dy}{dx}\right)$$

where y is a function of x, and f is a function of x, y, and $\frac{dy}{dx}$. The presence of the second derivative $\frac{d^2y}{dx^2}$ distinguishes these equations from their first-order counterparts, allowing them to capture more complex dynamics and oscillatory behavior.

The versatility of second-order differential equations is evident in their wide-ranging applications:

- **Physics:** These equations are fundamental in describing motion, vibrations, and waves. For instance, Newton's second law of motion, $F = ma$, leads to second-order differential equations when expressed in terms of position.
- **Engineering:** In electrical engineering, RLC circuits are modeled using second-order differential equations. In mechanical engineering, they describe the behavior of spring-mass systems and structural vibrations.
- **Biology:** Population dynamics involving predator-prey relationships or the spread of epidemics can be modeled using systems of second-order differential equations.
- **Economics:** Certain economic models, such as those describing price fluctuations or business cycles, employ second-order differential equations to capture complex market dynamics.

Subsection 10.1

Solving Second-Order Differential Equations

The methods for solving these equations vary depending on their specific form:

1. Linear equations with constant coefficients can often be solved using characteristic equations.
2. Non-linear equations may require numerical methods or specialized techniques like separation of variables.
3. Some equations can be reduced to systems of first-order equations for analysis.

Understanding and solving second-order differential equations is crucial for engineers, scientists, and researchers from all disciplines. Their ability to model complex, real-world phenomena makes them indispensable tools in modern scientific and technological advancements.

Remark A second-order differential equation is said to be *linear* if it can be written as

$$a(x)y'' + b(x)y' + c(x)y = g(x).$$

Or in the following format

$$y'' + p(x)y' + q(x)y = f(x). \tag{10.1}$$

We call the function f on the right a *forcing function*, since in physical applications it's often related to a force acting on some system modeled by the differential equation. We say that (10.1) is *homogeneous* if $f \equiv 0$ or *nonhomogeneous* if $f \not\equiv 0$. Since these definitions are like the corresponding definitions in Section 2.1 for the linear first order equation

$$y' + p(x)y = f(x), \tag{10.2}$$

In this course, we will be focusing on *linear* second-order differential equation, which is given in 10.1.

Remark It is important to look at the linear second-order differential equation given in 10.1 as two parts:
The lift-hand side of the equation, which represents the ***dynamics*** of a given system. It connects its internal properties and components.
However, the right-hand side $f(x)$ represents a ***force function*** that acts on the system from the outside.
For example, in the mass spring system given in the picture 4, the relationship between the spring and the mass is represented on the upper hand side of the equation, which represents the ***dynamics***. However, the force acting on the system represents the right-hand side $f(x)$ in the equation given in 10.1.

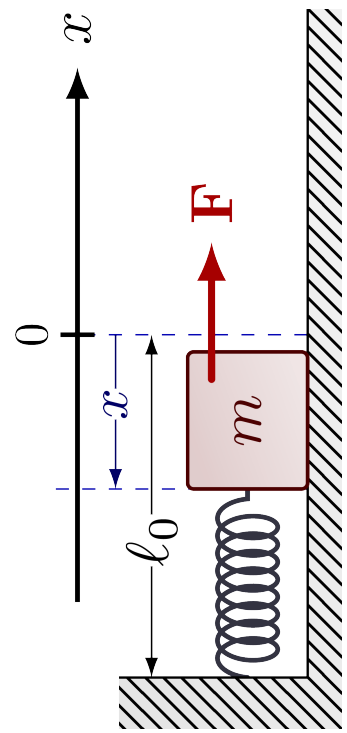


Figure 4. Mass spring system

The above representation of the linear second-order differential equation will allow us to classify the equation 10.1 into two categories:

A **homogeneous** second order linear differential equation:

$$y'' + p(x)y' + q(x)y = 0, \tag{10.3}$$

i.e., $f(x) = 0$, which means, no external force. A **non-homogeneous** second order linear differential equation:

$$y'' + p(x)y' + q(x)y = f(x), \tag{10.4}$$

i.e., $f(x) \neq 0$, which means, there is an external force.

In the next section, we are going to focus on the solution of the second-order linear differential equation, both, **homogeneous** and **non-homogeneous**.

SUBSECTION 10.2

On the Solution of Homogeneous SODE

Before we discuss the methods of solving a Homogeneous Second Order Linear Differential Equation, let us present the following important theorem about the solution concept of

the Homogeneous Second Order Linear Differential Equations.

Theorem 3 If y_1 and y_2 are both different solutions of the homogeneous equation

$$y'' + p(x)y' + q(x)y = 0 \tag{10.5}$$

on the interval, (a, b), then any **linear combination**

$$y = c_1 y_1 + c_2 y_2 \tag{10.6}$$

of y_1 and y_2 is also a solution of the Differential Equation.

PROOF If

$$y = c_1 y_1 + c_2 y_2$$

then

$$y' = c_1 y_1' + c_2 y_2' \quad \text{and} \quad y'' = c_1 y_1'' + c_2 y_2''.$$

Therefore

$$\begin{aligned} y'' + p(x)y' + q(x)y &= (c_1 y_1'' + c_2 y_2'') + p(x)(c_1 y_1' + c_2 y_2') + q(x)(c_1 y_1 + c_2 y_2) \\ &= c_1 \left(y_1'' + p(x)y_1' + q(x)y_1\right) + c_2 \left(y_2'' + p(x)y_2' + q(x)y_2\right) \\ &= c_1 \cdot 0 + c_2 \cdot 0 = 0, \end{aligned}$$

since y_1 and y_2 are solutions of (10.5). ■

We say that $\{y_1, y_2\}$ is a *fundamental set of solutions of* (10.5) *on* (a, b) if every solution of (10.5) on (a, b) can be written as a linear combination of y_1 and y_2 as in (10.6). In this case, we say that (10.6) is *general solution of* (10.5) *on* (a, b). □

Although the progression from the homogeneous to the nonhomogeneous case isn't that simple for the linear second order equation, it's still necessary to solve the homogeneous equation

$$y'' + p(x)y' + q(x)y = 0 \tag{10.7}$$

in order to solve the nonhomogeneous equation (10.1). This section is devoted to (10.7).

The next theorem gives sufficient conditions for existence and uniqueness of solutions of initial value problems for (10.7). We omit the proof.

Theorem 4 Suppose p and q are continuous on an open interval, (a, b), let x_0 be any point in (a, b), and let k_0 and k_1 be arbitrary real numbers. Then the initial value problem

$$y'' + p(x)y' + q(x)y = 0, \; y(x_0) = k_0, \; y'(x_0) = k_1$$

has a unique solution on (a, b).

Since $y \equiv 0$ is obviously a solution of (10.7) we call it the *trivial* solution. Any other solution is *nontrivial*. Under the assumptions of Theorem 4, the only solution of the initial value problem

$$y'' + p(x)y' + q(x)y = 0, \; y(x_0) = 0, \; y'(x_0) = 0$$

on (a, b) is the trivial solution.

The next three examples illustrate concepts that we'll develop later in this section. You shouldn't be concerned with how to *find* the given solutions of the equations in these

examples. This will be explained in later sections.

Example 32 The coefficients of y' and y in

$$y'' - y = 0 \tag{10.8}$$

are the constant functions $p \equiv 0$ and $q \equiv -1$, which are continuous on $(-\infty, \infty)$. Therefore Theorem 4 implies that every initial value problem for (10.8) has a unique solution on $(-\infty, \infty)$.

- Verify that $y_1 = e^x$ and $y_2 = e^{-x}$ are solutions of (10.8) on $(-\infty, \infty)$.
- Verify that if c_1 and c_2 are arbitrary constants, $y = c_1 e^x + c_2 e^{-x}$ is a solution of (10.8) on $(-\infty, \infty)$.
- Solve the initial value problem

$$y'' - y = 0, \quad y(0) = 1, \quad y'(0) = 3. \tag{10.9}$$

Solution Part(a) If $y_1 = e^x$ then $y_1' = e^x$ and $y_1'' = e^x = y_1$, so $y_1'' - y_1 = 0$. If $y_2 = e^{-x}$, then $y_2' = -e^{-x}$ and $y_2'' = e^{-x} = y_2$, so $y_2'' - y_2 = 0$.

Part (b) If

$$y = c_1 e^x + c_2 e^{-x} \tag{10.10}$$

then

$$y' = c_1 e^x - c_2 e^{-x} \tag{10.11}$$

and

$$y'' = c_1 e^x + c_2 e^{-x},$$

so

$$\begin{aligned} y'' - y &= (c_1 e^x + c_2 e^{-x}) - (c_1 e^x + c_2 e^{-x}) \\ &= c_1(e^x - e^x) + c_2(e^{-x} - e^{-x}) = 0 \end{aligned}$$

for all x. Therefore, $y = c_1 e^x + c_2 e^{-x}$ is a solution of (10.8) on $(-\infty, \infty)$.

Part(c)
We can solve (10.9) by choosing c_1 and c_2 in (10.10) so that $y(0) = 1$ and $y'(0) = 3$. Setting $x = 0$ in (10.10) and (10.11) shows that this is equivalent to

$$\begin{aligned} c_1 + c_2 &= 1 \\ c_1 - c_2 &= 3. \end{aligned}$$

Solving these equations yields $c_1 = 2$ and $c_2 = -1$. Therefore, $y = 2e^x - e^{-x}$ is the unique solution of (10.9) on $(-\infty, \infty)$.

Example 33 Let ω be a positive constant. The coefficients of y' and y in

$$y'' + \omega^2 y = 0 \tag{10.12}$$

are the constant functions $p \equiv 0$ and $q \equiv \omega^2$, which are continuous on $(-\infty, \infty)$. Therefore Theorem 4 implies that every initial value problem for (10.12) has a unique solution on $(-\infty, \infty)$.

- Verify that $y_1 = \cos \omega x$ and $y_2 = \sin \omega x$ are solutions of (10.12) on $(-\infty, \infty)$.
- Verify that if c_1 and c_2 are arbitrary constants then $y = c_1 \cos \omega x + c_2 \sin \omega x$ is a solution of (10.12) on $(-\infty, \infty)$.
- Solve the initial value problem

$$y'' + \omega^2 y = 0, \quad y(0) = 1, \quad y'(0) = 3. \tag{10.13}$$

Solution

(A)
If $y_1 = \cos \omega x$ then $y_1' = -\omega \sin \omega x$ and $y_1'' = -\omega^2 \cos \omega x = -\omega^2 y_1$, so $y_1'' + \omega^2 y_1 = 0$. If $y_2 = \sin \omega x$ then, $y_2' = \omega \cos \omega x$ and $y_2'' = -\omega^2 \sin \omega x = -\omega^2 y_2$, so $y_2'' + \omega^2 y_2 = 0$.

(B)
If

$$y = c_1 \cos \omega x + c_2 \sin \omega x \tag{10.14}$$

then

$$y' = \omega(-c_1 \sin \omega x + c_2 \cos \omega x) \tag{10.15}$$

and

$$y'' = -\omega^2(c_1 \cos \omega x + c_2 \sin \omega x),$$

so

$$\begin{aligned} y'' + \omega^2 y &= -\omega^2(c_1 \cos \omega x + c_2 \sin \omega x) + \omega^2(c_1 \cos \omega x + c_2 \sin \omega x) \\ &= c_1 \omega^2(-\cos \omega x + \cos \omega x) + c_2 \omega^2(-\sin \omega x + \sin \omega x) = 0 \end{aligned}$$

for all x. Therefore $y = c_1 \cos \omega x + c_2 \sin \omega x$ is a solution of (10.12) on $(-\infty, \infty)$.

(C)
To solve (10.13), we must be choosing c_1 and c_2 in (10.14) so that $y(0) = 1$ and $y'(0) = 3$. Setting $x = 0$ in (10.14) and (10.15) shows that $c_1 = 1$ and $c_2 = 3/\omega$. Therefore,

$$y = \cos \omega x + \frac{3}{\omega} \sin \omega x$$

is the unique solution of (10.13) on $(-\infty, \infty)$. ■

Theorem 4 implies that if k_0 and k_1 are arbitrary real numbers then the initial value problem

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = 0, \quad y(x_0) = k_0, \quad y'(x_0) = k_1 \tag{10.16}$$

has a unique solution on an interval (a, b) that contains x_0, provided that P_0, P_1, and P_2 are continuous and P_0 has no zeros on (a, b). To see this, we rewrite the differential

equation in (10.16) as

$$y'' + \frac{P_1(x)}{P_0(x)}y' + \frac{P_2(x)}{P_0(x)}y = 0$$

and apply Theorem 4 with $p = P_1/P_0$ and $q = P_2/P_0$.

SECTION 11

SODE with constant coefficient

In this section, we are going to focus on **constant coefficient** second order differential equations, i.e., the coefficient of the dependent variable y and its derivatives do not contain the independent variable x. In other words, we are going to solve the following equation:

Remark Constant coefficients means that the functions in front of y'', y', and y are constants, they do not depend on x.

$$ay'' + by' + cy = f(x). \tag{11.1}$$

$$\tag{11.2}$$

We will start with the homogeneous SODE with constant coefficient as given the following subsection.

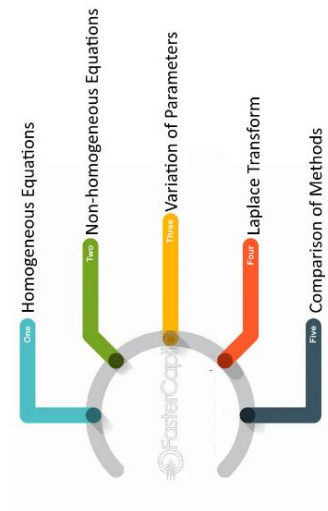


Figure 5. Methods of solving second order differential equations

SUBSECTION 11.1

Homogeneous SODE with constant coefficient

To start building towards the solution of the Homogeneous SODE with constant coefficient, consider the example:

$$y'' - 6y' + 8y = 0, \quad y(0) = -2, \quad y'(0) = 6.$$

This is a second order linear homogeneous equation with constant coefficients.
To guess a solution, think of a function that stays essentially the same when we differentiate it, so that we can take the function and its derivatives, add some multiples of these together, and end up with zero. Yes, we are talking about the **exponential**.
Let us try" a solution of the form $y = e^{rx}$. Then $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Plug in to get

$$\begin{aligned} y^{y''} - 6y' + 8y &= 0 \\ \underbrace{r^2e^{rx}}_{y''} - 6\underbrace{re^{rx}}_{y'} + 8\underbrace{e^{rx}}_{y} &= 0, \\ r^2 - 6r + 8 &= 0 \quad \text{(divide through by } e^{rx}\text{)}, \\ (r-2)(r-4) &= 0. \end{aligned}$$

Hence, if $r = 2$ or $r = 4$, then e^{rx} is a solution. So let $y_1 = e^{2x}$ and $y_2 = e^{4x}$.
The functions e^{2x} and e^{4x} are linearly independent. If they were not linearly independent, we could write $e^{4x} = Ce^{2x}$ for some constant C, implying that $e^{2x} = C$ for all x, which is clearly not possible. Hence, we can write the general solution as

$$y = C_1e^{2x} + C_2e^{4x}$$

We need to solve for C_1 and C_2. To apply the initial conditions, we first find $y' = 2C_1e^{2x} + 4C_2e^{4x}$. We plug $x = 0$ into y and y' and solve.

$$\begin{aligned} -2 = y(0) &= C_1 + C_2 \\ 6 = y'(0) &= 2C_1 + 4C_2 \end{aligned}$$

Either apply some matrix algebra, or just solve these by high school math. For example,

divide the second equation by 2 to obtain $3 = C_1 + 2C_2$, and subtract the two equations to get $5 = C_2$. Then $C_1 = -7$ as $-2 = C_1 + 5$. Hence, the solution we are looking for is

$$y = -7e^{2x} + 5e^{4x}$$

We can generalize the above example into a method. Suppose that we have an equation

$$ay'' + by' + cy = 0,$$

where a, b, c are constants. Try the solution $y = e^{rx}$ to obtain

$$ar^2e^{rx} + bre^{rx} + ce^{rx} = 0$$

Divide by e^{rx} to obtain the so-called **characteristic equation** of the ODE:

$$ar^2 + br + c = 0$$

Then we find the roots, r_1 and r_2, and hence find the general solution as:

$$y = C_1e^{r_1x} + C_2e^{r_2x}. \tag{11.3}$$

Remark Note that the above works only so far if r_1 and r_2 are real different. The good news is that we can generalize the above argument for all the cases of the roots r_1 and r_2, even when they are complex!

Theorem 5 Suppose that r_1 and r_2 are the roots of the characteristic equation. Then If r_1 and r_2 are:

- **Real Roots** $(r_1 \neq r_2)$: $y_c = C_1e^{r_1t} + C_2e^{r_2t}$
- **Repeated** $(r_1 = r_2)$: $y_c = (C_1 + C_2t)e^{r_1t}$
- **Complex** $(r = \alpha \pm i\beta)$: $y_c = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$

To summarize, if you have the following second order Homogeneous DEs,

$$ay'' + by' + cy = 0 \tag{11.4}$$
$$\tag{11.5}$$

Do the following steps:

1. Find the **Characteristic Equation** of the given Homogeneous DEs,

$$ar^2 + br + c = 0. \tag{11.6}$$

2. Solve the characteristic equation and find the quadratic roots,

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{11.7}$$

3. Substitute into the general form solution according to the roots as follows:
   - **Real Roots** $(r_1 \neq r_2)$: $y_c = C_1e^{r_1t} + C_2e^{r_2t}$
   - **Repeated** $(r_1 = r_2)$: $y_c = (C_1 + C_2t)e^{r_1t}$
   - **Complex** $(r = \alpha \pm i\beta)$: $y_c = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$

Let us do some examples:

Example 34 Solve the following differential equation

$$y'' + y' - 6y = 0 \tag{11.8}$$

Solution First, we get the Characteristic Equation of the given DEs,

$$r^2 + r - 6 = 0 \tag{11.9}$$

Second, we find the roots of the above equation using factorization as follows;

$$(r+3)(r-2) = 0. \tag{11.10}$$

It follows that we have two real roots, i.e., the solution is given as follows;

$$y_c = C_1 e^{2t} + C_2 e^{-3t} \tag{11.11}$$

Here are four different problems and their solutions in the same style:
—

Example 35 Solve the following differential equation

$$y'' - 4y' + 4y = 0 \tag{11.12}$$

Solution First, we get the characteristic equation of the given DE:

$$r^2 - 4r + 4 = 0 \tag{11.13}$$

Next, we find the roots of the characteristic equation. Notice that this can be factored as:

$$(r-2)^2 = 0 \tag{11.14}$$

This implies that there is a repeated root, $r = 2$. Thus, the general solution for the differential equation is:

$$y_c = (C_1 + C_2 t)e^{2t} \tag{11.15}$$

Example 36 Solve the following differential equation

$$y'' + 9y = 0 \tag{11.16}$$

Solution First, we write down the characteristic equation:

$$r^2 + 9 = 0 \tag{11.17}$$

Solving for r, we get:

$$r^2 = -9 \implies r = \pm 3i \tag{11.18}$$

Since the roots are complex, $r = \pm 3i$, the general solution is:

$$y_c = C_1 \cos(3t) + C_2 \sin(3t) \tag{11.19}$$

—

Example 37 Solve the following differential equation

$$y'' - 2y' + 5y = 0 \tag{11.20}$$

Solution First, we form the characteristic equation:

$$r^2 - 2r + 5 = 0 \tag{11.21}$$

Using the quadratic formula $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1, b = -2, c = 5$, we find:

$$r = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(5)}}{2(1)} = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} \tag{11.22}$$

So, the roots are $r = 1 + 2i$ and $r = 1 - 2i$. The general solution is:

$$y_c = e^t(C_1 \cos(2t) + C_2 \sin(2t)) \tag{11.23}$$

—

Example 38 Solve the following differential equation

$$y'' + 2y' + 2y = 0 \tag{11.24}$$

Solution First, write the characteristic equation:

$$r^2 + 2r + 2 = 0 \tag{11.25}$$

Using the quadratic formula $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 1, b = 2, c = 2$, we find:

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(2)}}{2(1)} = \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm \sqrt{-4}}{2} = \frac{-2 \pm 2i}{2} \tag{11.26}$$

Thus, $r = -1 + i$ and $r = -1 - i$. The general solution is:

$$y_c = e^{-t}(C_1 \cos(t) + C_2 \sin(t)) \tag{11.27}$$

—

Example 39 Solve the following differential equation

$$y'' + 4y' + 4y = 0 \tag{11.28}$$

Solution First, we get the Characteristic Equation of the given DEs,

$$r^2 + 4\,r + 4 = 0 \tag{11.29}$$

Second, we find the roots of the above equation using factorization as follows;

$$(r+2)\,(r+2) = 0. \tag{11.30}$$

It follows that we have one repeated root, i.e., the solution is given as follows;

$$y_c = (C_1 + C_2\,t)e^{-2\,t} \tag{11.31}$$

Example 40 Solve the following differential equation

$$y'' + 5y' + 6y = 0 \tag{11.32}$$

Solution First, we get the Characteristic Equation of the given DEs,

$$r^2 + 5\,r + 6 = 0 \tag{11.33}$$

Second, we find the roots of the above equation using factorization as follows;

$$(r+3)\,(r+2) = 0. \tag{11.34}$$

It follows that we have two real roots, i.e., the solution is given as follows;

$$y_c = C_1 e^{-2\,t} + C_2 e^{-3\,t} \tag{11.35}$$

Example 41 Solve the following differential equation

$$y'' - 9y = 0 \tag{11.36}$$

Solution First, we get the Characteristic Equation of the given DEs,

$$r^2 - 9 = 0 \tag{11.37}$$

Second, we find the roots of the above equation using factorization as follows;

$$(r+3)\,(r-3) = 0. \tag{11.38}$$

It follows that we have two real roots, i.e., the solution is given as follows;

$$y_c = C_1 e^{3\,t} + C_2 e^{-3\,t} \tag{11.39}$$

Example 42 Solve the following differential equation

$$y'' - 8y' + 17y = 0 \tag{11.40}$$

Solution First, we get the Characteristic Equation of the given DEs,

$$r^2 - 8\,r + 17 = 0 \tag{11.41}$$

Second, we find the roots of the above equation using the quadratic form as follows;

$$\begin{aligned} r &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{8 \pm \sqrt{8^2 - 4 \times 17}}{2} \\ &= 4 \pm j \end{aligned}$$

It follows that we have two real roots, i.e., the solution is given as follows;

$$y_c = e^{4\,t}(C_1 \cos(t) + C_2 \sin(t)) \tag{11.42}$$

SUBSECTION 11.2

Exercises

1. $y'' + 5y' - 6y = 0$
2. $y'' - 4y' + 5y = 0$
3. $y'' + 8y' + 7y = 0$
4. $y'' - 4y' + 4y = 0$
5. $y'' + 2y' + 10y = 0$
6. $y'' + 6y' + 10y = 0$
7. $y'' - 8y' + 16y = 0$
8. $y'' + y' = 0$
9. $y'' - 2y' + 3y = 0$
10. $y'' + 6y' + 13y = 0$
11. $4y'' + 4y' + 10y = 0$
12. $10y'' - 3y' - y = 0$
13. $y'' + 14y' + 50y = 0, \quad y(0) = 2, \quad y'(0) = -17$
14. $6y'' - y' - y = 0, \quad y(0) = 10, \quad y'(0) = 0$
15. $6y'' + y' - y = 0, \quad y(0) = -1, \quad y'(0) = 3$
16. $4y'' - 4y' - 3y = 0, \quad y(0) = \frac{13}{12}, \quad y'(0) = \frac{23}{24}$
17. $4y'' - 12y' + 9y = 0, \quad y(0) = 3, \quad y'(0) = \frac{5}{2}$

SUBSECTION 11.3

Non-Homogeneous SODE with constant coefficients

Now, we are ready to solve the second order non-homogeneous DEs, which takes the following form

$$ay'' + by' + cy = g(t) \tag{11.43}$$

As we explained before, the above problem can be separated into left-hand side, the **dynamics** and the right-hand side, the forcing function or the external force, $f(t)$. We will also construct the solution as two parts:

$$y_g = y_c(t) \; + \; y_p(t),$$

where $y_c(t)$ is called the **complementary solution, which is related to the dynamics** and $y_p(t)$ is called the **particular solution, which is related to external force** .

Remark

As you may guess, the way we obtain $y_c(t)$, the complementary solution, is by assuming that the given DE is **homogeneous** and find the solution exactly as before by following these steps:

1. Find the **Characteristic Equation** of the given Homogeneous DEs,

$$ar^2 + br + c = 0. \tag{11.44}$$

2. Solve the characteristic equation and find the quadratic roots,

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{11.45}$$

3. Substitute into the general form solution according to the roots as follows:

   - **Real Roots** $(r_1 \neq r_2)$: $y_c = C_1 e^{r_1 t} + C_2 e^{r_2 t}$
   - **Repeated** $(r_1 = r_2)$: $y_c = (C_1 + C_2 t) e^{r_1 t}$
   - **Complex** $(r = \alpha \pm i\beta)$: $y_c = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$

There are several ways to find a particular solution for a second-order differential equation. We are going to study two common ways:

- **Method of undetermined coefficient**
- **Method of Variation of parameters**

SUBSECTION 11.4

Method of undetermined coefficient

As you may expect, the particular solution will depend on the function $g(t)$ in the differential equation:

$$ay'' + by' + cy = g(t) \tag{11.46}$$

The **method of undetermined coefficients** is a technique used to find particular solutions to certain types of non-homogeneous linear differential equations with **constant coefficients**. This method is particularly useful when the non-homogeneous term (the forcing function) is a simple function such as a ***polynomial, exponential, sine, cosine, or a linear combination of these***.

1. **General Form of the Equation**: Consider a second-order non-homogeneous linear differential equation with constant coefficients:

$$ay'' + by' + cy = g(t),$$

where y'' is the second derivative of y with respect to t, a, b and c are constants, and $g(t)$ is a non-homogeneous term (a known function of t).

2. **Homogeneous Solution**: First, solve the corresponding **homogeneous equation**:

$$ay'' + by' + cy = 0.$$

The solution to this homogeneous equation, $y_h(t)$, is found by solving the characteristic equation associated with the differential equation:

$$ar^2 + br + c = 0.$$

The roots of this characteristic equation determine the form of the homogeneous solution $y_c(t)$.

3. **Particular Solution Guess**: The idea behind the method of undetermined coefficients is to find a **particular solution** $y_p(t)$ that satisfies the non-homogeneous equation. To do this, we make an **educated guess** for the form of $y_p(t)$ based on the form of $g(t)$. The guess is usually chosen to be of the same type as $g(t)$:

   - If $g(t)$ is a polynomial of degree n, guess a polynomial of the same degree.
   - If $g(t) = e^{kt}$, guess Ae^{kt}.
   - If $g(t) = \sin(kt)$ or $\cos(kt)$, guess $A\cos(kt) + B\sin(kt)$.
   - If $g(t)$ is a product or sum of these functions, guess a linear combination of functions.

4. **Determining the Coefficients**: The coefficients in the guess for $y_p(t)$ are called **undetermined coefficients** because they are initially unknown. To determine these coefficients, substitute the guess for $y_p(t)$ into the original non-homogeneous differential equation.

   This substitution will yield an equation where the left-hand side is expressed in terms of the guessed function (and its derivatives), and the right-hand side is the function $g(t)$. The coefficients are then determined by matching the coefficients of like terms on both sides of the equation.

5. **Handling Repeated Roots or Overlaps**: If the form of the guessed particular solution $y_p(t)$ is already a part of the homogeneous solution $y_h(t)$, or if it leads to zero after substitution, the guess must be modified. Typically, this is done by multiplying the guess by t enough times until it no longer duplicates any part of the homogeneous solution.

   For example:

   - If $g(t) = e^{2t}$ and the homogeneous solution already includes C_1e^{2t}, then the guess should be Ate^{2t}.

6. **Complete Solution**: The complete solution to the non-homogeneous differential equation is the sum of the homogeneous solution and the particular solution:

$$y(t) = y_c(t) + y_p(t).$$

Remark This method is efficient for differential equations where the non-homogeneous term $g(t)$ is of a form that allows a straightforward guess. However, it is not generally applicable to all types of non-homogeneous terms, especially if $g(t)$ is more complicated or does not fit the standard types listed above.

Here are a summary of the different cases you might have:

Case one:
If $g(t)$ is a polynomial with degree n, then

$$y_p = a_n t^n + a_{n-1} t^{n-1} ... a_1 t + a_0,$$

where non of t^0 and t do not belong to the y_c. If t^0 is in y_c, then

$$y_p = t(a_n t^n + a_{n-1} t^{n-1} ... a_1 t + a_0).$$

If t^0 and t are in y_c, then

$$y_p = t^2(a_n t^n + a_{n-1} t^{n-1} ... a_1 t + a_0).$$

Case two: If $g(t) = Ke^{\alpha t}$ and $e^{\alpha t}$ is not part of the complementary solutions, then

$$y_p = Ae^{\alpha t}.$$

If, $e^{\alpha t}$ is part of the complementary solutions but $te^{\alpha t}$ is not part of the complementary solutions, then

$$y_p = Ate^{\alpha t}.$$

If, $te^{\alpha t}$ is part of the complementary solutions, then

$$y_p = At^2 e^{\alpha t}.$$

Case three:
If $g(t) = K_1 \cos(\alpha t) \pm k_2 \sin(\alpha t)$ and $\cos(\alpha t)$ or $\sin(\alpha t)$ is not part of the complementary solutions, then

$$y_p = A\cos(\alpha t) + B\sin(\alpha t).$$

If, $\cos(\alpha t)$ or $\sin(\alpha t)$ is part of the complementary solutions. then

$$y_p = t(A\cos(\alpha t) + B\sin(\alpha t)).$$

Case four:
If $g(t)$ is a product of functions of the preceding types (Case 1, 2, 3), then the particular solution is a product of functions of the same type.
If $g(t)$ is a sum of functions of preceding types (Case 1, 2, 3), we use the easily verified principle of superposition, which says that if $y_{p1}(t)$ and $y_{p2}(t)$ are particular solutions of the non-homogenous DE,

$$ay'' + by' + cy = g_1(t) \quad ay'' + by' + cy = g_2(t) \tag{11.47}$$

then $y_p(t) = y_{p1}(t) + y_{p2}(t)$ is a particular solution of

$$ay'' + by' + cy = g_1(t) + g_2(t) \tag{11.48}$$

Now, let us do some examples to illustrate the different cases:

Example 43 Find the general solution of

$$y'' - 5y' + 6y = 2t.$$

Solution First, get the complementary solution, we note that the characteristic polynomial is given as follows;

$$y'' - 5y' + 6y = 0$$

is

$$r^2 - 5r + 6 = 0 = (r-6)(r+1),$$

so

$$y_c = C_1 e^{6t} + C_2 e^{-t}$$

is the complementary solution.
The particular solution will take the form

$$y_p = At + B.$$

Substitute $y_p = At + B$ for y will produce the following;

$$\begin{aligned} y_p'' - 5y_p' + 6y_p &= 2t \\ 0 - 5A + 6(Ax + B) &= 2x \end{aligned}$$

This implies that;

$$\begin{aligned} 6A = 2 &\Rightarrow A = 3 \\ -5A + 6B = 0 &\Rightarrow B = \frac{7}{3} \end{aligned}$$

Therefore $y_p = 3t + \frac{7}{3}$ is a particular solution. Therefore the general solution is

$$y = C_1 e^{6t} + C_2 e^{-t} + 3t + \frac{7}{3}.$$

Example 44 Find the general solution of

$$y'' - 7y' + 12y = 4e^{2t}.$$

Solution To find the general solution, we note that the characteristic polynomial of the complementary equation

$$y'' - 7y' + 12y = 0$$

is

$$r^2 - 7r + 12 = 0 = (r-3)(r-4),$$

so

$$y_c = C_1 e^{3t} + C_2 e^{4t}$$

is the complementary solution.
To find the particular solution, substitute $y_p = Ae^{2t}$ for y will produce a constant

multiple of Ae^{2t} on the left side, so it may be possible to choose A so that y_p is a solution. Let's try it; if $y_p = Ae^{2t}$ then

$$y_p'' - 7y_p' + 12y_p = 4Ae^{2t} - 14Ae^{2t} + 12Ae^{2t} = 2Ae^{2t} = 4e^{2t}$$

Then, $A = 2$. Therefore $y_p = 2e^{2t}$ is a particular solution. Therefore the general solution is

$$y = 2e^{2t} + C_1e^{3t} + C_2e^{4t}.$$

Example 45 Find the general solution of

$$y'' - 7y' + 12y = 5e^{4t}. \tag{11.49}$$

Solution To find the general solution, we note that the characteristic polynomial of the complementary equation

$$y'' - 7y' + 12y = 0$$

is

$$r^2 - 7r + 12 = 0 = (r-3)(r-4),$$

so

$$y_c = C_1e^{3t} + C_2e^{4t}$$

is the complementary solution.
Now, note that in this case, e^{4t} is part of the complementary solution. Therefore, we take $y_p = A\ t\ e^{4t}$, where A is a constant to be determined. Substituting

$$y_p = A\ t\ e^{4t}, \quad y' = A\ e^{4t} + 4A\ t\ e^{4t}, \quad \text{and} \quad y'' = 4Ae^{4x} + 4A\ e^{4x} + 16A\ t\ e^{4t}$$

in the main equation, we get;

$$\begin{aligned} 4Ae^{4x} + 4A\ e^{4x} + 16A\ t\ e^{4t} - 7(A\ e^{4t} + 4A\ t\ e^{4t}) + 12(A\ t\ e^{4t}) &= 5e^{4t} \\ Ae^{4x} &= 5e^{4t} \end{aligned}$$

This implies that $A = 5$ and therefore,

$$y_p = 5\ t\ e^{4t}$$

Therefore

$$y = 5te^{4t} + c_1e^{3t} + c_2e^{4t}$$

is the general solution.

Example 46 Find a particular solution of

$$y'' - 8y' + 16y = 2e^{4t}. \tag{11.50}$$

Solution Since the characteristic polynomial of the complementary equation

$$y'' - 8y' + 16y = 0$$

is $r^2 - 8r + 16 = (r-4)^2$, both $y_1 = e^{4t}$ and $y_2 = te^{4t}$ are solutions of the DE. Therefore $y_p = Ae^{4t}$ or $y_p = Ate^{4t}$ can not be a particular solution of the DE. The particular solution should take the form

$$y_p = A\ t^2\ e^{4t}$$

Differentiating twice

$$\begin{aligned} y_p &= A\ t^2\ e^{4t} \\ y_p' &= 2A\ t\ e^{4t} + 4A\ t^2\ e^{4t} \\ y_p'' &= 2A\ \ e^{4t} + 8A\ t\ e^{4t} + 8A\ t\ e^{4t} + 16A\ t^2\ e^{4t} \end{aligned}$$

and substitute in the DE, we find the following;

$$\begin{aligned} & 2A\ \ e^{4t} + 8A\ t\ e^{4t} + 8A\ t\ e^{4t} + 16A\ t^2\ e^{4t} \\ & \quad - 8(2A\ t\ e^{4t} + 4A\ t^2\ e^{4t}) + 16(A\ t^2\ e^{4t}) = 2e^{4t} \\ & 2A\ \ e^{4t} = 2e^{4t} \\ \Rightarrow \quad & A = 1 \end{aligned}$$

and therefore,

$$y = e^{4x}(x^2 + c_1 + c_2 x)$$

is the general solution.

Example 47 Find the general solution of

$$y'' - 2y' + y = 5\cos 2t + 10\sin 2t. \tag{11.51}$$

Solution First, let us find the complementary solution. The characteristic equation is given as follows;

$$\begin{aligned} r^2 - 2r + 1 &= 0 \\ (r-1)(r-1) &= 0 \end{aligned}$$

Therefore,

$$y_c = C_1 e^t + C_2\ t\ e^{-t}$$

The particular solution has the form,

$$y_p = A\cos 2t + B\sin 2t.$$

Since

$$y_p' = -2A\sin 2t + 2B\cos 2t \quad \text{and} \quad y_p'' = -4(A\cos 2t + B\sin 2t),$$

replacing y by y_p

$$\begin{aligned} y_p'' - 2y_p' + y_p &= -4(A\cos 2t + B\sin 2t) - 4(-A\sin 2t + B\cos 2t) \\ &\quad +(A\cos 2t + B\sin 2t) \\ &= (-3A - 4B)\cos 2t + (4A - 3B)\sin 2t. \end{aligned}$$

Equating the coefficients of $\cos 2x$ and $\sin 2x$ here with the corresponding coefficients on the right side of the DE, we get;

$$\begin{aligned} -3A - 4B &= 5 \\ 4A - 3B &= 10. \end{aligned}$$

Solving these equations yields $A = 1$, $B = -2$. Therefore

$$y_p = \cos 2t - 2\sin 2t$$

is a particular solution. This implies that the general solution is;

$$y_c = C_1 e^t + C_2\, t\, e^{-t} + \cos 2t - 2\sin 2t$$

Example 48 Find the general solution of

$$y'' + 4y = 8\cos 2t + 12\sin 2t. \tag{11.52}$$

Solution First, let us find the complementary solution. The characteristic equation is given as follows;

$$\begin{aligned} r^2 + 4 &= 0 \\ r &= \pm 2\, i \end{aligned}$$

Therefore,

$$y_c = C_1\cos(2t) + C_2\sin(2t)$$

The procedure used in Example 47 doesn't work here because $\cos 2t$ and $\sin 2t$ appear in the complementary solution. Therefore,

$$y_p = t(A\cos 2t + B\sin 2t).$$

Then

$$\begin{aligned} y_p' &= A\cos 2t + B\sin 2t + 2t(-A\sin 2t + B\cos 2t) \\ y_p'' &= -4A\sin 2t + 4B\cos 2t - 4t(A\cos 2t + B\sin 2t) \\ &= -4A\sin 2t + 4B\cos 2t - 4y_p. \end{aligned}$$

Substitute back in the DE, we get;

$$-4A\sin 2t + 4B\cos 2t = 8\cos 2t + 12\sin 2t,$$

which holds if $A = -3$ and $B = 2$. Therefore,

$$y_p = -t(3\cos 2t - 2\sin 2t)$$

is a particular solution.

Example 49

$$\text{Solve the equation } y'' - 4y' = te^t + \cos(2t) \tag{11.53}$$

Solution

First, $y_c(t)$: The auxiliary equation is $r^2 - 4r = 0$, with roots 0 and 4, so the solution of the complementary equation $y_C(t) = c_1 + c_2 e^{4t}$.
Second, $y_p(t) = y_{p_1}(t) + y_{p_2}(t) =$, where
y_{p_1} is a particular solution for $y'' - 4y' = te^t$.

$$y_{p_1} = (At + B)e^t \Rightarrow y'_{p_1} = (At + A + B)e^t \Rightarrow y''_{p_1} = (At + 2A + B)e^t$$

So, substituting into $y'' - 4y' = te^t$, we have

$$(At + 2A + B)e^t - 4(At + A + B)e^t = te^t$$

Thus, $A = -\frac{1}{3}$ and $B = -\frac{2}{9}$, and $y_{p_1}(t) = -\left(\frac{1}{3}t + \frac{2}{9}\right)e^t$.
y_{p_2} is a particular solution for $y'' - 4y' = \cos 2t$. Using Case 3 , we consider

$$y_{p_2}(t) = C\cos(2t) + D\sin(2t) \Rightarrow y'_{p_2}(t) = -C\sin(2t) + D\cos(2t)$$
$$\Rightarrow y''_{p_2}(t) = -C\cos(t) - D\sin(t).$$

So, substituting into $y'' - 4y' = \cos(2t)$, we have

$$-8\mathrm{C}\cos(2\mathrm{t}) - 8\mathrm{D}\sin(\mathrm{t}) = \cos(2\mathrm{t})$$

Thus, $\mathrm{C} = -\frac{1}{8}$ and $\mathrm{D} = 0$. A particular solution is therefore $y_{p_2}(t) = -\frac{1}{8}\cos(2t)$. 3y the principle of superposition, the general solution is

$$y(t) = y_c + y_{p_1}(t) + y_{p_2}(t) = c_1 + c_2 e^{4t} - \left(\frac{1}{3}t + \frac{2}{9}\right)e^t - \frac{1}{8}\cos(2t).$$

Example 50

Find a particular solution of

$$y'' + 4y = 8\cos 2x + 12\sin 2x. \tag{11.54}$$

Solution

The procedure used in the above example doesn't work here; substituting $y_p = A\cos 2x + B\sin 2x$ for y in (11.54) yields

$$y''_p + 4y_p = -4(A\cos 2x + B\sin 2x) + 4(A\cos 2x + B\sin 2x) = 0$$

for any choice of A and B, since $\cos 2x$ and $\sin 2x$ are both solutions of the complementary equation for (11.54).Therefore try a particular solution of the form

$$y_p = x(A\cos 2x + B\sin 2x). \tag{11.55}$$

Then

$$\begin{aligned} y_p' &= A\cos 2x + B\sin 2x + 2x(-A\sin 2x + B\cos 2x) \\ \text{and } y_p'' &= -4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x) \\ &= -4A\sin 2x + 4B\cos 2x - 4y_p, \end{aligned}$$

so

$$y_p'' + 4y_p = -4A\sin 2x + 4B\cos 2x.$$

Therefore y_p is a solution of (11.54) if

$$-4A\sin 2x + 4B\cos 2x = 8\cos 2x + 12\sin 2x,$$

which holds if $A = -3$ and $B = 2$. Therefore

$$y_p = -x(3\cos 2x - 2\sin 2x)$$

is a particular solution of (11.54).

Example 51 Find a particular solution of

$$y'' + 3y' + 2y = (16 + 20x)\cos x + 10\sin x. \tag{11.56}$$

Solution The coefficients of $\cos x$ and $\sin x$ in (11.56) are polynomials of degree one and zero, respectively. Therefore, a particular solution of (11.56) of the form

$$y_p = (A_0 + A_1 x)\cos x + (B_0 + B_1 x)\sin x. \tag{11.57}$$

Then

$$y_p' = (A_1 + B_0 + B_1 x)\cos x + (B_1 - A_0 - A_1 x)\sin x \tag{11.58}$$

and

$$y_p'' = (2B_1 - A_0 - A_1 x)\cos x - (2A_1 + B_0 + B_1 x)\sin x, \tag{11.59}$$

so

$$\begin{aligned} y_p'' + 3y_p' + 2y_p = {} & [A_0 + 3A_1 + 3B_0 + 2B_1 + (A_1 + 3B_1)x]\cos x \\ & + [B_0 + 3B_1 - 3A_0 - 2A_1 + (B_1 - 3A_1)x]\sin x. \end{aligned} \tag{11.60}$$

Comparing the coefficients of $x\cos x$, $x\sin x$, $\cos x$, and $\sin x$ here with the corresponding coefficients in (11.56) shows that y_p is a solution of (11.56) if

$$\begin{aligned} A_1 + 3B_1 &= 20 \\ -3A_1 + \ \ B_1 &= \ \ 0 \\ A_0 + 3B_0 + 3A_1 + 2B_1 &= 16 \\ -3A_0 + \ \ B_0 - 2A_1 + 3B_1 &= 10. \end{aligned}$$

Solving the first two equations yields $A_1 = 2$, $B_1 = 6$. Substituting these into the last two equations yields

$$\begin{aligned} A_0 + 3B_0 &= 16 - 3A_1 - 2B_1 = -2 \\ -3A_0 + \ \ B_0 &= 10 + 2A_1 - 3B_1 = -4. \end{aligned}$$

Solving these equations yields $A_0 = 1$, $B_0 = -1$. Substituting $A_0 = 1$, $A_1 = 2$,

$B_0 = -1$, $B_1 = 6$ into (11.57) shows that

$$y_p = (1 + 2x)\cos x - (1 - 6x)\sin x$$

is a particular solution of (11.56).

SUBSECTION 11.5

Exercises

find a particular solution of the equation. Then find the general solution and, where indicated, solve the initial value problem.

1. $y'' - 8y' + 16y = 23\cos x - 7\sin x$
2. $y'' + y' = -8\cos 2x + 6\sin 2x$
3. $y'' - 2y' + 3y = -6\cos 3x + 6\sin 3x$
4. $y'' + 6y' + 13y = 18\cos x + 6\sin x$
5. $y'' + 7y' + 12y = -2\cos 2x + 36\sin 2x, \quad y(0) = -3, \quad y'(0) = 3$
6. $y'' - 6y' + 9y = 18\cos 3x + 18\sin 3x, \quad y(0) = 2, \quad y'(0) = 2$

SUBSECTION 11.6

Some solutions

Solution of problem 1

The characteristic polynomial of the complementary equation is

$$p(r) = r^2 - 8r + 16 = (r - 4)^2$$

then,

$$y_c = C_1 e^{4x} + C_2 x e^{4x}$$

For the y_p

$y_p = A\cos x + B\sin x$. This implies that

$$\begin{aligned} y_p'' - 8y_p' + 16y_p &= -(A\cos x + B\sin x) - 8(-A\sin x + B\cos x) + 16(A\cos x + B\sin x) \\ &= (15A - 8B)\cos x + (8A + 15B)\sin x \\ 15A - 8B &= 23,\ 8A + 15B = -7 \end{aligned}$$

which implies that $A = 1$ and $B = -1$. Hence $y_p = \cos x - \sin x$ and $y = \cos x - \sin x + e^{4x}(c_1 + c_2 x)$ is the general solution.

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Solution of problem 3

The characteristic polynomial of the complementary equation is

$$p(r) = r^2 - 2r + 3 = (r - 1)^2 + 2$$

so $\{e^x \cos\sqrt{2}x, e^x \sin\sqrt{2}x\}$ is a fundamental set of solutions for the complementary equation.

If $y_p = A\cos 3x + B\sin 3x$, then $y_p'' - 2y_p' + 3y_p = -9(A\cos 3x + B\sin 3x) - 6(-A\sin 3x + B\cos 3x) + 3(A\cos 3x + B\sin 3x) = -(6A + 6B)\cos 3x + (6A - 6B)\sin 3x$, so $-6A - 6B = -6,\ 6A - 6B = 6$,

which implies that $A = 1$ and $B = 0$. Hence $y_p = \cos 3x$ is a particular solution and $y = \cos 3x + e^x(c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x)$ is the general solution.

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Solution of problem 5

The characteristic polynomial of the complementary equation is $p(r) = r^2 + 7r + 12 = (r+3)(r+4)$, so $\{e^{-4x}, e^{-3x}\}$ is a fundamental set of solutions for the complementary equation. If $y_p = A\cos 2x + B\sin 2x$, then $y_p'' + 7y_p' + 12y_p = -4(A\cos 2x + B\sin 2x) + 14(-A\sin 2x + B\cos 2x) + 12(A\cos x + B\sin x) = (8A + 14B)\cos 2x + (8B - 14a)\sin 2x$, so $8A + 14B = -2,\ -14A + 8B = 36$, which implies that $A = -2$ and $B = 1$. Hence, $y_p = -2\cos 2x + \sin 2x$ is a particular solution and (A) $y = -2\cos 2x + \sin 2x + c_1 e^{-4x} + c_2 e^{-3x}$ is the general solution. Differentiating (A) yields $y' = 2\sin 2x + 2\cos 2x - 4c_1 e^{-4x} - 3c_2 e^{-3x}$. Now $y(0) = -3 \Rightarrow -3 = -2 + c_1 + c_2$ and $y'(0) = 3 \Rightarrow 3 = 2 - 4c_1 - 3c_2$. Therefore, $c_1 = 2$ and $c_2 = -3$, so $y = -2\cos 2x + \sin 2x + 2e^{-4x} - 3e^{-3x}$ is the solution of the initial value problem.

SUBSECTION 11.7

More exercises

Find the particular solution of the following:

1. $y'' + 3y' + 2y = 7\cos x - \sin x$
2. $y'' + 3y' + y = (2 - 6x)\cos x - 9\sin x$
3. $y'' + 2y' + y = e^x(6\cos x + 17\sin x)$
4. $y'' + 3y' - 2y = -e^{2x}(5\cos 2x + 9\sin 2x)$
5. $y'' - y' + y = e^x(2 + x)\sin x$
6. $y'' + 3y' - 2y = e^{-2x}\left[(4 + 20x)\cos 3x + (26 - 32x)\sin 3x\right]$
7. $y'' + 4y = -12\cos 2x - 4\sin 2x$
8. $y'' + y = (-4 + 8x)\cos x + (8 - 4x)\sin x$
9. $4y'' + y = -4\cos x/2 - 8x\sin x/2$
10. $y'' + 2y' + 2y = e^{-x}(8\cos x - 6\sin x)$
11. $y'' - 2y' + 5y = e^x\left[(6 + 8x)\cos 2x + (6 - 8x)\sin 2x\right]$
12. $y'' + 2y' + y = 8x^2\cos x - 4x\sin x$
13. $y'' + 3y' + 2y = (12 + 20x + 10x^2)\cos x + 8x\sin x$
14. $y'' + 3y' + 2y = (1 - x - 4x^2)\cos 2x - (1 + 7x + 2x^2)\sin 2x$
15. $y'' - 5y' + 6y = -e^x\left[(4 + 6x - x^2)\cos x - (2 - 4x + 3x^2)\sin x\right]$
16. $y'' - 2y' + y = -e^x\left[(3 + 4x - x^2)\cos x + (3 - 4x - x^2)\sin x\right]$
17. $y'' - 2y' + 2y = e^x\left[(2 - 2x - 6x^2)\cos x + (2 - 10x + 6x^2)\sin x\right]$

In Exercises 1–17 find a particular solution.

18. $y'' + 2y' + y = e^{-x}\left[(5 - 2x)\cos x - (3 + 3x)\sin x\right]$
19. $y'' + 9y = -6\cos 3x - 12\sin 3x$
20. $y'' + 3y' + 2y = (1 - x - 4x^2)\cos 2x - (1 + 7x + 2x^2)\sin 2x$
21. $y'' + 4y' + 3y = e^{-x}\left[(2 + x + x^2)\cos x + (5 + 4x + 2x^2)\sin x\right]$

Solutions of some of the problems

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Solution for exercise 2 Let

$$\begin{aligned} y_p &= (A_0 + A_1x)\cos x + (B_0 + B_1x)\sin x;\ \text{then} \\ y_p' &= (A_1 + B_0 + B_1x)\cos x + (B_1 - A_0 - A_1x)\sin x \\ y_p'' &= (2B_1 - A_0 - A_1x)\cos x - (2A_1 + B_0 + B_1x)\sin x,\ \text{so} \end{aligned}$$

$$\begin{aligned} y_p'' + 3y_p' + y_p &= (3A_1 + 3B_0 + 2B_1 + 3B_1x)\cos x \\ &\quad +(3B_1 - 3A_0 - 2A_1 - 3A_1x)\sin x \\ &= (2 - 6x)\cos x - 9\sin x \end{aligned}$$

if $3B_1 = -6$, $-3A_1 = 0$, $3B_0+3A_1+2B_1 = 2$, $-3A_0+3B_1+2A_1 = -9$. Therefore,$A_1 = 0$, $B_1 = -2$, $A_0 = 1$, $B_0 = 2$, and $y_p = \cos x + (2-2x)\sin x$.

========== Solution for exercise 4

Let $y = ue^{2x}$. Then

$$\begin{aligned} y'' + 3y' - 2y &= e^{2x}\left[(u'' + 4u' + 4u) + 3(u' + 2u) - 2u\right] \\ &= e^{2x}(u'' + 7u' + 8u) = -e^{2x}(5\cos 2x + 9\sin 2x) \end{aligned}$$

if $u'' + 7u' + 8u = -5\cos 2x - 9\sin 2x$. Now let $u_p = A\cos 2x + B\sin 2x$. Then

$$\begin{aligned} u_p'' + 7u_p' + 8u_p &= -4(A\cos 2x + B\sin 2x) + 14(-A\sin 2x + B\cos 2x) \\ &\quad +8(A\cos 2x + B\sin 2x) \\ &= (4A + 14B)\cos 2x - (14A - 4B)\sin 2x \\ &= -5\cos 2x - 9\sin 2x \end{aligned}$$

if $4A + 14B = -5$, $-14A + 4B = -9$. Therefore,$A = \dfrac{1}{2}$, $B = -\dfrac{1}{2}$,

$$u_p = \frac{1}{2}(\cos 2x - \sin 2x), \text{ and } y_p = \frac{e^{2x}}{2}(\cos 2x - \sin 2x).$$

========== Solution for exercise 6

Let $y = ue^{-2x}$. Then

$$\begin{aligned} y'' + 3y' - 2y &= e^{-2x}\left[(u'' - 4u' + 4u) + 3(u' - 2u) - 2u\right] \\ &= e^{-2x}(u'' - u' - 4u) \\ &= e^{-2x}\left[(4 + 20x)\cos 3x + (26 - 32x)\sin 3x\right] \end{aligned}$$

if $u'' - u' - 4u = (4 + 20x)\cos 3x + (26 - 32x)\sin 3x$. Let

$$\begin{aligned} u_p &= (A_0 + A_1x)\cos 3x + (B_0 + B_1x)\sin 3x; \text{ then} \\ u_p' &= (A_1 + 3B_0 + 3B_1x)\cos 3x + (B_1 - 3A_0 - 3A_1x)\sin 3x \\ u_p'' &= (6B_1 - 9A_0 - 9A_1x)\cos 3x - (2A_1 + 9B_0 + 9B_1x)\sin 3x, \text{ so} \end{aligned}$$

$$\begin{aligned} u_p'' - u_p' - 4u_p &= -\left[13A_0 + A_1 + 3B_0 - 6B_1 + (13A_1 + 3B_1)x\right]\cos 3x \\ &\quad -\left[13B_0 + B_1 - 3A_0 + 6A_1 + (13B_1 - 3A_1)x\right]\sin 3x \\ &= (4 + 20x)\cos 3x + (26 - 32x)\sin 3x \text{ if} \end{aligned}$$

$$\begin{aligned} -13A_1 - 3B_1 &= 20 \\ 3A_1 - 13B_1 &= -32 \end{aligned} \quad \text{and} \quad \begin{aligned} -13A_0 - 3B_0 - A_1 + 6B_1 &= 4 \\ 3A_0 - 13B_0 - 6A_1 - B_1 &= 26. \end{aligned}$$

From the first two equations, $A_1 = -2$, $B_1 = 2$. Substituting these in the last two equations yields $-13A_0 - 3B_0 = -10$, $3A_0 - 13B_0 = 16$. Solving this pair yields $A_0 = 1$, $B_0 = -1$. Therefore,

$$u_p = (1 - 2x)(\cos 3x - \sin 3x) \text{ and } y_p = e^{-2x}(1 - 2x)(\cos 3x - \sin 3x).$$

========== Solution for exercise 8

Let

$$\begin{aligned}
y_p &= (A_0x + A_1x^2)\cos x + (B_0x + B_1x^2)\sin x;\ \text{then}\\
y_p' &= \left[A_0 + (2A_1 + B_0)x + B_1x^2\right]\cos x + \left[B_0 + (2B_1 - A_0)x - B_1x^2\right]\sin x\\
y_p'' &= \left[2A_1 + 2B_0 - (A_0 - 4B_1)x - A_1x^2\right]\cos x\\
&\quad + \left[2B_1 - 2A_0 - (B_0 + 4A_1)x - B_1x^2\right]\sin x,\ \text{so}
\end{aligned}$$

$$\begin{aligned}
y_p'' + y_p &= (2A_1 + 2B_0 + 4B_1x)\cos x + (2B_1 - 2A_0 - 4A_1x)\sin x\\
&= (-4 + 8x)\cos x + (8 - 4x)\sin x
\end{aligned}$$

if $4B_1 = 8$, $-4A_1 = -4$, $2B_0 + 2A_1 = -4$, $-2A_0 + 2B_1 = 8$. Therefore, $A_1 = 1$, $B_1 = 2$, $A_0 = -2$, $B_0 = -3$, and $y_p = -x\left[(2 - x)\cos x + (3 - 2x)\sin x\right]$.

==========

Solution for exercise 10

Let $y = ue^{-x}$. Then

$$\begin{aligned}
y'' + 2y' + 2y &= e^{-x}\left[(u'' - 2u' + u) + 2(u' - u) + 2u\right]\\
&= e^{-x}(u'' + u) = e^{-x}(8\cos x - 6\sin x)
\end{aligned}$$

if $u'' + u = 8\cos x - 6\sin x$. Now let

$$\begin{aligned}
u_p &= Ax\cos x + Bx\sin x;\ \text{then}\\
u_p' &= (A + Bx)\cos x + (B - Ax)\sin x\\
u_p'' &= (2B - Ax)\cos x - (2A + Bx)\sin x,\ \text{so}
\end{aligned}$$

$$u_p'' + u_p = 2B\cos x - 2A\sin x = 8\cos x - 6\sin x$$

if $2B = 8$, $-2A = -6$. Therefore,$A = 3$, $B = 4$, $u_p = x(3\cos x + 4\sin x)$, and $y_p = xe^{-x}(3\cos x + 4\sin x)$.

==========

Solution for exercise 12

Let

$$\begin{aligned}
y_p &= (A_0 + A_1x + A_2x^2)\cos x + (B_0 + B_1x + B_2x^2)\sin x;\ \text{then}\\
y_p' &= \left[A_1 + B_0 + (2A_2 + B_1)x + B_2x^2\right]\cos x\\
&\quad + \left[B_1 - A_0 + (2B_2 - A_1)x - A_2x^2\right]\sin x,\\
y_p'' &= \left[-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2\right]\cos x\\
&\quad + \left[-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2\right]\sin x,\ \text{so}
\end{aligned}$$

$$\begin{aligned}
y_p'' + 2y_p' + y_p &= 2\left[A_1 + A_2 + B_0 + B_1 + (2A_2 + B_1 + 2B_2)x + B_2x^2\right]\cos x\\
&\quad +2\left[B_1 + B_2 - A_0 - A_1 + (2B_2 - A_1 - 2A_2)x - A_2x^2\right]\sin x\\
&= 8x^2\cos x - 4x\sin x\ \text{if}
\end{aligned}$$

$$\text{(i)}\ \begin{aligned} 2B_2 &= 8\\ -2A_2 &= 0 \end{aligned},\ \text{(ii)}\ \begin{aligned} 2B_1 + 4A_2 + 4B_2 &= 0\\ -2A_1 - 4A_2 + 4B_2 &= -4 \end{aligned},$$

$$\text{(iii)}\ \begin{aligned} 2B_0 + 2A_1 + 2B_1 + 2A_2 &= 0\\ -2A_0 - 2A_1 + 2B_1 + 2B_2 &= 0 \end{aligned}.$$

From (i), $A_2 = 0$, $B_2 = 4$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 10$, $B_1 = -8$. Substituting the known coefficients into (iii) and solving for A_0 and

B_0 yields $A_0 = -14$, $B_0 = -2$. Therefore,$y_p = -(14 - 10x)\cos x - (2 + 8x - 4x^2)\sin x$.

==========

Solution for exercise 14

Let

$$\begin{aligned}
y_p &= (A_0 + A_1x + A_2x^2)\cos 2x + (B_0 + B_1x + B_2x^2)\sin 2x; \text{ then}\\
y_p' &= \left[A_1 + 2B_0 + (2A_2 + 2B_1)x + 2B_2x^2\right]\cos 2x\\
&\quad + \left[B_1 - 2A_0 + (2B_2 - 2A_1)x - 2A_2x^2\right]\sin 2x\\
y_p'' &= \left[-4A_0 + 2A_2 + 4B_1 - (4A_1 - 8B_2)x - 4A_2x^2\right]\cos 2x\\
&\quad + \left[-4B_0 - 2B_2 - 4A_1 - (4B_1 + 8A_2)x - 4B_2x^2\right]\sin 2x, \text{ so}
\end{aligned}$$

$$\begin{aligned}
y_p'' + 3y_p' + 2y_p &= [-2A_0 + 3A_1 + 4A_2 + 6B_0 + 4B_1\\
&\quad -(2A_1 - 6A_2 - 6B_1 - 8B_2)x - (2A_2 - 6B_2)x^2]\cos 2x\\
&\quad + [-2B_0 + 3B_1 + 4B_2 - 6A_0 - 4A_1\\
&\quad -(2B_1 - 6B_2 + 6A_1 + 8A_2)x - (2B_2 + 6A_2)x^2]\sin 2x\\
&= (1 - x - 4x^2)\cos 2x - (1 + 7x + 2x^2)\sin 2x \text{ if}
\end{aligned}$$

$$\text{(i)}\ \begin{aligned} -2A_2 + 6B_2 &= -4\\ -6A_2 - 2B_2 &= -2 \end{aligned},\quad \text{(ii)}\ \begin{aligned} -2A_1 + 6B_1 + 6A_2 + 8B_2 &= -1\\ -6A_1 - 2B_1 - 8A_2 + 6B_2 &= -7 \end{aligned},$$

$$\text{(iii)}\ \begin{aligned} -2A_0 + 6B_0 + 3A_1 + 4B_1 + 2A_2 &= 1\\ -6A_0 - 2B_0 - 4A_0 + 3B_1 + 2B_2 &= -1 \end{aligned}.$$

From (i), $A_2 = \dfrac{1}{2}$, $B_2 = -\dfrac{1}{2}$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 0$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = 0$. Therefore, $y_p = \dfrac{x^2}{2}(\cos 2x - \sin 2x)$.

==========

Solution for exercise 16

Let $y = ue^x$. Then

$$\begin{aligned}
y'' - 2y' + y &= e^x\left[(u'' + 2u' + u) - 2(u' + u) + u\right] = e^x u''\\
&= -e^x\left[(3 + 4x - x^2)\cos x + (3 - 4x - x^2)\sin x\right]
\end{aligned}$$

if $u'' = -(3 + 4x - x^2)\cos x - (3 - 4x - x^2)\sin x$. Now let

$$\begin{aligned}
u_p &= (A_0 + A_1x + A_2x^2)\cos x + (B_0 + B_1x + B_2x^2)\sin x; \text{ then}\\
u_p' &= \left[A_1 + B_0 + (2A_2 + B_1)x + B_2x^2\right]\cos x\\
&\quad + \left[B_1 - A_0 + (2B_2 - A_1)x - A_2x^2\right]\sin x,\\
u_p'' &= \left[-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2\right]\cos x\\
&\quad + \left[-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2\right]\sin x\\
&= -(3 + 4x - x^2)\cos x - (3 - 4x - x^2)\sin x \text{ if}
\end{aligned}$$

$$\text{(i)}\ \begin{aligned} -A_2 &= 1\\ -B_2 &= 1 \end{aligned},\quad \text{(ii)}\ \begin{aligned} -A_1 + 4B_2 &= -4\\ -B_1 - 4A_2 &= 4 \end{aligned},$$

$$\text{(iii)}\ \begin{aligned} -A_0 + 2B_1 + 2A_2 &= -3\\ -B_0 - 2A_1 + 2B_2 &= -3 \end{aligned}.$$

From (i), $A_2 = -1$, $B_2 = -1$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 0$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 1$, $B_0 = 1$. Therefore,$u_p = (1 - x^2)(\cos x + \sin x)$ and $y_p = e^x(1 - x^2)(\cos x + \sin x)$.

==========

Solution for exercise 18

Let $y = ue^{-x}$. Then

$$\begin{aligned} y'' + 2y' + y &= e^{-x}\left[(u'' - 2u' + u) + 2(u' - u) + u\right] \\ &= e^{-x}u'' = e^{-x}\left[(5 - 2x)\cos x - (3 + 3x)\sin x\right] \end{aligned}$$

if $u'' = (5 - 2x)\cos x - (3 + 3x)\sin x$. Let

$$\begin{aligned} u_p &= (A_0 + A_1x)\cos x + (B_0 + B_1x)\sin x;\ \text{then} \\ u_p' &= (A_1 + B_0 + B_1x)\cos x + (B_1 - A_0 - A_1x)\sin x \\ u_p'' &= (2B_1 - A_0 - A_1x)\cos x - (2A_1 + B_0 + B_1x)\sin x \\ &= (5 - 2x)\cos x - (3 + 3x)\sin x \end{aligned}$$

if $-A_1 = -2$, $-B_1 = -3$, $-A_0 + 2B_1 = 5$, $-B_0 - 2A_1$=-3. Therefore, $A_1 = 2$, $B_1 = 3$, $A_0 = 1$, $B_0 = -1$, $u_p = e^{-x}\left[(1 + 2x)\cos x - (1 - 3x)\sin x\right]$, and $y_p = e^{-x}\left[(1 + 2x)\cos x - (1 - 3x)\sin x\right]$.

==========

Solution for exercise 20

Let

$$\begin{aligned} y_p &= (A_0x + A_1x^2 + A_2x^3)\cos x + (B_0x + B_1x^2 + B_2x^3)\sin x;\ \text{then} \\ y_p' &= \left[A_0 + (2A_1 + B_0)x + (3A_2 + B_1)x^2 + B_2x^3\right]\cos x \\ &\quad + \left[B_0 + (2B_1 - A_0)x + (3B_2 - A_1)x^2 - A_2x^3\right]\sin x \\ y_p'' &= \left[2A_1 + 2B_0 - (A_0 - 6A_2 - 4B_1)x - (A_1 - 6B_2)x^2 - A_2x^3\right]\cos x \\ &\quad + \left[2B_1 - 2A_0 - (B_0 + 6B_2 + 4A_1)x - (B_1 + 6A_2)x^2 - B_2x^3\right]\sin x,\ \text{so} \end{aligned}$$

$$\begin{aligned} y_p'' + y_p &= \left[2A_1 + 2B_0 + (6A_2 + 4B_1)x + 6B_2x^2\right]\cos x \\ &\quad + \left[2B_1 - 2A_0 + (6B_2 - 4A_1)x - 6A_2x^2\right]\sin x \\ &= (2 + 2x)\cos x + (4 + 6x^2)\sin x\ \text{if} \end{aligned}$$

$$\text{(i)}\ \begin{aligned} 6B_2 &= 0 \\ -6A_2 &= 6 \end{aligned},\ \text{(ii)}\ \begin{aligned} 4B_1 + 6A_2 &= 2 \\ -4A_1 + 6B_2 &= 0 \end{aligned},\ \text{(iii)}\ \begin{aligned} 2B_0 + 2A_1 &= 2 \\ -2A_0 + 2B_1 &= 4 \end{aligned}.$$

From (i), $A_2 = -1$, $B_2 = 0$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 2$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = 2$. Therefore,$y_p = -x^3\cos x + (x + 2x^2)\sin x$.

Section 12

Method of Variation of Parameters

Variation of Parameters method is another method of finding the particular solution, y_p for the second order differential equation. It is a more general method compared to the method of undetermined coefficient. Unlike the method of undetermined coefficient, Variation of Parameters method can be used for differential equations with variable coefficient.

The variation of parameters method of finding $y_p(t)$ will be given in the following theorem:

Theorem 6 The differential equation,

$$ay'' + by' + cy = g(t),$$

with

$$y_c(t) = C_1 y_1(t) + C_2 y_2(t),$$

has $y_p(t)$ defined as

$$y_p(t) = u_1(t) y_1(t) + u_2(t) y_2(t),$$

as a particular solution, where

$$u_1(t) = -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)} dt$$

$$u_2(t) = \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)} dt,$$

where,

$$W(y_1, y_2) = y_1 y_2' - y_2 y_1' \neq 0.$$

Proof Suppose that the complementary solution $y_c(t)$ can be written as follows;

$$y_c(t) = C_1 y_1(t) + C_2 y_2(t) \tag{12.1}$$

Assume also that $y_p(t)$ is the particular solution that we looking for.

$$y_p(t) = u_1(t) y_1(t) + u_2(t) y_2(t), \tag{12.2}$$

where, u_1 and u_2 represents any two functions in t.
Since $y_p(t)$ is a solution for our differential equation, it follows that

$$y_p'(t) = u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'. \tag{12.3}$$

Now,let us choose u_1 and u_2 such that

$$u_1' y_1 + u_2' y_2 = 0. \tag{12.4}$$

It follows that the second derivative of y_p is given as follows;

$$y_p''(t) = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2'' \tag{12.5}$$

Now, substitute in

$$ay'' + by' + cy = g(t), \tag{12.6}$$

we get:

$$a\left(u'_1y'_1+u_1y''_1+u'_2y'_2+u_2y''_2\right)+b\left(u_1y'_1+u_2y'_2\right)+c\left(u_1y_1+u_2y_2\right)=g\left(t\right)$$
$$a\left(u'_1y'_1+u'_2y'_2\right)+u_1\left(t\right)\left(ay''_1+by'_1+cy_1\right)+u_2\left(t\right)\left(ay''_2+by'_2+cy_2\right)=g\left(t\right).$$

However, we know that $y_1(t)$ and $y_2(t)$ are solutions for the homogeneous differential equation. This implies that

$$ay''_1+by'_1+cy_1=0$$
$$ay''_2+by'_2+cy_2=0.$$

Then,

$$u_1'y_1'+u_2'y_2'=\frac{g\left(t\right)}{a} \tag{12.7}$$

Now, we have the following two equations;

$$u_1'y_1+u_2'y_2=0$$
$$u_1'y_1'+u_2'y_2'=\frac{g\left(t\right)}{a}.$$

$$u_1'=-\frac{u'_2y_2}{y_1}$$
$$\left(-\frac{u'_2y_2}{y_1}\right)y'_1+u'_2y'_2=\frac{g\left(t\right)}{a}$$
$$u'_2\left(y'_2-\frac{y_2y'_1}{y_1}\right)=\frac{g\left(t\right)}{a}$$
$$u'_2\left(\frac{y_1y'_2-y_2y'_1}{y_1}\right)=\frac{g\left(t\right)}{a}$$

Or,

$$u_2'=\frac{y_1\frac{g(t)}{a}}{y_1y'_2-y_2y'_1} \tag{12.8}$$

In other words,

$$u_1'=-\frac{y_2\frac{g(t)}{a}}{y_1y'_2-y_2y'_1} \tag{12.9}$$

$$u_1\left(t\right)=-\int\frac{y_2\frac{g(t)}{a}}{W\left(y_1,y_2\right)}\,dt \tag{12.10}$$

$$u_2\left(t\right)=\int\frac{y_1\frac{g(t)}{a}}{W\left(y_1,y_2\right)}\,dt, \tag{12.11}$$

where,

$$W\left(y_1,y_2\right)=y_1y_2'-y_2y_1'\neq 0. \tag{12.12}$$

□

The summary of the variation of parameters method can be given as follows:

For the following differential equation;

$$ay'' + by' + cy = g(t), \tag{12.13}$$

the solution is given as follows;

$$y_h = y_c(t) + y_p(t).$$

1 Find the $y_c(t)$ as before.

2 Determine $y_1(t)$ and $y_2(t)$ form the $y_c(t)$ in the above step.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0.$$

4 Find u_1 and u_2 as follows;

$$u_1(t) = -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt$$
$$u_2(t) = \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt,$$

5 Construct your y_p as follows;

$$y_p(t) = u_1(t)\, y_1(t) + u_2(t)\, y_2(t). \tag{12.14}$$

Remark Note that the variation of parameters method can work, in principle, with any forcing function, however, we might have difficulty with the integration in calculating u_1 and u_2.

Now let us do some examples:

Example 52 Find the general solution of

$$y'' + y = \tan(t).$$

Solution Following the above instructions, first we need to find the $y_c(t)$:
The characteristic equation is given as follows;

$$r^2 + 1 = 0$$

This implies that the roots are $\pm i$, i.e., the y_c is given as follows;

$$y_c = C_1 \cos(t) + C_2 \sin(t)$$

Now, to find the y_p using the variation of parameters method, we follow the above instructions as follows;

1 $y_1(t) = \cos(t)$ and $y_2(t) = \sin(t)$.

3 Find W as follows;

$$W\left(y_1, y_2\right)=\begin{vmatrix}\cos(t) & \sin(t)\\ -\sin(t) & \cos(t)\end{vmatrix}=\cos^2(t)+\sin^2(t)=1.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned}u_1\left(t\right)=-\int\frac{y_2\frac{g(t)}{a}}{W\left(y_1,y_2\right)}dt&=-\int\sin(t)\ \tan(t)dt\\&=-\int\frac{\sin^2(t)}{\cos(t)}dt=-\int\frac{1-\cos^2(t)}{\cos(t)}dt\\&=-\ln|\sec(t)+\tan(t)|+\sin(t)\\u_2\left(t\right)=\int\frac{y_1\frac{g(t)}{a}}{W\left(y_1,y_2\right)}dt&=\int\cos(t)\tan(t)dt=\int\sin(t)dt=-\cos(t),\end{aligned}$$

5 Construct your y_p as follows;

$$\begin{aligned}y_p\left(t\right)&=u_1\left(t\right)y_1\left(t\right)+u_2\left(t\right)y_2\left(t\right)\\&=(-\ln|\sec(t)+\tan(t)|+\sin(t))(\cos(t))\quad-\cos(t)\sin(t)\end{aligned}$$

Example 53 Find the general solution of

$$\frac{d^2y}{dt^2}+9y=9\sec^2 3t$$

Solution Similarly, first we need to find the $y_c(t)$:
The characteristic equation is given as follows;

$$r^2+9=0$$

This implies that the roots are $\pm 3i$, i.e., the y_c is given as follows;

$$y_c=C_1\cos(3t)+C_2\sin(3t)$$

Now, to find the y_p using the variation of parameters method, we follow the above instructions as follows;

1 $y_1(t)=\cos(3t)$ and $y_2(t)=\sin(3t)$.

3 Find W as follows;

$$W\left(y_1, y_2\right)=\begin{vmatrix}\cos(3t) & \sin(3t)\\ -3\sin(3t) & 3\cos(3t)\end{vmatrix}=3(\cos^2(3t)+\sin^2(3t))=3.$$

4 Find u_1 and u_2 as follows;

$$u_1(t) = -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = -\int 3\sin(3t)\ \sec^2(3t)dt$$
$$= -3\int \frac{\sin(3t)}{\cos^2(3t)}dt = ...$$
$$u_2(t) = \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = \int 3\cos(3t)\sec^(3t)dt = \int 3\cos^3(3t)dt = ...,$$

5 Construct your y_p as follows;

$$y_p(t) = u_1(t)\, y_1(t) + u_2(t)\, y_2(t)$$
$$= .....$$

Example 54 Find the general solution of the differential equation

$$y'' + 4y = \sec^2(2t).$$

Solution Let's solve this using the method of variation of parameters:
1) First, find $y_c(t)$: The characteristic equation is $r^2 + 4 = 0$. This gives $r = \pm 2i$. So, $y_c(t) = C_1\cos(2t) + C_2\sin(2t)$.
2) Let $y_1(t) = \cos(2t)$ and $y_2(t) = \sin(2t)$.
3) Calculate the Wronskian:

$$W(y_1, y_2) = \begin{vmatrix} \cos(2t) & \sin(2t) \\ -2\sin(2t) & 2\cos(2t) \end{vmatrix} = 2\cos^2(2t) + 2\sin^2(2t) = 2$$

4) Find u_1 and u_2:

$$u_1(t) = -\int \frac{y_2(t)g(t)}{W(y_1, y_2)}dt = -\frac{1}{2}\int \sin(2t)\sec^2(2t)dt$$
$$= -\frac{1}{2}\int \frac{\sin(2t)}{\cos^2(2t)}dt = \frac{1}{4}\int \sec(2t)d(\sec(2t))$$
$$= \frac{1}{8}\sec(2t)\tan(2t)$$
$$u_2(t) = \int \frac{y_1(t)g(t)}{W(y_1, y_2)}dt = \frac{1}{2}\int \cos(2t)\sec^2(2t)dt$$
$$= \frac{1}{2}\int \frac{1}{\cos(2t)}dt = \frac{1}{4}\ln|\sec(2t) + \tan(2t)|$$

5) The particular solution is:

$$y_p(t) = u_1(t)y_1(t) + u_2(t)y_2(t)$$
$$= \frac{1}{8}\sec(2t)\tan(2t)\cos(2t) + \frac{1}{4}\ln|\sec(2t) + \tan(2t)|\sin(2t)$$
$$= \frac{1}{8}\tan(2t) + \frac{1}{4}\ln|\sec(2t) + \tan(2t)|\sin(2t)$$

Therefore, the general solution is:

$$y(t) = C_1 \cos(2t) + C_2 \sin(2t) + \frac{1}{8}\tan(2t) + \frac{1}{4}\ln|\sec(2t) + \tan(2t)|\sin(2t)$$

Example 55 Solve the following DE

$$y'' - 2y' + 1 = (x+1)e^{2x}$$

Solution First, we find the y_c

$$\begin{aligned} &y'' - 2y' + 1 = 0 \\ &m^2 - 2m + 1 = 0 \\ &(m-1)(m-1) = 0 \\ &m = 1 \end{aligned} \tag{12.15}$$

Therefore one of the complimentary solutions is in the form,

$$\begin{aligned} &y_c = c_1 e^x + c_2 x e^x \\ &\text{ where} \\ &y_1 = e^x \\ &y_2 = x e^x \end{aligned} \tag{12.16}$$

$$\begin{aligned} W(y_1, y_2) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \\ W(e^x, xe^x) &= \begin{vmatrix} e^x & xe^x \\ e^x & xe^x + e^x \end{vmatrix} = e^{2x} \end{aligned} \tag{12.17}$$

Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) = -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dx &= -\int \frac{(x+1)e^{3x}}{e^{2x}} \mathrm{d}x \\ &= -\int (x+1)e^x \mathrm{d}x = \ldots\ldots\ldots\ldots \\ u_2(t) = \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dx &= \int \frac{x(x+1)e^{3x}}{e^{2x}} \mathrm{d}x \\ &= \int x(x+1)e^x \mathrm{d}x = \ldots\ldots\ldots\ldots, \end{aligned}$$

Construct your y_p as follows;

$$\begin{aligned} y_p(t) &= u_1(t)\, y_1(t) + u_2(t)\, y_2(t) \\ &= \ldots\ldots\ldots\ldots \end{aligned}$$

Certainly, I'll generate 5 different problems with their solutions using the same style of writing and method of solving as the example you provided. Here they are:

Example 56 Find the general solution of

$$y'' + 4y = \sec^2(2t).$$

Solution Following the instructions, first we need to find the $y_c(t)$: The characteristic equation is given as follows;

$$r^2 + 4 = 0$$

This implies that the roots are $\pm 2i$, i.e., the y_c is given as follows;

$$y_c = C_1 \cos(2t) + C_2 \sin(2t)$$

Now, to find the y_p using the variation of parameters method, we follow the instructions as follows;

1 $y_1(t) = \cos(2t)$ and $y_2(t) = \sin(2t)$.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} \cos(2t) & \sin(2t) \\ -2\sin(2t) & 2\cos(2t) \end{vmatrix} = 2\cos^2(2t) + 2\sin^2(2t) = 2.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) &= -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = -\int \frac{\sin(2t)\sec^2(2t)}{2} dt \\ &= -\frac{1}{4}\int \frac{\sin(4t)}{\cos^2(2t)} dt = \frac{1}{4}\tan(2t) \\ u_2(t) &= \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = \int \frac{\cos(2t)\sec^2(2t)}{2} dt = \frac{1}{4}\ln|\sec(2t) + \tan(2t)|, \end{aligned}$$

5 Construct your y_p as follows;

$$\begin{aligned} y_p(t) &= u_1(t)\, y_1(t) + u_2(t)\, y_2(t) \\ &= \frac{1}{4}\tan(2t)\cos(2t) + \frac{1}{4}\ln|\sec(2t) + \tan(2t)|\sin(2t) \end{aligned}$$

Example 57 Find the general solution of

$$y'' - y = e^t \sin(t).$$

Solution Following the instructions, first we need to find the $y_c(t)$: The characteristic equation is given as follows;

$$r^2 - 1 = 0$$

This implies that the roots are ± 1, i.e., the y_c is given as follows;

$$y_c = C_1 e^t + C_2 e^{-t}$$

Now, to find the y_p using the variation of parameters method, we follow the instructions as follows;

1 $y_1(t) = e^t$ and $y_2(t) = e^{-t}$.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{vmatrix} = -e^t e^{-t} - e^t e^{-t} = -2.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) &= -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = \frac{1}{2}\int e^{-t} e^{t} \sin(t) dt \\ &= \frac{1}{2}\int \sin(t) dt = -\frac{1}{2}\cos(t) \\ u_2(t) &= \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = -\frac{1}{2}\int e^{t} e^{t} \sin(t) dt = -\frac{1}{4} e^{2t}(\sin(t) - \cos(t)), \end{aligned}$$

5 Construct your y_p as follows;

$$\begin{aligned} y_p(t) &= u_1(t)\, y_1(t) + u_2(t)\, y_2(t) \\ &= -\frac{1}{2}\cos(t) e^{t} - \frac{1}{4} e^{2t}(\sin(t) - \cos(t)) e^{-t} \\ &= -\frac{1}{2} e^{t}\cos(t) - \frac{1}{4} e^{t}(\sin(t) - \cos(t)) \end{aligned}$$

Example 58 Find the general solution of

$$y'' + 2y' + y = t^2 e^{-t}.$$

Solution Following the instructions, first we need to find the $y_c(t)$: The characteristic equation is given as follows;

$$r^2 + 2r + 1 = 0$$

This implies that the root is -1 with multiplicity 2, i.e., the y_c is given as follows;

$$y_c = (C_1 + C_2 t) e^{-t}$$

Now, to find the y_p using the variation of parameters method, we follow the instructions as follows;

1 $y_1(t) = e^{-t}$ and $y_2(t) = te^{-t}$.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} e^{-t} & te^{-t} \\ -e^{-t} & e^{-t} - te^{-t} \end{vmatrix} = e^{-2t} - (-te^{-2t}) = e^{-2t}.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) &= -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = -\int te^{-t} t^2 e^{t} dt \\ &= -\int t^3 dt = -\frac{1}{4} t^4 \\ u_2(t) &= \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = \int e^{-t} t^2 e^{t} dt = \int t^2 dt = \frac{1}{3} t^3, \end{aligned}$$

5 Construct your y_p as follows;

$$\begin{aligned} y_p(t) &= u_1(t)\, y_1(t) + u_2(t)\, y_2(t) \\ &= -\frac{1}{4}t^4 e^{-t} + \frac{1}{3}t^4 e^{-t} \\ &= \frac{1}{12}t^4 e^{-t} \end{aligned}$$

Example 59 Find the general solution of

$$y'' + 4y' + 4y = \cosh(2t).$$

Solution Following the instructions, first we need to find the $y_c(t)$: The characteristic equation is given as follows;

$$r^2 + 4r + 4 = 0$$

This implies that the root is -2 with multiplicity 2, i.e., the y_c is given as follows;

$$y_c = (C_1 + C_2 t)e^{-2t}$$

Now, to find the y_p using the variation of parameters method, we follow the instructions as follows;

1 $y_1(t) = e^{-2t}$ and $y_2(t) = te^{-2t}$.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} e^{-2t} & te^{-2t} \\ -2e^{-2t} & e^{-2t} - 2te^{-2t} \end{vmatrix} = e^{-4t} - (-2te^{-4t}) = e^{-4t}.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) &= -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = -\int te^{-2t}\cosh(2t)e^{4t}dt \\ &= -\int te^{2t}\cosh(2t)dt = -\frac{1}{8}e^{2t}(2t\cosh(2t) - \sinh(2t)) \\ u_2(t) &= \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)}\, dt = \int e^{-2t}\cosh(2t)e^{4t}dt = \int e^{2t}\cosh(2t)dt = \frac{1}{8}e^{2t}\sinh(2t), \end{aligned}$$

5 Construct your y_p as follows;

$$\begin{aligned} y_p(t) &= u_1(t)\, y_1(t) + u_2(t)\, y_2(t) \\ &= -\frac{1}{8}e^{2t}(2t\cosh(2t) - \sinh(2t))e^{-2t} + \frac{1}{8}e^{2t}\sinh(2t)te^{-2t} \\ &= -\frac{1}{4}t\cosh(2t) + \frac{1}{8}\sinh(2t) \end{aligned}$$

Example 60 Find the general solution of

$$y'' - 2y' + y = te^t.$$

Solution Following the instructions, first we need to find the $y_c(t)$: The characteristic equation is given as follows;

$$r^2 - 2r + 1 = 0$$

This implies that the root is 1 with multiplicity 2, i.e., the y_c is given as follows;

$$y_c = (C_1 + C_2 t)e^t$$

Now, to find the y_p using the variation of parameters method, we follow the instructions as follows;

1 $y_1(t) = e^t$ and $y_2(t) = te^t$.

3 Find W as follows;

$$W(y_1, y_2) = \begin{vmatrix} e^t & te^t \\ e^t & e^t + te^t \end{vmatrix} = e^{2t} - te^{2t} = e^{2t}.$$

4 Find u_1 and u_2 as follows;

$$\begin{aligned} u_1(t) = -\int \frac{y_2 \frac{g(t)}{a}}{W(y_1, y_2)} dt &= -\int te^t te^t e^{-2t} dt \\ &= -\int t^2 dt = -\frac{1}{3}t^3 \\ u_2(t) = \int \frac{y_1 \frac{g(t)}{a}}{W(y_1, y_2)} dt &= \int e^t te^t e^{-2t} dt = \int t dt = \frac{1}{2}t^2, \end{aligned}$$

5 Construct your y_p as follows;

Section 13

exercise

In Exercises 1–6 use variation of parameters to find a particular solution.

1. $y'' + 9y = \tan 3x$
2. $y'' + 4y = \sin 2x \sec^2 2x$
3. $y'' - 3y' + 2y = \dfrac{4}{1 + e^{-x}}$
4. $y'' - 2y' + 2y = 3e^x \sec x$
5. $y'' - 2y' + y = 14x^{3/2}e^x$
6. $y'' - y = \dfrac{4e^{-x}}{1 - e^{-2x}}$

In Exercises 7–29 use variation of parameters to find a particular solution, given the solutions y_1, y_2 of the complementary equation.

7. $x^2y'' + xy' - y = 2x^2 + 2; \quad y_1 = x, \quad y_2 = \dfrac{1}{x}$

8. $xy'' + (2 - 2x)y' + (x - 2)y = e^{2x}; \quad y_1 = e^x, \quad y_2 = \dfrac{e^x}{x}$

9. $4x^2y'' + (4x - 8x^2)y' + (4x^2 - 4x - 1)y = 4x^{1/2}e^x, \quad x > 0;$
$y_1 = x^{1/2}e^x,\ y_2 = x^{-1/2}e^x$

10. $y'' + 4xy' + (4x^2 + 2)y = 4e^{-x(x+2)}; \quad y_1 = e^{-x^2}, \quad y_2 = xe^{-x^2}$

11. $x^2y'' - 4xy' + 6y = x^{5/2},\ x > 0; \quad y_1 = x^2,\ y_2 = x^3$

12. $x^2y'' - 3xy' + 3y = 2x^4 \sin x; \quad y_1 = x,\ y_2 = x^3$

13. $(2x + 1)y'' - 2y' - (2x + 3)y = (2x + 1)^2e^{-x}; \quad y_1 = e^{-x}, \quad y_2 = xe^x$

14. $4xy'' + 2y' + y = \sin\sqrt{x}; \quad y_1 = \cos\sqrt{x}, \quad y_2 = \sin\sqrt{x}$

15. $xy'' - (2x + 2)y' + (x + 2)y = 6x^3e^x; \quad y_1 = e^x, \quad y_2 = x^3e^x$

16. $x^2y'' - (2a - 1)xy' + a^2y = x^{a+1}; \quad y_1 = x^a, \quad y_2 = x^a \ln x$

17. $x^2y'' - 2xy' + (x^2 + 2)y = x^3 \cos x; \quad y_1 = x\cos x, \quad y_2 = x\sin x$

18. $xy'' - y' - 4x^3y = 8x^5; \quad y_1 = e^{x^2},\ y_2 = e^{-x^2}$

19. $(\sin x)y'' + (2\sin x - \cos x)y' + (\sin x - \cos x)y = e^{-x}; \quad y_1 = e^{-x}, \quad y_2 = e^{-x}\cos x$

20. $4x^2y'' - 4xy' + (3 - 16x^2)y = 8x^{5/2}; \quad y_1 = \sqrt{x}e^{2x},\ y_2 = \sqrt{x}e^{-2x}$

21. $4x^2y'' - 4xy' + (4x^2 + 3)y = x^{7/2}; \quad y_1 = \sqrt{x}\sin x,\ y_2 = \sqrt{x}\cos x$

22. $x^2y'' - 2xy' - (x^2 - 2)y = 3x^4; \quad y_1 = xe^x,\ y_2 = xe^{-x}$

23. $x^2y'' - 2x(x + 1)y' + (x^2 + 2x + 2)y = x^3e^x; \quad y_1 = xe^x, \quad y_2 = x^2e^x$

24. $x^2y'' - xy' - 3y = x^{3/2}; \quad y_1 = 1/x, \quad y_2 = x^3$

25. $x^2y'' - x(x + 4)y' + 2(x + 3)y = x^4e^x; \quad y_1 = x^2, \quad y_2 = x^2e^x$

26. $x^2y'' - 2x(x + 2)y' + (x^2 + 4x + 6)y = 2xe^x; \quad y_1 = x^2e^x, \quad y_2 = x^3e^x$

27. $x^2y'' - 4xy' + (x^2 + 6)y = x^4; \quad y_1 = x^2\cos x, \quad y_2 = x^2\sin x$

28. $(x - 1)y'' - xy' + y = 2(x - 1)^2e^x; \quad y_1 = x, \quad y_2 = e^x$

29. $4x^2y'' - 4x(x + 1)y' + (2x + 3)y = x^{5/2}e^x; \quad y_1 = \sqrt{x}, \quad y_2 = \sqrt{x}e^x$

In Exercises 30–32 use variation of parameters to solve the initial value problem, given y_1, y_2 are solutions of the complementary equation.

30. $(3x - 1)y'' - (3x + 2)y' - (6x - 8)y = (3x - 1)^2e^{2x}, \quad y(0) = 1,\ y'(0) = 2;$
$y_1 = e^{2x},\ y_2 = xe^{-x}$

31. $(x - 1)^2y'' - 2(x - 1)y' + 2y = (x - 1)^2, \quad y(0) = 3, \quad y'(0) = -6;$
$y_1 = x - 1,\ y_2 = x^2 - 1$

32. $(x - 1)^2y'' - (x^2 - 1)y' + (x + 1)y = (x - 1)^3e^x, \quad y(0) = 4, \quad y'(0) = -6;$
$y_1 = (x - 1)e^x, \quad y_2 = x - 1$

Subsection 13.1

Some solutions

5.7.2. (A) $y_p = u_1 \cos 2x + u_2 \sin 2x$;

$$\begin{aligned} u_1' \cos 2x + \ \ u_2' \sin 2x &= 0 \qquad \text{(B)} \\ -2u_1' \sin 2x + 2u_2' \cos 2x &= \sin 2x \sec^2 x. \qquad \text{(C)} \end{aligned}$$

Multiplying (B) by $2\sin 2x$ and (C) by $\cos 2x$ and adding the resulting equations yields $2u_2' = \tan 2x$, so $u_2' = \dfrac{\tan 2x}{2}$. Then (B) implies that $u_1' = -u_2' \tan(2x) = -\dfrac{\tan^2 2x}{2} = \dfrac{1-\sec^2 2x}{2}$. Therefore,$u_1 = \dfrac{x}{2} - \dfrac{\tan 2x}{4}$ and $u_2 = -\dfrac{\ln|\cos 2x|}{4}$. Now (A) yields $y_p = -\dfrac{\sin 2x \ln|\cos 2x|}{4} + \dfrac{x\cos 2x}{2} - \dfrac{\sin 2x}{4}$. Since $\sin 2x$ satisfies the complementary equation we redefine $y_p = -\dfrac{\sin 2x \ln|\cos 2x|}{4} + \dfrac{x\cos 2x}{2}$.

5.7.4. (A) $y_p = u_1 e^x \cos x + u_2 e^x \sin x$;

$$\begin{aligned} u_1' e^x \cos x + u_2' e^x \sin x &= 0 \qquad \text{(B)} \\ u_1'(e^x \cos x - e^x \sin x) + u_2'(e^x \sin x + e^x \cos x) &= 3e^x \sec x. \qquad \text{(C)} \end{aligned}$$

Subtracting (B) from (C) and cancelling e^x from the resulting equations yields

$$\begin{aligned} u_1' \cos x + u_2' \sin x &= 0 \qquad \text{(D)} \\ -u_1' \sin x + u_2' \cos x &= 3\sec x. \qquad \text{(E)} \end{aligned}$$

Multiplying (D) by $\sin x$ and (E) by $\cos x$ and adding the results yields $u_2' = 3$. From (D), $u_1' = -u_2' \tan x = -3\tan x$. Therefore $u_1 = 3\ln|\cos x|$, $u_2 = 3x$. Now (A) yields $y_p = 3e^x(\cos x \ln|\cos x| + x \sin x)$.

5.7.6. (A) $y_p = u_1 e^x + u_2 e^{-x}$;

$$\begin{aligned} u_1' e^x + u_2' e^{-x} &= 0 \qquad \text{(B)} \\ u_1' e^x - u_2' e^{-x} &= \frac{4e^{-x}}{1+e^{-2x}}. \qquad \text{(C)} \end{aligned}$$

Adding (B) to (C) yields $2u_1' e^x = \dfrac{4e^{-x}}{1+e^{-2x}}$, so $u_1' = \dfrac{2e^{-2x}}{1-e^{-2x}}$. From (B), $u_2' = -e^{2x} u_1' = -\dfrac{2}{1-e^{-2x}} = \dfrac{2e^{2x}}{1-e^{2x}}$. Using the substitution $v = e^{-2x}$ we integrate u_1' to obtain $u_1 = \ln\bigl(1-e^{-2x}\bigr)$. Using the substitution $v = e^{2x}$ we integrate u_2' to obtain $u_1 = \ln\bigl(1-e^{2x}\bigr)$. Now (A) yields $y_p = e^x \ln\bigl(1-e^{-2x}\bigr) - e^{-x}\ln\bigl(e^{2x}-1\bigr)$.

5.7.8. (A) $y_p = u_1 e^x + u_2 \dfrac{e^x}{x}$;

$$\begin{aligned} u_1' e^x + u_2' \frac{e^x}{x} &= 0 \qquad \text{(B)} \\ u_1' e^x + u_2' \left(\frac{e^x}{x} - \frac{e^x}{x^2}\right) &= \frac{e^{2x}}{x}. \qquad \text{(C)} \end{aligned}$$

Subtracting (B) from (C) yields $-\dfrac{u_2' e^x}{x^2} = \dfrac{e^{2x}}{x}$, so $u_2' = -xe^x$. From (B), $u_1' = \dfrac{u_2'}{x} = e^x$.

Therefore $u_1 = e^x$, $u_2 = -xe^x + e^x$. Now (A) yields $y_p = \dfrac{e^{2x}}{x}$.

5.7.10. (A) $y_p = u_1 e^{-x^2} + u_2 x e^{-x^2}$;

$$\begin{aligned} u_1' e^{-x^2} + u_2' x e^{-x^2} &= 0 \qquad \text{(B)} \\ -2xu_1' e^{-x^2} + u_2'(e^{-x^2} - 2x^2 e^{-x^2}) &= 4e^{-x(x+2)}. \qquad \text{(C)} \end{aligned}$$

Multiplying (B) by $2x$ and adding the result to (C) yields $u_2' e^{-x^2} = 4e^{-x(x+2)}$, so $u_2' = 4e^{-2x}$. From (B), $u_1' = -u_2' x = -4xe^{-2x}$. Therefore $u_1 = (2x+1)e^{-2x}$, $u_2 = -2e^{-2x}$. Now (A) yields $y_p = e^{-x(x+2)}$.

5.7.12. (A) $y_p = u_1 x + u_2 x^3$;

$$\begin{aligned} u_1' x + \ \ u_2' x^3 &= 0 \qquad \text{(B)} \\ u_1' \ \ + 3u_2' x^2 &= \frac{2x^4 \sin x}{x^2} = 2x^2 \sin x \qquad \text{(C)}. \end{aligned}$$

Multiplying (B) by $\dfrac{1}{x}$ and subtracting the result from (C) yields $2x^2 u_2' = 2x^2 \sin x$, so $u_2' = \sin x$. From (B), $u_1' = -u_2' x^2 = -x^2 \sin x$. Therefore $u_1 = (x^2 - 2)\cos x - 2x \sin x$, $u_2 = -\cos x$. Now (A) yields $y_p = -2x^2 \sin x - 2x \cos x$.

5.7.14. (A) $y_p = u_1 \cos\sqrt{x} + u_2 \sin\sqrt{x}$;

$$\begin{aligned} u_1' \cos\sqrt{x} + u_2' \sin\sqrt{x} &= 0 \qquad \text{(B)} \\ -u_1' \frac{\sin\sqrt{x}}{2\sqrt{x}} + u_2' \frac{\cos\sqrt{x}}{2\sqrt{x}} &= \frac{\sin\sqrt{x}}{4x} \qquad \text{(C)}. \end{aligned}$$

Multiplying (B) by $\dfrac{\sin\sqrt{x}}{2\sqrt{x}}$ and (C) by $\cos\sqrt{x}$ and adding the resulting equations yields $\dfrac{u_2'}{2\sqrt{x}} = \dfrac{\sin\sqrt{x}\cos\sqrt{x}}{4x}$, so $u_2' = \dfrac{\sin\sqrt{x}\cos\sqrt{x}}{2\sqrt{x}}$. From (B), $u_1' = -u_2' \tan\sqrt{x} = -\dfrac{\sin^2\sqrt{x}}{2\sqrt{x}}$. Therefore, $u_1 = \dfrac{\sin\sqrt{x}\cos\sqrt{x}}{2} - \dfrac{\sqrt{x}}{2}$, $u_2 = \dfrac{\sin^2\sqrt{x}}{2}$. Now (A) yields $y_p = \dfrac{\sin\sqrt{x}}{2} - \dfrac{\sqrt{x}\cos\sqrt{x}}{2}$. Since $\sin\sqrt{x}$ satisfies the complementary equation we redefine $y_p = -\dfrac{\sqrt{x}\cos\sqrt{x}}{2}$.

5.7.16. (A) $y_p = u_1 x^a + u_2 x^a \ln x$;

$$\begin{aligned} u_1' x^a + u_2' x^a \ln x &= 0 \qquad \text{(B)} \\ a u_1' x^{a-1} + u_2'(a x^{a-1} \ln x + x^{a-1}) &= \frac{x^{a+1}}{x^2} = x^{a-1} \qquad \text{(C)}. \end{aligned}$$

Multiplying (B) by $\dfrac{a}{x}$ and subtracting the result from (C) yields $u_2' x^{a-1} = x^{a-1}$, so $u_2' = 1$. From (B), $u_1' = -u_2' \ln x = -\ln x$. Therefore, $u_1 = x - \ln x$, $u_2 = x$. Now (A) yields $y_p = x^{a+1}$.

5.7.18. $y_p = u_1 e^{x^2} + u_2 e^{-x^2}$;

$$\begin{aligned} u_1' e^{x^2} + u_2' e^{-x^2} &= 0 \qquad \text{(B)} \\ 2u_1' x e^{x^2} - 2u_2' x e^{-x^2} &= \frac{8x^5}{x} = 8x^4. \qquad \text{(B)} \end{aligned}$$

Multiplying (B) by $2x$ and adding the result to (C) yields $4u_1' x e^{x^2} = 8x^4$, so $u_1' = 2x^3 e^{-x^2}$. From (B), $u_2' = -u_1' e^{2x^2} = -2x^3 e^{x^2}$. Therefore $u_1 = -e^{-x^2}(x^2+1)$, $u_2 = -e^{x^2}(x^2-1)$.

Now (A) yields $y_p = -2x^2$.

5.7.20. (A) $y_p = u_1\sqrt{x}e^{2x} + u_2\sqrt{x}e^{-2x}$;

$$\begin{aligned} u_1'\sqrt{x}e^{2x} + u_2'\sqrt{x}e^{-2x} &= 0 \qquad \text{(B)} \\ u_1'e^{2x}\left(2\sqrt{x} + \frac{1}{2\sqrt{x}}\right) - u_2'e^{-2x}\left(2\sqrt{x} - \frac{1}{2\sqrt{x}}\right) &= \frac{8x^{5/2}}{4x^2} = 2\sqrt{x} \qquad \text{(C)}. \end{aligned}$$

Multiplying (B) by $\dfrac{1}{2x}$, subtracting the result from (C), and cancelling common factors from the resulting equations yields

$$\begin{aligned} u_1'e^{2x} + u_2'e^{-2x} &= 0 \qquad \text{(D)} \\ u_1'e^{2x} - u_2'e^{-2x} &= 1. \qquad \text{(E)} \end{aligned}$$

Adding (D) to (E) yields $2u_1'e^{2x} = 1$, so $u_1' = \dfrac{e^{-2x}}{2}$. From (D), $u_2' = -u_1'e^{4x} = -\dfrac{e^{2x}}{2}$. Therefore, $u_1 = -\dfrac{e^{-2x}}{4}$, $u_2 = -\dfrac{e^{2x}}{4}$. Now (A) yields $y_p = -\dfrac{\sqrt{x}}{2}$.

5.7.22. (A)$y_p = u_1xe^x + u_2xe^{-x}$;

$$\begin{aligned} u_1'xe^x + u_2'xe^{-x} &= 0 \qquad \text{(B)} \\ u_1'(x+1)e^x - u_2'(x-1)e^{-x} &= \frac{3x^4}{x^2} = 3x^2 \qquad \text{(C)}. \end{aligned}$$

Multiplying (B) by $\dfrac{1}{x}$, subtracting the resulting equation from (C), and cancelling common factors yields

$$\begin{aligned} u_1'e^x + u_2'e^{-x} &= 0 \qquad \text{(D)} \\ u_1'e^x - u_2'e^{-x} &= 3x. \qquad \text{(E)} \end{aligned}$$

Adding (D) to (E) yields $2u_1'e^x = 3x$, so $u_1' = \dfrac{3xe^{-x}}{2}$. From (D), $u_2' = -u_1'e^{2x} = -\dfrac{3xe^x}{2}$. Therefore $u_1 = -\dfrac{3e^x(x+1)}{2}$, $u_2 = -\dfrac{3e^x(x-1)}{2}$. Now (A) yields $y_p = -3x^2$.

5.7.24. (A) $y_p = \dfrac{u_1}{x} + u_2x^3$;

$$\begin{aligned} \frac{u_1'}{x} + \ u_2'x^3 &= 0 \qquad \text{(B)} \\ -\frac{u_1'}{x^2} + 3u_2'x^2 &= \frac{x^{3/2}}{x^2} = x^{-1/2}. \qquad \text{(C)} \end{aligned}$$

Multiplying (B) by $\dfrac{1}{x}$ and adding the result to (C) yields $4u_2'x^2 = x^{-1/2}$, so $u_2' = \dfrac{x^{-5/2}}{4}$. From (B), $u_1' = -u_2'x^4 = -\dfrac{x^{3/2}}{4}$. Therefore $u_1 = -\dfrac{x^{5/2}}{10}$, $u_2 = -\dfrac{x^{-3/2}}{6}$. Now (A) yields $y_p = -\dfrac{4x^{3/2}}{15}$.

5.7.26. (A) $y_p = u_1x^2e^x + u_2x^3e^x$;

$$\begin{aligned} u_1'x^2e^x + u_2'x^3e^x &= 0 \qquad \text{(B)} \\ u_1'(x^2e^x + 2xe^x) + u_2'(x^3e^x + 3x^2e^x) &= \frac{2xe^x}{x^2} = \frac{2e^x}{x}. \qquad \text{(C)} \end{aligned}$$

Subtracting (B) from (C) and cancelling common factors in the resulting equations yields

$$\begin{aligned} u_1' + \ \ u_2'x &= 0 \qquad \text{(D)} \\ 2u_1'x + 3u_2'x^2 &= \frac{2}{x}. \qquad \text{(E)} \end{aligned}$$

Multiplying (D) by $2x$ and subtracting the result from (E) yields $x^2u_2' = \dfrac{2}{x}$, so $u_2' = \dfrac{2}{x^3}$. From (D), $u_1' = -u_2'x = -\dfrac{2}{x^2}$. Therefore $u_1 = \dfrac{2}{x}$, $u_2 = -\dfrac{1}{x^2}$. Now (A) yields $y_p = xe^x$.

5.7.28. (A) $y_p = u_1x + u_2e^x$;

$$\begin{aligned} u_1'x + u_2'e^x &= 0 \qquad \text{(B)} \\ u_1' + u_2'e^x &= \frac{2(x-1)^2e^x}{x-1} = 2(x-1)e^x. \qquad \text{(C)} \end{aligned}$$

Subtracting (B) from (C) yields $u_1'(1-x) = 2(x-1)e^x$, so $u_1' = -2e^x$. From (B), $u_2' = -u_1'xe^{-x} = 2x$. Therefore, $u_1 = -2e^x$, $u_2 = x^2$. Now (A) yields $y_p = xe^x(x-2)$.

5.7.30. (A) $y_p = u_1e^{2x} + u_2xe^{-x}$;

$$\begin{aligned} u_1'e^{2x} + u_2'xe^{-x} &= 0 \qquad \text{(B)} \\ 2u_1'e^{2x} + u_2'(e^{-x} - xe^{-x}) &= \frac{(3x-1)^2e^{2x}}{3x-1} = (3x-1)e^{2x}. \qquad \text{(C)} \end{aligned}$$

Multiplying (B) by 2 and subtracting the result from (C) yields $u_2'(1-3x)e^{-x} = (3x-1)e^{2x}$, so $u_2' = -e^{3x}$. From (B), $u_1' = -u_2'xe^{-3x} = x$. Therefore $u_1 = \dfrac{x^2}{2}$, $u_2 = -\dfrac{e^{3x}}{3}$. Now (A) yields $y_p = \dfrac{xe^{2x}(3x-2)}{3}$. The general solution of the given equation is $y = \dfrac{xe^{2x}(3x-2)}{3} + c_1e^{2x} + c_2xe^{-x}$. Differentiating this yields $y' = \dfrac{e^{2x}(3x^2+x-1)}{3} + 2c_1e^{2x} + c_2(1-x)e^{-x}$. Now $y(0) = 1,\ y'(0) = 2 \Rightarrow c_1 = 1,\ 2 = -\dfrac{1}{3} + 2c_1 + c_2$, so $c_2 = \dfrac{1}{3}$, and $y = \dfrac{e^{2x}(3x^2-2x+6)}{6} + \dfrac{xe^{-x}}{3}$.

5.7.32. (A) $y_p = u_1(x-1)e^x + u_2(x-1)$;

$$\begin{aligned} u_1'(x-1)e^x + u_2'(x-1) &= 0 \qquad \text{(B)} \\ u_1'xe^x + u_2' &= \frac{(x-1)^3e^x}{(x-1)^2} = (x-1)e^x. \qquad \text{(C)} \end{aligned}$$

From (B), $u_1' = -u_2'e^{-x}$. Substituting this into (C) yields $-u_2'(x-1) = (x-1)e^x$, so $u_2' = -e^x$, $u_1' = 1$. Therefore $u_1 = x$, $u_2 = e^x$. Now (A) yields $y_p = e^x(x-1)^2$. The general solution of the given equation is $y = (x-1)^2e^x + c_1(x-1)e^x + c_2(x-1)$. Differentiating this yields $y' = (x^2-1)e^x + c_1xe^x + c_2$. Now $y(0) = 4,\ y'(0) = -6 \Rightarrow 4 = 1 - c_1 - c_2,\ -6 = -1 + c_2$, so $c_1 = 2, c_2 = -5$ and $y = (x^2-1)e^x - 5(x-1)$.

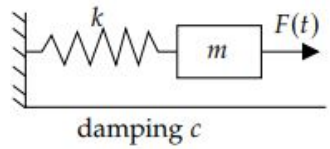
Section 14

Some Applications

Let us look at some applications of linear second order constant coefficient equations.

Subsection 14.1

Mechanical vibrations (mass on a spring)



Our first example is a mass on a spring. Suppose we have a mass $m > 0$ (in kilograms) connected by a spring with spring constant $k > 0$ (in newtons per meter) to a fixed wall. There may be some external force $F(t)$ (in newtons) acting on the mass. Here, t is time in seconds. Finally, there is some friction measured by $c \geq 0$ (in newton-seconds per meter) as the mass slides along the floor (or perhaps a damper is connected). Let x be the displacement of the mass in meters ($x = 0$ is the rest position), with x growing to the right (away from the wall). The force exerted by the spring is proportional to the compression of the spring by *Hooke's law*. Therefore, it is kx in the negative direction. Similarly, the force exerted by friction is proportional to the velocity of the mass. By Newton's second law we know that force equals mass times acceleration and hence $mx'' = F(t) - cx' - kx$ or

$$mx'' + cx' + kx = F(t).$$

As you can see, this is a linear second order constant coefficient ODE.

Remark We say the motion is

(i) *forced* if $F \not\equiv 0$ (if F is not identically zero),

(ii) *unforced* or *free* if $F \equiv 0$ (if F is identically zero),

(iii) *damped* if $c > 0$, and

(iv) *undamped* if $c = 0$.

This system appears in lots of applications even if it does not at first seem like it. Many real-world scenarios can be simplified to a mass on a spring. For example, a bungee jump setup is essentially a mass and spring system (you are the mass). It would be good if someone did the math before you jump off the bridge, right?

14.1.1 Free undamped motion

In this subsection, we only consider free or unforced motion, Let us start with undamped motion where $c = 0$. The equation is

$$mx'' + kx = 0.$$

We divide by m and let $\omega_0 = \sqrt{k/m}$ to rewrite the equation as

$$x'' + \omega_0^2 x = 0.$$

The general solution to this equation is

$$x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t).$$

By a trigonometric identity

$$A\cos(\omega_0 t) + B\sin(\omega_0 t) = C\cos(\omega_0 t - \gamma),$$

for two different constants C and γ. It is not hard to compute that $C = \sqrt{A^2 + B^2}$ and $\tan\gamma = {}^{B}\!/_{A}$. Therefore, we let C and γ be our arbitrary constants and write $x(t) = C\cos(\omega_0 t - \gamma)$.

Remark While it is generally easier to use the first form with A and B to solve for the initial conditions, the second form is much more natural. The constants C and γ have nice physical interpretation. Write the solution as

$$x(t) = C\cos(\omega_0 t - \gamma).$$

This is a pure-frequency oscillation (a sine wave). The *amplitude* is C, ω_0 is the (angular) *frequency*, and γ is the so-called *phase shift*. The phase shift just shifts the graph left or right. We call ω_0 the *natural (angular) frequency*. This entire setup is called *simple harmonic motion*.

Let us pause to explain the word *angular* before the word *frequency*. The units of ω_0 are radians per unit time, not cycles per unit time as is the usual measure of frequency. Because one cycle is 2π radians, the usual frequency is given by $\frac{\omega_0}{2\pi}$. It is simply a matter of where we put the constant 2π, and that is a matter of taste.
The *period* of the motion is one over the frequency (in cycles per unit time) and hence it is $\frac{2\pi}{\omega_0}$. That is the amount of time it takes to complete one full cycle.

Let us put this into some numbers to illustrate

Example 61 Suppose that $m = 2\,\mathrm{kg}$ and $k = 8\,{}^{\mathrm{N}}\!/_{\mathrm{m}}$. The whole mass and spring setup is sitting on a truck that was traveling at $1\,{}^{\mathrm{m}}\!/_{\mathrm{s}}$. The truck crashes and hence stops. The mass was held in place 0.5 meters forward from the rest position. During the crash the mass gets loose. That is, the mass is now moving forward at $1\,{}^{\mathrm{m}}\!/_{\mathrm{s}}$, while the other end of the spring is held in place. The mass therefore starts oscillating. What is the frequency of the resulting oscillation? What is the amplitude? The units are the mks units (meters-kilograms-seconds).
The setup means that the mass was at half a meter in the positive direction during the crash and relative to the wall the spring is mounted to, the mass was moving forward (in the positive direction) at $1\,{}^{\mathrm{m}}\!/_{\mathrm{s}}$. This gives us the initial conditions.
So the equation with initial conditions is

$$2x'' + 8x = 0, \qquad x(0) = 0.5, \qquad x'(0) = 1.$$

We directly compute $\omega_0 = \sqrt{{}^{k}\!/_{m}} = \sqrt{4} = 2$. Hence the angular frequency is 2. The usual frequency in Hertz (cycles per second) is ${}^{2}\!/_{2\pi} = {}^{1}\!/_{\pi} \approx 0.318$.
The general solution is

$$x(t) = A\cos(2t) + B\sin(2t).$$

Letting $x(0) = 0.5$ means $A = 0.5$. Then $x'(t) = -2(0.5)\sin(2t) + 2B\cos(2t)$. Letting $x'(0) = 1$ we get $B = 0.5$.

$$x(t) = 0.5\cos(2t) + 0.5\sin(2t).$$

A plot of $x(t)$ is shown in 6.

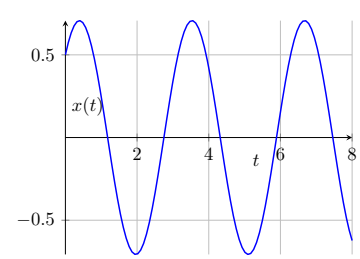


Figure 6. Simple undamped oscillation

Remark In general, for free undamped motion, a solution of the form

$$x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t),$$

corresponds to the initial conditions $x(0) = A$ and $x'(0) = \omega_0 B$. So it is easy to figure out A and B from the initial conditions. The amplitude and the phase shift are computed from A and B. In the example, we found the amplitude C. Let us compute the phase shift. We know $\tan\gamma = {}^{B}/_{A} = 1$. We take the arctangent of 1 and get ${}^{\pi}/_{4}$ or approximately 0.785. We need to check if ${}^{\pi}/_{4}$ gives the correct quadrant (same quadrant as the point (A, B)) and add π to get γ if it is not. Since both A and B are positive, then γ should be in the first quadrant, ${}^{\pi}/_{4}$ radians is in the first quadrant, so $\gamma = {}^{\pi}/_{4}$.

14.1.2 Free damped motion

Let us now focus on damped motion. We rewrite the equation

$$mx'' + cx' + kx = 0,$$

as

$$x'' + 2px' + \omega_0^2 x = 0,$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}, \qquad p = \frac{c}{2m}.$$

The characteristic equation is

$$r^2 + 2pr + \omega_0^2 = 0.$$

Using the quadratic formula, we get that the roots are

$$r = -p \pm \sqrt{p^2 - \omega_0^2}.$$

The form of the solution depends on whether we get complex or real roots. We get real roots if and only if the following number is nonnegative:

$$p^2 - \omega_0^2 = \left(\frac{c}{2m}\right)^2 - \frac{k}{m} = \frac{c^2 - 4km}{4m^2}.$$

The sign of $p^2 - \omega_0^2$ is the same as the sign of $c^2 - 4km$. Thus we get real roots if and only if $c^2 - 4km$ is nonnegative, or in other words if $c^2 \geq 4km$.

14.1.3 Underdamping

When $c^2 - 4km < 0$, the system is *underdamped*. In this case, the roots are complex.

Example 62 Consider a mass-spring-damper system where $m = 4\,\mathrm{kg}$, $k = 16\,{}^{\mathrm{N}}/_{\mathrm{m}}$, and the damping coefficient $c = 8\,{}^{\mathrm{N\cdot s}}/_{\mathrm{m}}$. The mass is initially displaced $0.3\,\mathrm{m}$ from its equilibrium position and given an initial velocity of $2\,{}^{\mathrm{m}}/_{\mathrm{s}}$ in the positive direction. Find the equation of motion for this system, determine if it's underdamped, critically damped, or overdamped, and sketch its graph. The units are the mks units (meters-kilograms-seconds).
The equation of motion with initial conditions is:

$$4x'' + 8x' + 16x = 0, \qquad x(0) = 0.3, \qquad x'(0) = 2.$$

We first calculate $\omega_0 = \sqrt{{}^{k}/_{m}} = \sqrt{{}^{16}/_{4}} = 2$ and $\zeta = \frac{c}{2\sqrt{km}} = \frac{8}{2\sqrt{16\cdot 4}} = \frac{1}{2}$.

Since $0 < \zeta < 1$, the system is underdamped. The damped natural frequency is:

$$\omega_d = \omega_0\sqrt{1-\zeta^2} = 2\sqrt{1-(\frac{1}{2})^2} = \sqrt{3} \approx 1.732$$

The general solution for an underdamped system is:

$$x(t) = e^{-\zeta\omega_0 t}(A\cos(\omega_d t) + B\sin(\omega_d t))$$

Substituting our values:

$$x(t) = e^{-t}(A\cos\left(\sqrt{3}t\right) + B\sin\left(\sqrt{3}t\right))$$

Using the initial conditions:

$$\begin{aligned} x(0) = 0.3 &\implies A = 0.3 \\ x'(0) = 2 &\implies -A + \sqrt{3}B = 2 \\ &\implies B = \frac{2+0.3}{\sqrt{3}} \approx 1.33 \end{aligned}$$

Therefore, the solution is:

$$x(t) = e^{-t}(0.3\cos\left(\sqrt{3}t\right) + 1.33\sin\left(\sqrt{3}t\right))$$

A plot of $x(t)$ is shown in 7.

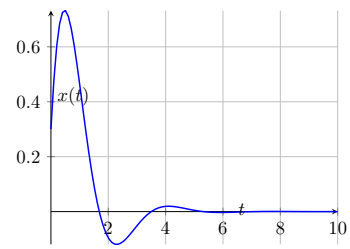


Figure 7. Underdamped oscillation

This example demonstrates a damped motion problem, showing how to determine the type of damping, solve for the equation of motion, and graph the result. The solution and graph show the characteristic behavior of an underdamped system, where the oscillations decrease in amplitude over time due to the damping force.

Subsection 14.2

Overdamping

When $c^2 - 4km > 0$, the system is *overdamped*. In this case, there are two distinct real roots r_1 and r_2. Both roots are negative: As $\sqrt{p^2-\omega_0^2}$ is always less than p, then $-p \pm \sqrt{p^2-\omega_0^2}$ is negative in either case.

The solution is

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}.$$

Since r_1, r_2 are negative, $x(t) \to 0$ as $t \to \infty$. Thus the mass will tend towards the rest position as time goes to infinity. For a few sample plots for different initial conditions, see 8.

Example 63

Consider a mass-spring-damper system where $m = 2\,\text{kg}$, $k = 8\,\text{N/m}$, and the damping coefficient $c = 12\,\text{N·s/m}$. The mass is initially displaced $0.4\,\text{m}$ from its equilibrium position and given an initial velocity of $1\,\text{m/s}$ in the negative direction. Find the equation of motion for this system, confirm that it's overdamped, and sketch its graph. The units are the mks units (meters-kilograms-seconds).
The equation of motion with initial conditions is:

$$2x'' + 12x' + 8x = 0, \qquad x(0) = 0.4, \qquad x'(0) = -1.$$

We first calculate $\omega_0 = \sqrt{k/m} = \sqrt{8/2} = 2$ and $\zeta = \frac{c}{2\sqrt{km}} = \frac{12}{2\sqrt{8\cdot 2}} = \frac{3}{2}$.
Since $\zeta > 1$, the system is indeed overdamped. We can proceed to solve the characteristic equation:

$$r^2 + 6r + 4 = 0$$

The roots are:

$$r_1 = -3 + \sqrt{5} \approx -0.764$$
$$r_2 = -3 - \sqrt{5} \approx -5.236$$

The general solution for an overdamped system is:

$$x(t) = Ae^{r_1 t} + Be^{r_2 t}$$

Using the initial conditions:

$$x(0) = 0.4 \implies A + B = 0.4$$
$$x'(0) = -1 \implies Ar_1 + Br_2 = -1$$

Solving this system of equations:

$$A \approx 0.446$$
$$B \approx -0.046$$

Therefore, the solution is:

$$x(t) \approx 0.446e^{-0.764t} - 0.046e^{-5.236t}$$

A plot of $x(t)$ is shown in 8.

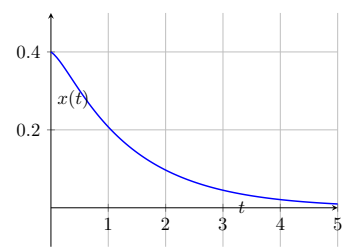


Figure 8. Overdamped motion

This example demonstrates an overdamped motion problem, showing how to determine that the system is overdamped, solve for the equation of motion, and graph the result. The solution and graph show the characteristic behavior of an overdamped system, where the displacement approaches zero without oscillating. The system returns to equilibrium more slowly than in a critically damped or underdamped system.

Subsection 14.3

Critical damping

When $c^2 - 4km = 0$, the system is *critically damped*. In this case, there is one root of multiplicity 2 and this root is $-p$. Our solution is

$$x(t) = C_1 e^{-pt} + C_2 te^{-pt}.$$

The behavior of a critically damped system is very similar to an overdamped system. After all a critically damped system is in some sense a limit of overdamped systems. Since these equations are really only an approximation to the real world, in reality we are never critically damped, it is a place we can only reach in theory. We are always a little bit underdamped or a little bit overdamped. It is better not to dwell on critical damping.

Example 64

Consider a mass-spring-damper system where $m = 1\,\text{kg}$, $k = 16\,\text{N/m}$, and the damping coefficient $c = 8\,\text{N·s/m}$. The mass is initially displaced $0.2\,\text{m}$ from its equilibrium position and given an initial velocity of $1\,\text{m/s}$ in the positive direction. Find the equation of motion for this system, confirm that it's critically damped, and sketch its graph. The

units are the mks units (meters-kilograms-seconds).
The equation of motion with initial conditions is:

$$x'' + 8x' + 16x = 0, \qquad x(0) = 0.2, \qquad x'(0) = 1.$$

We first calculate $\omega_0 = \sqrt{k/m} = \sqrt{16} = 4$ and $\zeta = \frac{c}{2\sqrt{km}} = \frac{8}{2\sqrt{16 \cdot 1}} = 1$.
Since $\zeta = 1$, the system is indeed critically damped. We can proceed to solve the characteristic equation:

$$r^2 + 8r + 16 = 0$$

The repeated root is:

$$r = -4$$

The general solution for a critically damped system is:

$$x(t) = (A + Bt)e^{-4t}$$

Using the initial conditions:

$$\begin{aligned} x(0) = 0.2 &\implies A = 0.2 \\ x'(0) = 1 &\implies -4A + B = 1 \\ &\implies B = 1 + 4A = 1.8 \end{aligned}$$

Therefore, the solution is:

$$x(t) = (0.2 + 1.8t)e^{-4t}$$

A plot of $x(t)$ is shown in 9.

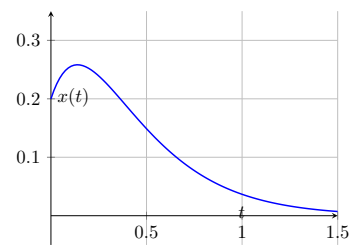


Figure 9. Critically damped motion

This example demonstrates a critically damped motion problem, showing how to determine that the system is critically damped, solve for the equation of motion, and graph the result. The solution and graph show the characteristic behavior of a critically damped system, where the displacement returns to equilibrium as quickly as possible without oscillating.

In a critically damped system, the damping coefficient is precisely tuned to bring the system to rest in the shortest time without overshooting. This is often the desired behavior in many engineering applications, such as door closers or some types of shock absorbers.

The graph shows that the displacement initially increases slightly due to the positive initial velocity, but then smoothly approaches zero without any oscillations. This is the fastest return to equilibrium possible without overshooting, which is the defining characteristic of critical damping.

Subsection 14.4

Exercises

Consider a mass and spring system with a mass $m = 2$, spring constant $k = 3$, and damping constant $c = 1$.

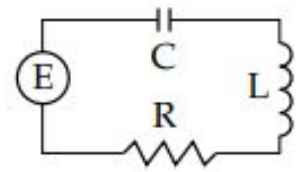
1. Set up and find the general solution of the system.
2. Is the system underdamped, overdamped or critically damped?
3. If the system is not critically damped, find a c that makes the system critically damped.

Do the above exercise for $m = 3$, $k = 12$, and $c = 12$.

Subsection 14.5

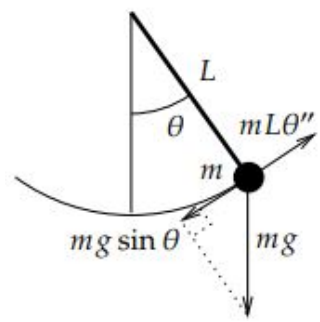
More Applications

Here is an example for electrical engineers. Consider the pictured RLC circuit. There is a resistor with a resistance of R ohms, an inductor with an inductance of L henries, and a capacitor with a capacitance of C farads. There is also an electric source (such as a battery) giving a voltage of $E(t)$ volts at time t (measured in seconds). Let $Q(t)$ be the charge (in coulombs) on the capacitor and $I(t)$ be the current (in amperes) in the circuit. The relation between the two is $Q' = I$. By elementary principles, we find $LI' + RI + Q/C = E$. We differentiate to get



$$LI''(t) + RI'(t) + \frac{1}{C}I(t) = E'(t).$$

This is a nonhomogeneous second order constant coefficient linear equation. As L, R, and C are all positive, this system behaves just like the mass and spring system. Position of the mass is replaced by current. Mass is replaced by inductance, damping is replaced by resistance, and the spring constant is replaced by one over the capacitance. The change in voltage becomes the forcing function—for constant voltage this is an unforced motion.



Our next example behaves like a mass and spring system only approximately. Suppose a mass m hangs on a pendulum of length L. We seek an equation for the angle $\theta(t)$ (in radians). Let g be the force of gravity. Elementary physics mandates that the equation is

$$\theta'' + \frac{g}{L}\sin\theta = 0.$$

Let us derive this equation using Newton's second law: force equals mass times acceleration. The acceleration is $L\theta''$ and mass is m. So $mL\theta''$ has to be equal to the tangential component of the force given by the gravity, which is $mg\sin\theta$ in the opposite direction. So $mL\theta'' = -mg\sin\theta$. The m curiously cancels from the equation.

Now we make our approximation. For small θ we have that approximately $\sin\theta \approx \theta$. This can be seen by looking at the graph. We can see that for approximately $-0.5 < \theta < 0.5$ (in radians) the graphs of $\sin\theta$ and θ are almost the same.

Therefore, when the swings are small, θ is small and we can model the behavior by the simpler linear equation

$$\theta'' + \frac{g}{L}\theta = 0.$$

The errors from this approximation build up. So after a long time, the state of the real-world system might be substantially different from our solution. Also we will see that in a mass-spring system, the amplitude is independent of the period. This is not true for a pendulum. Nevertheless, for reasonably short periods of time and small swings (that is, only small angles θ), the approximation is reasonably good.

In real-world problems it is often necessary to make these types of simplifications. We must understand both the mathematics and the physics of the situation to see if the simplification is valid in the context of the questions we are trying to answer.

Section 15

SODE with Variable coefficient

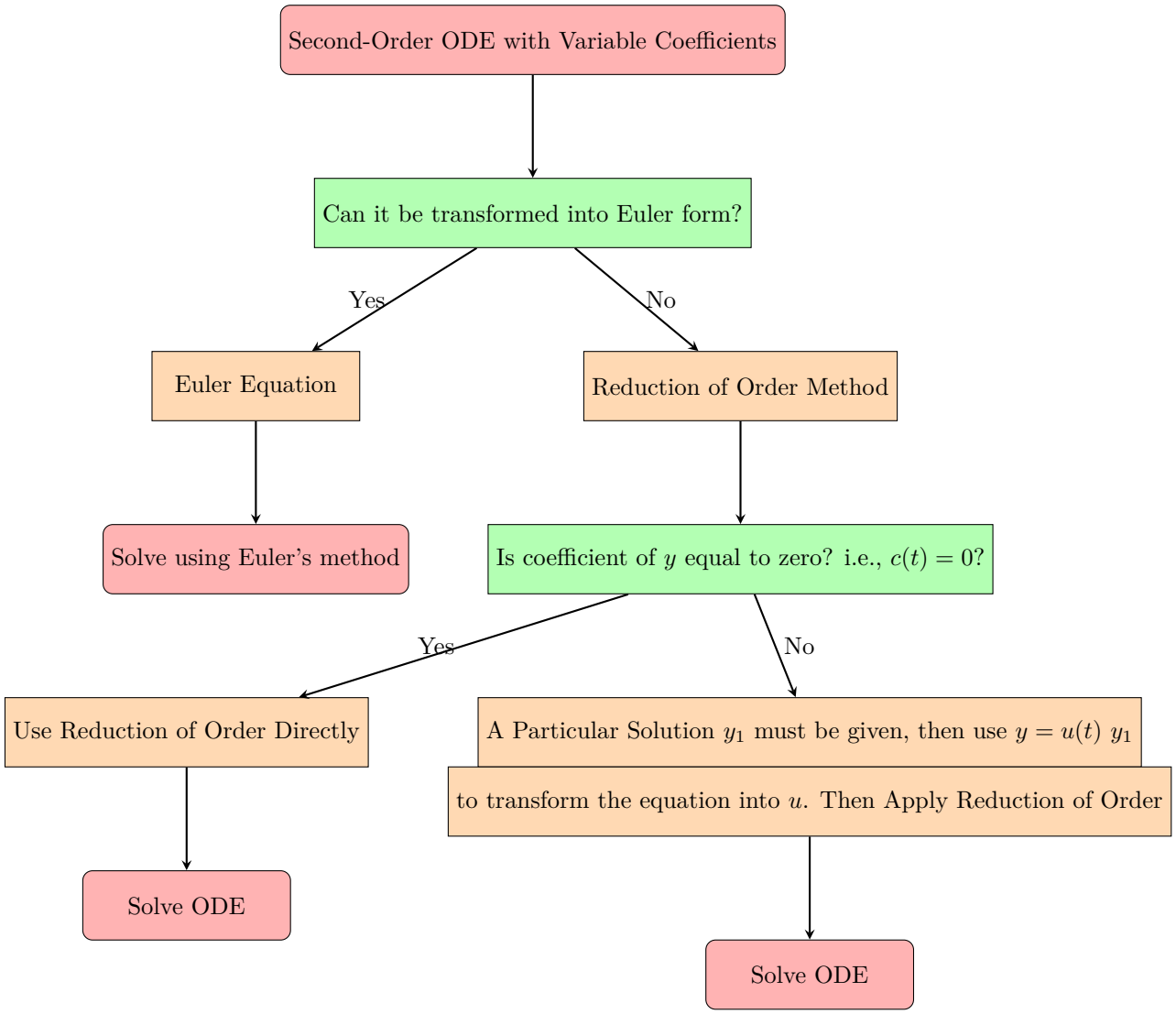
A second-order ordinary differential equation (ODE) with **variable coefficients** is a differential equation of the form:

$$a(t)\frac{d^2y}{dt^2} + b(t)\frac{dy}{dt} + c(t)y = f(t)$$

where, $a(t), b(t)$, and $c(t)$ are variable coefficients that depend on t.
Similarly, we can look at them as homogeneous and nonhomogeneous:
- If $f(t) = 0$, the ODE is called homogeneous.
- If $f(t) \neq 0$, the ODE is non-homogeneous.

Remark Note that solving the SODE with variable coefficient, particularly the y_c is not easy, and there is no one method that works for all kinds of equations. Normally, numerical methods are used to solve these kinds of differential equations.

The below tree decision will guide you to how to think about the A second-order ordinary differential equation (ODE) with **variable coefficients**



In the following sections, we will study the analytical solution of some special cases.

Subsection 15.1

Euler Equations

The simplest kind of equation with a regular singular point at $x_0 = 0$ is the Euler equation, defined as follows.

Definition 5 An *Euler equation* is an equation that can be written in the form

$$ax^2y'' + bxy' + cy = 0, \tag{15.1}$$

where a, b, and c are real constants and $a \neq 0$.

Theorem 4 implies that (15.1) has solutions defined on $(0, \infty)$ and $(-\infty, 0)$, since (15.1) can be rewritten as

$$ay'' + \frac{b}{x}y' + \frac{c}{x^2}y = 0.$$

Remark For convenience we'll restrict our attention to the interval $(0, \infty)$. The key to finding solutions on $(0, \infty)$ is that if $x > 0$ then x^r is defined as a real-valued function on $(0, \infty)$ for all values of r,

We will use the great idea of transformation again in order to convert the equation 15.1 into a constant coefficient. We have done this several times, and we will do it again and again during the course.

Theorem 7 The differential equation

$$ax^2y'' + bxy' + cy = 0, \tag{15.2}$$

is equivalent to the constant coefficient differential equation

$$a\frac{d^2Y}{dt^2} + (b-a)\frac{dY}{dt} + cY = 0, \tag{15.3}$$

under the transformation $Y(t) = y(e^t)$.

PROOF We first show that $y = y(x)$ satisfies (15.6) on $(0, \infty)$ if and only if $Y(t) = y(e^t)$ satisfies the constant coefficient equation

$$a\frac{d^2Y}{dt^2} + (b-a)\frac{dY}{dt} + cY = 0 \tag{15.4}$$

on $(-\infty, \infty)$. To do this, it's convenient to write $x = e^t$, or, equivalently, $t = \ln x$; thus, $Y(t) = y(x)$, where $x = e^t$. From the chain rule,

$$\frac{dY}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

and, since

$$\frac{dx}{dt} = e^t = x,$$

it follows that

$$\frac{dY}{dt} = x\frac{dy}{dx}. \tag{15.5}$$

Differentiating this with respect to t and using the chain rule again yields

$$\begin{aligned}\frac{d^2Y}{dt^2} &= \frac{d}{dt}\left(\frac{dY}{dt}\right) = \frac{d}{dt}\left(x\frac{dy}{dx}\right)\\ &= \frac{dx}{dt}\frac{dy}{dx} + x\frac{d^2y}{dx^2}\frac{dx}{dt}\\ &= x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} \quad \left(\text{since } \frac{dx}{dt} = x\right).\end{aligned}$$

From this and (15.5),

$$x^2\frac{d^2y}{dx^2} = \frac{d^2Y}{dt^2} - \frac{dY}{dt}.$$

Substituting this and (15.5) into (15.6) yields (15.4). □

Remark The above theorem facilitates a way of solving Euler equations easily.

Assume we have the Euler equation

$$ay'' + \frac{b}{x}y' + \frac{c}{x^2}y = 0.$$

Now, solve the corresponding constant coefficient differential equation

$$a\frac{d^2Y}{dt^2} + (b-a)\frac{dY}{dt} + cY = 0,$$

as follows:
Build the characteristic equation

$$ar(r-1) + br + c = 0,$$

then find the roots, r_1 and r_2. This implies that the solution will one of the following:

$$\begin{aligned} Y(t) &= c_1 e^{r_1 t} + c_2 e^{r_2 t} \text{ if } r_1 \text{ and } r_2 \text{ are distinct real numbers;} \\ Y(t) &= e^{r_1 t}(c_1 + c_2 t) \text{ if } r_1 = r_2; \\ Y(t) &= e^{\lambda t}\left(c_1 \cos \omega t + c_2 \sin \omega t\right) \text{ if } r_1, r_2 = \lambda \pm i\omega \text{ with } \omega \neq 0. \end{aligned}$$

Or equivalent, the solution of Euler equation

$$ax^2y'' + bxy' + cy = 0 \tag{15.6}$$

on $(0, \infty)$ is

$$\begin{aligned} y &= c_1 x^{r_1} + c_2 x^{r_2} \text{ if } r_1 \text{ and } r_2 \text{ are distinct real numbers ;} \\ y &= x^{r_1}(c_1 + c_2 \ln x) \text{ if } r_1 = r_2 \text{ ;} \\ y &= x^{\lambda}\left[c_1 \cos\left(\omega \ln x\right) + c_2 \sin\left(\omega \ln x\right)\right] \text{ if } r_1, r_2 = \lambda \pm i\omega \text{ with } \omega > 0. \end{aligned}$$

Here are some examples:

Example 65 Find the general solution of

$$x^2y'' - xy' - 8y = 0 \tag{15.7}$$

on $(0, \infty)$.

Solution The indicial polynomial of (15.7) is

$$p(r) = r(r-1) - r - 8 = (r-4)(r+2).$$

Therefore $y_1 = x^4$ and $y_2 = x^{-2}$ are solutions of (15.7) on $(0, \infty)$, and its general solution on $(0, \infty)$ is

$$y = c_1 x^4 + \frac{c_2}{x^2}.$$

Example 66 Find the general solution of

$$6x^2y'' + 5xy' - y = 0 \tag{15.8}$$

on $(0, \infty)$.

Solution The indicial polynomial of (15.8) is

$$p(r) = 6r(r-1) + 5r - 1 = (2r-1)(3r+1).$$

Therefore the general solution of (15.8) on $(0, \infty)$ is

$$y = c_1 x^{1/2} + c_2 x^{-1/3}. \blacksquare$$

If the indicial equation has a repeated root r_1, then $y_1 = x^{r_1}$ is a solution of

$$ax^2y'' + bxy' + cy = 0, \tag{15.9}$$

on $(0, \infty)$, but (15.9) has no other solution of the form $y = x^r$. If the indicial equation has complex conjugate zeros then (15.9) has no real–valued solutions of the form $y = x^r$. Fortunately we can use the results of Section 5.2 for constant coefficient equations to solve (15.9) in any case.

Example 67 Find the general solution of

$$x^2y'' - 5xy' + 9y = 0 \tag{15.10}$$

on $(0, \infty)$.

Solution The indicial polynomial of (15.10) is

$$p(r) = r(r-1) - 5r + 9 = (r-3)^2.$$

Therefore the general solution of (15.10) on $(0, \infty)$ is

$$y = x^3(c_1 + c_2 \ln x).$$

Example 68 Find the general solution of

$$x^2y'' + 3xy' + 2y = 0 \tag{15.11}$$

on $(0, \infty)$.

Solution The indicial polynomial of (15.11) is

$$p(r) = r(r-1) + 3r + 2 = (r+1)^2 + 1.$$

The roots of the indicial equation are $r = -1 \pm i$ and the general solution of (15.11) on $(0, \infty)$ is

$$y = \frac{1}{x}\left[c_1 \cos(\ln x) + c_2 \sin(\ln x)\right].$$

Example 69 Find the general solution of

$$x^2y'' - xy' + y = x^3 \tag{15.12}$$

on $(0, \infty)$.

Solution First, we solve the homogeneous equation:

$$x^2y'' - xy' + y = 0$$

The indicial polynomial is

$$p(r) = r(r-1) - r + 1 = r^2 - 2r + 1 = (r-1)^2.$$

The root of the indicial equation is $r = 1$ with multiplicity 2, so the general solution of the homogeneous equation on $(0, \infty)$ is

$$y_c = c_1 x + c_2 x \ln x.$$

Now, we find the particular solution using the method of variation of parameters. Let

$$y_p = u_1(x)x + u_2(x)x \ln x.$$

We need to solve:

$$\begin{aligned} xu_1' + (\ln x)xu_2' &= 0 \\ u_1 + u_2 + (\ln x)u_2' &= x^2 \end{aligned}$$

Solving this system:

$$\begin{aligned} u_1' &= -x \ln x \\ u_2' &= x \end{aligned}$$

Integrating:

$$\begin{aligned} u_1 &= -\frac{1}{2}x^2 \ln x + \frac{1}{4}x^2 \\ u_2 &= \frac{1}{2}x^2 \end{aligned}$$

Therefore, the particular solution is:

$$y_p = (-\frac{1}{2}x^2 \ln x + \frac{1}{4}x^2)x + (\frac{1}{2}x^2)x \ln x = \frac{1}{4}x^3$$

The general solution of (15.12) on $(0, \infty)$ is

$$y = c_1 x + c_2 x \ln x + \frac{1}{4}x^3.$$

Example 70 Find the general solution of

$$x^2 y'' + 5xy' + 4y = \frac{1}{x} \tag{15.13}$$

on $(0, \infty)$.

Solution First, we solve the homogeneous equation:

$$x^2 y'' + 5xy' + 4y = 0$$

The indicial polynomial is

$$p(r) = r(r-1) + 5r + 4 = r^2 + 4r + 4 = (r+2)^2.$$

The root of the indicial equation is $r = -2$ with multiplicity 2, so the general solution

of the homogeneous equation on $(0, \infty)$ is

$$y_c = \frac{c_1}{x^2} + \frac{c_2 \ln x}{x^2}.$$

Now, we find the particular solution using the method of variation of parameters. Let

$$y_p = u_1(x)\frac{1}{x^2} + u_2(x)\frac{\ln x}{x^2}.$$

We need to solve:

$$\begin{aligned} \frac{1}{x^2}u_1' + \frac{\ln x}{x^2}u_2' &= 0 \\ -\frac{2}{x^3}u_1 + \frac{1}{x^3}u_2 - \frac{2\ln x}{x^3}u_2' &= \frac{1}{x} \end{aligned}$$

Solving this system:

$$\begin{aligned} u_1' &= -x \ln x \\ u_2' &= x \end{aligned}$$

Integrating:

$$\begin{aligned} u_1 &= -\frac{1}{2}x^2 \ln x + \frac{1}{4}x^2 \\ u_2 &= \frac{1}{2}x^2 \end{aligned}$$

Therefore, the particular solution is:

$$y_p = (-\frac{1}{2}x^2 \ln x + \frac{1}{4}x^2)\frac{1}{x^2} + (\frac{1}{2}x^2)\frac{\ln x}{x^2} = -\frac{1}{2}\ln x + \frac{1}{4}$$

The general solution of (15.13) on $(0, \infty)$ is

$$y = \frac{c_1}{x^2} + \frac{c_2 \ln x}{x^2} - \frac{1}{2}\ln x + \frac{1}{4}.$$

Subsection 15.2

Reduction of Order Method

Now we are going to present another important method for solving a class of second-order differential equations, called the reduction-of-order method. As it was shown in the decision tree we presented above, the reduction-of-order method can be used directly if $c(t) = 0$.

A second-order differential equation can be reduced to a first-order differential equation, when the second-order equation is missing y, i.e;

$$y'' + P(x)y' = f(x), \tag{15.14}$$

In this case, we let, $u = y'$ which means $u' = y''$. This implies that

$$y'' + P(x)y' = f(x), \tag{15.15}$$
$$\rightarrow u' + p(x)u = f(x) \tag{15.16}$$

which is a linear first order DE with integration factor

$$I(x) = e^{\int p(x)dx};$$

and therefore, the solution of the first order DE takes the following form;

$$u = \frac{1}{I(x)}[\int I(x)f(x)dx + c].$$

Now, we can substitute $u = y'$, which leads to a separable FODE.

$$y = \int (\frac{1}{I(x)}[\int I(x)f(x)dx + c])dx.$$

Remark Note that the above method works with any coefficient of the second order differential equation as far as $c(x) = 0$.

We can generalize the above concept as in the case of $c(x) \neq 0$, however, we must have one of the particular solution of the homogenous part of the given differential equation.

Assume the following equation.

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = F(x), \tag{15.17}$$

where $y_1(x)$ is a given solution to

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = 0, \tag{15.18}$$

Then, we will obtain the solution as follows:

By now you shouldn't be surprised that we look for solutions of (15.17) in the form

$$y = uy_1 \tag{15.19}$$

where u is to be determined so that y satisfies (15.17). Substituting (15.19) and

$$\begin{aligned} y' &= u'y_1 + uy_1' \\ y'' &= u''y_1 + 2u'y_1' + uy_1'' \end{aligned}$$

into (15.17) yields

$$P_0(x)(u''y_1 + 2u'y_1' + uy_1'') + P_1(x)(u'y_1 + uy_1') + P_2(x)uy_1 = F(x).$$

Collecting the coefficients of u, u', and u'' yields

$$(P_0y_1)u'' + (2P_0y_1' + P_1y_1)u' + (P_0y_1'' + P_1y_1' + P_2y_1)u = F. \tag{15.20}$$

However, the coefficient of u is zero, since y_1 satisfies (15.17). Therefore (15.20) reduces to

$$Q_0(x)u'' + Q_1(x)u' = F, \tag{15.21}$$

with

$$Q_0 = P_0y_1 \quad \text{and} \quad Q_1 = 2P_0y_1' + P_1y_1.$$

As you can see, equation 15.21 can now be solved using reduction of order method to obtain the u, and hence obtain the general solution of the differential equation given in 15.17 as

$$y = u\dot{y}_1 \tag{15.22}$$

Now, lets do some examples

Example 71

- Find the general solution of

$$xy'' - (2x+1)y' + (x+1)y = x^2, \tag{15.23}$$

given that $y_1 = e^x$ is a solution of the complementary equation

$$xy'' - (2x+1)y' + (x+1)y = 0. \tag{15.24}$$

- As a byproduct of part (a), find a fundamental set of solutions of (15.24).

Solution

solutionparta If $y = ue^x$, then $y' = u'e^x + ue^x$ and $y'' = u''e^x + 2u'e^x + ue^x$, so

$$\begin{aligned} xy'' - (2x+1)y' + (x+1)y &= x(u''e^x + 2u'e^x + ue^x) \\ &\quad -(2x+1)(u'e^x + ue^x) + (x+1)ue^x \\ &= (xu'' - u')e^x. \end{aligned}$$

Therefore $y = ue^x$ is a solution of (15.23) if and only if

$$(xu'' - u')e^x = x^2,$$

which is a first order equation in u'. We rewrite it as

$$u'' - \frac{u'}{x} = xe^{-x}. \tag{15.25}$$

To focus on how we apply variation of parameters to this equation, we temporarily write $z = u'$, so that (15.25) becomes

$$z' - \frac{z}{x} = xe^{-x}. \tag{15.26}$$

We leave it to you to show (by separation of variables) that $z_1 = x$ is a solution of the complementary equation

$$z' - \frac{z}{x} = 0$$

for (15.26). By applying variation of parameters as in Section 1.2, we can now see that every solution of (15.26) is of the form

$$z = vx \quad \text{where} \quad v'x = xe^{-x}, \quad \text{so} \quad v' = e^{-x} \quad \text{and} \quad v = -e^{-x} + C_1.$$

Since $u' = z = vx$, u is a solution of (15.25) if and only if

$$u' = vx = -xe^{-x} + C_1 x.$$

Integrating this yields

$$u = (x+1)e^{-x} + \frac{C_1}{2}x^2 + C_2.$$

Therefore the general solution of (15.23) is

$$y = ue^x = x + 1 + \frac{C_1}{2}x^2e^x + C_2e^x. \tag{15.27}$$

solutionpartb By letting $C_1 = C_2 = 0$ in (15.27), we see that $y_{p_1} = x + 1$ is a solution of (15.23). By letting $C_1 = 2$ and $C_2 = 0$, we see that $y_{p_2} = x + 1 + x^2e^x$ is also a solution of (15.23). Since the difference of two solutions of (15.23) is a solution of (15.24), $y_2 = y_{p_1} - y_{p_2} = x^2e^x$ is a solution of (15.24). Since y_2/y_1 is nonconstant and we already know that $y_1 = e^x$ is a solution of (15.23), Theorem **??** implies that $\{e^x, x^2e^x\}$ is a fundamental set of solutions of (15.24). ■

Example 72

- Find the general solution of

$$x^2y'' + xy' - y = x^2 + 1,$$

given that $y_1 = x$ is a solution of the complementary equation

$$x^2y'' + xy' - y = 0. \tag{15.28}$$

As a byproduct of this result, find a fundamental set of solutions of (15.28).

- Solve the initial value problem

$$x^2y'' + xy' - y = x^2 + 1, \quad y(1) = 2,\ y'(1) = -3. \tag{15.29}$$

Solution

solutionparta If $y = ux$, then $y' = u'x + u$ and $y'' = u''x + 2u'$, so

$$\begin{aligned} x^2y'' + xy' - y &= x^2(u''x + 2u') + x(u'x + u) - ux \\ &= x^3u'' + 3x^2u'. \end{aligned}$$

Therefore $y = ux$ is a solution of (15.29) if and only if

$$x^3u'' + 3x^2u' = x^2 + 1,$$

which is a first order equation in u'. We rewrite it as

$$u'' + \frac{3}{x}u' = \frac{1}{x} + \frac{1}{x^3}. \tag{15.30}$$

To focus on how we apply variation of parameters to this equation, we temporarily write $z = u'$, so that (15.30) becomes

$$z' + \frac{3}{x}z = \frac{1}{x} + \frac{1}{x^3}. \tag{15.31}$$

We leave it to you to show by separation of variables that $z_1 = 1/x^3$ is a solution of the complementary equation

$$z' + \frac{3}{x}z = 0$$

for (15.31). By variation of parameters, every solution of (15.31) is of the form

$$z = \frac{v}{x^3} \quad \text{where} \quad \frac{v'}{x^3} = \frac{1}{x} + \frac{1}{x^3}, \quad \text{so} \quad v' = x^2 + 1 \quad \text{and} \quad v = \frac{x^3}{3} + x + C_1.$$

Since $u' = z = v/x^3$, u is a solution of (15.31) if and only if

$$u' = \frac{v}{x^3} = \frac{1}{3} + \frac{1}{x^2} + \frac{C_1}{x^3}.$$

Integrating this yields

$$u = \frac{x}{3} - \frac{1}{x} - \frac{C_1}{2x^2} + C_2.$$

Therefore the general solution of (15.29) is

$$y = ux = \frac{x^2}{3} - 1 - \frac{C_1}{2x} + C_2x. \tag{15.32}$$

Reasoning as in the solution of Example 71 part (a), we conclude that $y_1 = x$ and $y_2 = 1/x$ form a fundamental set of solutions for (15.28).
As we explained above, we rename the constants in (15.32) and rewrite it as

$$y = \frac{x^2}{3} - 1 + c_1x + \frac{c_2}{x}. \tag{15.33}$$

solution part b Differentiating (15.33) yields

$$y' = \frac{2x}{3} + c_1 - \frac{c_2}{x^2}. \tag{15.34}$$

Setting $x = 1$ in (15.33) and (15.34) and imposing the initial conditions $y(1) = 2$ and $y'(1) = -3$ yields

$$\begin{aligned} c_1 + c_2 &= \frac{8}{3} \\ c_1 - c_2 &= -\frac{11}{3}. \end{aligned}$$

Solving these equations yields $c_1 = -1/2$, $c_2 = 19/6$. Therefore the solution of (15.29) is

$$y = \frac{x^2}{3} - 1 - \frac{x}{2} + \frac{19}{6x}.$$

Using reduction of order to find the general solution of a homogeneous linear second order equation leads to a homogeneous linear first order equation in u' that can be solved by separation of variables. The next example illustrates this.

Example 73 Find the general solution and a fundamental set of solutions of

$$x^2y'' - 3xy' + 3y = 0, \tag{15.35}$$

given that $y_1 = x$ is a solution.

Solution If $y = ux$ then $y' = u'x + u$ and $y'' = u''x + 2u'$, so

$$\begin{aligned} x^2y'' - 3xy' + 3y &= x^2(u''x + 2u') - 3x(u'x + u) + 3ux \\ &= x^3u'' - x^2u'. \end{aligned}$$

Therefore $y = ux$ is a solution of (15.35) if and only if

$$x^3u'' - x^2u' = 0.$$

Separating the variables u' and x yields

$$\frac{u''}{u'} = \frac{1}{x},$$

so

$$\ln|u'| = \ln|x| + k, \quad \text{or, equivalently,} \quad u' = C_1x.$$

Therefore

$$u = \frac{C_1}{2}x^2 + C_2,$$

so the general solution of (15.35) is

$$y = ux = \frac{C_1}{2}x^3 + C_2x,$$

which we rewrite as

$$y = c_1x + c_2x^3.$$

Therefore $\{x, x^3\}$ is a fundamental set of solutions of (15.35).

Laplace Transforms

PART IV

Section 16

Introduction to Laplace Transforms

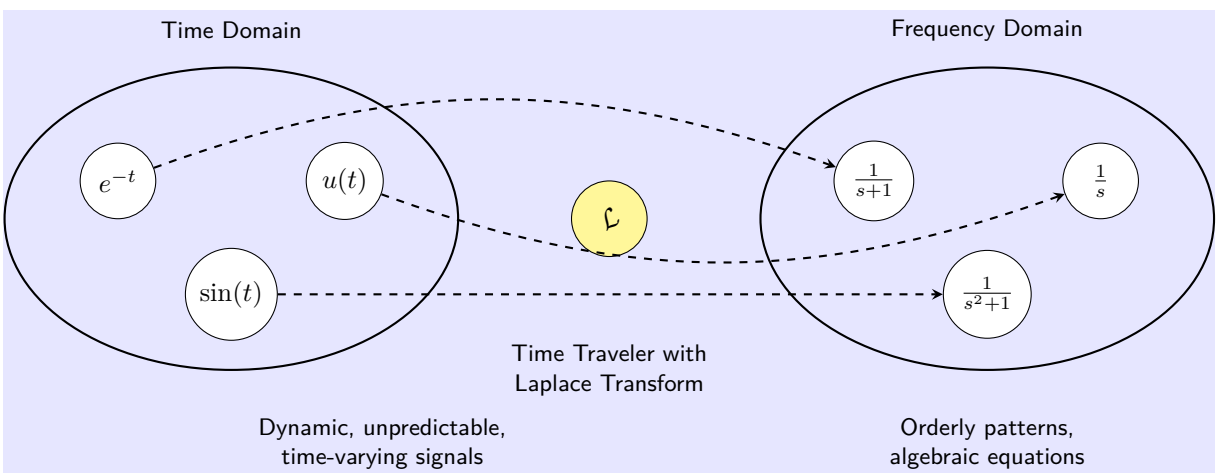
Imagine you're a time traveler, but with a twist. Instead of jumping between different points in history, you have the unique ability to leap between two parallel universes: **the Time Domain**" and the **Frequency Domain**." These universes are intrinsically linked, each offering a different perspective on the same reality.

Your mission? is to understand and translate the complex events of one universe into the language of the other.

Your Superpower: The Laplace Transform

As our intrepid time-and-dimension traveler, you possess a remarkable tool: the Laplace Transform. This mathematical "device" allows you to convert the intricate, time-varying signals of the Time Domain into the more analyzable forms of the Frequency Domain. With the Laplace Transform, you can:

1. Simplify complex time-based problems into algebraic equations
2. Predict future behaviors based on current patterns
3. Understand the underlying rhythms of seemingly chaotic systems

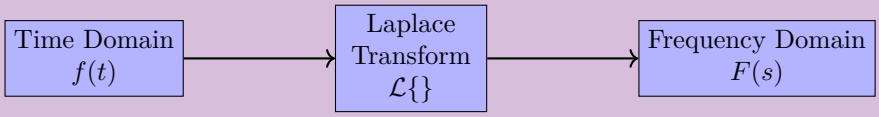


Remark Laplace transforms are a powerful mathematical tool used to simplify the analysis of linear time-invariant systems, particularly in engineering and physics. They convert complex differential equations, which describe systems in the time domain, into simpler algebraic equations in the Laplace domain, making them easier to solve. This transformation is especially useful for solving initial value problems and for analyzing systems' stability and response to various inputs. Additionally, Laplace transforms are essential in **control theory, signal processing, and electrical engineering** for designing and understanding systems and circuits, providing insights into their behavior over time.

Main Idea:
Laplace transform converts complex
time-domain problems into simpler
algebraic problems in the frequency domain

Time-varying signals

Complex frequency domain



Examples:
- Step function
- Exponential decay
- Sinusoidal signals

Properties:
- Algebraic operations
- Easier to solve
- Analysis of systems

Section 17

Definition of the Laplace Transform

To define the Laplace transform, we first recall the definition of an improper integral. If g is integrable over the interval $[a, T]$ for every $T > a$, then the *improper integral of g over* $[a, \infty)$ is defined as

$$\int_a^\infty g(t)\,dt = \lim_{T\to\infty} \int_a^T g(t)\,dt. \tag{17.1}$$

We say that the improper integral *converges* if the limit in (17.1) exists; otherwise, we say that the improper integral *diverges* or *does not exist*. Here's the definition of the Laplace transform of a function f.

Definition 6 Let f be defined for $t \geq 0$ and let s be a real number. Then the *Laplace transform* of f is the function F defined by

$$F(s) = \int_0^\infty e^{-st} f(t)\,dt, \tag{17.2}$$

for those values of s for which the improper integral converges.

Remark It is important to keep in mind that the variable of integration in (17.2) is t, while, s is a parameter independent of t. We use t as the independent variable for f because in applications the Laplace transform is usually applied to functions of time.
The Laplace transform can be viewed as an operator $\mathcal{L}$ that transforms the function $f = f(t)$ into the function $F = F(s)$. Thus, (17.2) can be expressed as

$$F = \mathcal{L}(f).$$

The functions f and F form a *transform pair*, which we'll sometimes denote by

$$f(t) \leftrightarrow F(s).$$

It can be shown that if $F(s)$ is defined for $s = s_0$ then it's defined for all $s > s_0$.

Subsection 17.1

Computation of Some Simple Laplace Transforms

Example 74 Find the Laplace transform of $f(t) = 1$.

Solution From (17.2) with $f(t) = 1$,

$$F(s) = \int_0^\infty e^{-st}\,dt = \lim_{T\to\infty} \int_0^T e^{-st}\,dt.$$

If $s \neq 0$ then

$$\int_0^T e^{-st} dt = -\frac{1}{s} e^{-st}\Big|_0^T = \frac{1 - e^{-sT}}{s}. \tag{17.3}$$

Therefore

$$\lim_{T\to\infty}\int_0^T e^{-st}dt = \begin{cases} \dfrac{1}{s}, & s>0, \\ \infty, & s<0. \end{cases} \tag{17.4}$$

If $s=0$ the integrand reduces to the constant 1, and

$$\lim_{T\to\infty}\int_0^T 1\,dt = \lim_{T\to\infty}\int_0^T 1\,dt = \lim_{T\to\infty} T = \infty.$$

Therefore $F(0)$ is undefined, and

$$F(s) = \int_0^\infty e^{-st}dt = \frac{1}{s}, \quad s>0.$$

This result can be written in operator notation as

$$\mathcal{L}(1) = \frac{1}{s}, \quad s>0,$$

or as the transform pair

$$1 \leftrightarrow \frac{1}{s}, \quad s>0.$$

Remark It is convenient to combine the steps of integrating from 0 to T and letting $T\to\infty$. Therefore, instead of writing (17.3) and (17.4) as separate steps we write

$$\int_0^\infty e^{-st}dt = -\frac{1}{s}e^{-st}\Big|_0^\infty = \begin{cases} \dfrac{1}{s}, & s>0, \\ \infty, & s<0. \end{cases}$$

We'll follow this practice throughout this chapter.

Example 75 Find the Laplace transform of $f(t)=t$.

Solution From (17.2) with $f(t)=t$,

$$F(s) = \int_0^\infty e^{-st}t\,dt. \tag{17.5}$$

If $s\neq 0$, integrating by parts yields

$$\begin{aligned}\int_0^\infty e^{-st}t\,dt &= -\frac{te^{-st}}{s}\Big|_0^\infty + \frac{1}{s}\int_0^\infty e^{-st}\,dt = -\left[\frac{t}{s}+\frac{1}{s^2}\right]e^{-st}\Big|_0^\infty \\ &= \begin{cases} \dfrac{1}{s^2}, & s>0, \\ \infty, & s<0. \end{cases}\end{aligned}$$

If $s=0$, the integral in (17.5) becomes

$$\int_0^\infty t\,dt = \frac{t^2}{2}\Big|_0^\infty = \infty.$$

Therefore $F(0)$ is undefined and

$$F(s) = \frac{1}{s^2}, \quad s>0.$$

This result can also be written as

$$\mathcal{L}(t) = \frac{1}{s^2}, \quad s > 0,$$

or as the transform pair

$$t \leftrightarrow \frac{1}{s^2}, \quad s > 0.$$

Example 76 Find the Laplace transform of $f(t) = e^{at}$, where a is a constant.

Solution From (17.2) with $f(t) = e^{at}$,

$$F(s) = \int_0^\infty e^{-st} e^{at}\, dt.$$

Combining the exponentials yields

$$F(s) = \int_0^\infty e^{-(s-a)t}\, dt.$$

However, we know from Example 74 that

$$\int_0^\infty e^{-st}\, dt = \frac{1}{s}, \quad s > 0.$$

Replacing s by $s - a$ here shows that

$$F(s) = \frac{1}{s-a}, \quad s > a.$$

This can also be written as

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}, \quad s > a, \quad \text{or} \quad e^{at} \leftrightarrow \frac{1}{s-a}, \quad s > a.$$

Example 77 Find the Laplace transforms of $f(t) = \sin \omega t$ and $g(t) = \cos \omega t$, where ω is a constant.

Solution Define

$$F(s) = \int_0^\infty e^{-st} \sin \omega t\, dt \tag{17.6}$$

and

$$G(s) = \int_0^\infty e^{-st} \cos \omega t\, dt. \tag{17.7}$$

If $s > 0$, integrating (17.6) by parts yields

$$F(s) = -\frac{e^{-st}}{s} \sin \omega t \Big|_0^\infty + \frac{\omega}{s} \int_0^\infty e^{-st} \cos \omega t\, dt,$$

so

$$F(s) = \frac{\omega}{s} G(s). \tag{17.8}$$

If $s > 0$, integrating (17.7) by parts yields

$$G(s) = -\frac{e^{-st}\cos\omega t}{s}\bigg|_0^\infty - \frac{\omega}{s}\int_0^\infty e^{-st}\sin\omega t\,dt,$$

so

$$G(s) = \frac{1}{s} - \frac{\omega}{s}F(s).$$

Now substitute from (17.8) into this to obtain

$$G(s) = \frac{1}{s} - \frac{\omega^2}{s^2}G(s).$$

Solving this for $G(s)$ yields

$$G(s) = \frac{s}{s^2+\omega^2}, \quad s > 0.$$

This and (17.8) imply that

$$F(s) = \frac{\omega}{s^2+\omega^2}, \quad s > 0.$$

Subsection 17.2

Tables of Laplace transforms

Extensive tables of Laplace transforms have been compiled and are commonly used in applications. The brief table of Laplace transforms is presented as follows:

Table of Laplace Transforms		
	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}$
2.	e^{at}	$\frac{1}{s-a}$
3.	$t^n, \quad n = 1, 2, 3, \ldots$	$\frac{n!}{s^{n+1}}$
4.	$t^a, \quad a > -1$	$\frac{\Gamma(a+1)}{s^{a+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$
6.	$t^{n-\frac{1}{2}}, \quad n = 1, 2, 3, \ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$
8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$
10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at) - at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$
12.	$\sin(at) + at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at) - at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$
14.	$\cos(at) + at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$
16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$
20.	$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$e^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$
22.	$e^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n e^{at}, \quad n = 1, 2, 3, \ldots$	$\frac{n!}{(s-a)^{n+1}}$
24.	$f(ct)$	$\frac{1}{c}F(\frac{s}{c})$
25.	$u_c(t) = u(t-c)$ Heaviside function	$\frac{e^{-cs}}{s}$
26.	$\delta(t-c)$ Dirac Delta function	e^{-cs}
27.	$u_c(t)f(t-c)$	$e^{-cs}F(s)$
28.	$u_c(t)g(t)$	$e^{-cs}\mathcal{L}\{g(t+c)\}$

29.	$e^{ct}f(t)$	$F(s-c)$
30.	$t^n f(t), \quad n = 1, 2, 3, \ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)du$
32.	$\int_0^t f(v)dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)d\tau$	$F(s)G(s)$
34.	$f(t+T) = f(t)$	$\frac{\int_0^T e^{-st}f(t)dt}{1-e^{-sT}}$
35.	$f'(t)$	$sF(s) - f(0)$
36.	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$

Example 78 Use the table of Laplace transforms to find $\mathcal{L}(t^3e^{4t})$.

Solution The table includes the transform pair

$$t^n e^{at} \leftrightarrow \frac{n!}{(s-a)^{n+1}}.$$

Setting $n = 3$ and $a = 4$ here yields

$$\mathcal{L}(t^3e^{4t}) = \frac{3!}{(s-4)^4} = \frac{6}{(s-4)^4}. \blacksquare$$

We'll sometimes write Laplace transforms of specific functions without explicitly stating how they are obtained. In such cases you should refer to the table of Laplace transforms.

Subsection 17.3

Linearity of the Laplace Transform

The next theorem presents an important property of the Laplace transform.

Theorem 8 [Linearity Property] Suppose $\mathcal{L}(f_i)$ is defined for $s > s_i$, $1 \le i \le n$). Let s_0 be the largest of the numbers s_1, s_2, $\ldots$,s_n, and let c_1, c_2,$\ldots$, c_n be constants. Then

$$\mathcal{L}(c_1f_1 + c_2f_2 + \cdots + c_nf_n) = c_1\mathcal{L}(f_1) + c_2\mathcal{L}(f_2) + \cdots + c_n\mathcal{L}(f_n) \text{ for } s > s_0.$$

Proof We give the proof for the case where $n = 2$. If $s > s_0$ then

$$\begin{aligned}\mathcal{L}(c_1f_1 + c_2f_2) &= \int_0^\infty e^{-st}\left(c_1f_1(t) + c_2f_2(t)\right)\, dt \\ &= c_1\int_0^\infty e^{-st}f_1(t)\, dt + c_2\int_0^\infty e^{-st}f_2(t)\, dt \\ &= c_1\mathcal{L}(f_1) + c_2\mathcal{L}(f_2).\end{aligned}$$

$\square$

Example 79 Use Theorem 8 and the known Laplace transform

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

to find $\mathcal{L}(\cosh bt)$ $(b \neq 0)$.

Solution By definition,

$$\cosh bt = \frac{e^{bt} + e^{-bt}}{2}.$$

Therefore

$$\begin{aligned} \mathcal{L}(\cosh bt) &= \mathcal{L}\left(\frac{1}{2}e^{bt} + \frac{1}{2}e^{-bt}\right) \\ &= \frac{1}{2}\mathcal{L}(e^{bt}) + \frac{1}{2}\mathcal{L}(e^{-bt}) \qquad \text{(linearity property)} \\ &= \frac{1}{2}\,\frac{1}{s-b} + \frac{1}{2}\,\frac{1}{s+b}, \end{aligned} \tag{17.9}$$

where the first transform on the right is defined for $s > b$ and the second for $s > -b$; hence, both are defined for $s > |b|$. Simplifying the last expression in (17.9) yields

$$\mathcal{L}(\cosh bt) = \frac{s}{s^2 - b^2}, \quad s > |b|.$$

Subsection 17.4

The First Shifting Theorem

The next theorem enables us to start with known transform pairs and derive others.

Theorem 9 [First Shifting Theorem] If

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt \tag{17.10}$$

is the Laplace transform of $f(t)$ for $s > s_0$, then $F(s-a)$ is the Laplace transform of $e^{at}f(t)$ for $s > s_0 + a$.

Proof Replacing s by $s - a$ in (17.10) yields

$$F(s-a) = \int_0^\infty e^{-(s-a)t} f(t)\, dt \tag{17.11}$$

if $s - a > s_0$; that is, if $s > s_0 + a$. However, (17.11) can be rewritten as

$$F(s-a) = \int_0^\infty e^{-st}\left(e^{at}f(t)\right)\, dt,$$

which implies the conclusion. □

Example 80 Use Theorem 9 and the known Laplace transforms of 1, t, $\cos\omega t$, and $\sin\omega t$ to find

$$\mathcal{L}(e^{at}), \quad \mathcal{L}(te^{at}), \quad \mathcal{L}(e^{\lambda t}\sin\omega t), \text{and } \mathcal{L}(e^{\lambda t}\cos\omega t).$$

Solution In the following table the known transform pairs are listed on the left and the required transform pairs listed on the right are obtained by applying Theorem 9.

$f(t) \leftrightarrow F(s)$	$e^{at}f(t) \leftrightarrow F(s-a)$
$1 \leftrightarrow \dfrac{1}{s}, \quad s > 0$	$e^{at} \leftrightarrow \dfrac{1}{(s-a)}, \quad s > a$
$t \leftrightarrow \dfrac{1}{s^2}, \quad s > 0$	$te^{at} \leftrightarrow \dfrac{1}{(s-a)^2}, \quad s > a$
$\sin \omega t \leftrightarrow \dfrac{\omega}{s^2+\omega^2}, \quad s > 0$	$e^{\lambda t} \sin \omega t \leftrightarrow \dfrac{\omega}{(s-\lambda)^2+\omega^2}, \; s > \lambda$
$\cos \omega t \leftrightarrow \dfrac{s}{s^2+\omega^2}, \quad s > 0$	$e^{\lambda t} \sin \omega t \leftrightarrow \dfrac{s-\lambda}{(s-\lambda)^2+\omega^2}, \; s > \lambda$

Subsection 17.5

Existence of Laplace Transforms

Not every function has a Laplace transform. For example,

$$\int_0^\infty e^{-st} e^{t^2}\, dt = \infty$$

for every real number s. Hence, the function $f(t) = e^{t^2}$ does not have a Laplace transform.

Our next objective is to establish conditions that ensure the existence of the Laplace transform of a function.

Remark We know from calculus that a definite integral isn't affected by changing the values of its integrand at isolated points. Therefore, redefining a function f to make it continuous at removable discontinuities does not change $\mathcal{L}(f)$.

Definition 7

- A function f is said to be *piecewise continuous* on a finite closed interval $[0, T]$ if $f(0+)$ and $f(T-)$ are finite and f is continuous on the open interval $(0, T)$ except possibly at finitely many points, where f may have jump discontinuities or removable discontinuities.
- A function f is said to be *piecewise continuous* on the infinite interval $[0, \infty)$ if it's piecewise continuous on $[0, T]$ for every $T > 0$.

It is shown in calculus that if a function is piecewise continuous on a finite closed interval then it's integrable on that interval. But if f is piecewise continuous on $[0, \infty)$, then so is $e^{-st}f(t)$, and therefore

$$\int_0^T e^{-st} f(t)\, dt$$

exists for every $T > 0$. However, piecewise continuity alone does not guarantee that the improper integral

$$\int_0^\infty e^{-st} f(t)\, dt = \lim_{T\to\infty} \int_0^T e^{-st} f(t)\, dt \tag{17.12}$$

converges for s in some interval (s_0, ∞). For example, we noted earlier that (17.12) diverges for all s if $f(t) = e^{t^2}$. Stated informally, this occurs because e^{t^2} increases too

rapidly as $t \to \infty$. The next definition provides a constraint on the growth of a function that guarantees convergence of its Laplace transform for s in some interval (s_0, ∞) .

Definition 8 A function f is said to be *of exponential order* s_0 if there are constants M and t_0 such that

$$|f(t)| \leq Me^{s_0 t}, \quad t \geq t_0. \tag{17.13}$$

In situations where the specific value of s_0 is irrelevant we say simply that f is *of exponential order*.

The next theorem gives useful sufficient conditions for a function f to have a Laplace transform.

Theorem 10 If f is piecewise continuous on $[0, \infty)$ and of exponential order s_0, then $\mathcal{L}(f)$ is defined for $s > s_0$.

Example 81 Find the Laplace transform of the piecewise continuous function

$$f(t) = \begin{cases} 1, & 0 \leq t < 1, \\ -3e^{-t}, & t \geq 1. \end{cases}$$

Solution Since f is defined by different formulas on $[0, 1)$ and $[1, \infty)$, we write

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt = \int_0^1 e^{-st}(1)\, dt + \int_1^\infty e^{-st}(-3e^{-t})\, dt.$$

Since

$$\int_0^1 e^{-st}\, dt = \begin{cases} \dfrac{1 - e^{-s}}{s}, & s \neq 0, \\ 1, & s = 0, \end{cases}$$

and

$$\int_1^\infty e^{-st}(-3e^{-t})\, dt = -3\int_1^\infty e^{-(s+1)t}\, dt = -\frac{3e^{-(s+1)}}{s+1}, \quad s > -1,$$

it follows that

$$F(s) = \begin{cases} \dfrac{1 - e^{-s}}{s} - 3\dfrac{e^{-(s+1)}}{s+1}, & s > -1,\ s \neq 0, \\ 1 - \dfrac{3}{e}, & s = 0. \end{cases}$$

This is consistent with Theorem 10, since

$$|f(t)| \leq 3e^{-t}, \quad t \geq 1,$$

and therefore f is of exponential order $s_0 = -1$.

Section 18

Inverse Laplace Transform

In the above section we defined the Laplace transform of f by

$$F(s) = \mathcal{L}(f) = \int_0^\infty e^{-st} f(t)\, dt.$$

We'll also say that f is an *inverse Laplace Transform* of F, and write

$$f = \mathcal{L}^{-1}(F).$$

To solve differential equations with the Laplace transform, we must be able to obtain f from its transform F. There's a formula for doing this, but we can't use it because it requires the theory of functions of a complex variable. Fortunately, we can use the table of Laplace transforms to find inverse transforms that we'll need.

Use the table of Laplace transforms to find

$$\mathcal{L}^{-1}\left(\frac{1}{s^2-1}\right)$$

$$\mathcal{L}^{-1}\left(\frac{s}{s^2+9}\right).$$

Setting $b = 1$ in the transform pair

$$\sinh bt \leftrightarrow \frac{b}{s^2-b^2}$$

shows that

$$\mathcal{L}^{-1}\left(\frac{1}{s^2-1}\right) = \sinh t.$$

Setting $\omega = 3$ in the transform pair

$$\cos \omega t \leftrightarrow \frac{s}{s^2+\omega^2}$$

shows that

$$\mathcal{L}^{-1}\left(\frac{s}{s^2+9}\right) = \cos 3t. \blacksquare$$

The next theorem enables us to find inverse transforms of linear combinations of transforms in the table. We omit the proof.

Theorem 11 [Linearity Property] If F_1, F_2, $\ldots$, F_n are Laplace transforms and c_1, c_2, $\ldots$, c_n are constants, then

$$\mathcal{L}^{-1}(c_1F_1 + c_2F_2 + \cdots + c_nF_n) = c_1\mathcal{L}^{-1}(F_1) + c_2\mathcal{L}^{-1}(F_2) + \cdots + c_n\mathcal{L}^{-1}F_n.$$

Example 82 Find

$$\mathcal{L}^{-1}\left(\frac{8}{s+5} + \frac{7}{s^2+3}\right).$$

Solution From the table of Laplace transforms in Section 8.8,,

$$e^{at} \leftrightarrow \frac{1}{s-a} \quad \text{and} \quad \sin \omega t \leftrightarrow \frac{\omega}{s^2+\omega^2}.$$

Theorem 11 with $a = -5$ and $\omega = \sqrt{3}$ yields

$$\begin{aligned}
\mathcal{L}^{-1}\left(\frac{8}{s+5} + \frac{7}{s^2+3}\right) &= 8\mathcal{L}^{-1}\left(\frac{1}{s+5}\right) + 7\mathcal{L}^{-1}\left(\frac{1}{s^2+3}\right) \\
&= 8\mathcal{L}^{-1}\left(\frac{1}{s+5}\right) + \frac{7}{\sqrt{3}}\mathcal{L}^{-1}\left(\frac{\sqrt{3}}{s^2+3}\right) \\
&= 8e^{-5t} + \frac{7}{\sqrt{3}}\sin\sqrt{3}t.
\end{aligned}$$

Example 83 Find

$$\mathcal{L}^{-1}\left(\frac{3s+8}{s^2+2s+5}\right).$$

Solution Completing the square in the denominator yields

$$\frac{3s+8}{s^2+2s+5} = \frac{3s+8}{(s+1)^2+4}.$$

Because of the form of the denominator, we consider the transform pairs

$$e^{-t}\cos 2t \leftrightarrow \frac{s+1}{(s+1)^2+4} \quad \text{and} \quad e^{-t}\sin 2t \leftrightarrow \frac{2}{(s+1)^2+4},$$

and write

$$\begin{aligned}
\mathcal{L}^{-1}\left(\frac{3s+8}{(s+1)^2+4}\right) &= \mathcal{L}^{-1}\left(\frac{3s+3}{(s+1)^2+4}\right) + \mathcal{L}^{-1}\left(\frac{5}{(s+1)^2+4}\right) \\
&= 3\mathcal{L}^{-1}\left(\frac{s+1}{(s+1)^2+4}\right) + \frac{5}{2}\mathcal{L}^{-1}\left(\frac{2}{(s+1)^2+4}\right) \\
&= e^{-t}(3\cos 2t + \frac{5}{2}\sin 2t).
\end{aligned}$$

Remark We'll often write inverse Laplace transforms of specific functions without explicitly stating how they are obtained. In such cases, you should refer to the table of Laplace transforms.

Subsection 18.1

Inverse Laplace Transforms of Rational Functions

Using the Laplace transform to solve differential equations often requires finding the inverse transform of a rational function

$$F(s) = \frac{P(s)}{Q(s)},$$

where P and Q are polynomials in s with no common factors. Since it can be shown that $\lim_{s\to\infty} F(s) = 0$ if F is a Laplace transform, we need only consider the case where degree(P) < degree(Q). To obtain $\mathcal{L}^{-1}(F)$, we find the partial fraction expansion of F, obtain inverse transforms of the individual terms in the expansion from the table of Laplace transforms, and use the linearity property of the inverse transform. The next two examples illustrate this.

Example 84 Find the inverse Laplace transform of

$$F(s) = \frac{3s+2}{s^2-3s+2}. \tag{18.1}$$

Solution (Method 1) Factoring the denominator in (18.1) yields

$$F(s) = \frac{3s+2}{(s-1)(s-2)}. \tag{18.2}$$

The form for the partial fraction expansion is

$$\frac{3s+2}{(s-1)(s-2)} = \frac{A}{s-1} + \frac{B}{s-2}. \tag{18.3}$$

Multiplying this by $(s-1)(s-2)$ yields

$$3s+2 = (s-2)A + (s-1)B.$$

Setting $s = 2$ yields $B = 8$ and setting $s = 1$ yields $A = -5$. Therefore

$$F(s) = -\frac{5}{s-1} + \frac{8}{s-2}$$

and

$$\mathcal{L}^{-1}(F) = -5\mathcal{L}^{-1}\left(\frac{1}{s-1}\right) + 8\mathcal{L}^{-1}\left(\frac{1}{s-2}\right) = -5e^t + 8e^{2t}.$$

(Method 2) We don't really have to multiply (18.3) by $(s-1)(s-2)$ to compute A and B. We can obtain A by simply ignoring the factor $s-1$ in the denominator of (18.2) and setting $s = 1$ elsewhere; thus,

$$A = \left.\frac{3s+2}{s-2}\right|_{s=1} = \frac{3\cdot 1+2}{1-2} = -5. \tag{18.4}$$

Similarly, we can obtain B by ignoring the factor $s-2$ in the denominator of (18.2) and setting $s = 2$ elsewhere; thus,

$$B = \left.\frac{3s+2}{s-1}\right|_{s=2} = \frac{3\cdot 2+2}{2-1} = 8. \tag{18.5}$$

To justify this, we observe that multiplying (18.3) by $s-1$ yields

$$\frac{3s+2}{s-2} = A + (s-1)\frac{B}{s-2},$$

and setting $s = 1$ leads to (18.4). Similarly, multiplying (18.3) by $s - 2$ yields

$$\frac{3s+2}{s-1} = (s-2)\frac{A}{s-2} + B$$

and setting $s = 2$ leads to (18.5). (It isn't necesary to write the last two equations. We wrote them only to justify the shortcut procedure indicated in (18.4) and (18.5).)

The shortcut employed in the second solution of Example 84 is *Heaviside's method*. The next theorem states this method formally.

Theorem 12 Suppose

$$F(s) = \frac{P(s)}{(s-s_1)(s-s_2)\cdots(s-s_n)}, \tag{18.6}$$

where $s_1, s_2, \ldots, s_n$ are distinct and P is a polynomial of degree less than n. Then

$$F(s) = \frac{A_1}{s-s_1} + \frac{A_2}{s-s_2} + \cdots + \frac{A_n}{s-s_n},$$

where A_i can be computed from (18.6) by ignoring the factor $s - s_i$ and setting $s = s_i$ elsewhere.

Example 85 Find the inverse Laplace transform of

$$F(s) = \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)}. \tag{18.7}$$

Solution The partial fraction expansion of (18.7) is of the form

$$F(s) = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s-2} + \frac{D}{s+1}. \tag{18.8}$$

To find A, we ignore the factor s in the denominator of (18.7) and set $s = 0$ elsewhere. This yields

$$A = \frac{6 + (1)(11)}{(-1)(-2)(1)} = \frac{17}{2}.$$

Similarly, the other coefficients are given by

$$B = \frac{6 + (2)(7)}{(1)(-1)(2)} = -10,$$

$$C = \frac{6 + 3(5)}{2(1)(3)} = \frac{7}{2},$$

and

$$D = \frac{6}{(-1)(-2)(-3)} = -1.$$

Therefore

$$F(s) = \frac{17}{2}\frac{1}{s} - \frac{10}{s-1} + \frac{7}{2}\frac{1}{s-2} - \frac{1}{s+1}$$

and

$$\begin{aligned}\mathcal{L}^{-1}(F) &= \frac{17}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right) - 10\mathcal{L}^{-1}\left(\frac{1}{s-1}\right) + \frac{7}{2}\mathcal{L}^{-1}\left(\frac{1}{s-2}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+1}\right)\\ &= \frac{17}{2} - 10e^{t} + \frac{7}{2}e^{2t} - e^{-t}.\end{aligned}$$

Example 86 Find the inverse Laplace transform of

$$F(s) = \frac{8-(s+2)(4s+10)}{(s+1)(s+2)^2}. \tag{18.9}$$

Solution The form for the partial fraction expansion is

$$F(s) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}. \tag{18.10}$$

Because of the repeated factor $(s+2)^2$ in (18.9), Heaviside's method doesn't work. Instead, we find a common denominator in (18.10). This yields

$$F(s) = \frac{A(s+2)^2 + B(s+1)(s+2) + C(s+1)}{(s+1)(s+2)^2}. \tag{18.11}$$

If (18.9) and (18.11) are to be equivalent, then

$$A(s+2)^2 + B(s+1)(s+2) + C(s+1) = 8-(s+2)(4s+10). \tag{18.12}$$

The two sides of this equation are polynomials of degree two. From a theorem of algebra, they will be equal for all s if they are equal for any three distinct values of s. We may determine A, B and C by choosing convenient values of s.
The left side of (18.12) suggests that we take $s=-2$ to obtain $C=-8$, and $s=-1$ to obtain $A=2$. We can now choose any third value of s to determine B. Taking $s=0$ yields $4A+2B+C=-12$. Since $A=2$ and $C=-8$ this implies that $B=-6$. Therefore

$$F(s) = \frac{2}{s+1} - \frac{6}{s+2} - \frac{8}{(s+2)^2}$$

and

$$\begin{aligned}\mathcal{L}^{-1}(F) &= 2\mathcal{L}^{-1}\left(\frac{1}{s+1}\right) - 6\mathcal{L}^{-1}\left(\frac{1}{s+2}\right) - 8\mathcal{L}^{-1}\left(\frac{1}{(s+2)^2}\right)\\ &= 2e^{-t} - 6e^{-2t} - 8te^{-2t}.\end{aligned}$$

Example 87 Find the inverse Laplace transform of

$$F(s) = \frac{s^2-5s+7}{(s+2)^3}.$$

Solution The form for the partial fraction expansion is

$$F(s) = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3}.$$

The easiest way to obtain A, B, and C is to expand the numerator in powers of $s+2$. This yields

$$s^2 - 5s + 7 = [(s+2)-2]^2 - 5[(s+2)-2] + 7 = (s+2)^2 - 9(s+2) + 21.$$

Therefore

$$\begin{aligned} F(s) &= \frac{(s+2)^2 - 9(s+2) + 21}{(s+2)^3} \\ &= \frac{1}{s+2} - \frac{9}{(s+2)^2} + \frac{21}{(s+2)^3} \end{aligned}$$

and

$$\begin{aligned} \mathcal{L}^{-1}(F) &= \mathcal{L}^{-1}\left(\frac{1}{s+2}\right) - 9\mathcal{L}^{-1}\left(\frac{1}{(s+2)^2}\right) + \frac{21}{2}\mathcal{L}^{-1}\left(\frac{2}{(s+2)^3}\right) \\ &= e^{-2t}\left(1 - 9t + \frac{21}{2}t^2\right). \end{aligned}$$

Example 88 Find the inverse Laplace transform of

$$F(s) = \frac{1 - s(5+3s)}{s\left[(s+1)^2+1\right]}. \tag{18.13}$$

Solution One form for the partial fraction expansion of F is

$$F(s) = \frac{A}{s} + \frac{Bs + C}{(s+1)^2+1}. \tag{18.14}$$

However, we see from the table of Laplace transforms that the inverse transform of the second fraction on the right of (18.14) will be a linear combination of the inverse transforms

$$e^{-t}\cos t \quad \text{and} \quad e^{-t}\sin t$$

of

$$\frac{s+1}{(s+1)^2+1} \quad \text{and} \quad \frac{1}{(s+1)^2+1}$$

respectively. Therefore, instead of (18.14) we write

$$F(s) = \frac{A}{s} + \frac{B(s+1) + C}{(s+1)^2+1}. \tag{18.15}$$

Finding a common denominator yields

$$F(s) = \frac{A\left[(s+1)^2+1\right] + B(s+1)s + Cs}{s\left[(s+1)^2+1\right]}. \tag{18.16}$$

If (18.13) and (18.16) are to be equivalent, then

$$A\left[(s+1)^2+1\right] + B(s+1)s + Cs = 1 - s(5+3s).$$

This is true for all s if it's true for three distinct values of s. Choosing $s = 0$, -1, and

1 yields the system

$$\begin{aligned} 2A &= 1 \\ A - C &= 3 \\ 5A + 2B + C &= -7. \end{aligned}$$

Solving this system yields

$$A = \frac{1}{2}, \quad B = -\frac{7}{2}, \quad C = -\frac{5}{2}.$$

Hence, from (18.15),

$$F(s) = \frac{1}{2s} - \frac{7}{2}\frac{s+1}{(s+1)^2+1} - \frac{5}{2}\frac{1}{(s+1)^2+1}.$$

Therefore

$$\begin{aligned} \mathcal{L}^{-1}(F) &= \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right) - \frac{7}{2}\mathcal{L}^{-1}\left(\frac{s+1}{(s+1)^2+1}\right) - \frac{5}{2}\mathcal{L}^{-1}\left(\frac{1}{(s+1)^2+1}\right) \\ &= \frac{1}{2} - \frac{7}{2}e^{-t}\cos t - \frac{5}{2}e^{-t}\sin t. \end{aligned}$$

Example 89

Find the inverse Laplace transform of

$$F(s) = \frac{8+3s}{(s^2+1)(s^2+4)}. \tag{18.17}$$

Solution

The form for the partial fraction expansion is

$$F(s) = \frac{A+Bs}{s^2+1} + \frac{C+Ds}{s^2+4}.$$

The coefficients A, B, C and D can be obtained by finding a common denominator and equating the resulting numerator to the numerator in (18.17). However, since there's no first power of s in the denominator of (18.17), there's an easier way: the expansion of

$$F_1(s) = \frac{1}{(s^2+1)(s^2+4)}$$

can be obtained quickly by using Heaviside's method to expand

$$\frac{1}{(x+1)(x+4)} = \frac{1}{3}\left(\frac{1}{x+1} - \frac{1}{x+4}\right)$$

and then setting $x = s^2$ to obtain

$$\frac{1}{(s^2+1)(s^2+4)} = \frac{1}{3}\left(\frac{1}{s^2+1} - \frac{1}{s^2+4}\right).$$

Multiplying this by $8+3s$ yields

$$F(s) = \frac{8+3s}{(s^2+1)(s^2+4)} = \frac{1}{3}\left(\frac{8+3s}{s^2+1} - \frac{8+3s}{s^2+4}\right).$$

Therefore

$$\mathcal{L}^{-1}(F) = \frac{8}{3}\sin t + \cos t - \frac{4}{3}\sin 2t - \cos 2t.$$

Section 19

Solution of Initial Value Problems

Subsection 19.1

Laplace Transforms of Derivatives

In the rest of this chapter, we'll use the Laplace transform to solve initial value problems for constant coefficient second order equations. To do this, we must know how the Laplace transform of f' is related to the Laplace transform of f. The next theorem answers this question.

Theorem 13 Suppose f is continuous on $[0,\infty)$ and of exponential order s_0, and f' is piecewise continuous on $[0,\infty)$. Then f and f' have Laplace transforms for $s > s_0$, and

$$\mathcal{L}(f') = s\mathcal{L}(f) - f(0). \tag{19.1}$$

Proof We know from Theorem 8.1.6 that $\mathcal{L}(f)$ is defined for $s > s_0$. We first consider the case where f' is continuous on $[0,\infty)$. Integration by parts yields

$$\begin{aligned}\int_0^T e^{-st} f'(t)\,dt &= e^{-st} f(t)\Big|_0^T + s\int_0^T e^{-st} f(t)\,dt \\ &= e^{-sT} f(T) - f(0) + s\int_0^T e^{-st} f(t)\,dt\end{aligned} \tag{19.2}$$

for any $T > 0$. Since f is of exponential order s_0, $\lim_{T\to\infty} e^{-sT} f(T) = 0$ and the last integral in (19.2) converges as $T \to \infty$ if $s > s_0$. Therefore

$$\begin{aligned}\int_0^\infty e^{-st} f'(t)\,dt &= -f(0) + s\int_0^\infty e^{-st} f(t)\,dt \\ &= -f(0) + s\mathcal{L}(f),\end{aligned}$$

which proves (19.1). Now suppose $T > 0$ and f' is only piecewise continuous on $[0,T]$, with discontinuities at $t_1 < t_2 < \cdots < t_{n-1}$. For convenience, let $t_0 = 0$ and $t_n = T$. Integrating by parts yields

$$\begin{aligned}\int_{t_{i-1}}^{t_i} e^{-st} f'(t)\,dt &= e^{-st} f(t)\Big|_{t_{i-1}}^{t_i} + s\int_{t_{i-1}}^{t_i} e^{-st} f(t)\,dt \\ &= e^{-st_i} f(t_i) - e^{-st_{i-1}} f(t_{i-1}) + s\int_{t_{i-1}}^{t_i} e^{-st} f(t)\,dt.\end{aligned}$$

Summing both sides of this equation from $i = 1$ to n and noting that

$$\begin{aligned}&\big(e^{-st_1} f(t_1) - e^{-st_0} f(t_0)\big) + \big(e^{-st_2} f(t_2) - e^{-st_1} f(t_1)\big) + \cdots + \big(e^{-st_N} f(t_N) - e^{-st_{N-1}} f(t_{N-1})\big) \\ &\qquad = e^{-st_N} f(t_N) - e^{-st_0} f(t_0) = e^{-sT} f(T) - f(0)\end{aligned}$$

yields (19.2), so (19.1) follows as before. □

Here are two examples to illustrate the above theorem.

In Example 77 we saw that

$$\mathcal{L}(\cos \omega t) = \frac{s}{s^2 + \omega^2}.$$

Applying (19.1) with $f(t) = \cos \omega t$ shows that

$$\mathcal{L}(-\omega \sin \omega t) = s\frac{s}{s^2 + \omega^2} - 1 = -\frac{\omega^2}{s^2 + \omega^2}.$$

Therefore

$$\mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2},$$

which agrees with the corresponding result obtained in 77.

Similarly, consider the simple DE:

$$y' = ay, \quad y(0) = y_0, \tag{19.3}$$

is $y = y_0 e^{at}$. We'll now obtain this result by using the Laplace transform.
Let $Y(s) = \mathcal{L}(y)$ be the Laplace transform of the unknown solution of (19.3). Taking Laplace transforms of both sides of (19.3) yields

$$\mathcal{L}(y') = \mathcal{L}(ay),$$

which, by Theorem 13, can be rewritten as

$$s\mathcal{L}(y) - y(0) = a\mathcal{L}(y),$$

or

$$sY(s) - y_0 = aY(s).$$

Solving for $Y(s)$ yields

$$Y(s) = \frac{y_0}{s - a},$$

so

$$y = \mathcal{L}^{-1}(Y(s)) = \mathcal{L}^{-1}\left(\frac{y_0}{s - a}\right) = y_0\mathcal{L}^{-1}\left(\frac{1}{s - a}\right) = y_0 e^{at},$$

which agrees with the known result.

We need the next theorem to solve second order differential equations using the Laplace transform.

Theorem 14 Suppose f and f' are continuous on $[0, \infty)$ and of exponential order s_0, and that f'' is piecewise continuous on $[0, \infty)$. Then f, f', and f'' have Laplace transforms for $s > s_0$,

$$\mathcal{L}(f') = s\mathcal{L}(f) - f(0), \tag{19.4}$$

and

$$\mathcal{L}(f'') = s^2\mathcal{L}(f) - f'(0) - sf(0). \tag{19.5}$$

Proof Theorem 13 implies that $\mathcal{L}(f')$ exists and satisfies (19.4) for $s > s_0$. To prove that $\mathcal{L}(f'')$ exists and satisfies (19.5) for $s > s_0$, we first apply Theorem 13 to $g = f'$. Since g satisfies the hypotheses of Theorem 13, we conclude that $\mathcal{L}(g')$ is defined and satisfies

$$\mathcal{L}(g') = s\mathcal{L}(g) - g(0)$$

for $s > s_0$. However, since $g' = f''$, this can be rewritten as

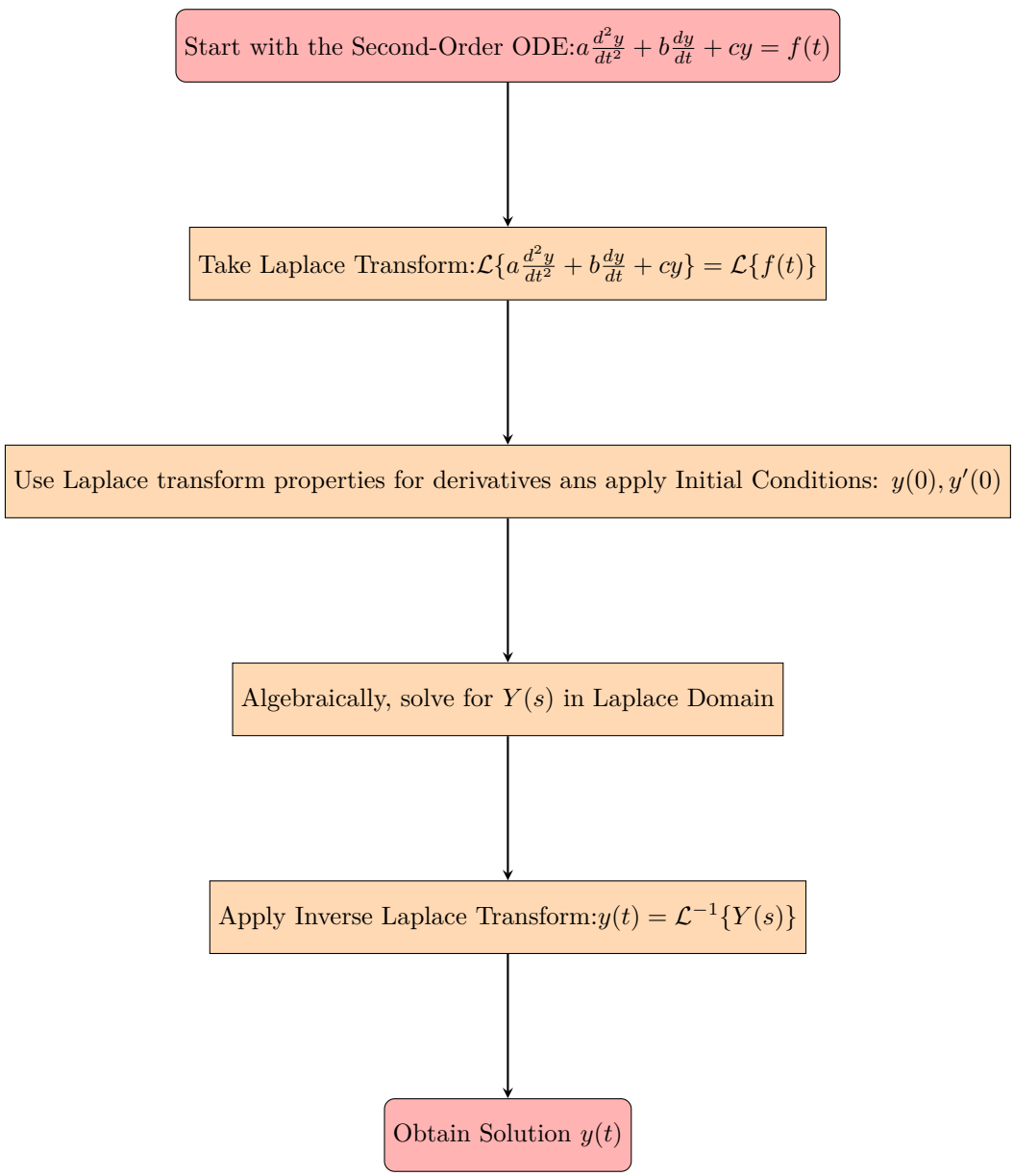
$$\mathcal{L}(f'') = s\mathcal{L}(f') - f'(0).$$

Substituting (19.4) into this yields (19.5). □

Subsection 19.2

Solving Second Order Equations with the Laplace Transform

Now we are ready to use what we learned in Laplace to solve deferential equations. The below digram explains the process of solving a DE using Laplace



Example 90 Use the Laplace transform to solve the initial value problem

$$y'' - 6y' + 5y = 3e^{2t}, \quad y(0) = 2, \quad y'(0) = 3. \tag{19.6}$$

Solution Taking Laplace transforms of both sides of the differential equation in (19.6) yields

$$\mathcal{L}(y'' - 6y' + 5y) = \mathcal{L}\left(3e^{2t}\right) = \frac{3}{s-2},$$

which we rewrite as

$$\mathcal{L}(y'') - 6\mathcal{L}(y') + 5\mathcal{L}(y) = \frac{3}{s-2}. \tag{19.7}$$

Now denote $\mathcal{L}(y) = Y(s)$. Theorem 14 and the initial conditions in (19.6) imply that

$$\mathcal{L}(y') = sY(s) - y(0) = sY(s) - 2$$

and

$$\mathcal{L}(y'') = s^2Y(s) - y'(0) - sy(0) = s^2Y(s) - 3 - 2s.$$

Substituting from the last two equations into (19.7) yields

$$\left(s^2Y(s) - 3 - 2s\right) - 6\left(sY(s) - 2\right) + 5Y(s) = \frac{3}{s-2}.$$

Therefore

$$(s^2 - 6s + 5)Y(s) = \frac{3}{s-2} + (3 + 2s) + 6(-2), \tag{19.8}$$

so

$$(s-5)(s-1)Y(s) = \frac{3 + (s-2)(2s-9)}{s-2},$$

and

$$Y(s) = \frac{3 + (s-2)(2s-9)}{(s-2)(s-5)(s-1)}.$$

Heaviside's method yields the partial fraction expansion

$$Y(s) = -\frac{1}{s-2} + \frac{1}{2}\frac{1}{s-5} + \frac{5}{2}\frac{1}{s-1},$$

and taking the inverse transform of this yields

$$y = -e^{2t} + \frac{1}{2}e^{5t} + \frac{5}{2}e^{t}$$

as the solution of (19.6). ■

It isn't necessary to write all the steps that we used to obtain (19.8). To see how to avoid this, let's apply the method of Example 90 to the general initial value problem

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1. \tag{19.9}$$

Taking Laplace transforms of both sides of the differential equation in (19.9) yields

$$a\mathcal{L}(y'') + b\mathcal{L}(y') + c\mathcal{L}(y) = F(s). \tag{19.10}$$

Now let $Y(s) = \mathcal{L}(y)$. Theorem 14 and the initial conditions in (19.9) imply that

$$\mathcal{L}(y') = sY(s) - k_0 \quad \text{and} \quad \mathcal{L}(y'') = s^2Y(s) - k_1 - k_0 s.$$

Substituting these into (19.10) yields

$$a\left(s^2Y(s) - k_1 - k_0 s\right) + b\left(sY(s) - k_0\right) + cY(s) = F(s). \tag{19.11}$$

The coefficient of $Y(s)$ on the left is the characteristic polynomial

$$p(s) = as^2 + bs + c$$

of the complementary equation for (19.9). Using this and moving the terms involving k_0 and k_1 to the right side of (19.11) yields

$$p(s)Y(s) = F(s) + a(k_1 + k_0 s) + bk_0. \tag{19.12}$$

This equation corresponds to (19.8) of Example 90. Having established the form of this equation in the general case, it is preferable to go directly from the initial value problem to this equation. You may find it easier to remember (19.12) rewritten as

$$p(s)Y(s) = F(s) + a\left(y'(0) + sy(0)\right) + by(0). \tag{19.13}$$

Example 91 Use the Laplace transform to solve the initial value problem

$$2y'' + 3y' + y = 8e^{-2t}, \quad y(0) = -4,\ y'(0) = 2. \tag{19.14}$$

Solution The characteristic polynomial is

$$p(s) = 2s^2 + 3s + 1 = (2s+1)(s+1)$$

and

$$F(s) = \mathcal{L}(8e^{-2t}) = \frac{8}{s+2},$$

so (19.13) becomes

$$(2s+1)(s+1)Y(s) = \frac{8}{s+2} + 2(2-4s) + 3(-4).$$

Solving for $Y(s)$ yields

$$Y(s) = \frac{4\left(1 - (s+2)(s+1)\right)}{(s+1/2)(s+1)(s+2)}.$$

Heaviside's method yields the partial fraction expansion

$$Y(s) = \frac{4}{3}\frac{1}{s+1/2} - \frac{8}{s+1} + \frac{8}{3}\frac{1}{s+2},$$

so the solution of (19.14) is

$$y = \mathcal{L}^{-1}(Y(s)) = \frac{4}{3}e^{-t/2} - 8e^{-t} + \frac{8}{3}e^{-2t}$$

Example 92 Solve the initial value problem

$$y'' + 2y' + 2y = 1, \quad y(0) = -3,\ y'(0) = 1. \tag{19.15}$$

Solution The characteristic polynomial is

$$p(s) = s^2 + 2s + 2 = (s+1)^2 + 1$$

and

$$F(s) = \mathcal{L}(1) = \frac{1}{s},$$

so (19.13) becomes

$$\left[(s+1)^2 + 1\right] Y(s) = \frac{1}{s} + 1 \cdot (1 - 3s) + 2(-3).$$

Solving for $Y(s)$ yields

$$Y(s) = \frac{1 - s(5+3s)}{s\left[(s+1)^2 + 1\right]}.$$

In Example 88 we found the inverse transform of this function to be

$$y = \frac{1}{2} - \frac{7}{2}e^{-t}\cos t - \frac{5}{2}e^{-t}\sin t$$

.

Remark In our examples, we applied Theorems 13 and 14 without verifying that the unknown function y satisfies their hypotheses. This is characteristic of the formal manipulative way in which the Laplace transform is used to solve differential equations. Any doubts about the validity of the method for solving a given equation can be resolved by verifying that the resulting function y is the solution of the given problem.

Section 20

Exercises

In Exercises 1–31 use the Laplace transform to solve the initial value problem.

1. $y'' + 3y' + 2y = e^t, \quad y(0) = 1, \quad y'(0) = -6$
2. $y'' - y' - 6y = 2, \quad y(0) = 1, \quad y'(0) = 0$
3. $y'' + y' - 2y = 2e^{3t}, \quad y(0) = -1, \quad y'(0) = 4$
4. $y'' - 4y = 2e^{3t}, \quad y(0) = 1, \quad y'(0) = -1$
5. $y'' + y' - 2y = e^{3t}, \quad y(0) = 1, \quad y'(0) = -1$
6. $y'' + 3y' + 2y = 6e^t, \quad y(0) = 1, \quad y'(0) = -1$
7. $y'' + y = \sin 2t, \quad y(0) = 0, \quad y'(0) = 1$
8. $y'' - 3y' + 2y = 2e^{3t}, \quad y(0) = 1, \quad y'(0) = -1$
9. $y'' - 3y' + 2y = e^{4t}, \quad y(0) = 1, \quad y'(0) = -2$
10. $y'' - 3y' + 2y = e^{3t}, \quad y(0) = -1, \quad y'(0) = -4$
11. $y'' + 3y' + 2y = 2e^t, \quad y(0) = 0, \quad y'(0) = -1$
12. $y'' + y' - 2y = -4, \quad y(0) = 2, \quad y'(0) = 3$
13. $y'' + 4y = 4, \quad y(0) = 0, \quad y'(0) = 1$

14. $y'' - y' - 6y = 2, \quad y(0) = 1, \quad y'(0) = 0$
15. $y'' + 3y' + 2y = e^t, \quad y(0) = 0, \quad y'(0) = 1$
16. $y'' - y = 1, \quad y(0) = 1, \quad y'(0) = 0$
17. $y'' + 4y = 3\sin t, \quad y(0) = 1, \quad y'(0) = -1$
18. $y'' + y' = 2e^{3t}, \quad y(0) = -1, \quad y'(0) = 4$
19. $y'' + y = 1, \quad y(0) = 2, \quad y'(0) = 0$
20. $y'' + y = t, \quad y(0) = 0, \quad y'(0) = 2$
21. $y'' + y = t - 3\sin 2t, \quad y(0) = 1, \quad y'(0) = -3$
22. $y'' + 5y' + 6y = 2e^{-t}, \quad y(0) = 1, \quad y'(0) = 3$
23. $y'' + 2y' + y = 6\sin t - 4\cos t, \quad y(0) = -1,\ y'(0) = 1$
24. $y'' - 2y' - 3y = 10\cos t, \quad y(0) = 2, \quad y'(0) = 7$
25. $y'' + y = 4\sin t + 6\cos t, \quad y(0) = -6,\ y'(0) = 2$
26. $y'' + 4y = 8\sin 2t + 9\cos t, \quad y(0) = 1, \quad y'(0) = 0$
27. $y'' - 5y' + 6y = 10e^t\cos t, \quad y(0) = 2, \quad y'(0) = 1$
28. $y'' + 2y' + 2y = 2t, \quad y(0) = 2, \quad y'(0) = -7$
29. $y'' - 2y' + 2y = 5\sin t + 10\cos t, \quad y(0) = 1,\ y'(0) = 2$
30. $y'' + 4y' + 13y = 10e^{-t} - 36e^t, \quad y(0) = 0,\ y'(0) = -16$
31. $y'' + 4y' + 5y = e^{-t}(\cos t + 3\sin t), \quad y(0) = 0, \quad y'(0) = 4$
32. $2y'' - 3y' - 2y = 4e^t, \quad y(0) = 1,\ y'(0) = -2$
33. $6y'' - y' - y = 3e^{2t}, \quad y(0) = 0,\ y'(0) = 0$
34. $2y'' + 2y' + y = 2t, \quad y(0) = 1,\ y'(0) = -1$
35. $4y'' - 4y' + 5y = 4\sin t - 4\cos t, \quad y(0) = 0,\ y'(0) = 11/17$
36. $4y'' + 4y' + y = 3\sin t + \cos t, \quad y(0) = 2,\ y'(0) = -1$
37. $9y'' + 6y' + y = 3e^{3t}, \quad y(0) = 0,\ y'(0) = -3$
38. Suppose a, b, and c are constants and $a \neq 0$. Let

$$y_1 = \mathcal{L}^{-1}\left(\frac{as+b}{as^2+bs+c}\right) \quad \text{and} \quad y_2 = \mathcal{L}^{-1}\left(\frac{a}{as^2+bs+c}\right).$$

Show that

$$y_1(0) = 1, \quad y_1'(0) = 0 \quad \text{and} \quad y_2(0) = 0, \quad y_2'(0) = 1.$$

Use the Laplace transform to solve the initial value problems

$$\begin{aligned} ay'' + by' + cy &= 0, \quad y(0) = 1, \quad y'(0) = 0 \\ ay'' + by' + cy &= 0, \quad y(0) = 0, \quad y'(0) = 1. \end{aligned}$$

Section 21

Piece-wise Continuous Functions

In the next section we'll consider initial value problems

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1,$$

where a, b, and c are constants and f is piecewise continuous. In this section we'll develop procedures for using the table of Laplace transforms to find Laplace transforms of piecewise continuous functions, and to find the piecewise continuous inverses of Laplace transforms.

Example 93 Use the table of Laplace transforms to find the Laplace transform of

$$f(t) = \begin{cases} 2t+1, & 0 \le t < 2, \\ 3t, & t \ge 2 \end{cases} \tag{21.1}$$

Solution Since the formula for f changes at $t = 2$, we write

$$\begin{aligned} \mathcal{L}(f) &= \int_0^\infty e^{-st} f(t)\,dt \\ &= \int_0^2 e^{-st}(2t+1)\,dt + \int_2^\infty e^{-st}(3t)\,dt. \end{aligned} \tag{21.2}$$

To relate the first term to a Laplace transform, we add and subtract

$$\int_2^\infty e^{-st}(2t+1)\,dt$$

in (21.2) to obtain

$$\begin{aligned} \mathcal{L}(f) &= \int_0^\infty e^{-st}(2t+1)\,dt + \int_2^\infty e^{-st}(3t-2t-1)\,dt \\ &= \int_0^\infty e^{-st}(2t+1)\,dt + \int_2^\infty e^{-st}(t-1)\,dt \\ &= \mathcal{L}(2t+1) + \int_2^\infty e^{-st}(t-1)\,dt. \end{aligned} \tag{21.3}$$

To relate the last integral to a Laplace transform, we make the change of variable $x = t - 2$ and rewrite the integral as

$$\begin{aligned} \int_2^\infty e^{-st}(t-1)\,dt &= \int_0^\infty e^{-s(x+2)}(x+1)\,dx \\ &= e^{-2s}\int_0^\infty e^{-sx}(x+1)\,dx. \end{aligned}$$

Since the symbol used for the variable of integration has no effect on the value of a definite integral, we can now replace x by the more standard t and write

$$\int_2^\infty e^{-st}(t-1)\,dt = e^{-2s}\int_0^\infty e^{-st}(t+1)\,dt = e^{-2s}\mathcal{L}(t+1).$$

This and (21.3) imply that

$$\mathcal{L}(f) = \mathcal{L}(2t+1) + e^{-2s}\mathcal{L}(t+1).$$

Now we can use the table of Laplace transforms to find that

$$\mathcal{L}(f) = \frac{2}{s^2} + \frac{1}{s} + e^{-2s}\left(\frac{1}{s^2} + \frac{1}{s}\right).$$

Subsection 21.1

Unit Step Function

We'll now develop the method of Example 93 into a systematic way to find the Laplace transform of a piecewise continuous function. It is convenient to introduce the *unit step function*, defined as

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0. \end{cases} \tag{21.4}$$

Thus, $u(t)$ "steps" from the constant value 0 to the constant value 1 at $t = 0$. If we replace t by $t - \tau$ in (21.4), then

$$u(t-\tau) = \begin{cases} 0, & t < \tau, \\ 1, & t \geq \tau \end{cases};$$

that is, the step now occurs at $t = \tau$.

Remark The step function enables us to represent piecewise continuous functions conveniently. For example, consider the function

$$f(t) = \begin{cases} f_0(t), & 0 \leq t < t_1, \\ f_1(t), & t \geq t_1, \end{cases} \tag{21.5}$$

where we assume that f_0 and f_1 are defined on $[0, \infty)$, even though they equal f only on the indicated intervals. This assumption enables us to rewrite (21.5) as

$$f(t) = f_0(t) + u(t-t_1)\left(f_1(t) - f_0(t)\right). \tag{21.6}$$

To verify this, note that if $t < t_1$ then $u(t - t_1) = 0$ and (21.6) becomes

$$f(t) = f_0(t) + (0)\left(f_1(t) - f_0(t)\right) = f_0(t).$$

If $t \geq t_1$ then $u(t - t_1) = 1$ and (21.6) becomes

$$f(t) = f_0(t) + (1)\left(f_1(t) - f_0(t)\right) = f_1(t).$$

We need the next theorem to show how (21.6) can be used to find $\mathcal{L}(f)$.

Theorem 15 Let g be defined on $[0, \infty)$. Suppose $\tau \geq 0$ and $\mathcal{L}\left(g(t+\tau)\right)$ exists for $s > s_0$. Then $\mathcal{L}\left(u(t-\tau)g(t)\right)$ exists for $s > s_0$, and

$$\mathcal{L}(u(t-\tau)g(t)) = e^{-s\tau}\mathcal{L}\left(g(t+\tau)\right).$$

Proof

By definition,

$$\mathcal{L}\left(u(t-\tau)g(t)\right) = \int_0^\infty e^{-st}u(t-\tau)g(t)\,dt.$$

From this and the definition of $u(t-\tau)$,

$$\mathcal{L}\left(u(t-\tau)g(t)\right) = \int_0^\tau e^{-st}(0)\,dt + \int_\tau^\infty e^{-st}g(t)\,dt.$$

The first integral on the right equals zero. Introducing the new variable of integration $x = t-\tau$ in the second integral yields

$$\mathcal{L}\left(u(t-\tau)g(t)\right) = \int_0^\infty e^{-s(x+\tau)}g(x+\tau)\,dx = e^{-s\tau}\int_0^\infty e^{-sx}g(x+\tau)\,dx.$$

Changing the name of the variable of integration in the last integral from x to t yields

$$\mathcal{L}\left(u(t-\tau)g(t)\right) = e^{-s\tau}\int_0^\infty e^{-st}g(t+\tau)\,dt = e^{-s\tau}\mathcal{L}(g(t+\tau)). \blacksquare$$

□

Example 94 Find

$$\mathcal{L}\left(u(t-1)(t^2+1)\right).$$

Solution Here $\tau = 1$ and $g(t) = t^2+1$, so

$$g(t+1) = (t+1)^2+1 = t^2+2t+2.$$

Since

$$\mathcal{L}\left(g(t+1)\right) = \frac{2}{s^3}+\frac{2}{s^2}+\frac{2}{s},$$

Theorem 15 implies that

$$\mathcal{L}\left(u(t-1)(t^2+1)\right) = e^{-s}\left(\frac{2}{s^3}+\frac{2}{s^2}+\frac{2}{s}\right).$$

Example 95 Use Theorem 15 to find the Laplace transform of the function

$$f(t) = \begin{cases} 2t+1, & 0 \le t < 2, \\ 3t, & t \ge 2, \end{cases}$$

from Example 93.

Solution We first write f in the form (21.6) as

$$f(t) = 2t+1+u(t-2)(t-1).$$

Therefore

$$\begin{aligned}\mathcal{L}(f) &= \mathcal{L}(2t+1) + \mathcal{L}\left(u(t-2)(t-1)\right)\\ &= \mathcal{L}(2t+1) + e^{-2s}\mathcal{L}(t+1) \quad \text{(from Theorem 15)}\\ &= \frac{2}{s^2} + \frac{1}{s} + e^{-2s}\left(\frac{1}{s^2} + \frac{1}{s}\right),\end{aligned}$$

which is the result obtained in Example 93. ■

Formula (21.6) can be extended to more general piecewise continuous functions. For example, we can write

$$f(t) = \begin{cases} f_0(t), & 0 \le t < t_1,\\ f_1(t), & t_1 \le t < t_2,\\ f_2(t), & t \ge t_2,\end{cases}$$

as

$$f(t) = f_0(t) + u(t-t_1)\left(f_1(t) - f_0(t)\right) + u(t-t_2)\left(f_2(t) - f_1(t)\right)$$

if f_0, f_1, and f_2 are all defined on $[0, \infty)$.

Example 96 Find the Laplace transform of

$$f(t) = \begin{cases} 1, & 0 \le t < 2,\\ -2t+1, & 2 \le t < 3,\\ 3t, & 3 \le t < 5,\\ t-1, & t \ge 5\end{cases} \tag{21.7}$$

Solution In terms of step functions,

$$\begin{aligned}f(t) &= 1 + u(t-2)(-2t+1-1) + u(t-3)(3t+2t-1)\\ &\quad + u(t-5)(t-1-3t),\end{aligned}$$

or

$$f(t) = 1 - 2u(t-2)t + u(t-3)(5t-1) - u(t-5)(2t+1).$$

Now Theorem 15 implies that

$$\begin{aligned}\mathcal{L}(f) &= \mathcal{L}(1) - 2e^{-2s}\mathcal{L}(t+2) + e^{-3s}\mathcal{L}\left(5(t+3)-1\right) - e^{-5s}\mathcal{L}\left(2(t+5)+1\right)\\ &= \mathcal{L}(1) - 2e^{-2s}\mathcal{L}(t+2) + e^{-3s}\mathcal{L}(5t+14) - e^{-5s}\mathcal{L}(2t+11)\\ &= \frac{1}{s} - 2e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right) + e^{-3s}\left(\frac{5}{s^2} + \frac{14}{s}\right) - e^{-5s}\left(\frac{2}{s^2} + \frac{11}{s}\right). \ ■\end{aligned}$$

The trigonometric identities

$$\sin(A+B) = \sin A\cos B + \cos A\sin B \tag{21.8}$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B \tag{21.9}$$

are useful in problems that involve shifting the arguments of trigonometric functions. We'll use these identities in the next example.

Example 97 Find the Laplace transform of

$$f(t) = \begin{cases} \sin t, & 0 \le t < \dfrac{\pi}{2}, \\ \cos t - 3\sin t, & \dfrac{\pi}{2} \le t < \pi, \\ 3\cos t, & t \ge \pi \end{cases} \tag{21.10}$$

Solution In terms of step functions,

$$f(t) = \sin t + u(t-\pi/2)(\cos t - 4\sin t) + u(t-\pi)(2\cos t + 3\sin t).$$

Now Theorem 15 implies that

$$\begin{aligned} \mathcal{L}(f) \;=\; & \mathcal{L}(\sin t) + e^{-\frac{\pi}{2}s}\mathcal{L}\left(\cos\left(t+\tfrac{\pi}{2}\right) - 4\sin\left(t+\tfrac{\pi}{2}\right)\right) \\ & + e^{-\pi s}\mathcal{L}\left(2\cos(t+\pi) + 3\sin(t+\pi)\right). \end{aligned} \tag{21.11}$$

Since

$$\cos\left(t+\frac{\pi}{2}\right) - 4\sin\left(t+\frac{\pi}{2}\right) = -\sin t - 4\cos t$$

and

$$2\cos(t+\pi) + 3\sin(t+\pi) = -2\cos t - 3\sin t,$$

we see from (21.11) that

$$\begin{aligned} \mathcal{L}(f) \;&=\; \mathcal{L}(\sin t) - e^{-\pi s/2}\mathcal{L}(\sin t + 4\cos t) - e^{-\pi s}\mathcal{L}(2\cos t + 3\sin t) \\ &=\; \frac{1}{s^2+1} - e^{-\frac{\pi}{2}s}\left(\frac{1+4s}{s^2+1}\right) - e^{-\pi s}\left(\frac{3+2s}{s^2+1}\right). \;\blacksquare \end{aligned}$$

Subsection 21.2

The Second Shifting Theorem

Replacing $g(t)$ by $g(t-\tau)$ in Theorem 15 yields the next theorem.

Theorem 16 [Second Shifting Theorem] If $\tau \ge 0$ and $\mathcal{L}(g)$ exists for $s > s_0$ then $\mathcal{L}\left(u(t-\tau)g(t-\tau)\right)$ exists for $s > s_0$ and

$$\mathcal{L}(u(t-\tau)g(t-\tau)) = e^{-s\tau}\mathcal{L}(g(t)),$$

or, equivalently,

$$\text{if } g(t) \leftrightarrow G(s), \text{ then } u(t-\tau)g(t-\tau) \leftrightarrow e^{-s\tau}G(s). \tag{21.12}$$

Remark Recall that the First Shifting Theorem (Theorem 9 states that multiplying a function by e^{at} corresponds to shifting the argument of its transform by a units. Theorem 16 states that multiplying a Laplace transform by the exponential $e^{-\tau s}$ corresponds to shifting the argument of the inverse transform by τ units.

Example 98 Use (21.12) to find

$$\mathcal{L}^{-1}\left(\frac{e^{-2s}}{s^2}\right).$$

Solution To apply (21.12) we let $\tau = 2$ and $G(s) = 1/s^2$. Then $g(t) = t$ and (21.12) implies that

$$\mathcal{L}^{-1}\left(\frac{e^{-2s}}{s^2}\right) = u(t-2)(t-2). \blacksquare$$

Example 99 Find the inverse Laplace transform h of

$$H(s) = \frac{1}{s^2} - e^{-s}\left(\frac{1}{s^2} + \frac{2}{s}\right) + e^{-4s}\left(\frac{4}{s^3} + \frac{1}{s}\right),$$

and find distinct formulas for h on appropriate intervals.

Solution Let

$$G_0(s) = \frac{1}{s^2}, \quad G_1(s) = \frac{1}{s^2} + \frac{2}{s}, \quad G_2(s) = \frac{4}{s^3} + \frac{1}{s}.$$

Then

$$g_0(t) = t,\ g_1(t) = t + 2,\ g_2(t) = 2t^2 + 1.$$

Hence, (21.12) and the linearity of $\mathcal{L}^{-1}$ imply that

$$\begin{aligned} h(t) &= \mathcal{L}^{-1}\left(G_0(s)\right) - \mathcal{L}^{-1}\left(e^{-s}G_1(s)\right) + \mathcal{L}^{-1}\left(e^{-4s}G_2(s)\right) \\ &= t - u(t-1)\left[(t-1)+2\right] + u(t-4)\left[2(t-4)^2+1\right] \\ &= t - u(t-1)(t+1) + u(t-4)(2t^2 - 16t + 33), \end{aligned}$$

which can also be written as

$$h(t) = \begin{cases} t, & 0 \le t < 1, \\ -1, & 1 \le t < 4, \\ 2t^2 - 16t + 32, & t \ge 4. \end{cases} \blacksquare$$

Example 100 Find the inverse transform of

$$H(s) = \frac{2s}{s^2+4} - e^{-\frac{\pi}{2}s}\frac{3s+1}{s^2+9} + e^{-\pi s}\frac{s+1}{s^2+6s+10}.$$

Solution Let

$$G_0(s) = \frac{2s}{s^2+4}, \quad G_1(s) = -\frac{(3s+1)}{s^2+9},$$

and

$$G_2(s) = \frac{s+1}{s^2+6s+10} = \frac{(s+3)-2}{(s+3)^2+1}.$$

Then

$$g_0(t) = 2\cos 2t, \quad g_1(t) = -3\cos 3t - \frac{1}{3}\sin 3t,$$

and

$$g_2(t) = e^{-3t}(\cos t - 2\sin t).$$

Therefore (21.12) and the linearity of $\mathcal{L}^{-1}$ imply that

$$\begin{aligned} h(t) &= 2\cos 2t - u(t-\pi/2)\left[3\cos 3(t-\pi/2) + \frac{1}{3}\sin 3\left(t-\frac{\pi}{2}\right)\right] \\ &\quad + u(t-\pi)e^{-3(t-\pi)}\left[\cos(t-\pi) - 2\sin(t-\pi)\right]. \end{aligned}$$

Using the trigonometric identities (21.8) and (21.9), we can rewrite this as

$$\begin{aligned} h(t) &= 2\cos 2t + u(t-\pi/2)\left(3\sin 3t - \tfrac{1}{3}\cos 3t\right) \\ &\quad - u(t-\pi)e^{-3(t-\pi)}(\cos t - 2\sin t) \end{aligned} \tag{21.13}$$

Section 22

Initial value problems with piecewise continuous fuction

We'll now consider initial value problems of the form

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1, \tag{22.1}$$

where a, b, and c are constants ($a \neq 0$) and f is piecewise continuous on $[0, \infty)$. Problems of this kind occur in situations where the input to a physical system undergoes instantaneous changes, as when a switch is turned on or off or the forces acting on the system change abruptly.

It can be shown that the differential equation in (22.1) has no solutions on an open interval that contains a jump discontinuity of f. Therefore we must define what we mean by a solution of (22.1) on $[0, \infty)$ in the case where f has jump discontinuities. The next theorem motivates our definition. We omit the proof.

Theorem 17 Suppose a, b, and c are constants ($a \neq 0$), and f is piecewise continuous on $[0, \infty)$. with jump discontinuities at $t_1, \ldots, t_n$, where

$$0 < t_1 < \cdots < t_n.$$

Let k_0 and k_1 be arbitrary real numbers. Then there is a unique function y defined on $[0, \infty)$ with these properties:

- $y(0) = k_0$ and $y'(0) = k_1$.
- y and y' are continuous on $[0, \infty)$.
- y'' is defined on every open subinterval of $[0, \infty)$ that does not contain any of the points $t_1, \ldots, t_n$, and
$$ay'' + by' + cy = f(t)$$
on every such subinterval.
- y'' has limits from the right and left at $t_1, \ldots, t_n$.

We define the function y of Theorem 17 to be the solution of the initial value problem (22.1).

We begin by considering initial value problems of the form

$$ay'' + by' + cy = \begin{cases} f_0(t),\ 0 \le t < t_1, \\ f_1(t),\ t \ge t_1, \end{cases} \qquad y(0) = k_0, \quad y'(0) = k_1, \tag{22.2}$$

where the forcing function has a single jump discontinuity at t_1.

We can summarize the solution of (22.2) in the following steps:

- **Step 1.** Find the solution y_0 of the initial value problem

$$ay'' + by' + cy = f_0(t), \quad y(0) = k_0, \quad y'(0) = k_1.$$

- **Step 2.** Compute $c_0 = y_0(t_1)$ and $c_1 = y_0'(t_1)$.
- **Step 3.** Find the solution y_1 of the initial value problem

$$ay'' + by' + cy = f_1(t), \quad y(t_1) = c_0, \quad y'(t_1) = c_1.$$

- **Step 4.** Obtain the solution y of (22.2) as

$$y = \begin{cases} y_0(t),\ 0 \le t < t_1 \\ y_1(t),\ t \ge t_1. \end{cases}$$

he next examples illustrates this procedure.

Example 101 Solve the initial value problem

$$y'' + y = f(t), \quad y(0) = 2,\ y'(0) = -1, \tag{22.3}$$

where

$$f(t) = \begin{cases} 1,\ 0 \le t < \dfrac{\pi}{2}, \\ -1,\ t \ge \dfrac{\pi}{2}. \end{cases}$$

Solution The initial value problem in Step 1 is

$$y'' + y = 1, \quad y(0) = 2, \quad y'(0) = -1.$$

We leave it to you to verify that its solution is

$$y_0 = 1 + \cos t - \sin t.$$

Doing Step 2 yields $y_0(\pi/2) = 0$ and $y_0'(\pi/2) = -1$, so the second initial value problem is

$$y'' + y = -1, \quad y\left(\frac{\pi}{2}\right) = 0,\ y'\left(\frac{\pi}{2}\right) = -1.$$

We leave it to you to verify that the solution of this problem is

$$y_1 = -1 + \cos t + \sin t.$$

Hence, the solution of (22.3) is

$$y = \begin{cases} 1 + \cos t - \sin t, \ 0 \le t < \dfrac{\pi}{2}, \\ -1 + \cos t + \sin t, \ t \ge \dfrac{\pi}{2} \end{cases} \tag{22.4}$$

If f_0 and f_1 are defined on $[0, \infty)$, we can rewrite (22.2) as

$$ay'' + by' + cy = f_0(t) + u(t - t_1)\left(f_1(t) - f_0(t)\right), \quad y(0) = k_0, \quad y'(0) = k_1,$$

and apply the method of Laplace transforms. We'll now solve the problem considered in Example 101 by this method.

Example 102 Use the Laplace transform to solve the initial value problem

$$y'' + y = f(t), \quad y(0) = 2, \ y'(0) = -1, \tag{22.5}$$

where

$$f(t) = \begin{cases} 1, \ 0 \le t < \dfrac{\pi}{2}, \\ -1, \ t \ge \dfrac{\pi}{2}. \end{cases}$$

Solution Here

$$f(t) = 1 - 2u\left(t - \frac{\pi}{2}\right),$$

so Theorem 15 (with $g(t) = 1$) implies that

$$\mathcal{L}(f) = \frac{1 - 2e^{-\pi s/2}}{s}.$$

Therefore, transforming (22.5) yields

$$(s^2 + 1)Y(s) = \frac{1 - 2e^{-\pi s/2}}{s} - 1 + 2s,$$

so

$$Y(s) = (1 - 2e^{-\pi s/2})G(s) + \frac{2s - 1}{s^2 + 1}, \tag{22.6}$$

with

$$G(s) = \frac{1}{s(s^2 + 1)}.$$

The form for the partial fraction expansion of G is

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1}. \tag{22.7}$$

Multiplying through by $s(s^2 + 1)$ yields

$$A(s^2 + 1) + (Bs + C)s = 1,$$

or

$$(A + B)s^2 + Cs + A = 1.$$

Equating coefficients of like powers of s on the two sides of this equation shows that

$A = 1$, $B = -A = -1$ and $C = 0$. Hence, from (22.7),

$$G(s) = \frac{1}{s} - \frac{s}{s^2+1}.$$

Therefore

$$g(t) = 1 - \cos t.$$

From this, (22.6), and Theorem 16,

$$y = 1 - \cos t - 2u\left(t - \frac{\pi}{2}\right)\left(1 - \cos\left(t - \frac{\pi}{2}\right)\right) + 2\cos t - \sin t.$$

Simplifying this (recalling that $\cos(t - \pi/2) = \sin t$) yields

$$y = 1 + \cos t - \sin t - 2u\left(t - \frac{\pi}{2}\right)(1 - \sin t),$$

or

$$y = \begin{cases} 1 + \cos t - \sin t, & 0 \le t < \dfrac{\pi}{2}, \\ -1 + \cos t + \sin t, & t \ge \dfrac{\pi}{2}, \end{cases}$$

which is the result obtained in Example 101.

Remark It isn't obvious that using the Laplace transform to solve (22.2) as we did in Example 102 yields a function y with the properties stated in Theorem 17; that is, such that y and y' are continuous on $[0, \infty)$ and y'' has limits from the right and left at t_1. However, this is true if f_0 and f_1 are continuous and of exponential order on $[0, \infty)$.

Example 103 Solve the initial value problem

$$y'' - y = f(t), \quad y(0) = -1,\ y'(0) = 2, \tag{22.8}$$

where

$$f(t) = \begin{cases} t, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$$

Solution Here

$$f(t) = t - u(t-1)(t-1),$$

so

$$\begin{aligned} \mathcal{L}(f) &= \mathcal{L}(t) - \mathcal{L}\left(u(t-1)(t-1)\right) \\ &= \mathcal{L}(t) - e^{-s}\mathcal{L}(t) \text{ (from Theorem 15)} \\ &= \frac{1}{s^2} - \frac{e^{-s}}{s^2}. \end{aligned}$$

Since transforming (22.8) yields

$$(s^2 - 1)Y(s) = \mathcal{L}(f) + 2 - s,$$

we see that

$$Y(s) = (1 - e^{-s})H(s) + \frac{2 - s}{s^2 - 1}, \tag{22.9}$$

where
$$H(s) = \frac{1}{s^2(s^2-1)} = \frac{1}{s^2-1} - \frac{1}{s^2};$$
therefore
$$h(t) = \sinh t - t. \tag{22.10}$$
Since
$$\mathcal{L}^{-1}\left(\frac{2-s}{s^2-1}\right) = 2\sinh t - \cosh t,$$
we conclude from (22.9), (22.10), and Theorem 15 that
$$y = \sinh t - t - u(t-1)\left(\sinh(t-1) - t + 1\right) + 2\sinh t - \cosh t,$$
or
$$y = 3\sinh t - \cosh t - t - u(t-1)\left(\sinh(t-1) - t + 1\right) \tag{22.11}$$
We leave it to you to verify that y and y' are continuous and y'' has limits from the right and left at $t_1 = 1$.

Example 104 Solve the initial value problem
$$y'' + y = f(t), \quad y(0) = 0,\ y'(0) = 0, \tag{22.12}$$
where
$$f(t) = \begin{cases} 0, & 0 \le t < \dfrac{\pi}{4}, \\ \cos 2t, & \dfrac{\pi}{4} \le t < \pi, \\ 0, & t \ge \pi. \end{cases}$$

Solution Here
$$f(t) = u(t-\pi/4)\cos 2t - u(t-\pi)\cos 2t,$$
so
$$\begin{aligned} \mathcal{L}(f) &= \mathcal{L}\left(u(t-\pi/4)\cos 2t\right) - \mathcal{L}\left(u(t-\pi)\cos 2t\right) \\ &= e^{-\pi s/4}\mathcal{L}\left(\cos 2(t+\pi/4)\right) - e^{-\pi s}\mathcal{L}\left(\cos 2(t+\pi)\right) \\ &= -e^{-\pi s/4}\mathcal{L}(\sin 2t) - e^{-\pi s}\mathcal{L}(\cos 2t) \\ &= -\frac{2e^{-\pi s/4}}{s^2+4} - \frac{se^{-\pi s}}{s^2+4}. \end{aligned}$$
Since transforming (22.12) yields
$$(s^2+1)Y(s) = \mathcal{L}(f),$$
we see that
$$Y(s) = e^{-\pi s/4}H_1(s) + e^{-\pi s}H_2(s), \tag{22.13}$$
where
$$H_1(s) = -\frac{2}{(s^2+1)(s^2+4)} \quad \text{and} \quad H_2(s) = -\frac{s}{(s^2+1)(s^2+4)}. \tag{22.14}$$

To simplify the required partial fraction expansions, we first write

$$\frac{1}{(x+1)(x+4)} = \frac{1}{3}\left[\frac{1}{x+1} - \frac{1}{x+4}\right].$$

Setting $x = s^2$ and substituting the result in (22.14) yields

$$H_1(s) = -\frac{2}{3}\left[\frac{1}{s^2+1} - \frac{1}{s^2+4}\right] \quad \text{and} \quad H_2(s) = -\frac{1}{3}\left[\frac{s}{s^2+1} - \frac{s}{s^2+4}\right].$$

The inverse transforms are

$$h_1(t) = -\frac{2}{3}\sin t + \frac{1}{3}\sin 2t \quad \text{and} \quad h_2(t) = -\frac{1}{3}\cos t + \frac{1}{3}\cos 2t.$$

From (22.13) and Theorem 8.4.2,

$$y = u\left(t - \frac{\pi}{4}\right) h_1\left(t - \frac{\pi}{4}\right) + u(t-\pi)h_2(t-\pi). \tag{22.15}$$

Since

$$\begin{aligned} h_1\left(t - \frac{\pi}{4}\right) &= -\frac{2}{3}\sin\left(t - \frac{\pi}{4}\right) + \frac{1}{3}\sin 2\left(t - \frac{\pi}{4}\right) \\ &= -\frac{\sqrt{2}}{3}(\sin t - \cos t) - \frac{1}{3}\cos 2t \end{aligned}$$

and

$$\begin{aligned} h_2(t-\pi) &= -\frac{1}{3}\cos(t-\pi) + \frac{1}{3}\cos 2(t-\pi) \\ &= \frac{1}{3}\cos t + \frac{1}{3}\cos 2t, \end{aligned}$$

(22.15) can be rewritten as

$$y = -\frac{1}{3}u\left(t - \frac{\pi}{4}\right)\left(\sqrt{2}(\sin t - \cos t) + \cos 2t\right) + \frac{1}{3}u(t-\pi)(\cos t + \cos 2t)$$

or

$$y = \begin{cases} 0, & 0 \le t < \frac{\pi}{4}, \\ -\frac{\sqrt{2}}{3}(\sin t - \cos t) - \frac{1}{3}\cos 2t, & \frac{\pi}{4} \le t < \pi, \\ -\frac{\sqrt{2}}{3}\sin t + \frac{1+\sqrt{2}}{3}\cos t, & t \ge \pi. \end{cases} \tag{22.16}$$

We leave it to you to verify that y and y' are continuous and y'' has limits from the right and left at $t_1 = \pi/4$ and $t_2 = \pi$

Section 23

Exercise

In Exercises 1–6 find the Laplace transform by the method of Example 93. Then express the given function f in terms of unit step functions as in Eqn. (21.6), and use Theorem 15 to find $\mathcal{L}(f)$. Where indicated by , graph f.

1. $f(t) = \begin{cases} 1, & 0 \le t < 4, \\ t, & t \ge 4. \end{cases}$

2. $f(t) = \begin{cases} t, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$

3. $f(t) = \begin{cases} 2t - 1, & 0 \le t < 2, \\ t, & t \ge 2. \end{cases}$

4. $f(t) = \begin{cases} 1, & 0 \le t < 1, \\ t + 2, & t \ge 1. \end{cases}$

5. $f(t) = \begin{cases} t - 1, & 0 \le t < 2, \\ 4, & t \ge 2. \end{cases}$

6. $f(t) = \begin{cases} t^2, & 0 \le t < 1, \\ 0, & t \ge 1. \end{cases}$

In Exercises 7–18 express the given function f in terms of unit step functions and use Theorem 15 to find $\mathcal{L}(f)$. Where indicated by , graph f.

7. $f(t) = \begin{cases} 0, & 0 \le t < 2, \\ t^2 + 3t, & t \ge 2. \end{cases}$

8. $f(t) = \begin{cases} t^2 + 2, & 0 \le t < 1, \\ t, & t \ge 1. \end{cases}$

9. $f(t) = \begin{cases} te^t, & 0 \le t < 1, \\ e^t, & t \ge 1. \end{cases}$

10. $f(t) = \begin{cases} e^{-t}, & 0 \le t < 1, \\ e^{-2t}, & t \ge 1. \end{cases}$

11. $f(t) = \begin{cases} -t, & 0 \le t < 2, \\ t - 4, & 2 \le t < 3, \\ 1, & t \ge 3. \end{cases}$

12. $f(t) = \begin{cases} 0, & 0 \le t < 1, \\ t, & 1 \le t < 2, \\ 0, & t \ge 2. \end{cases}$

13. $f(t) = \begin{cases} t, & 0 \le t < 1, \\ t^2, & 1 \le t < 2, \\ 0, & t \ge 2. \end{cases}$

14. $f(t) = \begin{cases} t, & 0 \le t < 1, \\ 2 - t, & 1 \le t < 2, \\ 6, & t > 2. \end{cases}$

15. $f(t) = \begin{cases} \sin t, & 0 \le t < \dfrac{\pi}{2}, \\ 2\sin t, & \dfrac{\pi}{2} \le t < \pi, \\ \cos t, & t \ge \pi. \end{cases}$

16. $f(t) = \begin{cases} 2, & 0 \le t < 1, \\ -2t + 2, & 1 \le t < 3, \\ 3t, & t \ge 3. \end{cases}$

17. $f(t) = \begin{cases} 3, & 0 \le t < 2, \\ 3t + 2, & 2 \le t < 4, \\ 4t, & t \ge 4. \end{cases}$

18. $f(t) = \begin{cases} (t+1)^2, & 0 \le t < 1, \\ (t+2)^2, & t \ge 1. \end{cases}$

In Exercises 19–28 use Theorem 16 to express the inverse transforms in terms of step functions, and then find distinct formulas the for inverse transforms on the

appropriate intervals, as in Example 99. Where indicated by , graph the inverse transform.

19. $H(s) = \dfrac{e^{-2s}}{s-2}$

20. $H(s) = \dfrac{e^{-s}}{s(s+1)}$

21. $H(s) = \dfrac{e^{-s}}{s^3} + \dfrac{e^{-2s}}{s^2}$

22. $H(s) = \left(\dfrac{2}{s} + \dfrac{1}{s^2}\right) + e^{-s}\left(\dfrac{3}{s} - \dfrac{1}{s^2}\right) + e^{-3s}\left(\dfrac{1}{s} + \dfrac{1}{s^2}\right)$

23. $H(s) = \left(\dfrac{5}{s} - \dfrac{1}{s^2}\right) + e^{-3s}\left(\dfrac{6}{s} + \dfrac{7}{s^2}\right) + \dfrac{3e^{-6s}}{s^3}$

24. $H(s) = \dfrac{e^{-\pi s}(1-2s)}{s^2+4s+5}$

25. $H(s) = \left(\dfrac{1}{s} - \dfrac{s}{s^2+1}\right) + e^{-\frac{\pi}{2}s}\left(\dfrac{3s-1}{s^2+1}\right)$

26. $H(s) = e^{-2s}\left[\dfrac{3(s-3)}{(s+1)(s-2)} - \dfrac{s+1}{(s-1)(s-2)}\right]$

27. $H(s) = \dfrac{1}{s} + \dfrac{1}{s^2} + e^{-s}\left(\dfrac{3}{s} + \dfrac{2}{s^2}\right) + e^{-3s}\left(\dfrac{4}{s} + \dfrac{3}{s^2}\right)$

28. $H(s) = \dfrac{1}{s} - \dfrac{2}{s^3} + e^{-2s}\left(\dfrac{3}{s} - \dfrac{1}{s^3}\right) + \dfrac{e^{-4s}}{s^2}$

29. Find $\mathcal{L}\left(u(t-\tau)\right)$.

Section 24

Some solutions

Solution of problem (2)

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t)\,dt = \int_0^1 e^{-st} t\,dt + \int_1^\infty e^{-st}\,dt. \tag{A}$$

To relate the first term to a Laplace transform we add and subtract $\int_1^\infty e^{-st} t\,dt$ in (A) to obtain

$$\mathcal{L}(f) = \int_0^\infty e^{-st} t\,dt + \int_1^\infty e^{-st}(1-t)\,dt = \mathcal{L}(t) - \int_1^\infty e^{-st}(t-1)\,dt. \tag{B}$$

Letting $t = x+1$ in the last integral yields

$$\int_1^\infty e^{-st}(t-1)\,dt = -\int_0^\infty e^{-s(x+1)} x\,dx = e^{-s}\mathcal{L}(t).$$

This and (B) imply that $\mathcal{L}(f) = (1-e^{-s})\mathcal{L}(t) = \dfrac{1-e^{-s}}{s^2}$.

Alternatively, $f(t) = t - u(t-1)(t-1) \leftrightarrow (1-e^{-s})\mathcal{L}(t) = \dfrac{1-e^{-s}}{s^2}$.

Solution of problem (4)

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t)\,dt = \int_0^1 e^{-st}\,dt + \int_1^\infty e^{-st}(t+2)\,dt. \tag{A}$$

To relate the first term to a Laplace transform we add and subtract $\int_1^\infty e^{-st}\,dt$ in (A) to obtain

$$\mathcal{L}(f) = \int_0^\infty e^{-st}\,dt + \int_1^\infty e^{-st}(t+1)\,dt = \mathcal{L}(t) + \int_1^\infty e^{-st}(t+1)\,dt. \tag{B}$$

Letting $t = x+1$ in the last integral yields

$$\int_1^\infty e^{-st}(t+1)\,dt = \int_0^\infty e^{-s(x+1)}(x+2)\,dx = e^{-s}\mathcal{L}(t+2).$$

This and (B) imply that $\mathcal{L}(f) = \mathcal{L}(1) + e^{-s}\mathcal{L}(t+2) = \dfrac{1}{s} + e^{-s}\left(\dfrac{1}{s^2} + \dfrac{2}{s}\right)$.

Alternatively,

$f(t) = 1 + u(t-1)(t+1) \leftrightarrow \mathcal{L}(1) + e^{-s}\mathcal{L}(t+2) = \dfrac{1}{s} + e^{-s}\left(\dfrac{1}{s^2} + \dfrac{2}{s}\right)$.

Solution of problem (6)

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t)\,dt = \int_0^1 e^{-st}t^2 = \mathcal{L}(t^2) - \int_1^\infty t^2\,dt. \tag{A}$$

Letting $t = x+1$ in the last integral yields

$$\int_1^\infty e^{-st}t^2\,dt = \int_0^\infty e^{-s(x+1)}(t^2+2t+1)\,dx = e^{-s}\mathcal{L}(t^2+2t+1).$$

This and (A) imply that

$$\mathcal{L}(f) = \mathcal{L}(t^2) + e^{-s}\mathcal{L}(t^2+2t+1) = \frac{2}{s^3} - e^{-s}\left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right).$$

Alternatively,

$$f(t) = t^2\left(1 - u(t-1)\right) \leftrightarrow \mathcal{L}(t^2) + e^{-s}\mathcal{L}(t^2+2t+1) = \frac{2}{s^3} - e^{-s}\left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right)$$

Solution of problem (8) $f(t) = t^2 + 2 + u(t-1)(t - t^2 - 2)$. Since $t^2 + 2 \leftrightarrow \dfrac{2}{s^3} + \dfrac{2}{s}$ and

$$\begin{aligned}\mathcal{L}\left(u(t-1)(t-t^2-2)\right) &= e^{-s}\mathcal{L}\left((t+1) - (t+1)^2 - 2\right)\\ &= -e^{-s}\mathcal{L}(t^2+t+2) = -e^{-s}\left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{2}{s}\right),\end{aligned}$$

it follows that $F(s) = \dfrac{2}{s^3} + \dfrac{2}{s} - e^{-s}\left(\dfrac{2}{s^3} + \dfrac{1}{s^2} + \dfrac{2}{s}\right)$.

Solution of problem (10) $f(t) = e^{-t} + u(t-1)(e^{-2t} - e^{-t}) \leftrightarrow \mathcal{L}(e^{-t}) + e^{-s}\mathcal{L}(e^{-2(t+1)}) - e^{-s}\mathcal{L}(e^{-t-1}) = \mathcal{L}(e^{-t}) + e^{-(s+2)}\mathcal{L}(e^{-2t}) - e^{-(s+1)}\mathcal{L}(e^{-t}) = \dfrac{1 - e^{-(s+1)}}{s+1} + \dfrac{e^{-(s+2)}}{s+2}$.

Solution of problem (12) $f(t) = [u(t-1) - u(t-2)]\,t \leftrightarrow e^{-s}\mathcal{L}(t+1) - e^{-2s}\mathcal{L}(t+2)$ $= e^{-s}\left(\dfrac{1}{s^2} + \dfrac{1}{s}\right) - e^{-2s}\left(\dfrac{1}{s^2} + \dfrac{2}{s}\right)$.

Solution of problem (14)

$$\begin{aligned} f(t) &= t - 2u(t-1)(t-1) + u(t-2)(t+4) \leftrightarrow \frac{1}{s^2} - 2e^{-s}\mathcal{L}(t) + e^{-2s}\mathcal{L}(t+6) \\ &= \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + e^{-2s}\left(\frac{1}{s^2} + \frac{6}{s}\right). \end{aligned}$$

Solution of problem (16) $f(t) = 2 - 2u(t-1)t + u(t-3)(5t-2) \leftrightarrow \mathcal{L}(2) - 2e^{-s}\mathcal{L}(t+1) + e^{-3s}\mathcal{L}(5t+13) = \frac{2}{s} - e^{-s}\left(\frac{2}{s^2} + \frac{2}{s}\right) + e^{-3s}\left(\frac{5}{s^2} + \frac{13}{s}\right)$.

Solution of problem (18) $f(t) = (t+1)^2 + u(t-1)\left((t+2)^2 - (t+1)^2\right) = t^2 + 2t + 1 + u(t-1)(2t+3) \leftrightarrow \mathcal{L}(t^2+2t+1) + e^{-s}\mathcal{L}(2t+5) = \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} + e^{-s}\left(\frac{2}{s^2} + \frac{5}{s}\right)$.

Solution of problem (20) $\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} \leftrightarrow 1 - e^{-t} \Rightarrow e^{-s}\frac{1}{s(s+1)} \leftrightarrow u(t-1)\left(1 - e^{-(t-1)}\right) = \begin{cases} 0, & 0 \le t < 1, \\ 1 - e^{-(t-1)}, & t \ge 1. \end{cases}$

Solution of problem (22)

$$\frac{3}{s} - \frac{1}{s^2} \leftrightarrow 3 - t \Rightarrow e^{-s}\left(\frac{3}{s} - \frac{1}{s^2}\right) \leftrightarrow u(t-1)\left(3 - (t-1)\right) = u(t-1)(4-t);$$

$$\frac{1}{s} + \frac{1}{s^2} \leftrightarrow 1 + t \Rightarrow e^{-3s}\left(\frac{1}{s} + \frac{1}{s^2}\right) \leftrightarrow u(t-3)\left(1 + (t-3)\right) = u(t-3)(t-2);$$

therefore

$$h(t) = 2 + t + u(t-1)(4-t) + u(t-3)(t-2) = \begin{cases} 2+t, & 0 \le t < 1, \\ 6, & 1 \le t < 3, \\ t+4, & t \ge 3. \end{cases}$$

Solution of problem (24)

$$\frac{1-2s}{s^2+4s+5} = \frac{5-2(s+2)}{(s+2)^2+1} \leftrightarrow e^{-2t}(5\sin t - 2\cos t);$$

therefore,

$$\begin{aligned} h(t) &= u(t-\pi)e^{-2(t-\pi)}\left(5\sin(t-\pi) - 2\cos(t-\pi)\right) \\ &= u(t-\pi)e^{-2(t-\pi)}(2\cos t - 5\sin t) \\ &= \begin{cases} 0, & 0 \le t < \pi, \\ e^{-2(t-\pi)}(2\cos t - 5\sin t), & t \ge \pi. \end{cases} \end{aligned}.$$

Solution of problem (26) Denote $F(s) = \frac{3(s-3)}{(s+1)(s-2)} - \frac{s+1}{(s-1)(s-2)}$. Since $\frac{3(s-3)}{(s+1)(s-2)} = \frac{4}{s+1} - \frac{1}{s-2}$ and $\frac{s+1}{(s-1)(s-2)} = \frac{3}{s-2} - \frac{2}{s-1}$, $F(s) = \frac{4}{s+1} - \frac{4}{s-2} + \frac{2}{s-1} \leftrightarrow 4e^{-t} - 4e^{2t} + 2e^{t}$. Therefore,$e^{-2s}F(s) \leftrightarrow u(t-2)\left(4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}\right) = \begin{cases} 0, & 0 \le t < 2, \\ 4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}, & t \ge 2. \end{cases}$

Solution of problem (28)

$$\frac{3}{s}-\frac{1}{s^3}\leftrightarrow 3-\frac{t^2}{2}\Rightarrow e^{-2s}\left(\frac{3}{s}-\frac{1}{s^3}\right)\leftrightarrow u(t-2)\left(3-\frac{(t-2)^2}{2}\right)=u(t-2)\left(-\frac{t^2}{2}+2t+1\right);$$

$$\frac{1}{s^2}\leftrightarrow t\Rightarrow\frac{e^{-4s}}{s^2}\leftrightarrow u(t-4)(t-4);$$

therefore

$$\begin{aligned} h(t) &= 1-t^2+u(t-2)\left(-\frac{t^2}{2}+2t+1\right)+u(t-4)(t-4)\\ &= \begin{cases} 1-t^2, & 0\le t<2\\ -\dfrac{3t^2}{2}+2t+2, & 2\le t<4,\\ -\dfrac{3t^2}{2}+3t-2, & t\ge 4. \end{cases}\end{aligned}$$

Section 25

Convolution Theorem

The Laplace transformation of a product is not the product of the transforms. All hope is not lost, however. We simply have to use a different type of *product.* Take two functions $f(t)$ and $g(t)$ defined for $t \geq 0$, and define the *convolution*[1] of $f(t)$ and $g(t)$ as

$$(f * g)(t) \stackrel{\text{def}}{=} \int_0^t f(\tau)g(t-\tau)\,d\tau. \tag{25.1}$$

As you can see, the convolution of two functions of t is another function of t.

To motivate our interest in this problem, consider the initial value problem

$$ay'' + by' + cy = f(t), \quad y(0) = 0, \quad y'(0) = 0.$$

Taking Laplace transforms yields

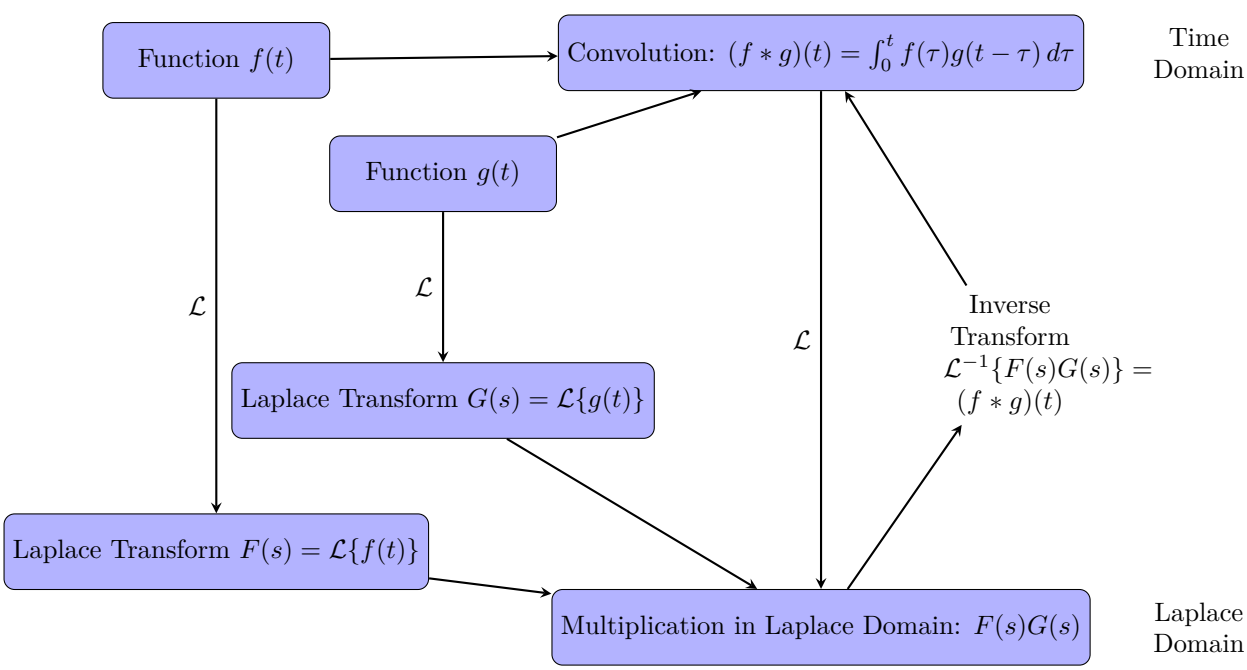
$$(as^2 + bs + c)Y(s) = F(s),$$

so

$$Y(s) = F(s)G(s), \tag{25.2}$$

where

$$G(s) = \frac{1}{as^2 + bs + c}.$$



[1]For those that have seen convolution before, you may have seen it defined as $(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau$. This definition agrees with (25.1) if you define $f(t)$ and $g(t)$ to be zero for $t < 0$. When discussing the Laplace transform, the definition we gave is sufficient. Convolution does occur in many other applications, however, where you may have to use the more general definition with infinities.

To motivate the formula for $\mathcal{L}^{-1}(FG)$, consider the initial value problem

$$y' - ay = f(t), \quad y(0) = 0, \tag{25.3}$$

which we first solve without using the Laplace transform. The solution of the differential equation in (25.3) is of the form $y = ue^{at}$ where

$$u' = e^{-at} f(t).$$

Integrating this from 0 to t and imposing the initial condition $u(0) = y(0) = 0$ yields

$$u = \int_0^t e^{-a\tau} f(\tau)\, d\tau.$$

Therefore

$$y(t) = e^{at} \int_0^t e^{-a\tau} f(\tau)\, d\tau = \int_0^t e^{a(t-\tau)} f(\tau)\, d\tau. \tag{25.4}$$

Now we'll use the Laplace transform to solve (25.3) and compare the result to (25.4). Taking Laplace transforms in (25.3) yields

$$(s-a)Y(s) = F(s),$$

so

$$Y(s) = F(s)\frac{1}{s-a},$$

which implies that

$$y(t) = \mathcal{L}^{-1}\left(F(s)\frac{1}{s-a}\right). \tag{25.5}$$

If we now let $g(t) = e^{at}$, so that

$$G(s) = \frac{1}{s-a},$$

then (25.4) and (25.5) can be written as

$$y(t) = \int_0^t f(\tau) g(t-\tau)\, d\tau$$

and

$$y = \mathcal{L}^{-1}(FG),$$

respectively. Therefore

$$\mathcal{L}^{-1}(FG) = \int_0^t f(\tau) g(t-\tau)\, d\tau \tag{25.6}$$

in this case.

Example 105 Take $f(t) = e^t$ and $g(t) = t$ for $t \geq 0$. Then

$$(f * g)(t) = \int_0^t e^{\tau}(t-\tau)\, d\tau = e^t - t - 1.$$

To solve the integral we did one integration by parts.

Example 106 Take $f(t) = \sin(\omega t)$ and $g(t) = \cos(\omega t)$ for $t \geq 0$. Then

$$(f * g)(t) = \int_0^t \sin(\omega\tau)\, \cos\big(\omega(t-\tau)\big)\, d\tau.$$

Apply the identity

$$\cos(\theta)\sin(\psi) = \frac{1}{2}\big(\sin(\theta+\psi) - \sin(\theta-\psi)\big),$$

to get

$$\begin{aligned}(f * g)(t) &= \int_0^t \frac{1}{2}\big(\sin(\omega t) - \sin(\omega t - 2\omega\tau)\big)\, d\tau \\ &= \left[\frac{1}{2}\tau\sin(\omega t) + \frac{1}{4\omega}\cos(2\omega\tau - \omega t)\right]_{\tau=0}^{t} \\ &= \frac{1}{2}t\sin(\omega t).\end{aligned}$$

The formula holds only for $t \geq 0$. The functions f, g, and $f * g$ are undefined for $t < 0$.

This motivates the next definition.

Definition 9 The *convolution* $f * g$ of two functions f and g is defined by

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\, d\tau.$$

It is easy to show that $f * g = g * f$; that is,

$$\int_0^t f(t-\tau)g(\tau)\, d\tau = \int_0^t f(\tau)g(t-\tau)\, d\tau.$$

Eqn. (25.6) shows that $\mathcal{L}^{-1}(FG) = f * g$ in the special case where $g(t) = e^{at}$. This next theorem states that this is true in general.

Theorem 18 [The Convolution Theorem] If $\mathcal{L}(f) = F$ and $\mathcal{L}(g) = G$, then

$$\mathcal{L}(f * g) = FG.$$

Example 107 Let

$$f(t) = e^{at} \quad \text{and} \quad g(t) = e^{bt} \qquad (a \neq b).$$

Verify that $\mathcal{L}(f * g) = \mathcal{L}(f)\mathcal{L}(g)$, as implied by the convolution theorem.

Solution We first compute

$$\begin{aligned}(f*g)(t) &= \int_0^t e^{a\tau}e^{b(t-\tau)}\,d\tau = e^{bt}\int_0^t e^{(a-b)\tau}d\tau \\ &= e^{bt}\frac{e^{(a-b)\tau}}{a-b}\bigg|_0^t = \frac{e^{bt}\left[e^{(a-b)t}-1\right]}{a-b} \\ &= \frac{e^{at}-e^{bt}}{a-b}.\end{aligned}$$

Since

$$e^{at} \leftrightarrow \frac{1}{s-a} \quad \text{and} \quad e^{bt} \leftrightarrow \frac{1}{s-b},$$

it follows that

$$\begin{aligned}\mathcal{L}(f*g) &= \frac{1}{a-b}\left[\frac{1}{s-a}-\frac{1}{s-b}\right] \\ &= \frac{1}{(s-a)(s-b)} \\ &= \mathcal{L}(e^{at})\mathcal{L}(e^{bt}) = \mathcal{L}(f)\mathcal{L}(g).\end{aligned}$$

Subsection 25.1

A Formula for the Solution of an Initial Value Problem

The convolution theorem provides a formula for the solution of an initial value problem for a linear constant coefficient second order equation with an unspecified. The next three examples illustrate this.

Example 108 Find a formula for the solution of the initial value problem

$$y'' - 2y' + y = f(t), \quad y(0) = k_0, \quad y'(0) = k_1. \tag{25.7}$$

Solution Taking Laplace transforms in (25.7) yields

$$(s^2 - 2s + 1)Y(s) = F(s) + (k_1 + k_0 s) - 2k_0.$$

Therefore

$$\begin{aligned}Y(s) &= \frac{1}{(s-1)^2}F(s) + \frac{k_1 + k_0 s - 2k_0}{(s-1)^2} \\ &= \frac{1}{(s-1)^2}F(s) + \frac{k_0}{s-1} + \frac{k_1 - k_0}{(s-1)^2}.\end{aligned}$$

From the table of Laplace transforms,

$$\mathcal{L}^{-1}\left(\frac{k_0}{s-1} + \frac{k_1-k_0}{(s-1)^2}\right) = e^t\left(k_0 + (k_1 - k_0)t\right).$$

Since

$$\frac{1}{(s-1)^2} \leftrightarrow te^t \quad \text{and} \quad F(s) \leftrightarrow f(t),$$

the convolution theorem implies that

$$\mathcal{L}^{-1}\left(\frac{1}{(s-1)^2}F(s)\right) = \int_0^t \tau e^\tau f(t-\tau)\,d\tau.$$

Therefore the solution of (25.7) is

$$y(t) = e^t\left(k_0 + (k_1 - k_0)t\right) + \int_0^t \tau e^\tau f(t-\tau)\,d\tau.$$

Example 109 Find a formula for the solution of the initial value problem

$$y'' + 4y = f(t), \quad y(0) = k_0, \quad y'(0) = k_1. \tag{25.8}$$

Solution Taking Laplace transforms in (25.8) yields

$$(s^2+4)Y(s) = F(s) + k_1 + k_0 s.$$

Therefore

$$Y(s) = \frac{1}{(s^2+4)}F(s) + \frac{k_1 + k_0 s}{s^2+4}.$$

From the table of Laplace transforms,

$$\mathcal{L}^{-1}\left(\frac{k_1 + k_0 s}{s^2+4}\right) = k_0\cos 2t + \frac{k_1}{2}\sin 2t.$$

Since

$$\frac{1}{(s^2+4)} \leftrightarrow \frac{1}{2}\sin 2t \quad \text{and} \quad F(s) \leftrightarrow f(t),$$

the convolution theorem implies that

$$\mathcal{L}^{-1}\left(\frac{1}{(s^2+4)}F(s)\right) = \frac{1}{2}\int_0^t f(t-\tau)\sin 2\tau\,d\tau.$$

Therefore the solution of (25.8) is

$$y(t) = k_0\cos 2t + \frac{k_1}{2}\sin 2t + \frac{1}{2}\int_0^t f(t-\tau)\sin 2\tau\,d\tau.$$

Example 110 Find a formula for the solution of the initial value problem

$$y'' + 2y' + 2y = f(t), \quad y(0) = k_0, \quad y'(0) = k_1. \tag{25.9}$$

Solution Taking Laplace transforms in (25.9) yields

$$(s^2+2s+2)Y(s) = F(s) + k_1 + k_0 s + 2k_0.$$

Therefore

$$\begin{aligned} Y(s) &= \frac{1}{(s+1)^2+1}F(s) + \frac{k_1 + k_0 s + 2k_0}{(s+1)^2+1} \\ &= \frac{1}{(s+1)^2+1}F(s) + \frac{(k_1+k_0) + k_0(s+1)}{(s+1)^2+1}. \end{aligned}$$

From the table of Laplace transforms,

$$\mathcal{L}^{-1}\left(\frac{(k_1+k_0)+k_0(s+1)}{(s+1)^2+1}\right) = e^{-t}\left((k_1+k_0)\sin t + k_0\cos t\right).$$

Since

$$\frac{1}{(s+1)^2+1} \leftrightarrow e^{-t}\sin t \quad \text{and} \quad F(s) \leftrightarrow f(t),$$

the convolution theorem implies that

$$\mathcal{L}^{-1}\left(\frac{1}{(s+1)^2+1}F(s)\right) = \int_0^t f(t-\tau)e^{-\tau}\sin\tau\, d\tau.$$

Therefore the solution of (25.9) is

$$y(t) = e^{-t}\left((k_1+k_0)\sin t + k_0\cos t\right) + \int_0^t f(t-\tau)e^{-\tau}\sin\tau\, d\tau. \tag{25.10}$$

Subsection 25.2

Evaluating Convolution Integrals

We'll say that an integral of the form $\int_0^t u(\tau)v(t-\tau)\,d\tau$ is a *convolution integral*. The convolution theorem provides a convenient way to evaluate convolution integrals.

Example 111 Evaluate the convolution integral

$$h(t) = \int_0^t (t-\tau)^5\tau^7 d\tau.$$

Solution We could evaluate this integral by expanding $(t-\tau)^5$ in powers of τ and then integrating. However, the convolution theorem provides an easier way. The integral is the convolution of $f(t) = t^5$ and $g(t) = t^7$. Since

$$t^5 \leftrightarrow \frac{5!}{s^6} \quad \text{and} \quad t^7 \leftrightarrow \frac{7!}{s^8},$$

the convolution theorem implies that

$$h(t) \leftrightarrow \frac{5!7!}{s^{14}} = \frac{5!7!}{13!}\frac{13!}{s^{14}},$$

where we have written the second equality because

$$\frac{13!}{s^{14}} \leftrightarrow t^{13}.$$

Hence,

$$h(t) = \frac{5!7!}{13!}\, t^{13}.$$

Example 112 Use the convolution theorem and a partial fraction expansion to evaluate the convolution integral

$$h(t) = \int_0^t \sin a(t-\tau)\cos b\tau \, d\tau \quad (|a| \neq |b|).$$

Solution Since

$$\sin at \leftrightarrow \frac{a}{s^2+a^2} \quad \text{and} \quad \cos bt \leftrightarrow \frac{s}{s^2+b^2},$$

the convolution theorem implies that

$$H(s) = \frac{a}{s^2+a^2}\frac{s}{s^2+b^2}.$$

Expanding this in a partial fraction expansion yields

$$H(s) = \frac{a}{b^2-a^2}\left[\frac{s}{s^2+a^2} - \frac{s}{s^2+b^2}\right].$$

Therefore

$$h(t) = \frac{a}{b^2-a^2}\left(\cos at - \cos bt\right).$$

Subsection 25.3

Volterra Integral Equations

An equation of the form

$$y(t) = f(t) + \int_0^t k(t-\tau)y(\tau)\, d\tau \tag{25.11}$$

is a *Volterra integral equation*. Here f and k are given functions and y is unknown. Since the integral on the right is a convolution integral, the convolution theorem provides a convenient formula for solving (25.11). Taking Laplace transforms in (25.11) yields

$$Y(s) = F(s) + K(s)Y(s),$$

and solving this for $Y(s)$ yields

$$Y(s) = \frac{F(s)}{1-K(s)}.$$

We then obtain the solution of (25.11) as $y = \mathcal{L}^{-1}(Y)$.

Example 113 Solve the integral equation

$$y(t) = 1 + 2\int_0^t e^{-2(t-\tau)}y(\tau)\, d\tau. \tag{25.12}$$

Solution Taking Laplace transforms in (25.12) yields

$$Y(s) = \frac{1}{s} + \frac{2}{s+2}Y(s),$$

and solving this for $Y(s)$ yields

$$Y(s) = \frac{1}{s} + \frac{2}{s^2}.$$

Hence,

$$y(t) = 1 + 2t.$$

SECTION 26

Exercises

1. Express the inverse transform as an integral.

- Part a $\dfrac{1}{s^2(s^2+4)}$
- Part b $\dfrac{s}{(s+2)(s^2+9)}$
- Part c $\dfrac{s}{(s^2+4)(s^2+9)}$
- Part d $\dfrac{s}{(s^2+1)^2}$
- Part e $\dfrac{1}{s(s-a)}$
- Part f $\dfrac{1}{(s+1)(s^2+2s+2)}$
- Part g $\dfrac{1}{(s+1)^2(s^2+4s+5)}$
- Part h $\dfrac{1}{(s-1)^3(s+2)^2}$
- Part i $\dfrac{s-1}{s^2(s^2-2s+2)}$
- Part j $\dfrac{s(s+3)}{(s^2+4)(s^2+6s+10)}$
- Part k $\dfrac{1}{(s-3)^5 s^6}$
- Part l $\dfrac{1}{(s-1)^3(s^2+4)}$
- Part m $\dfrac{1}{s^2(s-2)^3}$
- Part n $\dfrac{1}{s^7(s-2)^6}$

2. Find the Laplace transform.

- Part a $\displaystyle\int_0^t \sin a\tau \cos b(t-\tau)\, d\tau$
- Part b $\displaystyle\int_0^t e^{\tau} \sin a(t-\tau)\, d\tau$
- Part c $\displaystyle\int_0^t \sinh a\tau \cosh a(t-\tau)\, d\tau$
- Part d $\displaystyle\int_0^t \tau(t-\tau) \sin \omega\tau \cos \omega(t-\tau)\, d\tau$
- Part e $\displaystyle e^t \int_0^t \sin \omega\tau \cos \omega(t-\tau)\, d\tau$
- Part f $\displaystyle e^t \int_0^t \tau^2(t-\tau)e^{\tau}\, d\tau$
- Part g $\displaystyle e^{-t} \int_0^t e^{-\tau}\tau \cos \omega(t-\tau)\, d\tau$
- Part h $\displaystyle e^t \int_0^t e^{2\tau} \sinh(t-\tau)\, d\tau$
- Part i $\displaystyle\int_0^t \tau e^{2\tau} \sin 2(t-\tau)\, d\tau$
- Part j $\displaystyle\int_0^t (t-\tau)^3 e^{\tau}\, d\tau$
- Part k $\displaystyle\int_0^t \tau^6 e^{-(t-\tau)} \sin 3(t-\tau)\, d\tau$
- Part l $\displaystyle\int_0^t \tau^2(t-\tau)^3\, d\tau$
- Part m $\displaystyle\int_0^t (t-\tau)^7 e^{-\tau} \sin 2\tau\, d\tau$
- Part n $\displaystyle\int_0^t (t-\tau)^4 \sin 2\tau\, d\tau$

3. Find a formula for the solution of the initial value problem.

   - **Part** a $y'' + 3y' + y = f(t), \quad y(0) = 0, \quad y'(0) = 0$
   - **Part** b $y'' + 4y = f(t), \quad y(0) = 0, \quad y'(0) = 0$
   - **Part** c $y'' + 2y' + y = f(t), \quad y(0) = 0, \quad y'(0) = 0$
   - **Part** d $y'' + k^2 y = f(t), \quad y(0) = 1, \quad y'(0) = -1$
   - **Part** e $y'' + 6y' + 9y = f(t), \quad y(0) = 0, \quad y'(0) = -2$
   - **Part** f $y'' - 4y = f(t), \quad y(0) = 0, \quad y'(0) = 3$
   - **Part** g $y'' - 5y' + 6y = f(t), \quad y(0) = 1, \quad y'(0) = 3$
   - **Part** h $y'' + \omega^2 y = f(t), \quad y(0) = k_0, \quad y'(0) = k_1$

4. Solve the integral equation.

   - Part a $y(t) = t - \int_0^t (t-\tau)y(\tau)\,d\tau$
   - Part b $y(t) = \sin t - 2\int_0^t \cos(t-\tau)y(\tau)\,d\tau$
   - Part c $y(t) = 1 + 2\int_0^t y(\tau)\cos(t-\tau)\,d\tau$ Part d $y(t) = t + \int_0^t y(\tau)e^{-(t-\tau)}\,d\tau$
   - **Part** e $y'(t) = t + \int_0^t y(\tau)\cos(t-\tau)\,d\tau$, $y(0) = 4$
   - **Part** f $y(t) = \cos t - \sin t + \int_0^t y(\tau)\sin(t-\tau)\,d\tau$

5. Use the **convolution** theorem to evaluate the integral.

   - Part a $\int_0^t (t-\tau)^7\tau^8\,d\tau$ Part b $\int_0^t (t-\tau)^{13}\tau^7\,d\tau$
   - Part c $\int_0^t (t-\tau)^6\tau^7\,d\tau$ Part d $\int_0^t e^{-\tau}\sin(t-\tau)\,d\tau$
   - **Part** e $\int_0^t \sin\tau\cos 2(t-\tau)\,d\tau$

6. Show that
$$\int_0^t f(t-\tau)g(\tau)\,d\tau = \int_0^t f(\tau)g(t-\tau)\,d\tau$$
by introducing the new variable of integration $x = t - \tau$ in the first integral.

7. Use the convolution theorem to show that if $f(t) \leftrightarrow F(s)$ then
$$\int_0^t f(\tau)\,d\tau \leftrightarrow \frac{F(s)}{s}.$$

8. Show that if $p(s) = as^2 + bs + c$ has distinct real zeros r_1 and r_2 then the solution of
$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1$$

is

$$\begin{aligned} y(t) = \; & k_0 \frac{r_2 e^{r_1 t} - r_1 e^{r_2 t}}{r_2 - r_1} + k_1 \frac{e^{r_2 t} - e^{r_1 t}}{r_2 - r_1} \\ & + \frac{1}{a(r_2 - r_1)} \int_0^t (e^{r_2 \tau} - e^{r_1 \tau}) f(t - \tau)\, d\tau. \end{aligned}$$

9. Show that if $p(s) = as^2 + bs + c$ has a repeated real zero r_1 then the solution of

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1$$

is

$$y(t) = \; k_0(1 - r_1 t)e^{r_1 t} + k_1 t e^{r_1 t} + \frac{1}{a} \int_0^t \tau e^{r_1 \tau} f(t - \tau)\, d\tau.$$

10. Show that if $p(s) = as^2 + bs + c$ has complex conjugate zeros $\lambda \pm i\omega$ then the solution of

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1$$

is

$$\begin{aligned} y(t) = \; & e^{\lambda t} \left[k_0 (\cos \omega t - \frac{\lambda}{\omega} \sin \omega t) + \frac{k_1}{\omega} \sin \omega t \right] \\ & + \frac{1}{a\omega} \int_0^t e^{\lambda t} f(t - \tau) \sin \omega \tau \, d\tau. \end{aligned}$$

11. Let

$$w = \mathcal{L}^{-1} \left(\frac{1}{as^2 + bs + c} \right),$$

where a, b, and c are constants and $a \neq 0$.

- Show that w is the solution of

$$aw'' + bw' + cw = 0, \quad w(0) = 0, \quad w'(0) = \frac{1}{a}.$$

- Let f be continuous on $[0, \infty)$ and define

$$h(t) = \int_0^t w(t - \tau) f(\tau)\, d\tau.$$

  Use *Leibniz's rule* for differentiating an integral with respect to a parameter to show that h is the solution of

$$ah'' + bh' + ch = f, \quad h(0) = 0, \quad h'(0) = 0.$$

- Show that the function y in Eqn. (**??**) is the solution of Eqn. (**??**) provided that f is continuous on $[0, \infty)$; thus, it's not necessary to assume that f has a Laplace transform.

12. Consider the initial value problem

$$ay'' + by' + cy = f(t), \quad y(0) = 0, \quad y'(0) = 0, \tag{A}$$

where a, b, and c are constants, $a \neq 0$, and

$$f(t) = \begin{cases} f_0(t), & 0 \leq t < t_1, \\ f_1(t), & t \geq t_1. \end{cases}$$

Assume that f_0 is continuous and of exponential order on $[0, \infty)$ and f_1 is continuous and of exponential order on $[t_1, \infty)$. Let

$$p(s) = as^2 + bs + c.$$

- Show that the Laplace transform of the solution of (A) is

$$Y(s) = \frac{F_0(s) + e^{-st_1}G(s)}{p(s)}$$

  where $g(t) = f_1(t + t_1) - f_0(t + t_1)$.
- Let w be as in Exercise 11. Use Theorem 16 and the convolution theorem to show that the solution of (A) is

$$y(t) = \int_0^t w(t-\tau)f_0(\tau)\,d\tau + u(t-t_1)\int_0^{t-t_1} w(t-t_1-\tau)g(\tau)\,d\tau$$

  for $t > 0$.
- Henceforth, assume only that f_0 is continuous on $[0, \infty)$ and f_1 is continuous on $[t_1, \infty)$. Use Exercise 11**Part** a and Part b to show that

$$y'(t) = \int_0^t w'(t-\tau)f_0(\tau)\,d\tau + u(t-t_1)\int_0^{t-t_1} w'(t-t_1-\tau)g(\tau)\,d\tau$$

  for $t > 0$, and

$$y''(t) = \frac{f(t)}{a} + \int_0^t w''(t-\tau)f_0(\tau)\,d\tau + u(t-t_1)\int_0^{t-t_1} w''(t-t_1-\tau)g(\tau)\,d\tau$$

  for $0 < t < t_1$ and $t > t_1$. Also, show y satisfies the differential equation in (A) on$(0, t_1)$ and (t_1, ∞).
- Show that y and y' are continuous on $[0, \infty)$.

13. Suppose

$$f(t) = \begin{cases} f_0(t), & 0 \leq t < t_1, \\ f_1(t), & t_1 \leq t < t_2, \\ & \vdots \\ f_{k-1}(t), & t_{k-1} \leq t < t_k, \\ f_k(t), & t \geq t_k, \end{cases}$$

where f_m is continuous on $[t_m, \infty)$ for $m = 0, \ldots, k$ (let $t_0 = 0$), and define

$$g_m(t) = f_m(t + t_m) - f_{m-1}(t + t_m),\ m = 1, \ldots, k.$$

Extend the results of Exercise 12 to show that the solution of

$$ay'' + by' + cy = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

is

$$y(t) = \int_0^t w(t-\tau)f_0(\tau)\,d\tau + \sum_{m=1}^{k} u(t-t_m)\int_0^{t-t_m} w(t-t_m-\tau)g_m(\tau)\,d\tau.$$

14. Let $\{t_m\}_{m=0}^{\infty}$ be a sequence of points such that $t_0 = 0$, $t_{m+1} > t_m$, and $\lim_{m\to\infty} t_m = \infty$. For each nonegative integer m let f_m be continuous on $[t_m, \infty)$, and let f be defined on $[0, \infty)$ by

$$f(t) = f_m(t), \quad t_m \le t < t_{m+1} \quad m = 0, 1, 2 \ldots.$$

Let

$$g_m(t) = f_m(t+t_m) - f_{m-1}(t+t_m), \quad m = 1, \ldots, k.$$

Extend the results of Exercise 13 to show that the solution of

$$ay'' + by' + cy = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

is

$$y(t) = \int_0^t w(t-\tau)f_0(\tau)\,d\tau + \sum_{m=1}^{\infty} u(t-t_m)\int_0^{t-t_m} w(t-t_m-\tau)g_m(\tau)\,d\tau.$$

Section 27

Some solutions

8.6.2. Part a $\sin at \leftrightarrow \dfrac{a}{s^2+a^2}$ and $\cos bt \leftrightarrow \dfrac{s}{s^2+b^2}$, so $H(s) = \dfrac{as}{(s^2+a^2)(s^2+b^2)}$.

Part b $e^t \leftrightarrow \dfrac{1}{s-1}$ and $\sin at \leftrightarrow \dfrac{a}{s^2+a^2}$, so $H(s) = \dfrac{a}{(s-1)(s^2+a^2)}$.

Part c $\sinh at \leftrightarrow \dfrac{a}{s^2-a^2}$ and $\cosh at \leftrightarrow \dfrac{1}{s^2-a^2}$, so $H(s) = \dfrac{as}{(s^2-a^2)^2}$.

Part d $t\sin\omega t \leftrightarrow \dfrac{2\omega s}{(s^2+\omega^2)^2}$ and $t\cos\omega t \leftrightarrow \dfrac{s^2-\omega^2}{(s^2+\omega^2)^2}$, so $H(s) = \dfrac{2\omega s(s^2-\omega^2)}{(s^2+\omega^2)^4}$.

Part e $e^t\displaystyle\int_0^t \sin\omega\tau\cos\omega(t-\tau)\,d\tau = \int_0^t (e^\tau\sin\omega\tau)\left(e^{(t-\tau)}\cos\omega(t-\tau)\right)d\tau$; $e^t\sin\omega t \leftrightarrow \dfrac{\omega}{(s-1)^2+\omega^2}$ and $e^t\cos\omega t \leftrightarrow \dfrac{s-1}{(s-1)^2+\omega^2}$, so $H(s) = \dfrac{(s-1)\omega}{((s-1)^2+\omega^2)^2}$.

Part f $e^t\displaystyle\int_0^t \tau^2(t-\tau)e^\tau\,d\tau = \int_0^t \tau^2e^{2\tau}(t-\tau)e^{(t-\tau)}\,d\tau$; $t^2e^{2t} \leftrightarrow \dfrac{2}{(s-2)^3}$ and $te^t \leftrightarrow \dfrac{1}{(s-1)^2}$, so $H(s) = \dfrac{2}{(s-2)^3(s-1)^2}$.

Part g $e^{-t}\displaystyle\int_0^t e^{-\tau}\tau\cos\omega(t-\tau)\,d\tau = \int_0^t \tau e^{-2\tau}e^{-(t-\tau)}\cos\omega(t-\tau)\,d\tau$; $te^{-2t} \leftrightarrow \dfrac{1}{(s+2)^2}$ and $e^{-t}\cos\omega t \leftrightarrow \dfrac{s+1}{(s+1)^2+\omega^2}$, so $H(s) = \dfrac{s+1}{(s+2)^2\left[(s+1)^2+\omega^2\right]}$.

Part h $e^t\displaystyle\int_0^t e^{2\tau}\sinh(t-\tau)\,d\tau = \int_0^t e^{3\tau}\left(e^{(t-\tau)}\sinh(t-\tau)\right)d\tau$; $e^{3t} \leftrightarrow \dfrac{1}{s-3}$ and $e^t\sinh t \leftrightarrow \dfrac{1}{(s-1)^2-1}$, so $H(s) = \dfrac{1}{(s-3)\left((s-1)^2-1\right)}$.

Part i $te^{2t} \leftrightarrow \dfrac{1}{(s-2)^2}$ and $\sin 2t \leftrightarrow \dfrac{2}{s^2+4}$, so $H(s) = \dfrac{2}{(s-2)^2(s^2+4)}$.

Part j $t^3 \leftrightarrow \dfrac{6}{s^4}$ and $e^t \leftrightarrow \dfrac{1}{s-1}$, so $H(s) = \dfrac{6}{s^4(s-1)}$.

Part k $t^6 \leftrightarrow \dfrac{6!}{s^7}$ and $e^{-t}\sin 3t \leftrightarrow \dfrac{3}{(s+1)^2+9}$, so $H(s) = \dfrac{3\cdot 6!}{s^7\left[(s+1)^2+9\right]}$.

Part l $t^2 \leftrightarrow \dfrac{2}{s^3}$ and $t^3 \leftrightarrow \dfrac{6}{s^4}$, so $H(s) = \dfrac{12}{s^7}$.

Part m $t^7 \leftrightarrow \dfrac{7!}{s^8}$ and $e^{-t}\sin 2t \leftrightarrow \dfrac{2}{(s+1)^2+4}$, so $H(s) = \dfrac{2\cdot 7!}{s^8\left[(s+1)^2+4\right]}$.

Part n $t^4 \leftrightarrow \dfrac{24}{s^5}$ and $\sin 2t \leftrightarrow \dfrac{2}{s^2+4}$, so $H(s) = \dfrac{48}{s^5(s^2+4)}$.

8.6.4. Part a $Y(s) = \dfrac{1}{s^2} - \dfrac{Y(s)}{s^2}$; $Y(s)\left(1+\dfrac{1}{s^2}\right) = \dfrac{1}{s^2}$; $Y(s)\dfrac{s^2+1}{s^2} = \dfrac{1}{s^2}$; $Y(s) = \dfrac{1}{s^2+1}$, so $y = \sin t$.

Part b $Y(s) = \dfrac{1}{s^2+1} - \dfrac{2sY(s)}{s^2+1}$; $Y(s)\left(1+\dfrac{2s}{s^2+1}\right) = \dfrac{1}{s^2+1}$; $Y(s)\dfrac{(s+1)^2}{s^2+1} = \dfrac{1}{s^2+1}$; $Y(s) = \dfrac{1}{(s+1)^2}$, so $y = te^{-t}$.

Part c $Y(s) = \dfrac{1}{s} + \dfrac{2sY(s)}{s^2+1}$; $Y(s)\left(1-\dfrac{2s}{s^2+1}\right) = \dfrac{1}{s}$; $Y(s)\dfrac{(s-1)^2}{s^2+1} = \dfrac{1}{s}$; $Y(s) = \dfrac{(s^2+1)}{s(s-1)^2} = \dfrac{A}{s} + \dfrac{B}{s^2} + \dfrac{C}{(s-1)^2}$, where $A(s-1)^2 + Bs(s-1) + Cs = s^2+1$. Setting $s=0$ and $s=1$ shows that $A=1$ and $C=2$; equating coefficients of s^2 yields $A+B=1$, so $B=0$. Therefore, $Y(s) = \dfrac{1}{s} + \dfrac{1}{(s-1)^2}$, so $y = 1+2te^t$.

Part d $Y(s) = \dfrac{1}{s^2} + \dfrac{Y(s)}{s+1}$; $Y(s)\left(1-\dfrac{1}{s+1}\right) = \dfrac{1}{s^2}$; $Y(s)\left(\dfrac{s}{s+1}\right) = \dfrac{1}{s^2}$; $Y(s) = \dfrac{s+1}{s^3} = \dfrac{1}{s^2} + \dfrac{1}{s^3}$, so $y = t + \dfrac{t^2}{2}$.

Part e $sY(s) - 4 = \dfrac{1}{s^2} + \dfrac{sY(s)}{s^2+1}$; $Y(s)\left(s - \dfrac{s}{s^2+1}\right) = 4 + \dfrac{1}{s^2}$; $Y(s)\dfrac{s^3}{s^2+1} = \dfrac{4s^2+1}{s^2}$; $Y(s) = \dfrac{(4s^2+1)(s^2+1)}{s^5} = \dfrac{4s^4+5s^2+1}{s^5} = \dfrac{4}{s} + \dfrac{5}{s^3} + \dfrac{1}{s^5}$, so $y = 4 + \dfrac{5}{2}t^2 + \dfrac{1}{24}t^4$.

Part f $Y(s) = \dfrac{s-1}{s^2+1} + \dfrac{Y(s)}{s^2+1}$; $Y(s)\left(1-\dfrac{1}{s^2+1}\right) = \dfrac{s-1}{s^2+1}$; $Y(s)\dfrac{s^2}{s^2+1} = \dfrac{s-1}{s^2+1}$; $Y(s) = \dfrac{s-1}{s^2} = \dfrac{1}{s} - \dfrac{1}{s^2}$, so $y = 1-t$.

8.6.6. Substituting $x = t-\tau$ yields $\displaystyle\int_0^t f(t-\tau)g(\tau)\,d\tau = -\int_t^0 f(x)g(t-x)(-\,dx) = \int_0^t f(x)g(t-x)\,dx = \int_0^t f(\tau)g(t-\tau)\,d\tau$.

8.6.8. $p(s)Y(s) = F(s) + a(k_1+k_0s) + bk_0$, so (A) $Y(s) = \dfrac{F(s)}{p(s)} + \dfrac{k_0(as+b)+k_1a}{p(s)}$. Since $p(s) = a(s-r_1)(s-r_2)$ and therefore $b = -a(r_1+r_2)$, (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a(s-r_1)(s-r_2)} + \frac{k_0(s-r_1-r_2)}{(s-r_1)(s-r_2)} + \frac{k_1}{(s-r_1)(s-r_2)}.$$

$$\frac{1}{(s-r_1)(s-r_2)} = \frac{1}{r_2-r_1}\left(\frac{1}{s-r_2} - \frac{1}{s-r_1}\right) \leftrightarrow \frac{e^{r_2t}-e^{r_1t}}{r_2-r_1},$$

so the convolution theorem implies that

$$\frac{F(s)}{a(s-r_1)(s-r_2)} \leftrightarrow \frac{1}{a}\int_0^t \frac{e^{r_2\tau}-e^{r_1\tau}}{r_2-r_1} f(t-\tau)\,d\tau.$$

$$\frac{s-r_1-r_2}{(s-r_1)(s-r_2)} = \frac{r_2}{r_2-r_1}\frac{1}{s-r_1} - \frac{r_1}{r_2-r_1}\frac{1}{s-r_2} \leftrightarrow \frac{r_2e^{r_1t}-r_1e^{r_2}t}{r_2-r_1}.$$

Therefore,

$$y(t) = \; k_0\frac{r_2e^{r_1t}-r_1e^{r_2t}}{r_2-r_1} + k_1\frac{e^{r_2t}-e^{r_1t}}{r_2-r_1} + \frac{1}{a}a\int_0^t \frac{e^{r_2\tau}-e^{r_1\tau}}{r_2-r_1} f(t-\tau)\,d\tau.$$

8.6.10. $p(s)Y(s) = F(s) + a(k_1+k_0s) + bk_0$, so (A) $Y(s) = \dfrac{F(s)}{p(s)} + \dfrac{k_0(as+b)+k_1a}{p(s)}$. Since $p(s) = a(s-\lambda)^2+\omega^2$ and therefore $b = -2a\lambda$, (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a[(s-\lambda)^2+\omega^2]} + \frac{k_0(s-2\lambda)}{(s-\lambda)^2+\omega^2} + \frac{k_1}{(s-\lambda)^2+\omega^2}.$$

$\dfrac{1}{(s-\lambda)^2+\omega^2} \leftrightarrow \dfrac{1}{\omega}e^{\lambda t}\sin\omega t$, so the convolution theorem implies that

$$\frac{F(s)}{a[(s-\lambda)^2+\omega^2]} \leftrightarrow \frac{1}{a\omega}\int_0^t e^{\lambda t} f(t-\tau)\sin\omega\tau\,d\tau.$$

$$\frac{s-2\lambda}{(s-\lambda)^2+\omega^2} = \frac{(s-\lambda)-\lambda}{(s-\lambda)^2+\omega^2} \leftrightarrow e^{\lambda t}\left(\cos\omega t - \frac{\lambda}{\omega}\sin\omega t\right).$$

Therefore,

$$y(t) = e^{\lambda t}\left[k_0\left(\cos\omega t - \frac{\lambda}{\omega}\sin\omega t\right) + \frac{k_1}{\omega}\sin\omega t\right] + \frac{1}{a\omega}\int_0^t e^{\lambda t} f(t-\tau)\sin\omega\tau\,d\tau.$$

8.6.12. Part a

$$ay'' + by' + cy = f_0(t) + u(t-t_1)(f_1(t)-f_0(t)),\; y(0)=0,\; y'(0)=0;$$

$$p(s)Y(s) = F_0(s) + \mathcal{L}(u(t-t_1)(f_1(t)-f_0(t))) = F_0(s) + e^{-st_1}\mathcal{L}(g);$$

$$Y(s) = \frac{F_0(s)+e^{-st_1}G(s)}{p(s)}. \tag{B}$$

Part b Since $F_0(s) \leftrightarrow f_0(t)$, $G(s) \leftrightarrow g(t)$, and $\dfrac{1}{p(s)} \leftrightarrow w(t)$, the convolution theorem implies that

$$\frac{F_0(s)}{p(s)} \leftrightarrow \int_0^t w(t-\tau)f_0(\tau)\,d\tau \quad \text{and} \quad \frac{G(s)}{p(s)} \leftrightarrow \int_0^t w(t-\tau)g(\tau)\,d\tau.$$

Now Theorem 8.4.2 implies that $\dfrac{e^{-st_1}G(s)}{p(s)} \leftrightarrow u(t-t_1)\displaystyle\int_0^t w(t-t_1-\tau)g(\tau)\,d\tau$, and (B) implies that

$$y(t) = \int_0^t w(t-\tau)f_0(\tau)\,d\tau + u(t-t_1)\int_0^{t-t_1} w(t-t_1-\tau)g(\tau)\,d\tau.$$

Part c Let $z_0(t) = \int_0^t w(t-\tau)f_0(\tau)\,d\tau$ and $z_1(t) = \int_0^t w(t-\tau)g(\tau)\,d\tau$. Then $y(t) = z_0(t) + u(t-t_1)z_1(t-t_1)$. Using Leibniz's rule as in the solution of Exercise 8.6.11Part b shows that

$$z_0'(t) = \int_0^t w'(t-\tau)f_0(\tau)\,d\tau, \quad z_1'(t) = \int_0^t w'(t-\tau)g(\tau)\,d\tau, \quad t > 0,$$

$$z_0''(t) = \frac{f_0(t)}{a} + \int_0^t w''(t-\tau)f_0(\tau)\,d\tau, z_1''(t) = \frac{g(t)}{a} + \int_0^t w''(t-\tau)g(\tau)\,d\tau, \quad t > 0,$$

if $t > 0$, and that

$$az_0'' + bz_0' + cz_0 = f_0(t) \quad \text{and} \quad az_1'' + bz_1' + cz_1 = f_1(t+t_1) - f_0(t+t_1), \quad t > 0.$$

This implies the stated conclusion for y' and y'' on $(0,t)$ and (t,∞), and that $ay'' + by' + cy = f(t)$ on these intervals.

Part d Since the functions $z_0(t)$ and $h(t) = u(t-t_1)z_1(t-t_1)$ are both continuous on $[0,\infty)$ and $h(t) = 0$ if $0 \le t \le t_1$, y is continuous on $[0,\infty)$. From**Part** c, y' is continuous on $[0,t_1)$ and (t_1,∞), so we need only show that y' is continuous at t_1. For this it suffices to show that $h'(t_1) = 0$. Since $h(t_1) = 0$ if $t \le t_1$, (B) $\lim_{t\to t_1-} \dfrac{h(t)-h(t_1)}{t-t_1} = 0$. If $t > t_1$, then $h(t) = \displaystyle\int_0^{t-t_1} w(t-t_1\tau)g(\tau)\,d\tau$. Since $h(t_1) = 0$,

$$\left|\frac{h(t)-h(t_1)}{t-t_1}\right| \le \int_0^{t-t_1} |w(t-t_1-\tau)g(\tau)|\,d\tau. \tag{B}$$

Since g is continuous from the right at 0, we can choose constants $T > 0$ and $M > 0$ so that $|g(\tau)| < M$ if $0 \le \tau \le T$. Then (B) implies that

$$\left|\frac{h(t)-h(t_1)}{t-t_1}\right| \le M\int_0^{t-t_1} |w(t-t_1-\tau)|\,d\tau, \quad t_1 < t < t_1 + T. \tag{C}$$

Now suppose $\epsilon > 0$. Since $w(0) = 0$, we can choose T_1 such that $0 < T_1 < T$ and $|w(x)| < \epsilon/M$ if $0 \le x < T_1$. If $t_1 < t < t_1+T_1$ and $0 \le \tau \le t-t_1$, then $0 \le t-t_1-\tau < T_1$, so (C) implies that

$$\left|\frac{h(t)-h(t_1)}{t-t_1}\right| < \epsilon, \quad t_1 < t < t_1 + T.$$

Therefore, $\lim_{t\to t_1+} \dfrac{h(t)-h(t_1)}{t-t_1} = 0$. This and (B) imply that $h'(t_1) = 0$.

Section 28

Basic Theory of Homogeneous Linear Systems

We consider homogeneous linear systems $\mathbf{y}' = A(t)\mathbf{y}$, where $A = A(t)$ is a continuous $n \times n$ matrix function on an interval (a, b). The theory of linear homogeneous systems has much in common with the theory of linear homogeneous scalar equations.

Whenever we refer to solutions of $\mathbf{y}' = A(t)\mathbf{y}$ we'll mean solutions on (a, b). Since $\mathbf{y} \equiv \mathbf{0}$ is obviously a solution of $\mathbf{y}' = A(t)\mathbf{y}$, we call it the *trivial* solution. Any other solution is *nontrivial*.

Subsection 28.1

Constant Coefficient Homogeneous Systems

In this section, we assume that all the eigenvalues of A are real and that A has a set of n linearly independent eigenvectors. In the next two sections, we consider the cases where some eigenvalues of A are complex, or where A does not have n linearly independent eigenvectors.

Theorem 19 Suppose the $n \times n$ constant matrix A has n real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (which need not be distinct) with associated linearly independent eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$. Then the functions

$$\mathbf{y}_1 = \mathbf{x}_1 e^{\lambda_1 t},\ \mathbf{y}_2 = \mathbf{x}_2 e^{\lambda_2 t},\ \ldots,\ \mathbf{y}_n = \mathbf{x}_n e^{\lambda_n t}$$

form a fundamental set of solutions of $\mathbf{y}' = A\mathbf{y}$; that is, the general solution of this system is

$$\mathbf{y} = c_1\mathbf{x}_1 e^{\lambda_1 t} + c_2\mathbf{x}_2 e^{\lambda_2 t} + \cdots + c_n\mathbf{x}_n e^{\lambda_n t}.$$

Proof Differentiating $\mathbf{y}_i = \mathbf{x}_i e^{\lambda_i t}$ and recalling that $A\mathbf{x}_i = \lambda_i \mathbf{x}_i$ yields

$$\mathbf{y}_i' = \lambda_i \mathbf{x}_i e^{\lambda_i t} = A\mathbf{x}_i e^{\lambda_i t} = A\mathbf{y}_i.$$

This shows that $\mathbf{y}_i$ is a solution of $\mathbf{y}' = A\mathbf{y}$.

The Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n\}$ is

$$\begin{vmatrix} x_{11}e^{\lambda_1 t} & x_{12}e^{\lambda_2 t} & \cdots & x_{1n}e^{\lambda_n t} \\ x_{21}e^{\lambda_1 t} & x_{22}e^{\lambda_2 t} & \cdots & x_{2n}e^{\lambda_n t} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1}e^{\lambda_1 t} & x_{n2}e^{\lambda_2 t} & \cdots & x_{nn}e^{\lambda x_n t} \end{vmatrix} = e^{\lambda_1 t}e^{\lambda_2 t}\cdots e^{\lambda_n t} \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{vmatrix}.$$

Since the columns of the determinant on the right are $\mathbf{x}_1$, $\mathbf{x}_2$, ..., $\mathbf{x}_n$, which are assumed to be linearly independent, the determinant is nonzero. Therefore Theorem **??** implies that $\{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n\}$ is a fundamental set of solutions of $\mathbf{y}' = A\mathbf{y}$.

Example 114

- Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}\mathbf{y}. \tag{28.1}$$

- Solve the initial value problem

$$\mathbf{y}' = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} \mathbf{y}, \quad \mathbf{y}(0) = \begin{bmatrix} 5 \\ -1 \end{bmatrix}. \tag{28.2}$$

Solution

Solution(a) The characteristic polynomial of the coefficient matrix A in (28.1) is

$$\begin{aligned} \begin{vmatrix} 2-\lambda & 4 \\ 4 & 2-\lambda \end{vmatrix} &= (\lambda-2)^2 - 16 \\ &= (\lambda-2-4)(\lambda-2+4) \\ &= (\lambda-6)(\lambda+2). \end{aligned}$$

Hence, $\lambda_1 = 6$ and $\lambda_2 = -2$ are eigenvalues of A. To obtain the eigenvectors, we must solve the system

$$\begin{bmatrix} 2-\lambda & 4 \\ 4 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{28.3}$$

with $\lambda = 6$ and $\lambda = -2$. Setting $\lambda = 6$ in (28.3) yields

$$\begin{bmatrix} -4 & 4 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which implies that $x_1 = x_2$. Taking $x_2 = 1$ yields the eigenvector

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{6t}$$

is a solution of (28.1). Setting $\lambda = -2$ in (28.3) yields

$$\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which implies that $x_1 = -x_2$. Taking $x_2 = 1$ yields the eigenvector

$$\mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

so

$$\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}$$

is a solution of (28.1). From Theorem 19, the general solution of (28.1) is

$$\mathbf{y} = c_1\mathbf{y}_1 + c_2\mathbf{y}_2 = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{6t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}. \tag{28.4}$$

Solution(b) To satisfy the initial condition in (28.2), we must choose c_1 and c_2 in (28.4) so that

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}.$$

This is equivalent to the system

$$\begin{aligned} c_1 - c_2 &= 5 \\ c_1 + c_2 &= -1, \end{aligned}$$

so $c_1 = 2, c_2 = -3$. Therefore the solution of (28.2) is

$$\mathbf{y} = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{6t} - 3\begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t},$$

or, in terms of components,

$$y_1 = 2e^{6t} + 3e^{-2t}, \quad y_2 = 2e^{6t} - 3e^{-2t}.$$

Example 115

- Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} 3 & -1 & -1 \\ -2 & 3 & 2 \\ 4 & -1 & -2 \end{bmatrix} \mathbf{y}. \tag{28.5}$$

- Solve the initial value problem

$$\mathbf{y}' = \begin{bmatrix} 3 & -1 & -1 \\ -2 & 3 & 2 \\ 4 & -1 & -2 \end{bmatrix} \mathbf{y}, \quad \mathbf{y}(0) = \begin{bmatrix} 2 \\ -1 \\ 8 \end{bmatrix}. \tag{28.6}$$

Solution

Solution(**a**) The characteristic polynomial of the coefficient matrix A in (28.5) is

$$\begin{vmatrix} 3-\lambda & -1 & -1 \\ -2 & 3-\lambda & 2 \\ 4 & -1 & -2-\lambda \end{vmatrix} = -(\lambda-2)(\lambda-3)(\lambda+1).$$

Hence, the eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = 3$, and $\lambda_3 = -1$. To find the eigenvectors, we must solve the system

$$\begin{bmatrix} 3-\lambda & -1 & -1 \\ -2 & 3-\lambda & 2 \\ 4 & -1 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{28.7}$$

with $\lambda = 2, 3, -1$. With $\lambda = 2$, the augmented matrix of (28.7) is

$$\left[\begin{array}{ccc:c} 1 & -1 & -1 & 0 \\ -2 & 1 & 2 & 0 \\ 4 & -1 & -4 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $x_1 = x_3$ and $x_2 = 0$. Taking $x_3 = 1$ yields

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

as a solution of (28.5). With $\lambda = 3$, the augmented matrix of (28.7) is

$$\left[\begin{array}{ccc:c} 0 & -1 & -1 & 0 \\ -2 & 0 & 2 & 0 \\ 4 & -1 & -5 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $x_1 = x_3$ and $x_2 = -x_3$. Taking $x_3 = 1$ yields

$$\mathbf{y}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^{3t}$$

as a solution of (28.5). With $\lambda = -1$, the augmented matrix of (28.7) is

$$\left[\begin{array}{ccc:c} 4 & -1 & -1 & 0 \\ -2 & 4 & 2 & 0 \\ 4 & -1 & -1 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & \frac{3}{7} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $x_1 = x_3/7$ and $x_2 = -3x_3/7$. Taking $x_3 = 7$ yields

$$\mathbf{y}_3 = \begin{bmatrix} 1 \\ -3 \\ 7 \end{bmatrix} e^{-t}$$

as a solution of (28.5). By Theorem 19, the general solution of (28.5) is

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 1 \\ -3 \\ 7 \end{bmatrix} e^{-t},$$

which can also be written as

$$\mathbf{y} = \begin{bmatrix} e^{2t} & e^{3t} & e^{-t} \\ 0 & -e^{3t} & -3e^{-t} \\ e^{2t} & e^{3t} & 7e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}. \tag{28.8}$$

Solution(b) To satisfy the initial condition in (28.6) we must choose c_1, c_2, c_3 in (28.8) so that

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 1 & 1 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 8 \end{bmatrix}.$$

Solving this system yields $c_1 = 3$, $c_2 = -2$, $c_3 = 1$. Hence, the solution of (28.6) is

$$\begin{aligned} \mathbf{y} &= \begin{bmatrix} e^{2t} & e^{3t} & e^{-t} \\ 0 & -e^{3t} & -3e^{-t} \\ e^{2t} & e^{3t} & 7e^{-t} \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \\ &= 3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} - 2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^{3t} + \begin{bmatrix} 1 \\ -3 \\ 7 \end{bmatrix} e^{-t}. \end{aligned}$$

Example 116 Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix} \mathbf{y}. \tag{28.9}$$

Solution The characteristic polynomial of the coefficient matrix A in (28.9) is

$$\begin{vmatrix} -3-\lambda & 2 & 2 \\ 2 & -3-\lambda & 2 \\ 2 & 2 & -3-\lambda \end{vmatrix} = -(\lambda - 1)(\lambda + 5)^2.$$

Hence, $\lambda_1 = 1$ is an eigenvalue of multiplicity 1, while $\lambda_2 = -5$ is an eigenvalue of multiplicity 2. Eigenvectors associated with $\lambda_1 = 1$ are solutions of the system with augmented matrix

$$\left[\begin{array}{ccc:c} -4 & 2 & 2 & 0 \\ 2 & -4 & 2 & 0 \\ 2 & 2 & -4 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $x_1 = x_2 = x_3$, and we choose $x_3 = 1$ to obtain the solution

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t \tag{28.10}$$

of (28.9). Eigenvectors associated with $\lambda_2 = -5$ are solutions of the system with augmented matrix

$$\left[\begin{array}{ccc:c} 2 & 2 & 2 & 0 \\ 2 & 2 & 2 & 0 \\ 2 & 2 & 2 & 0 \end{array}\right].$$

Hence, the components of these eigenvectors need only satisfy the single condition

$$x_1 + x_2 + x_3 = 0.$$

Since there's only one equation here, we can choose x_2 and x_3 arbitrarily. We obtain one eigenvector by choosing $x_2 = 0$ and $x_3 = 1$, and another by choosing $x_2 = 1$ and $x_3 = 0$. In both cases $x_1 = -1$. Therefore

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

are linearly independent eigenvectors associated with $\lambda_2 = -5$, and the corresponding solutions of (28.9) are

$$\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-5t} \quad \text{and} \quad \mathbf{y}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-5t}.$$

Because of this and (28.10), Theorem 19 implies that the general solution of (28.9) is

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-5t} + c_3 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-5t}.$$

Subsection 28.2

Linearly independent eigenvectors

We saw that if an $n \times n$ constant matrix A has n real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (which need not be distinct) with associated linearly independent eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$, then the general solution of $\mathbf{y}' = A\mathbf{y}$ is

$$\mathbf{y} = c_1\mathbf{x}_1 e^{\lambda_1 t} + c_2\mathbf{x}_2 e^{\lambda_2 t} + \cdots + c_n\mathbf{x}_n e^{\lambda_n t}.$$

In this section we consider the case where A has n real eigenvalues, but does not have n linearly independent eigenvectors. It is shown in linear algebra that this occurs if and only

if A has at least one eigenvalue of multiplicity $r > 1$ such that the associated eigenspace has dimension less than r. In this case A is said to be *defective*. Since it's beyond the scope of this book to give a complete analysis of systems with defective coefficient matrices, we will restrict our attention to some commonly occurring special cases.

Example 117 Show that the system

$$\mathbf{y}' = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix} \mathbf{y} \tag{28.11}$$

does not have a fundamental set of solutions of the form $\{\mathbf{x}_1 e^{\lambda_1 t}, \mathbf{x}_2 e^{\lambda_2 t}\}$, where λ_1 and λ_2 are eigenvalues of the coefficient matrix A of (28.11) and $\mathbf{x}_1$, and $\mathbf{x}_2$ are associated linearly independent eigenvectors.

Solution The characteristic polynomial of A is

$$\begin{aligned} \begin{vmatrix} 11-\lambda & -25 \\ 4 & -9-\lambda \end{vmatrix} &= (\lambda - 11)(\lambda + 9) + 100 \\ &= \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2. \end{aligned}$$

Hence, $\lambda = 1$ is the only eigenvalue of A. The augmented matrix of the system $(A - I)\mathbf{x} = \mathbf{0}$ is

$$\left[\begin{array}{cc:c} 10 & -25 & 0 \\ 4 & -10 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{cc:c} 1 & -\frac{5}{2} & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Hence, $x_1 = 5x_2/2$ where x_2 is arbitrary. Therefore all eigenvectors of A are scalar multiples of $\mathbf{x}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, so A does not have a set of two linearly independent eigenvectors. ■

From Example 117, we know that all scalar multiples of $\mathbf{y}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^t$ are solutions of (28.11); however, to find the general solution we must find a second solution $\mathbf{y}_2$ such that $\{\mathbf{y}_1, \mathbf{y}_2\}$ is linearly independent. Based on your recollection of the procedure for solving a constant coefficient scalar equation

$$ay'' + by' + cy = 0$$

in the case where the characteristic polynomial has a repeated root, you might expect to obtain a second solution of (28.11) by multiplying the first solution by t. However, this yields $\mathbf{y}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix} te^t$, which doesn't work, since

$$\mathbf{y}_2' = \begin{bmatrix} 5 \\ 2 \end{bmatrix} (te^t + e^t), \quad \text{while} \quad \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix} \mathbf{y}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix} te^t.$$

The next theorem shows what to do in this situation.

Theorem 20 Suppose the $n \times n$ matrix A has an eigenvalue λ_1 of multiplicity ≥ 2 and the associated eigenspace has dimension 1; that is, all λ_1-eigenvectors of A are scalar multiples of an eigenvector $\mathbf{x}$. Then there are infinitely many vectors $\mathbf{u}$ such that

$$(A - \lambda_1 I)\mathbf{u} = \mathbf{x}. \tag{28.12}$$

Moreover, if $\mathbf{u}$ is any such vector then

$$\mathbf{y}_1 = \mathbf{x}e^{\lambda_1 t} \quad \text{and} \quad \mathbf{y}_2 = \mathbf{u}e^{\lambda_1 t} + \mathbf{x}te^{\lambda_1 t} \tag{28.13}$$

are linearly independent solutions of $\mathbf{y}' = A\mathbf{y}$.

Example 118 Use Theorem 20 to find the general solution of the system

$$\mathbf{y}' = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix} \mathbf{y} \tag{28.14}$$

considered in Example 117.

Solution In Example 117 we saw that $\lambda_1 = 1$ is an eigenvalue of multiplicity 2 of the coefficient matrix A in (28.14), and that all of the eigenvectors of A are multiples of

$$\mathbf{x} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

Therefore

$$\mathbf{y}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^t$$

is a solution of (28.14). From Theorem 20, a second solution is given by $\mathbf{y}_2 = \mathbf{u}e^t + \mathbf{x}te^t$, where $(A - I)\mathbf{u} = \mathbf{x}$. The augmented matrix of this system is

$$\left[\begin{array}{cc:c} 10 & -25 & 5 \\ 4 & -10 & 2 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{cc:c} 1 & -\frac{5}{2} & \frac{1}{2} \\ 0 & 0 & 0 \end{array}\right].$$

Therefore the components of $\mathbf{u}$ must satisfy

$$u_1 - \frac{5}{2}u_2 = \frac{1}{2},$$

where u_2 is arbitrary. We choose $u_2 = 0$, so that $u_1 = 1/2$ and

$$\mathbf{u} = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}.$$

Thus,

$$\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^t}{2} + \begin{bmatrix} 5 \\ 2 \end{bmatrix} te^t.$$

Since $\mathbf{y}_1$ and $\mathbf{y}_2$ are linearly independent by Theorem 20, they form a fundamental set

of solutions of (28.14). Therefore the general solution of (28.14) is

$$\mathbf{y} = c_1 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^t}{2} + \begin{bmatrix} 5 \\ 2 \end{bmatrix} te^t \right). \blacksquare$$

Example 119

Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} 3 & 4 & -10 \\ 2 & 1 & -2 \\ 2 & 2 & -5 \end{bmatrix} \mathbf{y}. \tag{28.15}$$

Solution

The characteristic polynomial of the coefficient matrix A in (28.15) is

$$\begin{vmatrix} 3-\lambda & 4 & -10 \\ 2 & 1-\lambda & -2 \\ 2 & 2 & -5-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+1)^2.$$

Hence, the eigenvalues are $\lambda_1 = 1$ with multiplicity 1 and $\lambda_2 = -1$ with multiplicity 2. Eigenvectors associated with $\lambda_1 = 1$ must satisfy $(A - I)\mathbf{x} = \mathbf{0}$. The augmented matrix of this system is

$$\begin{bmatrix} 2 & 4 & -10 & \vdots & 0 \\ 2 & 0 & -2 & \vdots & 0 \\ 2 & 2 & -6 & \vdots & 0 \end{bmatrix},$$

which is row equivalent to

$$\begin{bmatrix} 1 & 0 & -1 & \vdots & 0 \\ 0 & 1 & -2 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}.$$

Hence, $x_1 = x_3$ and $x_2 = 2x_3$, where x_3 is arbitrary. Choosing $x_3 = 1$ yields the eigenvector

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^t$$

is a solution of (28.15).
Eigenvectors associated with $\lambda_2 = -1$ satisfy $(A + I)\mathbf{x} = \mathbf{0}$. The augmented matrix of this system is

$$\begin{bmatrix} 4 & 4 & -10 & \vdots & 0 \\ 2 & 2 & -2 & \vdots & 0 \\ 2 & 2 & -4 & \vdots & 0 \end{bmatrix},$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $x_3 = 0$ and $x_1 = -x_2$, where x_2 is arbitrary. Choosing $x_2 = 1$ yields the eigenvector

$$\mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix},$$

so

$$\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-t}$$

is a solution of (28.15).
Since all the eigenvectors of A associated with $\lambda_2 = -1$ are multiples of $\mathbf{x}_2$, we must now use Theorem 20 to find a third solution of (28.15) in the form

$$\mathbf{y}_3 = \mathbf{u}e^{-t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-t}, \tag{28.16}$$

where $\mathbf{u}$ is a solution of $(A + I)\mathbf{u} = \mathbf{x}_2$. The augmented matrix of this system is

$$\left[\begin{array}{ccc:c} 4 & 4 & -10 & -1 \\ 2 & 2 & -2 & 1 \\ 2 & 2 & -4 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Hence, $u_3 = 1/2$ and $u_1 = 1 - u_2$, where u_2 is arbitrary. Choosing $u_2 = 0$ yields

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ \frac{1}{2} \end{bmatrix},$$

and substituting this into (28.16) yields the solution

$$\mathbf{y}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \frac{e^{-t}}{2} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-t}$$

of (28.15).

Since the Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ at $t = 0$ is

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & \frac{1}{2} \end{vmatrix} = \frac{1}{2},$$

$\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ is a fundamental set of solutions of (28.15). Therefore the general solution of (28.15) is

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-t} + c_3 \left(\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \frac{e^{-t}}{2} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-t} \right).$$

Subsection 28.3

Complex eigenvalues

We now consider the system $\mathbf{y}' = A\mathbf{y}$, where A has a complex eigenvalue $\lambda = \alpha + i\beta$ with $\beta \neq 0$. We continue to assume that A has real entries, so the characteristic polynomial of A has real coefficients. This implies that $\overline{\lambda} = \alpha - i\beta$ is also an eigenvalue of A.

An eigenvector $\mathbf{x}$ of A associated with $\lambda = \alpha + i\beta$ will have complex entries, so we'll write

$$\mathbf{x} = \mathbf{u} + i\mathbf{v}$$

where $\mathbf{u}$ and $\mathbf{v}$ have real entries; that is, $\mathbf{u}$ and $\mathbf{v}$ are the real and imaginary parts of $\mathbf{x}$. Since $A\mathbf{x} = \lambda\mathbf{x}$,

$$A(\mathbf{u} + i\mathbf{v}) = (\alpha + i\beta)(\mathbf{u} + i\mathbf{v}). \tag{28.17}$$

Taking complex conjugates here and recalling that A has real entries yields

$$A(\mathbf{u} - i\mathbf{v}) = (\alpha - i\beta)(\mathbf{u} - i\mathbf{v}),$$

which shows that $\mathbf{x} = \mathbf{u} - i\mathbf{v}$ is an eigenvector associated with $\overline{\lambda} = \alpha - i\beta$. The complex conjugate eigenvalues λ and $\overline{\lambda}$ can be separately associated with linearly independent solutions $\mathbf{y}' = A\mathbf{y}$; however, we won't pursue this approach, since solutions obtained in this way turn out to be complex–valued. Instead, we'll obtain solutions of $\mathbf{y}' = A\mathbf{y}$ in the form

$$\mathbf{y} = f_1\mathbf{u} + f_2\mathbf{v} \tag{28.18}$$

where f_1 and f_2 are real–valued scalar functions.

The next theorem shows how to do this.

Theorem 21 Let A be an $n \times n$ matrix with real entries. Let $\lambda = \alpha + i\beta$ ($\beta \neq 0$) be a complex eigenvalue of A and let $\mathbf{x} = \mathbf{u} + i\mathbf{v}$ be an associated eigenvector, where $\mathbf{u}$ and $\mathbf{v}$ have real components. Then $\mathbf{u}$ and $\mathbf{v}$ are both nonzero and

$$\mathbf{y}_1 = e^{\alpha t}(\mathbf{u}\cos\beta t - \mathbf{v}\sin\beta t) \quad \text{and} \quad \mathbf{y}_2 = e^{\alpha t}(\mathbf{u}\sin\beta t + \mathbf{v}\cos\beta t),$$

which are the real and imaginary parts of

$$e^{\alpha t}(\cos\beta t + i\sin\beta t)(\mathbf{u} + i\mathbf{v}), \tag{28.19}$$

are linearly independent solutions of $\mathbf{y}' = A\mathbf{y}$.

Example 120 Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} 4 & -5 \\ 5 & -2 \end{bmatrix} \mathbf{y}. \tag{28.20}$$

Solution The characteristic polynomial of the coefficient matrix A in (28.20) is

$$\begin{vmatrix} 4-\lambda & -5 \\ 5 & -2-\lambda \end{vmatrix} = (\lambda - 1)^2 + 16.$$

Hence, $\lambda = 1 + 4i$ is an eigenvalue of A. The associated eigenvectors satisfy $(A - (1+4i)\,I)\,\mathbf{x} = \mathbf{0}$. The augmented matrix of this system is

$$\left[\begin{array}{cc:c} 3-4i & -5 & 0 \\ 5 & -3-4i & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{cc:c} 1 & -\frac{3+4i}{5} & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1 = (3+4i)x_2/5$. Taking $x_2 = 5$ yields $x_1 = 3+4i$, so

$$\mathbf{x} = \begin{bmatrix} 3+4i \\ 5 \end{bmatrix}$$

is an eigenvector. The real and imaginary parts of

$$e^t(\cos 4t + i \sin 4t) \begin{bmatrix} 3+4i \\ 5 \end{bmatrix}$$

are

$$\mathbf{y}_1 = e^t \begin{bmatrix} 3\cos 4t - 4\sin 4t \\ 5\cos 4t \end{bmatrix} \quad \text{and} \quad \mathbf{y}_2 = e^t \begin{bmatrix} 3\sin 4t + 4\cos 4t \\ 5\sin 4t \end{bmatrix},$$

which are linearly independent solutions of (28.20). The general solution of (28.20) is

$$\mathbf{y} = c_1 e^t \begin{bmatrix} 3\cos 4t - 4\sin 4t \\ 5\cos 4t \end{bmatrix} + c_2 e^t \begin{bmatrix} 3\sin 4t + 4\cos 4t \\ 5\sin 4t \end{bmatrix}.$$

Example 121 Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} -14 & 39 \\ -6 & 16 \end{bmatrix} \mathbf{y}. \tag{28.21}$$

Solution The characteristic polynomial of the coefficient matrix A in (28.21) is

$$\begin{vmatrix} -14-\lambda & 39 \\ -6 & 16-\lambda \end{vmatrix} = (\lambda - 1)^2 + 9.$$

Hence, $\lambda = 1 + 3i$ is an eigenvalue of A. The associated eigenvectors satisfy

$(A-(1+3i)I)\,\mathbf{x}=\mathbf{0}$. The augmented augmented matrix of this system is

$$\left[\begin{array}{cc:c} -15-3i & 39 & 0 \\ -6 & 15-3i & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{cc:c} 1 & \frac{-5+i}{2} & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1=(5-i)/2$. Taking $x_2=2$ yields $x_1=5-i$, so

$$\mathbf{x}=\begin{bmatrix} 5-i \\ 2 \end{bmatrix}$$

is an eigenvector. The real and imaginary parts of

$$e^t(\cos 3t+i\sin 3t)\begin{bmatrix} 5-i \\ 2 \end{bmatrix}$$

are

$$\mathbf{y}_1=e^t\begin{bmatrix} \sin 3t+5\cos 3t \\ 2\cos 3t \end{bmatrix} \quad \text{and} \quad \mathbf{y}_2=e^t\begin{bmatrix} -\cos 3t+5\sin 3t \\ 2\sin 3t \end{bmatrix},$$

which are linearly independent solutions of (28.21). The general solution of (28.21) is

$$\mathbf{y}=c_1e^t\begin{bmatrix} \sin 3t+5\cos 3t \\ 2\cos 3t \end{bmatrix}+c_2e^t\begin{bmatrix} -\cos 3t+5\sin 3t \\ 2\sin 3t \end{bmatrix}.$$

Example 122 Find the general solution of

$$\mathbf{y}'=\begin{bmatrix} -5 & 5 & 4 \\ -8 & 7 & 6 \\ 1 & 0 & 0 \end{bmatrix}\mathbf{y}. \tag{28.22}$$

Solution The characteristic polynomial of the coefficient matrix A in (28.22) is

$$\begin{vmatrix} -5-\lambda & 5 & 4 \\ -8 & 7-\lambda & 6 \\ 1 & 0 & -\lambda \end{vmatrix}=-(\lambda-2)(\lambda^2+1).$$

Hence, the eigenvalues of A are $\lambda_1=2$, $\lambda_2=i$, and $\lambda_3=-i$. The augmented matrix of $(A-2I)\mathbf{x}=\mathbf{0}$ is

$$\left[\begin{array}{ccc:c} -7 & 5 & 4 & 0 \\ -8 & 5 & 6 & 0 \\ 1 & 0 & -2 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1 = x_2 = 2x_3$. Taking $x_3 = 1$ yields

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix},$$

so

$$\mathbf{y}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{2t}$$

is a solution of (28.22).
The augmented matrix of $(A - iI)\mathbf{x} = \mathbf{0}$ is

$$\left[\begin{array}{ccc:c} -5-i & 5 & 4 & 0 \\ -8 & 7-i & 6 & 0 \\ 1 & 0 & -i & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -i & 0 \\ 0 & 1 & 1-i & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1 = ix_3$ and $x_2 = -(1-i)x_3$. Taking $x_3 = 1$ yields the eigenvector

$$\mathbf{x}_2 = \begin{bmatrix} i \\ -1+i \\ 1 \end{bmatrix}.$$

The real and imaginary parts of

$$(\cos t + i \sin t) \begin{bmatrix} i \\ -1+i \\ 1 \end{bmatrix}$$

are

$$\mathbf{y}_2 = \begin{bmatrix} -\sin t \\ -\cos t - \sin t \\ \cos t \end{bmatrix} \quad \text{and} \quad \mathbf{y}_3 = \begin{bmatrix} \cos t \\ \cos t - \sin t \\ \sin t \end{bmatrix},$$

which are solutions of (28.22). Since the Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ at $t = 0$ is

$$\begin{vmatrix} 2 & 0 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 1,$$

$\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ is a fundamental set of solutions of (28.22). The general solution of (28.22) is

$$\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -\sin t \\ -\cos t - \sin t \\ \cos t \end{bmatrix} + c_3 \begin{bmatrix} \cos t \\ \cos t - \sin t \\ \sin t \end{bmatrix}.$$

Example 123 Find the general solution of

$$\mathbf{y}' = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 3 & 2 \\ 1 & -1 & 2 \end{bmatrix} \mathbf{y}. \tag{28.23}$$

Solution The characteristic polynomial of the coefficient matrix A in (28.23) is

$$\begin{vmatrix} 1-\lambda & -1 & -2 \\ 1 & 3-\lambda & 2 \\ 1 & -1 & 2-\lambda \end{vmatrix} = -(\lambda-2)\left((\lambda-2)^2+4\right).$$

Hence, the eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = 2+2i$, and $\lambda_3 = 2-2i$. The augmented matrix of $(A-2I)\mathbf{x} = \mathbf{0}$ is

$$\left[\begin{array}{ccc:c} -1 & -1 & -2 & 0 \\ 1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1 = x_2 = -x_3$. Taking $x_3 = 1$ yields

$$\mathbf{x}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

so

$$\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$$

is a solution of (28.23).
The augmented matrix of $(A-(2+2i)I)\,\mathbf{x} = \mathbf{0}$ is

$$\left[\begin{array}{ccc:c} -1-2i & -1 & -2 & 0 \\ 1 & 1-2i & 2 & 0 \\ 1 & -1 & -2i & 0 \end{array}\right],$$

which is row equivalent to

$$\left[\begin{array}{ccc:c} 1 & 0 & -i & 0 \\ 0 & 1 & i & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Therefore $x_1 = ix_3$ and $x_2 = -ix_3$. Taking $x_3 = 1$ yields the eigenvector

$$\mathbf{x}_2 = \begin{bmatrix} i \\ -i \\ 1 \end{bmatrix}$$

The real and imaginary parts of

$$e^{2t}(\cos 2t + i \sin 2t) \begin{bmatrix} i \\ -i \\ 1 \end{bmatrix}$$

are

$$\mathbf{y}_2 = e^{2t} \begin{bmatrix} -\sin 2t \\ \sin 2t \\ \cos 2t \end{bmatrix} \quad \text{and} \quad \mathbf{y}_2 = e^{2t} \begin{bmatrix} \cos 2t \\ -\cos 2t \\ \sin 2t \end{bmatrix},$$

which are solutions of (28.23). Since the Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ at $t = 0$ is

$$\begin{vmatrix} -1 & 0 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix} = -2,$$

$\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ is a fundamental set of solutions of (28.23). The general solution of (28.23) is

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t} + c_2 e^{2t} \begin{bmatrix} -\sin 2t \\ \sin 2t \\ \cos 2t \end{bmatrix} + c_3 e^{2t} \begin{bmatrix} \cos 2t \\ -\cos 2t \\ \sin 2t \end{bmatrix}.$$

Section 29

Complex Numbers

Subsection 29.1

Algebraic approach

Real numbers, while being extremely useful, are not complete - they can't solve all equations involving numbers. For example, the equation

$$x^2 + 1 = 0 \tag{29.1}$$

has no real solutions, since there can be no real number x such that $x^2 = -1$. However, we can choose to define a new number, $= \sqrt{-1}$ and using it to build a new number system. This system is of course the set of complex numbers, $\mathbb{C}$. It is defined as the set of all z such that

$$z = a + b, \tag{29.2}$$

where $a, b \in \mathbb{R}$ and $= \sqrt{-1}$. We call a the *real component* of z or $\mathrm{Re}(z)$, and b its *imaginary component* or $\mathrm{Im}(z)$[2]. These numbers appear a lot all throughout the exact sciences (but especially in physics and engineering), so we must at the very least learn their basic properties.

It is not so obvious that we can add two different kinds of numbers together, but it works (the linear algebra chapter sheds more light on this idea). What is important is that we always keep these two parts separated. We see this when we add together two complex numbers z_1, z_2:

$$z = z_1 + z_2 = (a_1 + b_1) + (a_2 + b_2) = (a_1 + a_2) + (b_1 + b_2)\,. \tag{29.3}$$

The real part of z is therefore $a_1 + b_1$, and its imaginary part is $b_1 + b_2$.

What happens when we multiply two complex numbers? Let's check:

$$\begin{aligned} z = z_1 z_2 &= (a_1 + b_1)(a_2 + b_2) \\ &= a_1 a_2 + a_1 b_2 + a_2 b_1 +^2 b_1 b_2 \\ &= a_1 a_2 + a_1 b_2 + a_2 b_1 - b_1 b_2 \\ &= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)\,. \end{aligned} \tag{29.4}$$

We see that we can still separate the real part and imaginary part of the result. What happens in the case of two real numbers? For real numbers $b = 0$, and thus Equation 29.4 devolves to $z = a_1 a_2 \in \mathbb{R}$, which is exactly what we expect: multiplying two real numbers yields their product, which is a real number. Notice that this doesn't happen with purely imaginary numbers: multiplying together two imaginary numbers (i.e. numbers for which $a = 0$) results in a real number. Will get to understand why this happens very soon.

When discussing real numbers sometimes we like to refer to their *magnitude*, i.e. their absolute value. With complex numbers this is defined as

$$|z| = \sqrt{a^2 + b^2}, \tag{29.5}$$

i.e. in a sense, to get the magnitude of a complex number we imagine its two components as being perpendicular and calculate the length of the resulting hypotenuse (cf. the Pythagorean theorem). In fact, this is one very useful interpretation of complex numbers, which we will explore in depth in the next subsection.

[2] There is nothing more "real" about real numbers than imaginary numbers, but unfortunately that's the terminology we're stuck with

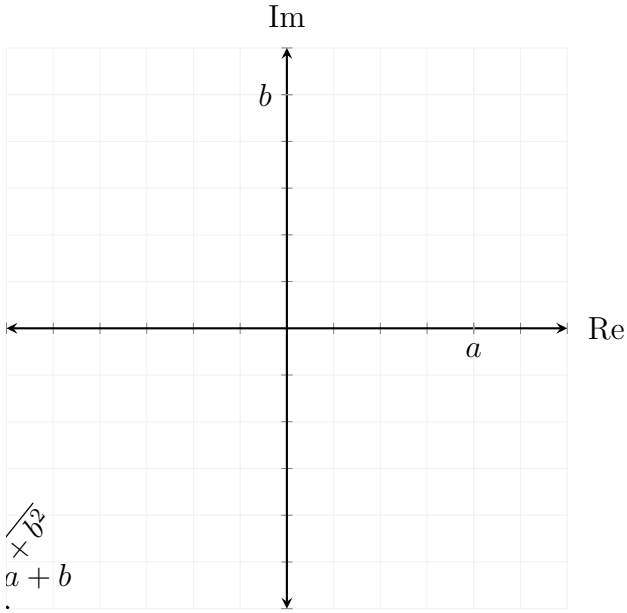


Figure 10. A complex number $z = a + b$ shown on the complex plane: the horizontal and vertical axes represent the real and imaginary components, respectively.

A very important operation that can be applied to complex numbers is *conjugation*. The conjugate of a complex number $z = a + b$ is defined as

$$z = a - b, \tag{29.6}$$

i.e. conjugating a number is simply negating its imaginary part. When we multiply a complex number by its own complex conjugate we get

$$zz = (a + b)(a - b) = a^2 + ab - ab - b^{22} = a^2 + b^2, \tag{29.7}$$

i.e. $zz = |z|^2$. The inverse of a complex number can be expressed as

$$z^{-1} = \frac{z}{|z|^2}. \tag{29.8}$$

Subsection 29.2

Geometric approach

As alluded to in the previous subsection, we can interpret a complex number $z = a + b$ as two components in a 2-dimensional space (called the *complex plane*), in which the horizontal axis represents real components, and the vertical access represents imaginary components:

Drawing a line from z to a (on the real axis) creates a right triangle. We can then define θ to be the angle near the origin and r the length of the hypotenuse: We call r the *magnitude* of z, and θ its *argument*. The ranges for r and θ are, respectively, $[0, \infty)$ and $[0, 2\pi)$.

Using **??** the real and imaginary components of z are

$$\begin{aligned} a &= r\cos(\theta), \\ b &= r\sin(\theta), \end{aligned} \tag{29.9}$$

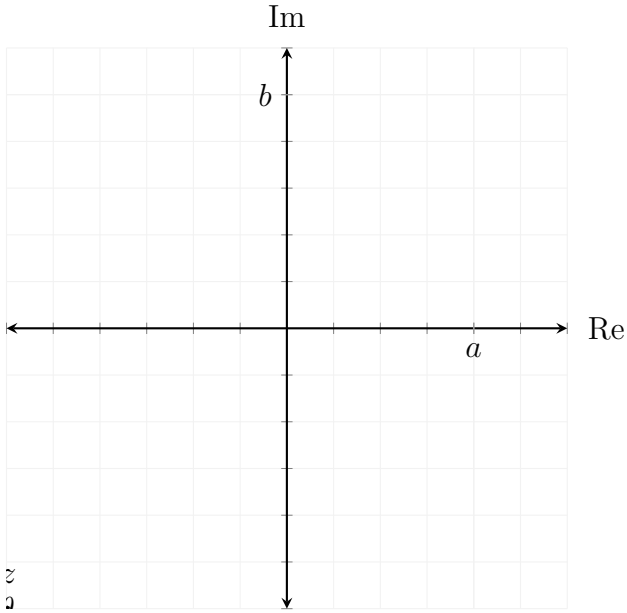


Figure 11. The same complex number z from Figure 10 shown with its polar components $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \arctan\left(\frac{b}{a}\right)$.

and z can be re-written as

$$z = r\left(\cos(\theta) + \sin(\theta)\right). \tag{29.10}$$

Which we call the *polar form* of z (contrasted with $z = a + b$ being the *Cartesian form* of z).

Inverting the relations in Equation 29.9 yields the relations

$$\begin{aligned} r &= a^2 + b^2, \\ \theta &= \arctan\left(\frac{b}{a}\right). \end{aligned} \tag{29.11}$$

Let's examine the same properties of complex numbers shown in Equations 29.3, 29.4 and 29.6, and verify that they work in the polar form of complex numbers. We start with addition (Equation 29.3):

$$\begin{aligned} z_1 + z_2 &= r_1\left[\cos\left(\theta_1\right) + \sin\left(\theta_1\right)\right] + r_2\left[\cos\left(\theta_2\right) + \sin\left(\theta_2\right)\right] \\ &= \underbrace{r_1\cos\left(\theta_1\right)}_{a_1} + \underbrace{r_2\cos\left(\theta_2\right)}_{a_2} + \underbrace{r_1\sin\left(\theta_1\right)}_{b_1} + \underbrace{r_2\sin\left(\theta_2\right)}_{b_2} \\ &= \left(a_1 + a_2\right) + \left(b_1 + b_2\right). \end{aligned} \tag{29.12}$$

We see that indeed, the polar form of complex numbers adheres to the addition rule in Equation 29.3. Next is the product rule:

$$\begin{aligned} z_1 z_2 &= r_1\left[\cos\left(\theta_1\right) + \sin\left(\theta_1\right)\right] \cdot r_2\left[\cos\left(\theta_2\right) + \sin\left(\theta_2\right)\right] \\ &= r_1 r_2\left[\cos\left(\theta_1\right)\cos\left(\theta_2\right) + \cos\left(\theta_1\right)\sin\left(\theta_2\right) + \sin\left(\theta_1\right)\cos\left(\theta_2\right) - \sin\left(\theta_1\right)\sin\left(\theta_2\right)\right] \\ &= r_1\cos\left(\theta_1\right) r_2\cos\left(\theta_2\right) - r_1\sin\left(\theta_1\right) r_2\sin\left(\theta_2\right) + \left[r_1\cos\left(\theta_1\right) r_2\sin\left(\theta_2\right) + r_1\sin\left(\theta_1\right) r_2\cos\left(\theta_2\right)\right] \\ &= \left(a_1 a_2 - b_1 b_2\right) + \left(a_1 b_2 + a_2 b_1\right), \end{aligned} \tag{29.13}$$

which is indeed the result seen in Equation 29.4. We can also develop further the second

row of Equation 29.13 using some trigonometry (specifically the trigonometric identities in **??**):

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \left[\cos(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2) + \sin(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)\right] \\ &= r_1 r_2 \left[\cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2) + i\left[\cos(\theta_1)\sin(\theta_2) + \sin(\theta_1)\cos(\theta_2)\right]\right] \\ &= r_1 r_2 \left[\cos(\theta_1+\theta_2) + \sin(\theta_1+\theta_2)\right]. \end{aligned} \tag{29.14}$$

This is a very important result: it shows that multiplying a complex number z_1 by another complex number z_2 gives a complex number with magnitude $r_1 r_2$, i.e. the product of the magnitudes of the two complex numbers, and argument $\theta_1 + \theta_2$, i.e. the argument of z_1 rotated by the argument of z_2 (or vice-versa). We will consider this result in more detail soon.

In the polar form the complex conjugate of a number $z = r\left[\cos(\theta) + \sin(\theta)\right]$ can be brought about by substituting $-\theta$ into the arguments of the trigonometric functions:

$$\begin{aligned} z &= r\left[\cos(-\theta) + \sin(-\theta)\right] \\ &= r\left[\cos(\theta) - \sin(\theta)\right] \\ &= r\cos(\theta) - r\sin(\theta) \\ &= a - b. \end{aligned} \tag{29.15}$$

Lastly, let's show that Equation 29.7 can be derived in the polar form:

$$\begin{aligned} zz &= r\left[\cos(\theta) + \sin(\theta)\right] \cdot r\left[\cos(\theta) - \sin(\theta)\right] \\ &= r^2\left[\cos^2(\theta) - \cos(\theta)\sin(\theta) + \sin(\theta)\cos(\theta) + \sin^2(\theta)\right] \\ &= r^2\left[\sin^2(\theta) + \cos^2(\theta)\right] \\ &= r^2 = a^2 + b^2. \end{aligned} \tag{29.16}$$

In 1748 Leonhard Euler published his famous work *Introduction to analysis of the infinite*[3]. In it he introduced the following relation, called *Euler's formula*:

$$^{x} = \sin(x) + \cos(x). \tag{29.17}$$

Using Euler's formula a complex number z can be written as

$$z = r^{\theta}. \tag{29.18}$$

In Table 2 we can see some useful complex exponentials ^{x}. Specifically, setting $x = \pi$ yields the famous *Eurler's identity*, considered by many to be one of the most beautiful equations in mathematics, as it binds together five important numbers, namely $0, 1, \pi, e$ and :

$$^{\pi} + 1 = 0. \tag{29.19}$$

Table 2 also shows us the integer behaviours of :

$$^{2} = -1,\ ^{3} = -,^{4} = 1,\ ^{5} =,\ ^{6} = -1,\ ^{7} = -,\ \ldots \tag{29.20}$$

Subsection 29.3

Roots of complex numbers

What is the n-th order roots of a complex number z, i.e. nz? An illuminating way to approach this problem is by looking at the polar form of z. As an example, we start with the number $z = 1$ and find its 3rd order roots, i.e. all number w such that $w^3 = 1$ (spoiler

[3] Latin for **Introduction to the Analysis of the Infinite**.

Table 2. Values of ^{ix} for some useful values of x (cf. **??** for the values of $\sin(\theta)$ and $\cos(\theta)$). lccl[cell-space-limits=5pt, code-before=1!15!10 !501]

[bold=true]x	$\cos(x)$	$\sin(x)$	z
$\frac{\pi}{2}$	0	1	
π	-1	0	-1
$\frac{3\pi}{2}$	0	-1	$-$
$\frac{\pi}{3}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}\left(1+\sqrt{3}\right)$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}\,(1+)$
$\frac{\pi}{6}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{1}{2}\left(\sqrt{3}+\right)$

alert: there are three such numbers).

Equation 29.14 taught us that complex numbers not only scale other numbers, but also rotate them: with real numbers the product $x \cdot y$ is equivalent to a scaling of x by y. With complex numbers the product has two components: its magnitude is the scale of x by y, and its argument is the argument of x rotated by the argument of y[4].

Example 124 Rotation using complex numbers Let $z = 2 + 2$ and $w = 3$. Their polar forms are $z = 2\sqrt{2}\,[\cos(\pi/4) + \sin(\pi/4)]$ and $w = 3\,[\cos(\pi/2) + \sin(\pi/2)]$. Their product is

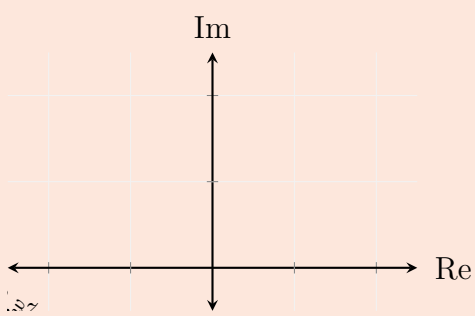
$$z \cdot w = (2+2) \cdot 3 = 6 + 6^2 = -6 + 6,$$

which in polar form is $6\sqrt{2}\,[\cos(3\pi/4) + \sin(3\pi/4)]$. Note that

$$2\sqrt{2} \cdot 3 = 6\sqrt{2}, \text{ and}$$
$$\frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4},$$

i.e. $z \cdot w$ has magnitude which is the **product** of the magnitudes of z and w, and an argument which is the **sum** of the arguments of z and w.

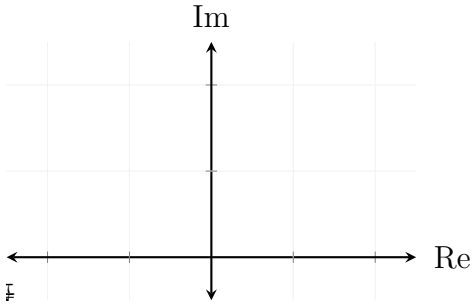
The figure below depict the arguments of the three numbers $z, w, z \cdot w$. Note that $\boldsymbol{\theta_{z \cdot w} = \theta_z + \theta_w}$.



In polar coordinates has magnitude 1 and argument $\frac{\pi}{2}$, i.e. multiplying by is equivalent to rotation by $\frac{\pi}{2}$ (90) counter clockwise. Therefore, multiplying by itself, i.e. ^{2}, rotates itself by $\frac{\pi}{2}$ counter clockwise, bringing it to -1.

[4]Due to the commutativity of the complex product we can switch the order of z and w and get the same result.

[5]$\boldsymbol{2 = -1}$ from a geometric (polar) viewpoint



In polar form $1 = \cos(0) + \sin(0)$. Finding the arguments of the cube roots of 1 is therefore done by answering the following question: what angles θ will equal 0 (or its equivalent angles $2\pi, 4\pi, 6\pi, \dots$) when multiplied by 3? The answer is very simple: the only possible solutions are

$$\begin{aligned} \theta_1 &= 0, \\ \theta_2 &= \frac{2\pi}{3}, \\ \theta_3 &= \frac{4\pi}{3}. \end{aligned} \tag{29.21}$$

(any other number in the range $[0, 2\pi)$ will give the same angles)

Therefore, the three cube roots of 1 are [6] (Figure 12)

$$\begin{aligned} z_1 &= \cos(\theta_1) + \sin(\theta_1) = \cos(0) + \sin(0) = 1, \\ z_2 &= \cos(\theta_2) + \sin(\theta_2) = \cos\left(\frac{2\pi}{3}\right) + \sin\left(\frac{2\pi}{3}\right) = -0.5 + \frac{\sqrt{3}}{2}, \\ z_3 &= \cos(\theta_3) + \sin(\theta_1) = \cos\left(\frac{4\pi}{3}\right) + \sin\left(\frac{4\pi}{3}\right) = -0.5 - \frac{\sqrt{3}}{2}. \end{aligned} \tag{29.22}$$

The n-th degree roots of 1 will follow the same pattern for $n \in \mathbb{N}$ (see Figure 17): their magnitude is always 1, and the argument of the k-th root is

$$\theta_k = \frac{2\pi}{n} k. \tag{29.23}$$

Finding the n-th degree roots of a general complex number $z = r\,[\cos(\theta) + \sin(\theta)]$ can be done in a similar fashion: all roots will have the magnitude nr, and their argument θ_k will be such that multiplying it by n gives $\theta + 2\pi m$ for some integer value m, i.e.

$$\theta_k = \frac{\theta + 2\pi m}{n}. \tag{29.24}$$

[6] recall that $31 = 1$, and therefore all roots have magnitude 1.

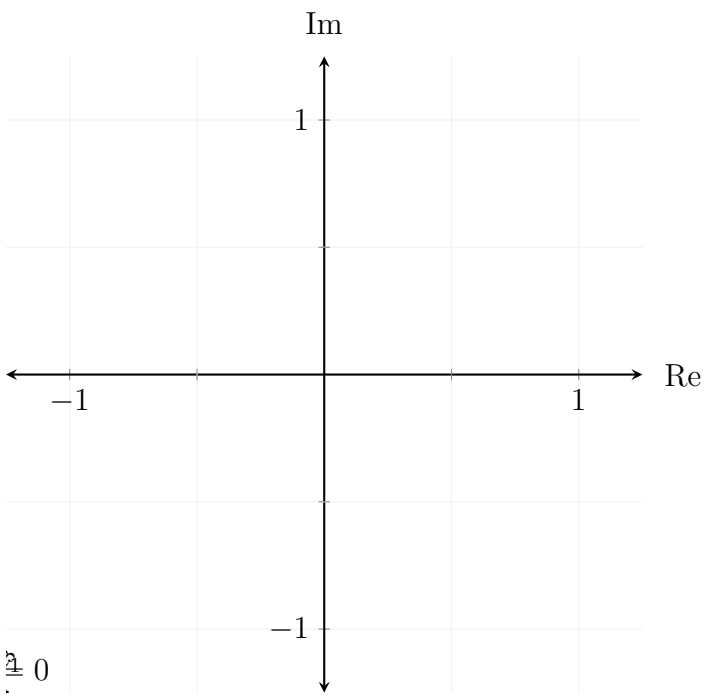
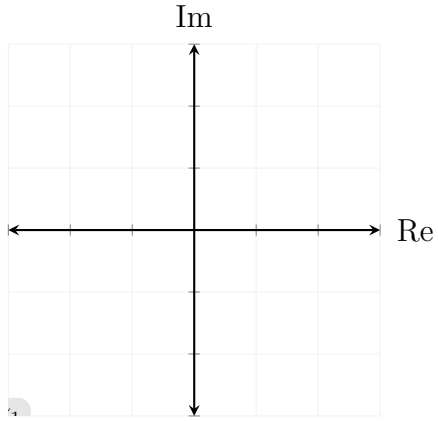
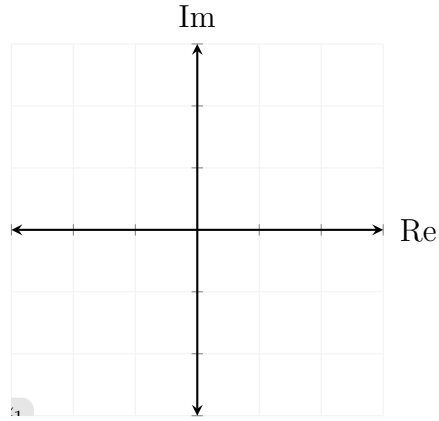


Figure 12. The three cube roots of $z = 1$.



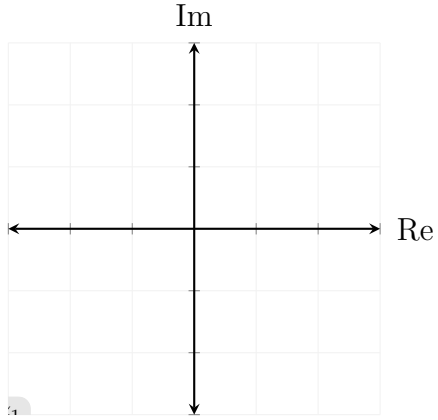
[b]0.475

Figure 13. $n = 4,\ \theta_k = \frac{\pi}{2}k$



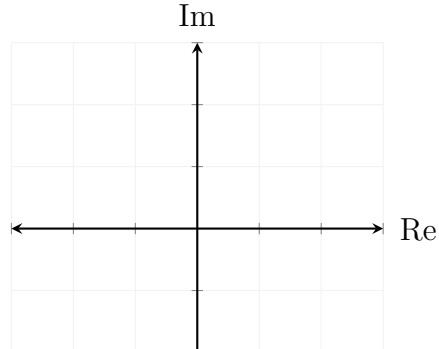
[b]0.475

Figure 14. $n = 5,\ \theta_k = \frac{2\pi}{5}k$



[b]0.475

Figure 15. $n = 6,\ \theta_k = \frac{\pi}{3}k$



Section 30

Introduction: The Imaginary Unit

We begin with the problem of solving the polynomial equation $x^2 + 1 = 0$. In the real number system $\mathbb{R}$, this equation has no solution. We introduce a new number, denoted by i, with the defining property:

The Imaginary Unit

$$i^2 = -1 \quad \text{or} \quad i = \sqrt{-1}$$

This allows us to write the solutions to $x^2+1 = 0$ as $x = \pm i$. We call i the **imaginary unit**.

Section 31

Rectangular Form and Basic Operations

A **complex number** z is a number of the form

Rectangular Form

$$z = a + bi$$

where a and b are real numbers.

- $a = \mathrm{Re}(z)$ is the **real part** of z.
- $b = \mathrm{Im}(z)$ is the **imaginary part** of z.

The set of all complex numbers is denoted by $\mathbb{C}$. Two complex numbers are equal if and only if their real parts and imaginary parts are equal.

Subsection 31.1

Operations

Let $z_1 = a + bi$ and $z_2 = c + di$.

1. **Addition/Subtraction:** We add/subtract the real and imaginary parts separately.

$$z_1 \pm z_2 = (a \pm c) + (b \pm d)i$$

2. **Multiplication:** We treat i like any variable, but replace i^2 with -1.

$$\begin{aligned} z_1 z_2 &= (a + bi)(c + di) \\ &= ac + adi + bci + bdi^2 \\ &= ac + (ad + bc)i + bd(-1) \end{aligned}$$

$$z_1 z_2 = (ac - bd) + (ad + bc)i$$

Basic Operations

Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$.

- **Addition:** $z_1 + z_2 = (3 + 1) + (2 - 4)i = 4 - 2i$
- **Subtraction:** $z_1 - z_2 = (3 - 1) + (2 - (-4))i = 2 + 6i$
- **Multiplication:**

$$\begin{aligned} z_1 z_2 &= (3 + 2i)(1 - 4i) \\ &= 3(1) - 3(4i) + 2i(1) - (2i)(4i) \\ &= 3 - 12i + 2i - 8i^2 \\ &= 3 - 10i - 8(-1) \\ &= (3 + 8) - 10i = 11 - 10i \end{aligned}$$

Section 32

The Complex Conjugate and Division

Subsection 32.1

Complex Conjugate

The **complex conjugate** of $z = a + bi$ is denoted by $\bar{z}$ and defined as:

Complex Conjugate

$$\bar{z} = a - bi$$

A key property is the product of a complex number and its conjugate:

$$z\bar{z} = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - b^2 i^2 = a^2 + b^2$$

This result is always a non-negative real number.

Subsection 32.2

Modulus

The **modulus** (or absolute value) of $z = a + bi$ is denoted by $|z|$ and is the non-negative real number:

Modulus

$$|z| = \sqrt{a^2 + b^2}$$

Notice that $|z|^2 = z\bar{z}$.

Subsection 32.3

Division

To divide z_1 by z_2 (where $z_2 \neq 0$), we multiply the numerator and denominator by the conjugate of the denominator. This process is called "realizing the denominator."

Division

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z_2}}{\bar{z_2}} = \frac{z_1 \bar{z_2}}{|z_2|^2}$$

Writing this out fully:

$$\frac{z_1}{z_2} = \frac{a+bi}{c+di} = \frac{(a+bi)}{(c+di)} \cdot \frac{(c-di)}{(c-di)} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2}$$

Conjugate Modulus and Division

Let $z_1 = 2+5i$ and $z_2 = 3-i$.

- **Conjugates:** $\bar{z_1} = 2-5i$ and $\bar{z_2} = 3+i$.
- **Moduli:** $|z_1| = \sqrt{2^2+5^2} = \sqrt{4+25} = \sqrt{29}$. $|z_2| = \sqrt{3^2+(-1)^2} = \sqrt{9+1} = \sqrt{10}$.
- **Division:**

$$\begin{aligned}
\frac{z_1}{z_2} &= \frac{2+5i}{3-i} = \frac{(2+5i)}{(3-i)} \cdot \frac{(3+i)}{(3+i)} \\
&= \frac{2(3)+2(i)+5i(3)+5i(i)}{3^2+(-1)^2} \\
&= \frac{6+2i+15i+5i^2}{9+1} \\
&= \frac{6+17i-5}{10} = \frac{1+17i}{10} = \frac{1}{10} + i\frac{17}{10}
\end{aligned}$$

Section 33

The Complex Plane (Argand Diagram)

We can visualize complex numbers as points (or vectors) in a 2D plane, called the **complex plane** or **Argand diagram**.

- The horizontal axis is the **real axis** (Re).
- The vertical axis is the **imaginary axis** (Im).

The number $z = a+bi$ corresponds to the point (a,b). The modulus $|z|$ is the distance from the origin $(0,0)$ to the point (a,b). The conjugate $\bar{z}$ is the reflection of z across the real axis. Addition $z_1 + z_2$ follows the parallelogram law for vector addition.

Section 34

Polar and Exponential Forms

Subsection 34.1

Polar Form

Instead of using rectangular coordinates (a,b), we can use polar coordinates (r,θ).

- $r = |z| = \sqrt{a^2 + b^2}$ (the modulus).
- $\theta = \arg(z)$ is the **argument** of z, the angle measured counter-clockwise from the positive real axis.

From trigonometry, we have $a = r\cos\theta$ and $b = r\sin\theta$. Substituting this into the rectangular form gives the **polar form**:

Polar Form

$$z = r\cos\theta + i(r\sin\theta) = r(\cos\theta + i\sin\theta)$$

The argument θ is not unique; we can add any integer multiple of 2π. We define the **Principal Argument**, $\mathrm{Arg}(z)$, as the unique angle θ such that $-\pi < \theta \leq \pi$. Then $\arg(z) = \mathrm{Arg}(z) + 2k\pi$ for $k \in \mathbb{Z}$.

Subsection 34.2

Euler's Formula and Exponential Form

One of the most important formulas in mathematics is **Euler's Formula**:

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

This connects the exponential function to trigonometry. A famous special case is $e^{i\pi} = \cos\pi + i\sin\pi = -1 + 0$, which gives the beautiful identity $e^{i\pi} + 1 = 0$.

Using this, we can write the polar form in a much more compact way, called the **exponential form**:

Exponential Form

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$$

Rectangular to Exponential Form

Find the exponential form of $z = -1 - i$.

1. **Find the modulus r:**

$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2}$$

2. **Find the argument θ:** The point $(-1, -1)$ is in the third quadrant.

$$\tan\theta = \frac{b}{a} = \frac{-1}{-1} = 1$$

The angle in the third quadrant whose tangent is 1 is $\theta = 5\pi/4$. However, the *Principal Argument* $\text{Arg}(z)$ must be in $(-\pi, \pi]$. The angle is $\theta = -3\pi/4$. Both are valid arguments. Let's use the principal argument.

So, $z = \sqrt{2}e^{i(-3\pi/4)}$ or $z = \sqrt{2}e^{-i3\pi/4}$. (Using $\theta = 5\pi/4$ would give $z = \sqrt{2}e^{i5\pi/4}$, which is the same number.)

Subsection 34.3

Multiplication and Division in Exponential Form

This form is extremely useful for multiplication, division, powers, and roots. Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$.

Multiplication and Division (Exponential Form)

Multiplication: $z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)}$

- To multiply, **multiply the moduli** and **add the arguments**.

Division: $\dfrac{z_1}{z_2} = \dfrac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \dfrac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$

- To divide, **divide the moduli** and **subtract the arguments**.

Example

Let $z_1 = 2e^{i\pi/3}$ and $z_2 = 5e^{i\pi/6}$.

- $z_1 z_2 = (2)(5)e^{i(\pi/3+\pi/6)} = 10e^{i(2\pi/6+\pi/6)} = 10e^{i(3\pi/6)} = 10e^{i\pi/2}$ (This is $10(0 + i(1)) = 10i$)
- $\dfrac{z_1}{z_2} = \dfrac{2}{5}e^{i(\pi/3-\pi/6)} = \dfrac{2}{5}e^{i(\pi/6)}$

Section 35

Powers: De Moivre's Theorem

Let's find z^n for an integer n. Using the exponential form $z = re^{i\theta}$:

$$z^n = (re^{i\theta})^n = r^n(e^{i\theta})^n = r^n e^{i(n\theta)}$$

If we write this out in polar form (with $r = 1$ for simplicity), we get **De Moivre's Theorem**:

De Moivre's Theorem

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

This theorem is incredibly useful for deriving trigonometric identities.

Calculating Powers

Find $(1+i)^8$. First, convert $z = 1+i$ to exponential form.

- $r = |z| = \sqrt{1^2+1^2} = \sqrt{2}$.
- $\theta = \text{Arg}(z) = \arctan(1/1) = \frac{\pi}{4}$ (since (1,1) is in Q1).

So, $z = \sqrt{2}e^{i\pi/4}$. Now we take the 8th power:

$z^8 = (\sqrt{2}e^{i\pi/4})^8 = (\sqrt{2})^8 e^{i(8\pi/4)} = 2^{8/2}e^{i(2\pi)} = 2^4 e^{i2\pi} = 16(\cos(2\pi)+i\sin(2\pi)) = 16(1+0) = 16$

Calculating $(1+i)(1+i)(1+i)$... 8 times would be much harder!

Section 36

Roots of Complex Numbers

We want to find the n-th roots of a complex number z. That is, we seek all w such that $w^n = z$. Let $z = re^{i\theta}$ and $w = \rho e^{i\phi}$. We must account for the multi-valued argument of z:

$$z = re^{i(\theta+2k\pi)}, \quad k \in \mathbb{Z}$$

Now, set $w^n = z$:

$$(\rho e^{i\phi})^n = re^{i(\theta+2k\pi)} \implies \rho^n e^{in\phi} = re^{i(\theta+2k\pi)}$$

Equating the moduli and arguments:

1. $\rho^n = r \implies \rho = \sqrt[n]{r}$ (the unique positive real n-th root)
2. $n\phi = \theta + 2k\pi \implies \phi = \frac{\theta+2k\pi}{n}$

So the n-th roots of z are given by:

Formula for n-th Roots

$$w_k = \sqrt[n]{r}\exp\left(i\left(\frac{\theta+2k\pi}{n}\right)\right), \quad \text{for } k = 0, 1, 2, \ldots, n-1$$

We only need n consecutive values of k (e.g., $k = 0$ to $n-1$) to find all n distinct roots. For $k = n$, the angle $\frac{\theta+2n\pi}{n} = \frac{\theta}{n} + 2\pi$, which gives the same root as $k = 0$. Geometrically, the n roots of z are equally spaced on a circle of radius $\sqrt[n]{r}$.

Cube Roots of Unity

Find the cube roots of $z = 1$. Here $r = 1$ and $\theta = \text{Arg}(z) = 0$. So $z = 1e^{i(0+2k\pi)}$. The formula is $w_k = \sqrt[3]{1}\exp\left(i\left(\frac{0+2k\pi}{3}\right)\right) = e^{i(2k\pi/3)}$ for $k = 0, 1, 2$.

- $k = 0 : w_0 = e^{i(0)} = 1$
- $k = 1 : w_1 = e^{i(2\pi/3)} = \cos(2\pi/3) + i\sin(2\pi/3) = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$
- $k = 2 : w_2 = e^{i(4\pi/3)} = \cos(4\pi/3) + i\sin(4\pi/3) = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

These three roots form an equilateral triangle on the unit circle.

Square Roots of i

Find the square roots of $z = i$. Here $n = 2$. First, write $z = i$ in exponential form. $r = |i| = 1$. $\theta = \text{Arg}(i) = \pi/2$. So $z = 1e^{i(\pi/2+2k\pi)}$. The roots are $w_k = \sqrt{1}\exp\left(i\left(\frac{\pi/2+2k\pi}{2}\right)\right) = \exp\left(i\left(\frac{\pi}{4} + k\pi\right)\right)$ for $k = 0, 1$.

- $k = 0 : w_0 = e^{i(\pi/4)} = \cos(\pi/4) + i\sin(\pi/4) = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$
- $k = 1 : w_1 = e^{i(\pi/4+\pi)} = e^{i(5\pi/4)} = \cos(5\pi/4) + i\sin(5\pi/4) = -\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}$

Note that $w_1 = -w_0$, as expected for a square root.

Section 37

The Exponential Function $f(z) = e^z$

We now define the exponential function for a complex variable $z = x + iy$.

Complex Exponential Function

Definition:

$$e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos y + i\sin y)$$

Subsection 37.1

Properties

1. $e^{z_1}e^{z_2} = e^{z_1+z_2}$
2. $\frac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$
3. $|e^z| = |e^x(\cos y + i\sin y)| = |e^x||\cos y + i\sin y| = e^x \cdot 1 = e^x$
4. $\arg(e^z) = y + 2k\pi$
5. $e^z \neq 0$ for all $z \in \mathbb{C}$.

6. **Periodicity:** The exponential function is periodic with a pure imaginary period of $2\pi i$.

$$e^{z+2\pi i} = e^{x+i(y+2\pi)} = e^x(\cos(y+2\pi) + i\sin(y+2\pi)) = e^x(\cos y + i\sin y) = e^z$$

Evaluating e^z

Find the value of $e^{2+i\pi/2}$. Here $x = 2$ and $y = \pi/2$.

$$e^{2+i\pi/2} = e^2 e^{i\pi/2} = e^2\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right) = e^2(0 + i(1)) = ie^2$$

Section 38

The Logarithmic Function

The complex logarithm $w = \log z$ is defined as the inverse of the exponential function, $e^w = z$. Let $w = u + iv$ and $z = re^{i\theta}$.

$$e^w = z \implies e^{u+iv} = re^{i\theta} \implies e^u e^{iv} = re^{i\theta}$$

Equating moduli and arguments:

- $e^u = r \implies u = \ln r = \ln|z|$ (where ln is the real natural log)
- $v = \theta = \arg(z)$

So, we define the **complex logarithm** as:

Complex Logarithm (Multi-valued)

$$\log z = \ln|z| + i\arg(z)$$

Subsection 38.1

Multi-valued Nature

Since $\arg(z) = \mathrm{Arg}(z) + 2k\pi$ for $k \in \mathbb{Z}$, the logarithm is a **multi-valued function**:

$$\log z = \ln|z| + i(\mathrm{Arg}(z) + 2k\pi)$$

Subsection 38.2

Principal Value

We define the **principal value of the logarithm**, $\log z$, by using the principal argument $\mathrm{Arg}(z)$:

Principal Value $\log z$

$$\log z = \ln|z| + i\,\mathrm{Arg}(z), \quad (-\pi < \mathrm{Arg}(z) \leq \pi)$$

This makes the function single-valued, but it has a "jump" (a discontinuity) along the negative real axis, known as a **branch cut**.

Evaluating $\log z$

Find $\log(-1)$ and the principal value $\log(-1)$.

- $z = -1$.
- Modulus: $|z| = |-1| = 1$. So $\ln|z| = \ln(1) = 0$.
- Argument: $\arg(-1) = \pi + 2k\pi$ (it's on the negative real axis).
- Principal Argument: $\operatorname{Arg}(-1) = \pi$.

Therefore:

- $\log(-1) = \ln(1) + i(\pi + 2k\pi) = i\pi(1 + 2k)$ for $k \in \mathbb{Z}$. This gives $\ldots, -3i\pi, -i\pi, i\pi, 3i\pi, 5i\pi, \ldots$
- $\log(-1) = \ln(1) + i\operatorname{Arg}(-1) = 0 + i\pi = i\pi$.

Section 39

Trigonometric Functions

We can extend the definitions of sin and cos to the complex plane using Euler's formula. Recall:

$$\begin{aligned} e^{ix} &= \cos x + i\sin x \quad \text{(for real } x\text{)} \\ e^{-ix} &= \cos x - i\sin x \end{aligned}$$

Adding these gives $\cos x = \frac{e^{ix}+e^{-ix}}{2}$. Subtracting gives $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$. We use these as our **definitions** for complex z:

Complex Trig Functions

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Note: Unlike their real counterparts, $\sin z$ and $\cos z$ are **unbounded** in the complex plane.

Evaluating $\cos z$

Find the value of $\cos(i)$. Using the definition with $z = i$:

$$\begin{aligned} \cos(i) &= \frac{e^{i(i)} + e^{-i(i)}}{2} = \frac{e^{i^2} + e^{-i^2}}{2} \\ &= \frac{e^{-1} + e^{1}}{2} = \frac{e + e^{-1}}{2} \\ &= \cosh(1) \end{aligned}$$

This is a real number, approximately 1.543.

Section 40

Hyperbolic Functions

The real hyperbolic functions are $\cosh x = \frac{e^x+e^{-x}}{2}$ and $\sinh x = \frac{e^x-e^{-x}}{2}$. We extend this to the complex plane:

Complex Hyperbolic Functions

$$\cosh z = \frac{e^z + e^{-z}}{2} \qquad \sinh z = \frac{e^z - e^{-z}}{2}$$

Subsection 40.1

Relationship to Trigonometric Functions

We can relate the two sets of functions by replacing z with iz:

$$\cos(iz) = \frac{e^{i(iz)} + e^{-i(iz)}}{2} = \frac{e^{-z} + e^z}{2} = \cosh z$$
$$\sin(iz) = \frac{e^{i(iz)} - e^{-i(iz)}}{2i} = \frac{e^{-z} - e^z}{2i} = \frac{-(e^z - e^{-z})}{2i} = i\frac{e^z - e^{-z}}{2} = i\sinh z$$

Osborn's Rules

$$\cos(iz) = \cosh(z) \qquad \sin(iz) = i\sinh(z)$$
$$\cosh(iz) = \cos(z) \qquad \sinh(iz) = i\sin(z)$$

Section 41

Functions of a Complex Variable

In general, a **function of a complex variable** z is a map f from a subset S of $\mathbb{C}$ to $\mathbb{C}$. We write $w = f(z)$. We can split both z and w into their real and imaginary parts:

- $z = x + iy$
- $w = u + iv$

Since w depends on z, the real part u and imaginary part v of the output must depend on the real part x and imaginary part y of the input.

$$w = f(z) \implies w(x, y) = u(x, y) + iv(x, y)$$

Thus, a complex function $f : \mathbb{C} \to \mathbb{C}$ is equivalent to two real-valued functions of two real variables: $u(x, y)$ and $v(x, y)$.

Finding u and v

Let $f(z) = z^2$. Find $u(x, y)$ and $v(x, y)$.

$$\begin{aligned} f(z) &= (x + iy)^2 \\ &= x^2 + 2(x)(iy) + (iy)^2 \\ &= x^2 + 2xyi - y^2 \\ &= (x^2 - y^2) + i(2xy) \end{aligned}$$

So, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

Finding u and v for

Let $f(z) = 1/z$ for $z \neq 0$. Find $u(x, y)$ and $v(x, y)$. We "realize the denominator" just as in division:

$$\begin{aligned} f(z) &= \frac{1}{x + iy} = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} \\ &= \frac{x - iy}{x^2 + y^2} \\ &= \left(\frac{x}{x^2 + y^2}\right) + i\left(\frac{-y}{x^2 + y^2}\right) \end{aligned}$$

So, $u(x, y) = \frac{x}{x^2+y^2}$ and $v(x, y) = \frac{-y}{x^2+y^2}$.

Section 42

Sets in the Complex Plane

To study limits, continuity, and derivatives (i.e., complex analysis), we need a basic understanding of the topology of $\mathbb{C}$.

- **ϵ-neighborhood:** An **open disk** centered at z_0 with radius $\epsilon > 0$ is the set $B(z_0, \epsilon) = \{z \in \mathbb{C} : |z - z_0| < \epsilon\}$.
- **Interior Point:** A point z_0 is an **interior point** of a set S if there exists some ϵ-neighborhood of z_0 that is completely contained in S.
- **Open Set:** A set S is **open** if every point in S is an interior point. (e.g., $|z| < 1$).
- **Boundary Point:** A point z_0 is a **boundary point** of S if every ϵ-neighborhood of z_0 contains at least one point in S and at least one point not in S.
- **Boundary:** The set of all boundary points is the **boundary** of S. (e.g., for $|z| < 1$, the boundary is the circle $|z| = 1$).
- **Closed Set:** A set S is **closed** if it contains all of its boundary points. (e.g., $|z| \leq 1$).
- **Connected Set:** An open set S is **connected** if any two points in S can be joined by a path (a continuous line) that lies entirely within S.
- **Domain:** In complex analysis, a **domain** is an **open, connected set**. This is the most common type of set we will work with when defining functions.

Example 4: $f(z)$

Let
$$f(z) = u(x, y) + iv(x, y) = x^2y + i(xy^2).$$
Compute the Cauchy–Riemann equations and determine whether f is analytic.

We have
$$u(x, y) = x^2y, \qquad v(x, y) = xy^2.$$
Compute partial derivatives:
$$u_x = 2xy, \qquad u_y = x^2, \qquad v_x = y^2, \qquad v_y = 2xy.$$
CR equations require:
$$u_x = v_y \quad \Rightarrow \quad 2xy = 2xy \quad \text{(satisfied)},$$
$$u_y = -v_x \quad \Rightarrow \quad x^2 = -y^2.$$
The equation $x^2 = -y^2$ holds only when $x = y = 0$.

Thus the CR equations hold only at the point $(0, 0)$, not on a neighborhood. Therefore, the function is **not analytic anywhere**.

Example 5: $f(z)$

Consider
$$f(z) = e^x(\cos y + i \sin y).$$
Verify the CR equations and determine analyticity.

We identify
$$u(x, y) = e^x \cos y, \qquad v(x, y) = e^x \sin y.$$
Then
$$u_x = e^x \cos y, \qquad u_y = -e^x \sin y,$$
$$v_x = e^x \sin y, \qquad v_y = e^x \cos y.$$
Check CR:
$$u_x = v_y \quad \Rightarrow \quad e^x \cos y = e^x \cos y,$$
$$u_y = -v_x \quad \Rightarrow \quad -e^x \sin y = -e^x \sin y.$$
Both are satisfied everywhere. Therefore, the function is **analytic on all of $\mathbb{C}$**. Indeed, this is the standard decomposition of the analytic function
$$f(z) = e^z.$$

Example 6: $f(z)$

Verify analytic behavior of the function

$$f(z) = x^3 - 3xy^2 + i(3x^2y - y^3).$$

We recognize this as the real/imaginary form of z^3, but we verify directly.

$$u(x, y) = x^3 - 3xy^2, \qquad v(x, y) = 3x^2y - y^3.$$

Compute partial derivatives:

$$u_x = 3x^2 - 3y^2, \qquad u_y = -6xy,$$

$$v_x = 6xy, \qquad v_y = 3x^2 - 3y^2.$$

Check CR:

$$u_x = v_y \quad \Rightarrow \quad 3x^2 - 3y^2 = 3x^2 - 3y^2,$$

$$u_y = -v_x \quad \Rightarrow \quad -6xy = -6xy.$$

CR holds everywhere, so the function is **entire**, consistent with

$$f(z) = z^3.$$

Example 7: $f(z)$

Consider

$$f(z) = u(x, y) + iv(x, y) = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2},$$

defined on $\mathbb{C} \setminus \{0\}$. Determine analyticity.

This is the real/imaginary form of

$$f(z) = \frac{1}{z}.$$

Compute derivatives:

$$u = \frac{x}{x^2 + y^2}, \quad v = -\frac{y}{x^2 + y^2}.$$

$$u_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \qquad u_y = \frac{-2xy}{(x^2 + y^2)^2},$$

$$v_x = \frac{2xy}{(x^2 + y^2)^2}, \qquad v_y = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

Check CR:

$$u_x = v_y, \qquad u_y = -v_x.$$

Both equations hold for all $(x, y) \neq (0, 0)$.
Thus the function is **analytic on its domain**

$$\mathbb{C} \setminus \{0\}.$$

Box for important formulas and definitions

Section 44

Introduction: The Imaginary Unit

We begin with the problem of solving the polynomial equation $x^2 + 1 = 0$. In the real number system $\mathbb{R}$, this equation has no solution. We introduce a new number, denoted by i, with the defining property:

The Imaginary Unit

$$i^2 = -1 \quad \text{or} \quad i = \sqrt{-1}$$

This allows us to write the solutions to $x^2+1 = 0$ as $x = \pm i$. We call i the **imaginary unit**.

Historical Context

The term "imaginary" was coined by René Descartes in the 17th century, originally as a derogatory term. However, complex numbers are just as "real" as real numbers in mathematics, and they have profound applications in physics, engineering, and many other fields.

Subsection 44.1

Powers of i

Understanding the powers of i is fundamental:

Powers of i

$$\begin{aligned} i^0 &= 1 \\ i^1 &= i \\ i^2 &= -1 \\ i^3 &= i^2 \cdot i = -i \\ i^4 &= (i^2)^2 = (-1)^2 = 1 \end{aligned}$$

The pattern repeats with period 4: $i^n = i^{n \bmod 4}$

Computing Powers of i

Compute i^{37} and i^{-5}.
Solution:

- For i^{37}: Since $37 = 4 \cdot 9 + 1$, we have $i^{37} = i^1 = i$.
- For i^{-5}: First, $i^{-5} = \frac{1}{i^5} = \frac{1}{i^{4+1}} = \frac{1}{i}$.
  To simplify, multiply by $\frac{i}{i}$:
  $$\frac{1}{i} \cdot \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i$$
  Alternatively, $-5 \equiv 3 \pmod 4$, so $i^{-5} = i^3 = -i$.

Section 45

Rectangular Form and Basic Operations

A **complex number** z is a number of the form

Rectangular Form

$$z = a + bi$$

where a and b are real numbers.

- $a = \mathrm{Re}(z)$ is the **real part** of z.
- $b = \mathrm{Im}(z)$ is the **imaginary part** of z.

The set of all complex numbers is denoted by $\mathbb{C}$. Two complex numbers are equal if and only if their real parts and imaginary parts are equal.

Important Convention

The imaginary part b is a real number. We write $\mathrm{Im}(3 + 4i) = 4$, not $4i$.

Subsection 45.1

Operations

Let $z_1 = a + bi$ and $z_2 = c + di$.

1. **Addition/Subtraction:** We add/subtract the real and imaginary parts separately.

$$z_1 \pm z_2 = (a \pm c) + (b \pm d)i$$

2. **Multiplication:** We treat i like any variable, but replace i^2 with -1.

$$\begin{aligned} z_1 z_2 &= (a+bi)(c+di) \\ &= ac + adi + bci + bdi^2 \\ &= ac + (ad+bc)i + bd(-1) \end{aligned}$$

$$z_1 z_2 = (ac - bd) + (ad + bc)i$$

Basic Operations

Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$.

- **Addition:** $z_1 + z_2 = (3+1) + (2-4)i = 4 - 2i$
- **Subtraction:** $z_1 - z_2 = (3-1) + (2-(-4))i = 2 + 6i$
- **Multiplication:**

$$\begin{aligned} z_1 z_2 &= (3+2i)(1-4i) \\ &= 3(1) - 3(4i) + 2i(1) - (2i)(4i) \\ &= 3 - 12i + 2i - 8i^2 \\ &= 3 - 10i - 8(-1) \\ &= (3+8) - 10i = 11 - 10i \end{aligned}$$

Solving Equations with Complex Numbers

Solve for z in the equation $(2+i)z = 5 + 3i$.
Solution: We need to divide both sides by $(2+i)$. We'll use the method of realizing the denominator:

$$\begin{aligned} z &= \frac{5+3i}{2+i} = \frac{(5+3i)(2-i)}{(2+i)(2-i)} \\ &= \frac{10 - 5i + 6i - 3i^2}{4 - i^2} \\ &= \frac{10 + i + 3}{4+1} = \frac{13+i}{5} = \frac{13}{5} + \frac{1}{5}i \end{aligned}$$

Section 46

The Complex Conjugate and Division

Subsection 46.1

Complex Conjugate

The **complex conjugate** of $z = a + bi$ is denoted by $\bar{z}$ and defined as:

Complex Conjugate

$$\bar{z} = a - bi$$

A key property is the product of a complex number and its conjugate:

$$z\bar{z} = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - b^2 i^2 = a^2 + b^2$$

This result is always a non-negative real number.

Subsection 46.2

Properties of the Conjugate

Conjugate Properties

For complex numbers $z, w \in \mathbb{C}$:

1. $\overline{z + w} = \bar{z} + \bar{w}$
2. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$
3. $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$ (when $w \neq 0$)
4. $\bar{\bar{z}} = z$
5. $z + \bar{z} = 2\,\mathrm{Re}(z)$
6. $z - \bar{z} = 2i\,\mathrm{Im}(z)$
7. z is real if and only if $z = \bar{z}$

Subsection 46.3

Modulus

The **modulus** (or absolute value) of $z = a + bi$ is denoted by $|z|$ and is the non-negative real number:

Modulus

$$|z| = \sqrt{a^2 + b^2}$$

Notice that $|z|^2 = z\bar{z}$.

Subsection 46.4

Properties of the Modulus

Modulus Properties

For complex numbers $z, w \in \mathbb{C}$:

1. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$
2. $|\bar{z}| = |z|$
3. $|zw| = |z||w|$
4. $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ (when $w \neq 0$)
5. **Triangle Inequality:** $|z + w| \leq |z| + |w|$
6. **Reverse Triangle Inequality:** $||z| - |w|| \leq |z - w|$

Subsection 46.5

Division

To divide z_1 by z_2 (where $z_2 \neq 0$), we multiply the numerator and denominator by the conjugate of the denominator. This process is called "realizing the denominator."

Division

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z_2}}{\bar{z_2}} = \frac{z_1 \bar{z_2}}{|z_2|^2}$$

Writing this out fully:

$$\frac{z_1}{z_2} = \frac{a+bi}{c+di} = \frac{(a+bi)}{(c+di)} \cdot \frac{(c-di)}{(c-di)} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2}$$

Conjugate

Let $z_1 = 2 + 5i$ and $z_2 = 3 - i$.

- **Conjugates:** $\bar{z_1} = 2 - 5i$ and $\bar{z_2} = 3 + i$.
- **Moduli:** $|z_1| = \sqrt{2^2 + 5^2} = \sqrt{4 + 25} = \sqrt{29}$. $|z_2| = \sqrt{3^2 + (-1)^2} = \sqrt{9 + 1} = \sqrt{10}$.
- **Division:**

$$\begin{aligned}
\frac{z_1}{z_2} &= \frac{2+5i}{3-i} = \frac{(2+5i)}{(3-i)} \cdot \frac{(3+i)}{(3+i)} \\
&= \frac{2(3) + 2(i) + 5i(3) + 5i(i)}{3^2 + (-1)^2} \\
&= \frac{6 + 2i + 15i + 5i^2}{9+1} \\
&= \frac{6 + 17i - 5}{10} = \frac{1 + 17i}{10} = \frac{1}{10} + i\frac{17}{10}
\end{aligned}$$

Triangle Inequality Application

Verify the triangle inequality for $z = 3 + 4i$ and $w = -1 + 2i$.
Solution:

$$\begin{aligned} |z| &= \sqrt{3^2 + 4^2} = 5 \\ |w| &= \sqrt{(-1)^2 + 2^2} = \sqrt{5} \approx 2.236 \\ z + w &= (3 - 1) + (4 + 2)i = 2 + 6i \\ |z + w| &= \sqrt{2^2 + 6^2} = \sqrt{40} = 2\sqrt{10} \approx 6.325 \end{aligned}$$

Now check: $|z + w| \leq |z| + |w|$ becomes $6.325 \leq 5 + 2.236 = 7.236$

Section 47

The Complex Plane (Argand Diagram)

We can visualize complex numbers as points (or vectors) in a 2D plane, called the **complex plane** or **Argand diagram**.

- The horizontal axis is the **real axis** (Re).
- The vertical axis is the **imaginary axis** (Im).

The number $z = a + bi$ corresponds to the point (a, b). The modulus $|z|$ is the distance from the origin $(0, 0)$ to the point (a, b). The conjugate $\bar{z}$ is the reflection of z across the real axis. Addition $z_1 + z_2$ follows the parallelogram law for vector addition.

Geometric Interpretation

- $|z_1 - z_2|$ represents the distance between points z_1 and z_2
- $|z - z_0| = r$ represents a circle centered at z_0 with radius r
- $|z - z_0| < r$ represents the interior of such a circle (an open disk)

Section 48

Polar and Exponential Forms

Subsection 48.1

Polar Form

Instead of using rectangular coordinates (a, b), we can use polar coordinates (r, θ).

- $r = |z| = \sqrt{a^2 + b^2}$ (the modulus).
- $\theta = \arg(z)$ is the **argument** of z, the angle measured counter-clockwise from the positive real axis.

From trigonometry, we have $a = r\cos\theta$ and $b = r\sin\theta$. Substituting this into the rectangular form gives the **polar form**:

Polar Form

$$z = r\cos\theta + i(r\sin\theta) = r(\cos\theta + i\sin\theta)$$

The argument θ is not unique; we can add any integer multiple of 2π. We define the **Principal Argument**, $\mathrm{Arg}(z)$, as the unique angle θ such that $-\pi < \theta \leq \pi$. Then $\arg(z) = \mathrm{Arg}(z) + 2k\pi$ for $k \in \mathbb{Z}$.

Finding the Argument

For $z = a + bi$ with $z \neq 0$:

$$\theta = \begin{cases} \arctan\left(\frac{b}{a}\right) & \text{if } a > 0 \\ \arctan\left(\frac{b}{a}\right) + \pi & \text{if } a < 0, b \geq 0 \\ \arctan\left(\frac{b}{a}\right) - \pi & \text{if } a < 0, b < 0 \\ \frac{\pi}{2} & \text{if } a = 0, b > 0 \\ -\frac{\pi}{2} & \text{if } a = 0, b < 0 \end{cases}$$

Subsection 48.2

Euler's Formula and Exponential Form

One of the most important formulas in mathematics is **Euler's Formula**:

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

This connects the exponential function to trigonometry. A famous special case is $e^{i\pi} = \cos\pi + i\sin\pi = -1 + 0$, which gives the beautiful identity $e^{i\pi} + 1 = 0$.

Using this, we can write the polar form in a much more compact way, called the **exponential form**:

Exponential Form

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$$

Rectangular to Exponential Form

Find the exponential form of $z = -1 - i$.

1. **Find the modulus r:**

$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2}$$

2. **Find the argument θ:** The point $(-1, -1)$ is in the third quadrant.

$$\tan\theta = \frac{b}{a} = \frac{-1}{-1} = 1$$

Since $a < 0$ and $b < 0$, we have:

$$\theta = \arctan(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

So, $z = \sqrt{2}e^{-i3\pi/4}$.

More Conversions

Convert the following to exponential form:

1. $z_1 = -3$
2. $z_2 = 2i$
3. $z_3 = 1 + \sqrt{3}i$

Solutions:

1. $r = |-3| = 3$, $\theta = \pi$ (negative real axis), so $z_1 = 3e^{i\pi}$
2. $r = |2i| = 2$, $\theta = \pi/2$ (positive imaginary axis), so $z_2 = 2e^{i\pi/2}$
3. $r = \sqrt{1^2 + (\sqrt{3})^2} = 2$, $\theta = \arctan\left(\sqrt{3}/1\right) = \pi/3$, so $z_3 = 2e^{i\pi/3}$

Subsection 48.3

Multiplication and Division in Exponential Form

This form is extremely useful for multiplication, division, powers, and roots. Let $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$.

Multiplication and Division (Exponential Form)

Multiplication: $z_1z_2 = (r_1e^{i\theta_1})(r_2e^{i\theta_2}) = r_1r_2e^{i(\theta_1+\theta_2)}$

- To multiply, **multiply the moduli** and **add the arguments**.

Division: $\dfrac{z_1}{z_2} = \dfrac{r_1e^{i\theta_1}}{r_2e^{i\theta_2}} = \dfrac{r_1}{r_2}e^{i(\theta_1-\theta_2)}$

- To divide, **divide the moduli** and **subtract the arguments**.

Operations in Exponential Form

Let $z_1 = 2e^{i\pi/3}$ and $z_2 = 5e^{i\pi/6}$.

- $z_1 z_2 = (2)(5)e^{i(\pi/3+\pi/6)} = 10e^{i(2\pi/6+\pi/6)} = 10e^{i(3\pi/6)} = 10e^{i\pi/2}$
  (This is $10(\cos(\pi/2) + i\sin(\pi/2)) = 10(0 + i) = 10i$)
- $\dfrac{z_1}{z_2} = \dfrac{2}{5}e^{i(\pi/3-\pi/6)} = \dfrac{2}{5}e^{i(\pi/6)}$

Section 49

Powers: De Moivre's Theorem

Let's find z^n for an integer n. Using the exponential form $z = re^{i\theta}$:

$$z^n = (re^{i\theta})^n = r^n(e^{i\theta})^n = r^n e^{i(n\theta)}$$

If we write this out in polar form, we get **De Moivre's Theorem**:

De Moivre's Theorem

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

Or for $r = 1$:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

This theorem is incredibly useful for deriving trigonometric identities and computing powers of complex numbers.

Calculating Powers

Find $(1 + i)^8$. First, convert $z = 1 + i$ to exponential form.

- $r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}$.
- $\theta = \text{Arg}(z) = \arctan(1/1) = \frac{\pi}{4}$ (since $(1, 1)$ is in Q1).

So, $z = \sqrt{2}e^{i\pi/4}$. Now we take the 8th power:

$$z^8 = (\sqrt{2}e^{i\pi/4})^8 = (\sqrt{2})^8 e^{i(8\pi/4)} = 2^{8/2}e^{i(2\pi)} = 2^4 e^{i2\pi} = 16(\cos(2\pi)+i\sin(2\pi)) = 16(1+0) = 16$$

Calculating $(1 + i)(1 + i)(1 + i)$... 8 times would be much harder!

Deriving Trigonometric Identities

Use De Moivre's theorem to find formulas for $\cos(3\theta)$ and $\sin(3\theta)$.
Solution: By De Moivre's theorem:

$$(\cos\theta + i\sin\theta)^3 = \cos(3\theta) + i\sin(3\theta)$$

Expanding the left side using the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^3 &= \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3 \\ &= \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta \\ &= (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta)\end{aligned}$$

Equating real and imaginary parts:

$$\begin{aligned}\cos(3\theta) &= \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta \\ \sin(3\theta) &= 3\cos^2\theta\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta\end{aligned}$$

Negative Powers

Compute $(1 - i\sqrt{3})^{-4}$.
Solution: First convert to exponential form:

- $r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$
- $\theta = \arctan\left(-\sqrt{3}/1\right) = -\pi/3$ (fourth quadrant)
- So $z = 2e^{-i\pi/3}$

Then:

$$z^{-4} = (2e^{-i\pi/3})^{-4} = 2^{-4}e^{i4\pi/3} = \frac{1}{16}e^{i4\pi/3} = \frac{1}{16}\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = -\frac{1}{32} - i\frac{\sqrt{3}}{32}$$

Section 50

Roots of Complex Numbers

We want to find the n-th roots of a complex number z. That is, we seek all w such that $w^n = z$. Let $z = re^{i\theta}$ and $w = \rho e^{i\phi}$. We must account for the multi-valued argument of z:

$$z = re^{i(\theta+2k\pi)}, \quad k \in \mathbb{Z}$$

Now, set $w^n = z$: $(\rho e^{i\phi})^n = re^{i(\theta+2k\pi)} \implies \rho^n e^{in\phi} = re^{i(\theta+2k\pi)}$ Equating the moduli and arguments:

1. $\rho^n = r \implies \rho = \sqrt[n]{r}$ (the unique positive real n-th root)
2. $n\phi = \theta + 2k\pi \implies \phi = \frac{\theta+2k\pi}{n}$

So the n-th roots of z are given by:

Formula for n-th Roots

$$w_k = \sqrt[n]{r}\exp\left(i\left(\frac{\theta+2k\pi}{n}\right)\right), \quad \text{for } k = 0, 1, 2, \ldots, n-1$$

We only need n consecutive values of k (e.g., $k = 0$ to $n-1$) to find all n distinct roots. For $k = n$, the angle $\frac{\theta+2n\pi}{n} = \frac{\theta}{n} + 2\pi$, which gives the same root as $k = 0$. Geometrically, the n roots of z are equally spaced on a circle of radius $\sqrt[n]{r}$.

Geometric Pattern

The n-th roots of any complex number form a regular n-gon inscribed in a circle centered at the origin. The angular spacing between consecutive roots is $\frac{2\pi}{n}$.

Cube Roots of Unity

Find the cube roots of $z = 1$. Here $r = 1$ and $\theta = \text{Arg}(z) = 0$. So $z = 1e^{i(0+2k\pi)}$. The formula is $w_k = \sqrt[3]{1}\exp\left(i\left(\frac{0+2k\pi}{3}\right)\right) = e^{i(2k\pi/3)}$ for $k = 0, 1, 2$.

- $k = 0 : w_0 = e^{i(0)} = 1$
- $k = 1 : w_1 = e^{i(2\pi/3)} = \cos(2\pi/3) + i\sin(2\pi/3) = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$
- $k = 2 : w_2 = e^{i(4\pi/3)} = \cos(4\pi/3) + i\sin(4\pi/3) = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

These three roots form an equilateral triangle on the unit circle.

Square Roots of i

Find the square roots of $z = i$. Here $n = 2$. First, write $z = i$ in exponential form. $r = |i| = 1$. $\theta = \text{Arg}(i) = \pi/2$. So $z = 1e^{i(\pi/2+2k\pi)}$. The roots are $w_k = \sqrt{1}\exp\left(i\left(\frac{\pi/2+2k\pi}{2}\right)\right) = \exp\left(i\left(\frac{\pi}{4} + k\pi\right)\right)$ for $k = 0, 1$.

- $k = 0 : w_0 = e^{i(\pi/4)} = \cos(\pi/4) + i\sin(\pi/4) = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}(1+i)$
- $k = 1 : w_1 = e^{i(\pi/4+\pi)} = e^{i(5\pi/4)} = \cos(5\pi/4) + i\sin(5\pi/4) = -\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}$

Note that $w_1 = -w_0$, as expected for a square root.

***Fourth Roots of* -16**

Find all fourth roots of $z = -16$.
Solution: First convert to exponential form:

- $r = |-16| = 16$
- $\theta = \text{Arg}(-16) = \pi$ (negative real axis)
- So $z = 16e^{i\pi}$

The fourth roots are: $w_k = \sqrt[4]{16}e^{i(\pi+2k\pi)/4} = 2e^{i\pi(1+2k)/4}, \quad k = 0, 1, 2, 3$

- $k = 0 : w_0 = 2e^{i\pi/4} = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2}(1+i)$
- $k = 1 : w_1 = 2e^{i3\pi/4} = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2}(-1+i)$
- $k = 2 : w_2 = 2e^{i5\pi/4} = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2}(-1-i)$
- $k = 3 : w_3 = 2e^{i7\pi/4} = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2}(1-i)$

Solving Polynomial Equations

Solve $z^3 + 8 = 0$.
Solution: We need to find the cube roots of -8.

- $r = |-8| = 8$
- $\theta = \pi$
- $z = 8e^{i\pi}$

The cube roots are: $w_k = 2e^{i\pi(1+2k)/3}, \quad k = 0, 1, 2$

- $k = 0 : w_0 = 2e^{i\pi/3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$
- $k = 1 : w_1 = 2e^{i\pi} = -2$
- $k = 2 : w_2 = 2e^{i5\pi/3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}$

Section 51

The Exponential Function $f(z) = e^z$

We now define the exponential function for a complex variable $z = x + iy$.

Complex Exponential Function

Definition: $e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y)$

Subsection 51.1

Properties

1. $e^{z_1}e^{z_2} = e^{z_1+z_2}$
2. $\frac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$
3. $|e^z| = |e^x(\cos y + i\sin y)| = |e^x||\cos y + i\sin y| = e^x \cdot 1 = e^x = e^{\mathrm{Re}(z)}$
4. $\arg(e^z) = y + 2k\pi = \mathrm{Im}(z) + 2k\pi$
5. $e^z \neq 0$ for all $z \in \mathbb{C}$.
6. **Periodicity:** The exponential function is periodic with a pure imaginary period of $2\pi i$. $e^{z+2\pi i} = e^{x+i(y+2\pi)} = e^x(\cos(y+2\pi) + i\sin(y+2\pi)) = e^x(\cos y + i\sin y) = e^z$

Key Difference from Real Exponential

Unlike the real exponential function which is strictly increasing and never repeats values, the complex exponential is periodic and takes every nonzero complex value infinitely many times.

Evaluating e^z

Find the value of $e^{2+i\pi/2}$. Here $x = 2$ and $y = \pi/2$. $e^{2+i\pi/2} = e^2e^{i\pi/2} = e^2\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right) = e^2(0 + i(1)) = ie^2$

More Exponential Evaluations

Evaluate the following:

1. $e^{i\pi}$
2. $e^{1-i\pi/4}$
3. e^{-2+3i}

Solutions:

1. $e^{i\pi} = e^0(\cos\pi + i\sin\pi) = 1(-1+0) = -1$ (Euler's identity)
2. $e^{1-i\pi/4} = e^1(\cos(-\pi/4) + i\sin(-\pi/4)) = e\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \frac{e\sqrt{2}}{2}(1-i)$
3. $e^{-2+3i} = e^{-2}(\cos 3 + i\sin 3)$ (angles in radians)

Solving Exponential Equations

Solve $e^z = -1$.
Solution: We need $e^{x+iy} = -1$. Write -1 in exponential form: $-1 = e^{i\pi}$ (using principal argument).
More generally, $-1 = e^{i(\pi+2k\pi)}$ for $k \in \mathbb{Z}$.
So we need: $e^x(\cos y + i\sin y) = \cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi)$
Comparing moduli: $e^x = 1 \implies x = 0$
Comparing arguments: $y = \pi + 2k\pi$ for $k \in \mathbb{Z}$
Therefore: $z = i\pi(1+2k) = i\pi, \pm 3i\pi, \pm 5i\pi, \ldots$

Section 52

The Logarithmic Function

The complex logarithm $w = \log z$ is defined as the inverse of the exponential function, $e^w = z$. Let $w = u + iv$ and $z = re^{i\theta}$. $e^w = z \implies e^{u+iv} = re^{i\theta} \implies e^u e^{iv} = re^{i\theta}$ Equating moduli and arguments:

- $e^u = r \implies u = \ln r = \ln |z|$ (where ln is the real natural log)
- $v = \theta = \arg(z)$

So, we define the **complex logarithm** as:

Complex Logarithm (Multi-valued)

$$\log z = \ln |z| + i \arg(z)$$

Subsection 52.1

Multi-valued Nature

Since $\arg(z) = \operatorname{Arg}(z) + 2k\pi$ for $k \in \mathbb{Z}$, the logarithm is a **multi-valued function**: $\log z = \ln |z| + i(\operatorname{Arg}(z) + 2k\pi)$

Subsection 52.2

Principal Value

We define the **principal value of the logarithm**, denoted $\operatorname{Log} z$, by using the principal argument $\operatorname{Arg}(z)$:

Principal Value $\operatorname{Log} z$

$$\operatorname{Log} z = \ln |z| + i \operatorname{Arg}(z), \quad (-\pi < \operatorname{Arg}(z) \leq \pi)$$

This makes the function single-valued, but it has a "jump" (a discontinuity) along the negative real axis, known as a **branch cut**.

Subsection 52.3

Properties of the Complex Logarithm

Logarithm Properties

For $z, w \in \mathbb{C}$ (nonzero):

1. $e^{\operatorname{Log} z} = z$ (but $\operatorname{Log}(e^z) = z$ only if $\operatorname{Im}(z) \in (-\pi, \pi]$)
2. $\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2 + 2k\pi i$ for some integer k
3. $\operatorname{Log}(z_1/z_2) = \operatorname{Log} z_1 - \operatorname{Log} z_2 + 2k\pi i$ for some integer k
4. $\operatorname{Log}(z^n) = n \operatorname{Log} z + 2k\pi i$ for some integer k
5. $\operatorname{Log}(\bar{z}) = \overline{\operatorname{Log} z}$ only when $\operatorname{Arg}(z) \neq \pi$

Warning: The logarithm laws from real analysis require care with the multi-valued nature!

Evaluating $\log z$

Find $\log(-1)$ and the principal value $\operatorname{Log}(-1)$.

- $z = -1$.
- Modulus: $|z| = |-1| = 1$. So $\ln|z| = \ln(1) = 0$.
- Argument: $\arg(-1) = \pi + 2k\pi$ (it's on the negative real axis).
- Principal Argument: $\operatorname{Arg}(-1) = \pi$.

Therefore:

- $\log(-1) = \ln(1) + i(\pi + 2k\pi) = i\pi(1 + 2k)$ for $k \in \mathbb{Z}$. This gives $\ldots, -3i\pi, -i\pi, i\pi, 3i\pi, 5i\pi, \ldots$
- $\operatorname{Log}(-1) = \ln(1) + i \operatorname{Arg}(-1) = 0 + i\pi = i\pi$.

More Logarithm Evaluations

Find the principal values:

1. $\operatorname{Log}(i)$
2. $\operatorname{Log}(-3i)$
3. $\operatorname{Log}(1 + i)$

Solutions:

1. $|i| = 1$, $\operatorname{Arg}(i) = \pi/2$, so $\operatorname{Log}(i) = \ln(1) + i\pi/2 = i\pi/2$
2. $|-3i| = 3$, $\operatorname{Arg}(-3i) = -\pi/2$, so $\operatorname{Log}(-3i) = \ln(3) - i\pi/2$
3. $|1+i| = \sqrt{2}$, $\operatorname{Arg}(1+i) = \pi/4$, so $\operatorname{Log}(1+i) = \ln\left(\sqrt{2}\right) + i\pi/4 = \frac{1}{2}\ln(2) + i\pi/4$

Solving Logarithmic Equations

Solve $\text{Log}(z^2) = 2 + i\pi/3$.
Solution: Taking exponentials of both sides: $z^2 = e^{2+i\pi/3} = e^2 e^{i\pi/3} = e^2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
Now find the square roots. First convert to exponential form:

- $r = e^2$
- $\theta = \pi/3$

The square roots are: $z_k = \sqrt{e^2}e^{i(\pi/3+2k\pi)/2} = e \cdot e^{i\pi(1+6k)/6}, \quad k = 0, 1$

- $z_0 = e \cdot e^{i\pi/6} = e\left(\frac{\sqrt{3}}{2} + i\frac{1}{2}\right)$
- $z_1 = e \cdot e^{i7\pi/6} = e\left(-\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)$

Section 53

Trigonometric Functions

We can extend the definitions of sin and cos to the complex plane using Euler's formula. Recall:

$$e^{ix} = \cos x + i\sin x \quad \text{(for real } x\text{)}$$
$$e^{-ix} = \cos x - i\sin x$$

Adding these gives $\cos x = \frac{e^{ix}+e^{-ix}}{2}$. Subtracting gives $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$. We use these as our **definitions** for complex z:

Complex Trig Functions

$\cos z = \frac{e^{iz}+e^{-iz}}{2}$ $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$ $\tan z = \frac{\sin z}{\cos z}$ $\cot z = \frac{\cos z}{\sin z}$ $\sec z = \frac{1}{\cos z}$ $\csc z = \frac{1}{\sin z}$

Unbounded Functions

Unlike their real counterparts, $\sin z$ and $\cos z$ are **unbounded** in the complex plane. For example, $|\sin(iy)| = \frac{e^y+e^{-y}}{2}$ which grows exponentially as $y \to \infty$.

Subsection 53.1

Properties of Complex Trigonometric Functions

Many familiar properties from real trigonometry extend to the complex case:

Trigonometric Identities

1. $\sin^2 z + \cos^2 z = 1$
2. $\sin(-z) = -\sin z$, $\cos(-z) = \cos z$ (odd/even)
3. $\sin(z + 2\pi) = \sin z$, $\cos(z + 2\pi) = \cos z$ (periodic)
4. $\sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2$
5. $\cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2$
6. $\frac{d}{dz} \sin z = \cos z$, $\frac{d}{dz} \cos z = -\sin z$

Evaluating $\cos z$

Find the value of $\cos(i)$. Using the definition with $z = i$:

$$\begin{aligned}\cos(i) &= \frac{e^{i(i)} + e^{-i(i)}}{2} = \frac{e^{i^2} + e^{-i^2}}{2} \\ &= \frac{e^{-1} + e^{1}}{2} = \frac{e + e^{-1}}{2} \\ &= \cosh(1)\end{aligned}$$

This is a real number, approximately 1.543.

Evaluating $\sin z$ ***for Complex Argument***

Evaluate $\sin(1 + i)$.
Solution: Using the definition:

$$\begin{aligned}\sin(1+i) &= \frac{e^{i(1+i)} - e^{-i(1+i)}}{2i} \\ &= \frac{e^{i-1} - e^{-i+1}}{2i} \\ &= \frac{e^{-1}e^{i} - e^{1}e^{-i}}{2i} \\ &= \frac{e^{-1}(\cos 1 + i \sin 1) - e(\cos 1 - i \sin 1)}{2i} \\ &= \frac{(e^{-1} - e)\cos 1 + i(e^{-1} + e)\sin 1}{2i}\end{aligned}$$

Multiplying numerator and denominator by $-i$: $= \frac{(e^{-1}+e)\sin 1}{2} + i\frac{(e-e^{-1})\cos 1}{2} = \sin 1 \cosh 1 + i \cos 1 \sinh 1$
Using the identity: $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$

Solving Trigonometric Equations

Solve $\cos z = 2$.
Solution: This equation has no real solutions, but has complex solutions. Using $\cos z = \frac{e^{iz}+e^{-iz}}{2} = 2$:
$e^{iz} + e^{-iz} = 4$
Let $w = e^{iz}$. Then: $w + \frac{1}{w} = 4 \implies w^2 - 4w + 1 = 0$
Using the quadratic formula: $w = \frac{4\pm\sqrt{16-4}}{2} = \frac{4\pm2\sqrt{3}}{2} = 2 \pm \sqrt{3}$
So $e^{iz} = 2 + \sqrt{3}$ or $e^{iz} = 2 - \sqrt{3}$.
Taking logarithms: $iz = \operatorname{Log}(2 + \sqrt{3}) + 2k\pi i$ or $iz = \operatorname{Log}(2 - \sqrt{3}) + 2k\pi i$
$z = -i\operatorname{Log}(2 + \sqrt{3}) + 2k\pi = 2k\pi - i\ln\left(2 + \sqrt{3}\right)$ $z = -i\operatorname{Log}(2 - \sqrt{3}) + 2k\pi = 2k\pi - i\ln\left(2 - \sqrt{3}\right)$
for $k \in \mathbb{Z}$.

Section 54

Hyperbolic Functions

The real hyperbolic functions are $\cosh x = \frac{e^x+e^{-x}}{2}$ and $\sinh x = \frac{e^x-e^{-x}}{2}$. We extend this to the complex plane:

Complex Hyperbolic Functions

$\cosh z = \frac{e^z+e^{-z}}{2}$ $\sinh z = \frac{e^z-e^{-z}}{2}$ $\tanh z = \frac{\sinh z}{\cosh z}$ $\coth z = \frac{\cosh z}{\sinh z}$ $\operatorname{sech} z = \frac{1}{\cosh z}$ $\operatorname{csch} z = \frac{1}{\sinh z}$

Subsection 54.1

Relationship to Trigonometric Functions

We can relate the two sets of functions by replacing z with iz:

$$\cos(iz) = \frac{e^{i(iz)} + e^{-i(iz)}}{2} = \frac{e^{-z} + e^{z}}{2} = \cosh z$$
$$\sin(iz) = \frac{e^{i(iz)} - e^{-i(iz)}}{2i} = \frac{e^{-z} - e^{z}}{2i} = \frac{-(e^{z} - e^{-z})}{2i} = i\frac{e^{z} - e^{-z}}{2} = i\sinh z$$

Osborn's Rules

$\cos(iz) = \cosh(z)$ $\sin(iz) = i\sinh(z)$ $\cosh(iz) = \cos(z)$ $\sinh(iz) = i\sin(z)$

Subsection 54.2

Hyperbolic Identities

Hyperbolic Identities

1. $\cosh^2 z - \sinh^2 z = 1$
2. $\sinh(-z) = -\sinh z$, $\cosh(-z) = \cosh z$ (odd/even)
3. $\sinh(z+w) = \sinh z \cosh w + \cosh z \sinh w$
4. $\cosh(z+w) = \cosh z \cosh w + \sinh z \sinh w$
5. $\frac{d}{dz}\sinh z = \cosh z$, $\frac{d}{dz}\cosh z = \sinh z$

Evaluating Hyperbolic Functions

Evaluate $\sinh(i\pi/2)$ and $\cosh(1+i)$.
Solutions:

1. Using Osborn's rule: $\sinh(i\pi/2) = i\sin(\pi/2) = i(1) = i$
2. For $\cosh(1+i)$:

$$\begin{aligned}\cosh(1+i) &= \frac{e^{1+i} + e^{-1-i}}{2} \\ &= \frac{e \cdot e^{i} + e^{-1} \cdot e^{-i}}{2} \\ &= \frac{e(\cos 1 + i\sin 1) + e^{-1}(\cos 1 - i \sin 1)}{2} \\ &= \frac{(e + e^{-1})\cos 1 + i(e - e^{-1})\sin 1}{2} \\ &= \cosh 1 \cos 1 + i \sinh 1 \sin 1\end{aligned}$$

Separation into Real and Imaginary Parts

Express $\sinh(x+iy)$ in the form $u(x,y) + iv(x,y)$.
Solution:

$$\begin{aligned}\sinh(x+iy) &= \frac{e^{x+iy} - e^{-x-iy}}{2} \\ &= \frac{e^{x}(\cos y + i\sin y) - e^{-x}(\cos y - i \sin y)}{2} \\ &= \frac{(e^{x} - e^{-x})\cos y + i(e^{x} + e^{-x})\sin y}{2} \\ &= \sinh x \cos y + i \cosh x \sin y\end{aligned}$$

Therefore: $u(x,y) = \sinh x \cos y$ and $v(x,y) = \cosh x \sin y$.

Section 55

Functions of a Complex Variable

In general, a **function of a complex variable** z is a map f from a subset S of $\mathbb{C}$ to $\mathbb{C}$. We write $w = f(z)$. We can split both z and w into their real and imaginary parts:

- $z = x + iy$
- $w = u + iv$

Since w depends on z, the real part u and imaginary part v of the output must depend on the real part x and imaginary part y of the input. $w = f(z) \implies w(x, y) = u(x, y) + iv(x, y)$ Thus, a complex function $f : \mathbb{C} \to \mathbb{C}$ is equivalent to two real-valued functions of two real variables: $u(x, y)$ and $v(x, y)$.

Finding u and v for $f(z)$

Let $f(z) = z^2$. Find $u(x, y)$ and $v(x, y)$.

$$\begin{aligned} f(z) &= (x + iy)^2 \\ &= x^2 + 2(x)(iy) + (iy)^2 \\ &= x^2 + 2xyi - y^2 \\ &= (x^2 - y^2) + i(2xy) \end{aligned}$$

So, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

Finding u and v for $f(z)$

Let $f(z) = 1/z$ for $z \neq 0$. Find $u(x, y)$ and $v(x, y)$. We "realize the denominator" just as in division:

$$\begin{aligned} f(z) &= \frac{1}{x + iy} = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} \\ &= \frac{x - iy}{x^2 + y^2} \\ &= \left(\frac{x}{x^2 + y^2}\right) + i\left(\frac{-y}{x^2 + y^2}\right) \end{aligned}$$

So, $u(x, y) = \frac{x}{x^2+y^2}$ and $v(x, y) = \frac{-y}{x^2+y^2}$.

Finding u and v for $f(z)$

Let $f(z) = e^z$. Find $u(x, y)$ and $v(x, y)$.
Solution:

$$\begin{aligned} f(z) &= e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y) \\ &= e^x \cos y + ie^x \sin y \end{aligned}$$

Therefore: $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$.

Finding u and v for $f(z)$

Let $f(z) = \sin z$. Find $u(x, y)$ and $v(x, y)$.
Solution:

$$\begin{aligned}\sin z &= \sin(x + iy)\\ &= \frac{e^{i(x+iy)} - e^{-i(x+iy)}}{2i}\\ &= \frac{e^{ix-y} - e^{-ix+y}}{2i}\\ &= \frac{e^{-y}(\cos x + i\sin x) - e^{y}(\cos x - i\sin x)}{2i}\end{aligned}$$

Separating real and imaginary parts: $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$
Therefore: $u(x, y) = \sin x \cosh y$ and $v(x, y) = \cos x \sinh y$.

Subsection 55.1

Mappings and Transformations

Complex functions can be viewed as transformations or mappings of the complex plane. Given $w = f(z)$, each point z in the domain is mapped to a point w in the range.

Mapping under $f(z)$

Describe how the vertical line $x = 1$ is mapped under $f(z) = z^2$.
Solution: Points on the line $x = 1$ have the form $z = 1 + iy$ for $y \in \mathbb{R}$.
Under $f(z) = z^2$:

$$w = f(1 + iy) = (1 + iy)^2 = 1 + 2iy - y^2 = (1 - y^2) + 2iy$$

So $u = 1 - y^2$ and $v = 2y$.
From the second equation: $y = v/2$. Substituting into the first: $u = 1 - (v/2)^2 = 1 - v^2/4$
This gives: $v^2 = 4(1 - u)$ or $v^2 = -4(u - 1)$.
This is a parabola in the w-plane opening to the left with vertex at $(1, 0)$.

Mapping of Circles

Where does the circle $|z| = 2$ map under $f(z) = 1/z$?
Solution: For $|z| = 2$, we can write $z = 2e^{i\theta}$ for $\theta \in [0, 2\pi)$.
Then: $w = \frac{1}{z} = \frac{1}{2e^{i\theta}} = \frac{1}{2}e^{-i\theta}$
This describes a circle centered at the origin with radius $|w| = 1/2$.
In general, $f(z) = 1/z$ maps circles centered at the origin to circles centered at the origin, with radius inverted: $r \to 1/r$.

Section 56

Sets in the Complex Plane

To study limits, continuity, and derivatives (i.e., complex analysis), we need a basic understanding of the topology of $\mathbb{C}$.

- **ϵ-neighborhood:** An **open disk** centered at z_0 with radius $\epsilon > 0$ is the set $B(z_0, \epsilon) = \{z \in \mathbb{C} : |z - z_0| < \epsilon\}$.
- **Interior Point:** A point z_0 is an **interior point** of a set S if there exists some ϵ-neighborhood of z_0 that is completely contained in S.
- **Open Set:** A set S is **open** if every point in S is an interior point. (e.g., $|z| < 1$).
- **Boundary Point:** A point z_0 is a **boundary point** of S if every ϵ-neighborhood of z_0 contains at least one point in S and at least one point not in S.
- **Boundary:** The set of all boundary points is the **boundary** of S. (e.g., for $|z| < 1$, the boundary is the circle $|z| = 1$).
- **Closed Set:** A set S is **closed** if it contains all of its boundary points. (e.g., $|z| \leq 1$).
- **Bounded Set:** A set S is **bounded** if there exists a positive real number M such that $|z| < M$ for all $z \in S$.
- **Compact Set:** A set is **compact** if it is both closed and bounded.
- **Connected Set:** An open set S is **connected** if any two points in S can be joined by a path (a continuous line) that lies entirely within S.
- **Domain:** In complex analysis, a **domain** is an **open, connected set**. This is the most common type of set we will work with when defining functions.
- **Region:** A **region** is a domain together with some, all, or none of its boundary points.

Identifying Set Properties

Classify the following sets as open, closed, bounded, compact, and/or connected:

1. $S_1 = \{z : |z| < 1\}$
2. $S_2 = \{z : |z| \leq 1\}$
3. $S_3 = \{z : 1 < |z| < 2\}$
4. $S_4 = \{z : \text{Re}(z) > 0\}$
5. $S_5 = \{z : |z - 1| < 1 \text{ or } |z + 1| < 1\}$

Solutions:

1. S_1: Open (no boundary), bounded, connected. Not closed, not compact.
2. S_2: Closed (contains boundary $|z| = 1$), bounded, compact, connected. Not open.
3. S_3: Open (annulus), bounded, connected. Not closed, not compact.
4. S_4: Open (right half-plane), unbounded, connected. Not closed, not compact, not bounded.
5. S_5: Open (union of two disks), bounded, NOT connected (two separate components). Not closed, not compact.

Finding Boundaries

Find the boundary of each set:

1. $A = \{z : |z - i| < 2\}$
2. $B = \{z : 0 < \text{Arg}(z) < \pi/4\}$
3. $C = \{z : \text{Im}(z) > 1\}$

Solutions:

1. Boundary of A: the circle $|z - i| = 2$ (centered at i with radius 2)
2. Boundary of B: the two rays $\text{Arg}(z) = 0$ and $\text{Arg}(z) = \pi/4$ (both emanating from origin)
3. Boundary of C: the horizontal line $\text{Im}(z) = 1$

Subsection 56.1

Special Regions

Common Regions in $\mathbb{C}$

1. **Unit Disk:** $\mathbb{D} = \{z : |z| < 1\}$
2. **Upper Half-Plane:** $\mathbb{H} = \{z : \mathrm{Im}(z) > 0\}$
3. **Right Half-Plane:** $\{z : \mathrm{Re}(z) > 0\}$
4. **Annulus:** $\{z : r_1 < |z - z_0| < r_2\}$ for $0 \leq r_1 < r_2$
5. **Punctured Disk:** $\{z : 0 < |z - z_0| < r\}$ (disk with center removed)

Describing Regions

Describe geometrically and sketch the following sets:

1. $S_1 = \{z : |z - 2| > 1\}$
2. $S_2 = \{z : |z - 1| = |z + 1|\}$
3. $S_3 = \{z : |z| < 2, \mathrm{Re}(z) > 1\}$

Solutions:

1. S_1: The exterior of a circle centered at $z = 2$ with radius 1 (all points more than distance 1 from point 2).
2. S_2: Points equidistant from 1 and -1. This is the imaginary axis $\mathrm{Re}(z) = 0$ (the perpendicular bisector of the segment from -1 to 1).
3. S_3: The intersection of the disk $|z| < 2$ and the right half-plane $\mathrm{Re}(z) > 1$. This is a "lens-shaped" region.

Section 57

Additional Topics and Applications

Subsection 57.1

Complex Exponentiation

For $a, b \in \mathbb{C}$ with $a \neq 0$, we define a^b using the exponential and logarithm:

Complex Power

$a^b = e^{b \log a}$

Since log is multi-valued, a^b is generally multi-valued unless b is an integer.

Computing Complex Powers

Compute i^i.
Solution: $i^i = e^{i \log i} = e^{i(\ln|i| + i\arg(i))} = e^{i(0+i(\pi/2+2k\pi))} = e^{-(\pi/2+2k\pi)} = e^{-\pi/2} \cdot e^{-2k\pi}$
The principal value (using $k = 0$) is: $i^i = e^{-\pi/2} \approx 0.2078$
This is a real number!

Another Complex Power

Find the principal value of $(1+i)^{2-i}$.
Solution: First find $\text{Log}(1+i)$: $\text{Log}(1+i) = \ln\sqrt{2} + i\frac{\pi}{4} = \frac{1}{2}\ln 2 + i\frac{\pi}{4}$
Then:

$$\begin{aligned}
(1+i)^{2-i} &= e^{(2-i)\,\text{Log}(1+i)} \\
&= e^{(2-i)(\frac{1}{2}\ln 2 + i\frac{\pi}{4})} \\
&= e^{(2-i)(\frac{1}{2}\ln 2) + (2-i)(i\frac{\pi}{4})} \\
&= e^{\ln 2 - \frac{i}{2}\ln 2 + \frac{i\pi}{2} + \frac{\pi}{4}} \\
&= e^{\ln 2 + \pi/4} \cdot e^{i(\pi/2 - \ln 2/2)} \\
&= 2e^{\pi/4} \cdot e^{i(\pi/2 - \ln 2/2)}
\end{aligned}$$

Subsection 57.2

Applications

Physical Applications

Complex numbers and functions have numerous applications in:

- **Electrical Engineering:** AC circuit analysis using impedance
- **Quantum Mechanics:** Wave functions and probability amplitudes
- **Fluid Dynamics:** Potential flow and conformal mappings
- **Signal Processing:** Fourier transforms and filter design
- **Control Theory:** Transfer functions and stability analysis

Application: AC Circuits

In AC circuit analysis, impedance is represented as a complex number. For a circuit with resistance $R = 30\,\Omega$, inductance $L = 0.1\,H$, and capacitance $C = 100\,\mu F$ at frequency $f = 60\,Hz$:
The impedances are:

- Resistor: $Z_R = R = 30$
- Inductor: $Z_L = i\omega L = i(2\pi \cdot 60)(0.1) = 37.7i$
- Capacitor: $Z_C = \frac{1}{i\omega C} = \frac{1}{i(2\pi\cdot 60)(100\times 10^{-6})} = -26.5i$

Total impedance: $Z = 30 + 37.7i - 26.5i = 30 + 11.2i\,\Omega$
Magnitude: $|Z| = \sqrt{30^2 + 11.2^2} \approx 31.0\,\Omega$

Section 58

Summary

Key Concepts Summary

Forms of Complex Numbers:

- Rectangular: $z = a + bi$
- Polar: $z = r(\cos\theta + i\sin\theta)$
- Exponential: $z = re^{i\theta}$

Key Operations:

- Conjugate: $\bar{z} = a - bi$
- Modulus: $|z| = \sqrt{a^2 + b^2}$
- Argument: $\theta = \arg(z)$

De Moivre's Theorem: $z^n = r^n e^{in\theta}$
Roots: $w_k = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}$, $k = 0, 1, \ldots, n-1$
Special Functions:

- $e^z = e^x(\cos y + i\sin y)$
- $\operatorname{Log} z = \ln|z| + i\operatorname{Arg}(z)$
- $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$, $\cos z = \frac{e^{iz}+e^{-iz}}{2}$
- $\sinh z = \frac{e^{z}-e^{-z}}{2}$, $\cosh z = \frac{e^{z}+e^{-z}}{2}$

Section 59

Cauchy–Riemann (CR) equations

The Cauchy–Riemann (CR) equations form the foundational criterion for determining whether a complex-valued function is differentiable in the complex sense. They link the real and imaginary parts of a function and encode the structural condition for conformality and analyticity.

Let

$$f(z) = u(x, y) + iv(x, y), \qquad z = x + iy.$$

Complex differentiability of f at a point is much stronger than real differentiability and requires that the limit

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists and is independent of the direction of approach.

Example 4: $f(z)$

Let

$$f(z) = u(x, y) + iv(x, y) = x^2 y + i(xy^2).$$

Compute the Cauchy–Riemann equations and determine whether f is analytic.

We have

$$u(x, y) = x^2 y, \qquad v(x, y) = xy^2.$$

Compute partial derivatives:

$$u_x = 2xy, \qquad u_y = x^2, \qquad v_x = y^2, \qquad v_y = 2xy.$$

CR equations require:

$$u_x = v_y \quad \Rightarrow \quad 2xy = 2xy \quad \text{(satisfied)},$$

$$u_y = -v_x \quad \Rightarrow \quad x^2 = -y^2.$$

The equation $x^2 = -y^2$ holds only when $x = y = 0$.

Thus the CR equations hold only at the point $(0, 0)$, not on a neighborhood. Therefore, the function is **not analytic anywhere**.

Example 5: $f(z)$

Consider

$$f(z) = e^x(\cos y + i \sin y).$$

Verify the CR equations and determine analyticity.

We identify

$$u(x, y) = e^x \cos y, \qquad v(x, y) = e^x \sin y.$$

Then

$$u_x = e^x \cos y, \qquad u_y = -e^x \sin y,$$

$$v_x = e^x \sin y, \qquad v_y = e^x \cos y.$$

Check CR:

$$u_x = v_y \quad \Rightarrow \quad e^x \cos y = e^x \cos y,$$

$$u_y = -v_x \quad \Rightarrow \quad -e^x \sin y = -e^x \sin y.$$

Both are satisfied everywhere. Therefore, the function is **analytic on all of $\mathbb{C}$**. Indeed, this is the standard decomposition of the analytic function

$$f(z) = e^z.$$

Example 6: $f(z)$

Verify analytic behavior of the function

$$f(z) = x^3 - 3xy^2 + i(3x^2y - y^3).$$

We recognize this as the real/imaginary form of z^3, but we verify directly.

$$u(x, y) = x^3 - 3xy^2, \qquad v(x, y) = 3x^2y - y^3.$$

Compute partial derivatives:

$$u_x = 3x^2 - 3y^2, \qquad u_y = -6xy,$$

$$v_x = 6xy, \qquad v_y = 3x^2 - 3y^2.$$

Check CR:

$$u_x = v_y \quad \Rightarrow \quad 3x^2 - 3y^2 = 3x^2 - 3y^2,$$

$$u_y = -v_x \quad \Rightarrow \quad -6xy = -6xy.$$

CR holds everywhere, so the function is **entire**, consistent with

$$f(z) = z^3.$$

Example 7: $f(z)$

Consider

$$f(z) = u(x, y) + iv(x, y) = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2},$$

defined on $\mathbb{C} \setminus \{0\}$. Determine analyticity.

This is the real/imaginary form of

$$f(z) = \frac{1}{z}.$$

Compute derivatives:

$$u = \frac{x}{x^2 + y^2}, \quad v = -\frac{y}{x^2 + y^2}.$$

$$u_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \qquad u_y = \frac{-2xy}{(x^2 + y^2)^2},$$

$$v_x = \frac{2xy}{(x^2 + y^2)^2}, \qquad v_y = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

Check CR:

$$u_x = v_y, \qquad u_y = -v_x.$$

Both equations hold for all $(x, y) \neq (0, 0)$.
Thus the function is **analytic on its domain**

$$\mathbb{C} \setminus \{0\}.$$